

ISC Class XII Mathematics

Chapter 13 Probability

189 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 13.1 – Addition Theorem of Probability
- II. 13.2 – Conditional Probability
- III. 13.3 – Multiplication Theorem & Independent Events
- IV. 13.4 – Law of Total Probability
- V. 13.5 – Bayes' Theorem
- VI. 13.6 – Probability Distribution (Random Variables)
- VII. 13.7 – Mean and Variance of a Random Variable

Compiled Answer Key – Step-by-step ISC board-style solutions

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Index of Problems

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
13.1 – Addition Theorem of Probability				
1	Given $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$, find $P(A \text{ or } B)$, given that A and B are mutually exclusive events.	1	Numerical	13.1
2	The probability of an event A occurring is 0.5 and of B is 0.3. If A and B are mutually exclusive events, then find the probability of...	2	Numerical	13.1
3	If A and B are mutually exclusive events, $P(A) = 0.35$ and $P(B) = 0.45$ – see full data/parts in body	1	Numerical	13.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
4	If E and F are events such that $P(E) = \frac{1}{4}$, $P(F) = \frac{1}{2}$ and $P(E \text{ and } F) = \frac{1}{8}$ – see full data/parts in body	3	Numerical	13.1
5	A and B are events such that $P(A) = 0.42$, $P(B) = 0.48$ and $P(A \text{ and } B) = 0.16$. – see full data/parts in body	2	Numerical	13.1
6	A and B are two mutually exclusive events of an experiment. If $P(\text{not } A) = 0.65$, $P(A \cup B) = 0.65$ and $P(B) = p$, find the value of p ...	1	Numerical	13.1
7	E and F are two events associated with a random experiment for which $P(E) = 0.60$, $P(E \text{ or } F) = 0.85$, $P(E \text{ and } F) = 0.42$...	1	Numerical	13.1
8	If A, B and C are mutually exclusive and exhaustive events and it is known that $P(A \cup B) = 0.63$, calculate $P(C)$.	4	Numerical	13.1
9	A, B, C are three mutually exclusive and exhaustive events associated with a random experiment. Find $P(A)$, it being given that $P(B) = \frac{3}{2}P(A)$...	4	Numerical	13.1
10	A die is thrown once. Let A be the event that the number obtained is greater than 3 and B be the event that the number obtained is less than 5...	5	Numerical	13.1
11	A and B are two candidates seeking admission in an engineering college. The probability that A is selected is 0.5 and the probability that...	6	Proof	13.1
12	The probability of an event A occurring is $\frac{2}{3}$ and the probability of event B not occurring is $\frac{5}{9}$. If the probability of...	2	Numerical	13.1
13	Tickets numbered from 1 to 20 are mixed up together and then a ticket is drawn at random. What is the probability that the ticket has a number...	5	Numerical	13.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
14	The probability that a student will pass the final examination in both English and Hindi is 0.5 and the probability of passing neither is 0.1. If...	3	Numerical	13.1
15	In a class of 60 students, 30 opted for Mathematics, 32 opted for Biology and 24 opted for both Mathematics and Biology. If one of these...	7	Numerical	13.1
16	A bag contains 150 nuts and 50 bolts. Half of the bolts and half of the nuts are rusted. One item is drawn at random from the bag, find the...	7	Numerical	13.1
17	A card is drawn from a well shuffled pack of playing cards. What is the probability that it is either a diamond or a king or both?	5	Numerical	13.1
18	A card is drawn at random from a pack of 52 playing cards. What is the probability that the card drawn is neither a spade nor a queen?	2	Numerical	13.1
19	A card is drawn at random from well shuffled pack of 52 playing cards. Find the probability that it is neither a king nor a red card.	2	Numerical	13.1
20	Find the probability of getting an odd number on the first die or a total of 8 in a single throw of two dice.	5	Numerical	13.1
21	If two dice are thrown simultaneously, find the probability of getting – see full data/parts in body	5	Numerical	13.1
22	The probabilities that a student will get A, B, C or D grade are 0.4, 0.35, 0.15 and 0.1 respectively. Find the probability that she will get –...	1	Numerical	13.1
23	Two cards are drawn at random from a pack of 52 cards. What is probability that the cards are either both aces or both black cards?	5	Numerical	13.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
24	A basket contains 20 apples and 10 oranges out of which 5 apples and 3 oranges are defective. If a person takes out 2 fruits at random...	5	Numerical	13.1
13.2 – Conditional Probability				
1	Multi-part / tabulated question – see full data in body	8	Proof	13.2
2	If A and B are events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$ – see full data/parts in body	9	Numerical	13.2
3	If A and B are two events such that $P(A) = \frac{5}{8}$, $P(B) = \frac{1}{2}$ and $P(A B) = \frac{3}{4}$, then find the value of $P(B A)$.	10	Numerical	13.2
4	If the events A and B are such that $P(A) = \frac{1}{4}$, $P(B A) = \frac{2}{3}$, $P(A B) = \frac{1}{2}$, then find $P(B)$.	9	Numerical	13.2
5	Multi-part / tabulated question – see full data in body	11	Numerical	13.2
6	Compute $P(A B)$ when $P(A) = \frac{1}{5}$, $P(B) = \frac{2}{5}$, $P(A \cup B) = \frac{3}{5}$.	11	Numerical	13.2
7	A die is thrown twice and the sum of numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at...	10	Numerical	13.2
8	A pair of dice is thrown and the sum of the numbers is observed to be even. What is the probability that both dice have come up with even numbers?	10	Numerical	13.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
9	Given that the numbers appearing on rolling two dice together are different. What is the probability that the sum of numbers appearing on two dice is...	10	Numerical	13.2
10	A family has two children. – see full data/parts in body	12	Numerical	13.2
11	12 cards numbered 1 to 12 are placed in a box, mixed up thoroughly and then a card is drawn at random from the box. If it is known that the number on...	13	Numerical	13.2
12	Two digits are selected at random from the digits 1 to 9. If the sum is even, then find the probability that both digits are odd.	12	Numerical	13.2
13	One card is drawn from a well-shuffled pack of 52 cards. If E is the event 'the card drawn is a king or a queen' and F is the event 'the card...	13	Numerical	13.2
14	Multi-part / tabulated question – see full data in body	10	Numerical	13.2
15	In a certain school, 20% of the students failed in English, 15% of the students failed in Mathematics and 10% of the students failed in both...	14	Numerical	13.2
16	60% students read Hindi newspaper, 40% students read Tamil newspaper and 20% students read both Hindi and Tamil newspaper. Find the probability...	14	Numerical	13.2
17	A committee of 4 students is selected from a group consisting of 7 boys and 4 girls. Find the probability that there are exactly 2 boys in the...	15	Numerical	13.2
18	Three friends A, B and C got their photograph clicked. Find the probability that B is standing at the central position, given that A is standing at...	15	Numerical	13.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
13.3 – Multi-plication Theorem & Independent Events				
1	Fill in the blanks: – see full data/parts in body	16	Numerical	13.3
2	What is the probability of obtaining an even prime number on each die when a pair of dice is rolled?	17	Numerical	13.3
3	Multi-part / tabulated question – see full data in body	18	Numerical	13.3
4	Multi-part / tabulated question – see full data in body	19	Numerical	13.3
5	Multi-part / tabulated question – see full data in body	18	Numerical	13.3
6	Multi-part / tabulated question – see full data in body	18	Numerical	13.3
7	Bhuvan is preparing chapati and curry for a dinner party at his home. Let event A = Bhuvan prepares chapati well. Let event B = Bhuvan cooks...	16	Numerical	13.3
8	A coin is tossed thrice and all the eight outcomes are assumed to be equally likely. In which of the cases are the events E_1 and E_2 ...	17	Numerical	13.3
9	Multi-part / tabulated question – see full data in body	17	Numerical	13.3
10	A bag contains 19 tickets, numbered from 1 to 19. Two tickets are drawn randomly in succession with replacement. Find the probability that both the...	17	Numerical	13.3
11	A company has estimated that the probabilities of success for three products introduced in the market are $\frac{1}{3}$, $\frac{2}{5}$ and $\frac{2}{3}$...	20	Numerical	13.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
12	A bag contains 4 white, 3 red and 2 blue balls. Four balls are drawn one by one. Find the probability that all balls are white when the drawing is...	17	Numerical	13.3
13	A bag contains 4 white, 3 red and 2 blue balls. Four balls are drawn one by one. Find the probability that no ball is white when the drawing is done:...	17	Numerical	13.3
14	Multi-part / tabulated question – see full data in body	17	Numerical	13.3
15	A bag contains 5 white, 7 red and 4 black balls. Four balls are drawn one by one with replacement. What is the probability that none is white?	17	Numerical	13.3
16	A bag contains 1 white, 5 red and 4 black balls. If three balls are drawn one by one without replacement, then find the probability of drawing a...	17	Numerical	13.3
17	From a bag containing 20 tickets, numbered from 1 to 20, two tickets are drawn at random. Find the probability that – see full data/parts in body	17	Numerical	13.3
18	An urn contains 4 red and 7 blue balls. Two balls are drawn at random with replacement. Find the probability of getting – see full data/parts in body	17	Numerical	13.3
19	Three balls are drawn one by one without replacement from an urn containing 3 white, 3 red and 3 black balls. What is the probability that at least...	20	Numerical	13.3
20	Three distinct numbers are chosen randomly from the first 50 natural numbers. Find the probability that all the three numbers are divisible by 2 and...	17	Numerical	13.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
21	The probability of finding a green signal on a busy crossing X is 30%. What is the probability of finding a green signal on two consecutive days out...	17	Numerical	13.3
22	Multi-part / tabulated question – see full data in body	17	Numerical	13.3
23	An urn contains 7 white, 5 black and 3 red balls. Two balls are drawn at random. Find the probability that – see full data/parts in body	17	Numerical	13.3
24	In a group of students, there are 3 boys and 3 girls. Four students are to be selected at random from the group. Find the probability that either 3...	17	Numerical	13.3
25	Three cards are drawn from a pack of well-shuffled 52 cards. Find the probability that – see full data/parts in body	17	Numerical	13.3
26	A bag contains 5 white and 4 black balls and another bag contains 7 white and 9 black balls. A ball is drawn from the first bag and two balls drawn...	17	Numerical	13.3
27	Two dice are rolled once. Find the probability that – see full data/parts in body	17	Numerical	13.3
28	In a single throw of three dice, find the probability of getting – see full data/parts in body	17	Numerical	13.3
29	The probabilities of two students A and B coming to school in time are $\frac{2}{7}$ and $\frac{4}{7}$ respectively. Assuming that the events, 'A...	17	Numerical	13.3
30	Two horses are considered for race. The probability of selection of first horse is $\frac{1}{5}$ and that of second is $\frac{2}{3}$. Find the...	17	Numerical	13.3
31	A husband and a wife appear in an interview for two vacancies in the same post. $P(H) = \frac{1}{7}$, $P(W) = \frac{1}{5}$ – find (i), (ii), (iii), (iv).	21	Numerical	13.3
32	Multi-part / tabulated question – see full data in body	17	Numerical	13.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
33	Three persons A, B and C shoot to hit a target. Their probabilities of hitting the target are $\frac{5}{6}$, $\frac{4}{5}$ and $\frac{3}{4}$...	17	Numerical	13.3
34	A machine operates if all of its three components function. The probability that the first component fails during the year is 0.14, the probability...	20	Numerical	13.3
35	A speaks truth in 60% of the cases, while B in 40% of the cases. In what percent of cases are they likely to contradict each other in stating...	17	Numerical	13.3
36	Ten Tickets are numbered 1 to 10. Two tickets are drawn one after the other at random. Find the probability that the number on one of the tickets is...	17	Numerical	13.3
37	A bag contains 5 red discs and 4 blue discs. If 3 discs are drawn from the bag without replacement, then find the probability that – see full...	17	Numerical	13.3
38	A bag contains 5 white and 3 black balls. Four balls are successively drawn out without replacement. What is the probability that they are...	17	Numerical	13.3
39	A bag contains one black and two white balls. A drawing from the bag consists of taking a ball from the bag and keeping it out if it is white but...	17	Numerical	13.3
40	A bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. If one ball is drawn from each bag, then find the probability...	17	Numerical	13.3
41	A bag contains 2 black, 4 white and 3 red balls. One ball is drawn at random and kept aside. Then another ball is drawn and kept aside. This process...	17	Numerical	13.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
42	A policeman fires four bullets on a dacoit. The probability that the dacoit will be killed by one bullet is 0.6. What is the probability that the...	20	Numerical	13.3
43	A and B toss a coin alternately till one of them gets a head and wins the game. If A starts first, then find the probability that B will win the game.	21	Numerical	13.3
44	Two aeroplanes I and II bomb a target in succession. The probabilities of aeroplanes I and II scoring a hit correctly are 0.3 and 0.2...	17	Numerical	13.3
45	Aman and Bhuvan throw a pair of dice alternately. In order to win, they have to get a sum of 8. Find their respective probabilities of winning if...	21	Numerical	13.3
46	A and B throw a pair of dice alternately till one of them gets the sum of numbers as multiples of 6 and wins the game. If A starts first, find the...	21	Numerical	13.3
47	A and B cut a pack of 52 playing cards alternately and every time the card is put back. The pack is well-shuffled after each cut. A starts the game...	21	Numerical	13.3
48	A and B throw a pair of dice alternately. A wins the game if he gets a total of 7 and B wins the game if he gets a total of 10. If A starts the game...	21	Numerical	13.3
49	A, B and C in order toss a fair coin. The first one to throw a head wins. What are their respective chances of winning? Assume that the game may...	21	Numerical	13.3
50	A, B and C throw a die one after the other in the same order till one of them gets a '6' and wins the game. Find their respective probability of...	21	Numerical	13.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
51	In a hockey match, both teams A and B scored same number of goals up to the end of the game, so to decide the winner, the referee asked both the...	21	Numerical	13.3
52	A and B throw a die alternately till one of them gets a '6' and wins the game. Find their respective probabilities of winning, if A starts the game...	21	Numerical	13.3
53	A and B are two independent events such that $P(A \cap \bar{B}) = \frac{1}{4}$ and $P(\bar{A} \cap B) = \frac{1}{6}$. Find $P(A)$ and $P(B)$.	19	Numerical	13.3
54	X and Y are the two events such that $P(X Y) = 0.2$ and $P(Y) = 0.5$. Find the value of $P(X' \cap Y)$. Show your steps.	22	Numerical	13.3
55	Shown below is a Venn diagram representing the number of people in a music class who play different instruments. Find the probability that a...	22	Numerical	13.3
13.4 – Law of Total Probability				
1	A bag contains 3 white and 6 black balls while another bag contains 6 white and 3 black balls. A bag is selected at random and a ball is drawn. Find...	23	Numerical	13.4
2	A purse contains 4 silver and 5 copper coins. A second purse contains 3 silver and 7 copper coins. If a coin is taken out at random from one of the...	23	Numerical	13.4
3	Suppose that 5 men out of 100 and 25 women out of 1000 are good orators. Assuming that there are equal number of men and women, find the probability...	23	Numerical	13.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
4	A person has undertaken a construction job. The probabilities are 0.65 that there will be strike, 0.80 that the construction job will be completed on...	23	Numerical	13.4
5	There are two bags. One bag contains six green and three red balls. The second bag contains five green and four red balls. One ball is transferred...	23a	Numerical	13.4
6	An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and is then returned to the urn. Moreover, 2 additional balls...	23a	Numerical	13.4
7	There are two bags, one of which contains 3 black and 4 white balls while the other contains 4 black and 3 white balls. A die is thrown. If it shows...	23	Numerical	13.4
8	Urn I has 2 white and 3 black balls, urn II has 4 white and 1 black ball and urn III has 3 white and 2 black balls. An urn is selected at random and...	23	Numerical	13.4
9	In a factory, a product is manufactured by any of three machines A, B and C. They produce respectively 25%, 35% and 40% of the total products. A...	23	Numerical	13.4
10	Bag A contains 1 white, 2 blue and 3 red balls. Bag B contains 3 white, 3 blue and 2 red balls. Bag C contains 2 white, 3 blue and 4 red balls. One...	23	Numerical	13.4
11	Bag A contains 4 white balls and 3 black balls, while bag B contains 3 white balls and 5 black balls. Two balls are drawn from bag A and placed in...	23a	Numerical	13.4
13.5 – Bayes’ Theorem				

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
1	Bag I contains 3 red and 4 black balls while another bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it...	24	Numerical	13.5
2	Suppose 5% of men and 0.25% of women have grey hairs. A grey-haired person is selected at random. What is the probability of this person being...	24	Numerical	13.5
3	Two groups are competing for the positions on Board of Directors of a corporation. The probabilities that the first group and the second group will...	24	Numerical	13.5
4	Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous years result report that 70% of...	24	Numerical	13.5
5	In a certain college, 4% of boys and 1% of girls are taller than 1.75 metres. Furthermore, 60% of the students in the college are girls. A student...	24	Numerical	13.5
6	A company has two plants to manufacture scooters. Plant I manufactures 80% of the scooters and plant II manufactures 20%. At plant I, 85 out of 100...	24	Numerical	13.5
7	In a class, having 60% boys, 5% of the boys and 10% of the girls have an I.Q. of more than 150. A student is selected at random and found to have...	24	Numerical	13.5
8	In a company, 15% of the employees are graduates and 85% of the employees are non-graduates. As per the annual report of the company, 80% of the...	24	Numerical	13.5
9	A man is known to speak truth 4 out of 5 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.	24	Numerical	13.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
10	There are two boxes, namely box-I and box-II. Box-I contains 3 red and 6 black balls. Box-II contains 5 red and 5 black balls. One of the two boxes...	24	Numerical	13.5
11.	There are three urns containing 2 white and 3 black balls, 3 white and 2 black balls, and 4 white and 1 black ball, respectively. There is an equal...	24	Numerical	13.5
(i)				
(ii)	A bag X contains 4 white balls and 2 black balls, while another bag Y contains 3 white balls and 3 black balls. Two balls are drawn (without...	24	Numerical	13.5
12	A bag contains 5 red and 4 black balls, a second bag contains 3 red and 6 black balls. One of the two bags is selected at random and two balls are...	24	Numerical	13.5
13	Urn I has 2 white and 3 black balls, urn II has 4 white and 1 black ball and urn III has 3 white and 4 black balls. An urn is selected at random and...	24a	Numerical	13.5
14	A fair die is rolled. If face 1 turns up, a ball is drawn from bag A. If face 2 or 3 turn up, a ball is drawn from bag B. If face 4 or 5 or 6 turn...	24b	Numerical	13.5
15	There are two bags, I and II. Bag I contains 3 red and 5 black balls and bag II contains 4 red and 3 black balls. One ball is transferred randomly...	24b	Numerical	13.5
16	There are two boxes, box I and box II. Each box has 8 red balls, 5 green balls and 7 blue balls. Sahil takes out one ball from box I and places it in...	24b	Numerical	13.5
17	Bag I contains 5 red and 4 white balls and Bag II contains 3 red and 3 white balls. Two balls are transferred from Bag I to Bag II and then one ball...	24b	Numerical	13.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
18	Each of three identical jewellery boxes has two drawers. In each drawer of the first box there is a gold watch. In each drawer of the second box...	24c	Numerical	13.5
19	A manufacturer has three machine operators A, B and C. The first operator A produces 1% of defective items, where as the other two operators B and C...	24	Numerical	13.5
20	Given three identical boxes A, B and C, box A contains 2 gold and 1 silver coin, box B contains 1 gold and 2 silver coins and box C contains 3 silver...	24	Numerical	13.5
21.				
(i)	In a bolt factory, three machines A, B and C manufacture 25%, 35% and 40% of the total production respectively. Of their respective outputs, 5%...	24	Numerical	13.5
(ii)	In a factory, machine A produces 30% of total output, machine B produces 25% and machine C produces the remaining output. The defective items...	24	Numerical	13.5
22	A factory has three machines A, B and C producing 1500, 2500 and 3000 bulbs per day respectively. Machine A produces 1.5% defective bulbs, machine B...	24	Numerical	13.5
23	An insurance company insured 4000 doctors, 8000 teachers and 12000 engineers. The probability of a doctor, a teacher and an engineer dying before the...	24	Numerical	13.5
24	In an automobile factory, certain parts are to be fixed into the chassis in a section before it moves to the next section. On a given day, one of the...	24	Numerical	13.5
25.				

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
(i)	There are three coins. One is a two-headed coin, another is a biased coin that comes up heads 75% of the time and third is an unbiased coin. One of...	24	Numerical	13.5
(ii)	There are three coins. One is a coin having tail on both faces, another is a biased coin that comes up tails 70% of the time and the third is an...	24	Numerical	13.5
(iii)	There are two coins. One of them is a biased coin such that $P(\text{head}) : P(\text{tail})$ is 1 : 3 and other coin is a fair coin. A coin is...	24	Numerical	13.5
26	By examining the chest X-ray, the probability that TB is detected when a person is actually suffering from it is 0.99. The probability of a healthy...	24	Numerical	13.5
27	A doctor is to visit a patient. From the past experience, it is known that the probability that he will come by public transport, scooter, taxi or...	24	Numerical	13.5
28	Bag A contains 2 white, 1 black and 3 red balls; Bag B contains 3 white, 2 black and 4 red balls; Bag C contains 4 white, 3 black and 2 red balls...	24	Numerical	13.5
29	A coin is tossed. If it turns up heads, 2 balls will be drawn from urn A, otherwise 2 balls will be drawn from urn B. Urn A contains 3 black and 5...	24	Numerical	13.5
30	Redo the above problem assuming that the coin is biased and turns up heads 2 times out of 3.	24	Numerical	13.5
31	There are three urns. Urn I contains 1 white, 2 black and 3 red balls. Urn II contains 2 white, 1 black and 1 red balls. Urn III contains 4 white, 5...	24	Numerical	13.5
32	A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn randomly one-by-one without replacement and are...	24	Numerical	13.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
33	A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be hearts. Find the probability of...	24	Numerical	13.5
34	A jewellery seller has precious gems in white and red colour which he has put in three boxes. The distribution of these gems is shown in the table...	24	Numerical	13.5
35	Dobleron, a chocolate is known for its distinctive shape and its taste. The quality assurance team of the brand undertakes quality test in two steps:...	24	Numerical	13.5
36	Three companies viz. Amber, Bonzi and Comet are manufacturing cars which run on petrol and on battery as well. Their production share in the market...	24a	Numerical	13.5
13.6 – Probability Distribution (Random Variables)				
1	A random variable X has the following probability distribution: – see full data/parts in body	25	Numerical	13.6
2	A random variable X has the probability distribution: – see full data/parts in body	25b	Numerical	13.6
3	A random variable X has the following probability distribution where k is some number: – see full data/parts in body	25	Numerical	13.6
4	Let X denote the number of colleges where you will apply after your results and $P(X = x)$ denote your probability of getting admission in x ...	25	Numerical	13.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
5	A coin is tossed twice (or two coins are tossed simultaneously). – see full data/parts in body	25b	Numerical	13.6
6	A die is rolled. If a random variable X is defined as the number on the upper face, then find its probability distribution.	25b	Numerical	13.6
7	Two balls are drawn at random from a bag containing 2 red balls and 3 blue balls, without replacement. Let the variable X denote the number of...	25a	Numerical	13.6
8	An urn contains 5 white and 3 black balls. If two balls are drawn at random without replacement and X denotes the number of white balls drawn...	25a	Numerical	13.6
9	Two cards are drawn simultaneously from a well-shuffled pack of 52 cards. Find the probability distribution of the number of jacks.	25b	Numerical	13.6
10	Four bad oranges are mixed accidentally with 16 good oranges. Find the probability distribution of the number of bad oranges in a draw of two...	25b	Numerical	13.6
11	Out of a group of 60 architects, 40 are qualified and co-operative while the remaining are qualified but remain reserved. Two architects are...	25a	Numerical	13.6
12	Three balls are drawn one by one without replacement from a bag containing 5 white and 4 red balls. Find the probability distribution of the...	25a	Numerical	13.6
13	A box contains 12 bulbs of which 3 are defective. A sample of 3 bulbs is selected from the box. Let X denote the number of defective bulbs in...	25a	Numerical	13.6
14	Four bad eggs are mixed with 10 good ones. If 3 eggs are drawn one by one without replacement, then find the probability distribution of the...	25b	Numerical	13.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
15	A pair of dice is thrown simultaneously. If X denotes the absolute difference of numbers obtained on the pair of dice, then find the probability...	25b	Numerical	13.6
16	A group of class XI students designed a game for class XII students to be played at the farewell programme. They kept two sets of cards on the stage...	25	Numerical	13.6
13.7 – Mean and Variance of a Random Variable				
1	A boy throws a coin. He is to receive two rupees for getting a tail and ten rupees for getting a head. Find his expectation (mean receipt).	26	Numerical	13.7
2	A game at a school fair costs Rs. 10, with a prize of Rs. 50. If the probability of winning is $\frac{1}{10}$, is it a fair game?	26a	Numerical	13.7
3	In a certain village, families are strictly limited to two children. The probability distribution of the number of children of any given individual...	26	Numerical	13.7
4	Find the mean for the following probability distribution: – see full data/parts in body	26	Numerical	13.7
5	Find the mean μ for the following probability distribution: – see full data/parts in body	26	Numerical	13.7
6	A random variable X has the following probability distribution: – see full data/parts in body	26e	Numerical	13.7
7	The probability distribution of a random variable X is given as below: – see full data/parts in body	26e	Numerical	13.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
8	A discrete random variable X has the following probability distribution: – see full data/parts in body	26e	Numerical	13.7
9	The probability distribution of a random variable X is given as below: – see full data/parts in body	26	Numerical	13.7
10	The probability distribution of a random variable X is given below: – see full data/parts in body	26e	Numerical	13.7
11	Two fair dice are thrown simultaneously. If X denotes the number of sixes, find the mean of X .	26c	Numerical	13.7
12	Find the mean of number of heads in three tosses of a fair coin.	26c	Numerical	13.7
13	Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the mean of the number of kings.	26b	Numerical	13.7
14	An urn contains 5 red and 2 black balls. Two balls are randomly drawn without replacement. Find the probability distribution of the black balls...	26b	Numerical	13.7
15	3 defective bulbs are mixed with 7 good ones. Find the probability distribution of the number of defective bulbs, if 3 bulbs are drawn at random...	26	Numerical	13.7
16	2 bad eggs are accidentally mixed with 10 good ones. If three eggs are drawn at random, find the probability distribution of bad eggs drawn. Also...	26b	Numerical	13.7
17	A box contains 4 red and 5 black marbles. Find the probability distribution of the red marbles in a random draw of three marbles. Also find the mean...	26	Numerical	13.7
18	From a lot of 6 items containing 2 defective items, a sample of 4 items is drawn at random (without replacement). If the random variable X denotes...	26	Numerical	13.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Ex.
19	Two positive numbers are selected at random (without replacement) from the first five positive integers. Let X denote the larger of the two numbers...	26d	Numerical	13.7
20	Two numbers are selected at random (without replacement) from positive integers 2, 3, 4, 5, 6 and 7. Let X denote the larger of the two numbers...	26d	Numerical	13.7
21	Three numbers are selected at random (without replacement) from the first six positive integers. Let X denote the largest of the three numbers...	26d	Numerical	13.7
22	From a lot of 30 bulbs which include 6 defective bulbs, a sample of 2 bulbs is drawn at random one by one with replacement. Find the probability...	26c	Numerical	13.7
23	A box contains 10 tickets, 2 of which carry a prize of Rs. 80 each, 5 of which carry a prize of Rs. 40 each and remaining 3 carry a prize of Rs. 20 each...	26	Numerical	13.7
24	Two balls are drawn at random one by one with replacement from an urn containing equal number of red balls and green balls. Find the probability...	26c	Numerical	13.7
25	Kiran plays a game of throwing a fair die 3 times but to quit as and when she gets a six. Kiran gets +1 point for a six and -1 for any other...	26	Numerical	13.7

Formula & Standard Results Sheet

13.1 – Addition Theorem of Probability

General Addition Theorem

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Holds for any two events A, B .

Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B)$$

Valid only when $P(A \cap B) = 0$.

Complement Rule

$$P(A') = 1 - P(A)$$

A' denotes the complementary event of A .

Event “A only”

$$P(A \cap B') = P(A) - P(A \cap B)$$

Probability that A occurs but B does not.

De Morgan's Laws

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

Used to convert between union/intersection complements.

Exhaustive Events

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = 1$$

When $A_1, \dots, A_n$ together cover the entire sample space.

13.2 – Conditional Probability

Definition of Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Probability of A given that B has occurred.

Multiplication Rule (General)

$$P(A \cap B) = P(B) P(A | B) = P(A) P(B | A)$$

Rearrangement of the conditional-probability definition.

Conditional Complement

$$P(A' | B) = 1 - P(A | B)$$

Complement rule applied within the reduced sample space B .

Conditional Addition Theorem

$$P((A \cup B) | C) = P(A | C) + P(B | C) - P((A \cap B) | C)$$

Addition theorem applied to a conditional sample space.

13.3 – Multiplication Theorem & Independent Events**Independent Events**

$$P(A \cap B) = P(A) \cdot P(B)$$

Holds if and only if A and B are independent.

Mutual Independence (Three Events)

$$P(A \cap B \cap C) = P(A) P(B) P(C)$$

Requires pairwise AND three-way independence.

At-Least-One / All-Fail

$$P(\text{at least one occurs}) = 1 - \prod_i (1 - P(A_i))$$

Complement of “none of the independent events occur”.

Infinite Alternating-Turns Game

$$S_\infty = \frac{a}{1 - r}, \quad |r| < 1$$

Sum of an infinite geometric series; used when a game can in principle continue forever.

13.4 – Law of Total Probability**Law of Total Probability**

$$P(A) = \sum_{i=1}^n P(E_i) P(A | E_i)$$

$E_1, E_2, \dots, E_n$ must form a partition of the sample space S .

13.5 – Bayes’ Theorem

Bayes’ Theorem

$$P(E_i | A) = \frac{P(E_i) P(A | E_i)}{\sum_{j=1}^n P(E_j) P(A | E_j)}$$

Gives the “reverse” conditional probability (posterior) given a partition $E_1, \dots, E_n$ and an observed event A .

13.6 – Probability Distribution (Random Variables)

PMF Normalization

$$\sum_i P(X = x_i) = 1$$

Every valid probability mass function must sum to 1.

Validity of a PMF

$$0 \leq P(X = x_i) \leq 1 \text{ for every } x_i$$

Each individual probability must be a valid probability.

Hypergeometric-Type Distribution

$$P(X = r) = \frac{\binom{K}{r} \binom{N-K}{n-r}}{\binom{N}{n}}$$

Sampling n items without replacement from a population of size N containing K “successes”.

13.7 – Mean and Variance of a Random Variable

Mean / Expectation

$$E(X) = \sum_i x_i P(X = x_i)$$

The expected (average) value of the random variable X .

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2, \quad E(X^2) = \sum_i x_i^2 P(X = x_i)$$

Measures the spread of X about its mean.

Standard Deviation

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Square root of the variance; same units as X .

Fair Game Condition

$$E(\text{Gain}) = 0$$

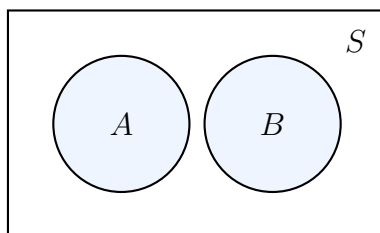
A game is “fair” exactly when the player’s expected net gain is zero.

13.1 – Addition Theorem of Probability

Question 1

Given $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$, find $P(A \text{ or } B)$, given that A and B are mutually exclusive events.

TYPE 1 Addition Theorem – Direct Application



Venn diagram of two mutually exclusive events A and B in sample space S

Given: $P(A) = \frac{3}{5}$, $P(B) = \frac{1}{5}$, $A \cap B = \emptyset$

Concept / Formula Used: $P(A \cap B) = 0$ $P(A \cup B) = P(A) + P(B)$

Solution:

By the addition theorem of probability for any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since A and B are mutually exclusive events, we have:

$$P(A \cap B) = 0$$

Substituting $P(A \cap B) = 0$ into the addition theorem:

$$\begin{aligned} P(A \text{ or } B) &= P(A \cup B) \\ &= P(A) + P(B) \\ &= \frac{3}{5} + \frac{1}{5} \\ &= \frac{3+1}{5} \\ &= \frac{4}{5} \end{aligned}$$

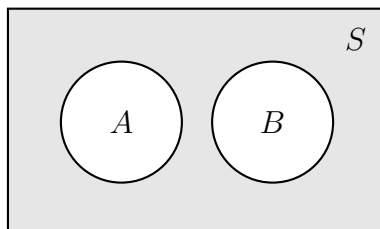
Final Answer

$$P(A \text{ or } B) = \frac{4}{5}$$

Question 2

The probability of an event A occurring is 0.5 and of B is 0.3. If A and B are mutually exclusive events, then find the probability of neither A nor B occurring.

TYPE 2 Addition Theorem + Complement (Find $P(\text{neither})$ / $P(\text{complement of union})$)



Venn diagram illustrating neither A nor B occurring (region outside $A \cup B$)

Given: $P(A) = 0.5$, $P(B) = 0.3$, A and B are mutually exclusive events

Concept / Formula Used: $A' \cap B' = (A \cup B)'$ $P(E') = 1 - P(E)$

Solution:

First, we find the probability that either A or B occurs, i.e., $P(A \cup B)$.

Since A and B are mutually exclusive events, $P(A \cap B) = 0$.

Using the addition theorem:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= 0.5 + 0.3 \\ &= 0.8 \end{aligned}$$

The event "neither A nor B occurring" is represented by $A' \cap B'$.

Using De Morgan's Law:

$$A' \cap B' = (A \cup B)'$$

Applying the complement rule of probability:

$$\begin{aligned} P(A' \cap B') &= P((A \cup B)') \\ &= 1 - P(A \cup B) \\ &= 1 - 0.8 \\ &= 0.2 \end{aligned}$$

Final Answer

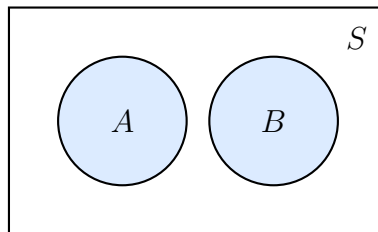
$$P(\text{neither } A \text{ nor } B) = 0.2$$

Question 3

If A and B are mutually exclusive events, $P(A) = 0.35$ and $P(B) = 0.45$, then find

- (i) $P(A')$
- (ii) $P(B')$
- (iii) $P(A \cup B)$
- (iv) $P(A \cap B)$
- (v) $P(A \cap B')$
- (vi) $P(A' \cap B')$

TYPE 1 Addition Theorem – Direct Application



Disjoint events A and B inside sample space S

Given: $P(A) = 0.35$, $P(B) = 0.45$, A and B are mutually exclusive

Concept / Formula Used: $P(E') = 1 - P(E)$ $P(A \cap B') = P(A) - P(A \cap B)$

Solution:

Since A and B are mutually exclusive, $A \cap B = \emptyset$.

(i) Using the complement formula for event A :

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - 0.35 \\ &= 0.65 \end{aligned}$$

(ii) Using the complement formula for event B :

$$\begin{aligned} P(B') &= 1 - P(B) \\ &= 1 - 0.45 \\ &= 0.55 \end{aligned}$$

(iii) By addition theorem for mutually exclusive events:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= 0.35 + 0.45 \\ &= 0.80 \end{aligned}$$

(iv) Since A and B are mutually exclusive:

$$P(A \cap B) = 0$$

(v) The event $A \cap B'$ represents $A - B$ (elements in A but not in B):

$$\begin{aligned} P(A \cap B') &= P(A) - P(A \cap B) \\ &= 0.35 - 0 \\ &= 0.35 \end{aligned}$$

(vi) By De Morgan's Law:

$$\begin{aligned} P(A' \cap B') &= P((A \cup B)') \\ &= 1 - P(A \cup B) \\ &= 1 - 0.80 \\ &= 0.20 \end{aligned}$$

Final Answer

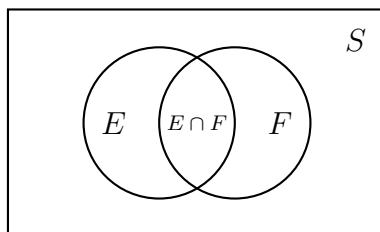
- (i) $P(A') = 0.65$
- (ii) $P(B') = 0.55$
- (iii) $P(A \cup B) = 0.8$
- (iv) $P(A \cap B) = 0$
- (v) $P(A \cap B') = 0.35$
- (vi) $P(A' \cap B') = 0.2$

Question 4

If E and F are events such that $P(E) = \frac{1}{4}$, $P(F) = \frac{1}{2}$ and $P(E \text{ and } F) = \frac{1}{8}$, find

- (i) $P(E \text{ or } F)$
- (ii) $P(\text{not } E \text{ and not } F)$

TYPE 3 Addition Theorem + De Morgan's Law



Venn diagram of two non-disjoint events E and F

Given: $P(E) = \frac{1}{4}$, $P(F) = \frac{1}{2}$, $P(E \cap F) = \frac{1}{8}$

Concept / Formula Used: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ $P(E' \cap F') = 1 - P(E \cup F)$

Solution:

(i) To find $P(E \text{ or } F)$, we calculate $P(E \cup F)$ using the general addition theorem:

$$\begin{aligned} P(E \cup F) &= P(E) + P(F) - P(E \cap F) \\ &= \frac{1}{4} + \frac{1}{2} - \frac{1}{8} \end{aligned}$$

Taking the common denominator 8:

$$\begin{aligned} P(E \cup F) &= \frac{2}{8} + \frac{4}{8} - \frac{1}{8} \\ &= \frac{2+4-1}{8} \\ &= \frac{5}{8} \end{aligned}$$

(ii) To find $P(\text{not } E \text{ and not } F)$, we evaluate $P(E' \cap F')$.

Using De Morgan's Law:

$$E' \cap F' = (E \cup F)'$$

Therefore:

$$\begin{aligned} P(E' \cap F') &= 1 - P(E \cup F) \\ &= 1 - \frac{5}{8} \\ &= \frac{8-5}{8} \\ &= \frac{3}{8} \end{aligned}$$

Final Answer

(i) $P(E \text{ or } F) = \frac{5}{8}$

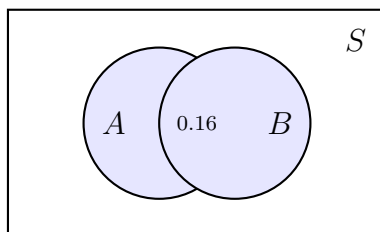
(ii) $P(\text{not } E \text{ and not } F) = \frac{3}{8}$

Question 5

A and B are events such that $P(A) = 0.42$, $P(B) = 0.48$ and $P(A \text{ and } B) = 0.16$. Determine

- (i) $P(\text{not } A)$
- (ii) $P(\text{not } B)$
- (iii) $P(A \text{ or } B)$

TYPE 2 Addition Theorem + Complement (Find $P(\text{neither})$ / $P(\text{complement of union})$)



Venn diagram showing events A and B with non-zero intersection

Given: $P(A) = 0.42$, $P(B) = 0.48$, $P(A \cap B) = 0.16$

Concept / Formula Used: $P(A') = 1 - P(A)$ $P(B') = 1 - P(B)$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Solution:

(i) $P(\text{not } A)$ is the complement of A :

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - 0.42 \\ &= 0.58 \end{aligned}$$

(ii) $P(\text{not } B)$ is the complement of B :

$$\begin{aligned} P(B') &= 1 - P(B) \\ &= 1 - 0.48 \\ &= 0.52 \end{aligned}$$

(iii) $P(A \text{ or } B)$ is calculated using the addition theorem:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.42 + 0.48 - 0.16 \\ &= 0.90 - 0.16 \\ &= 0.74 \end{aligned}$$

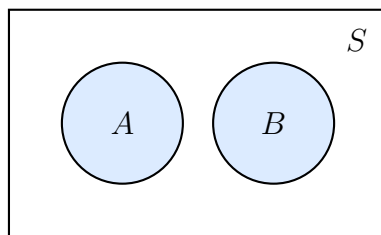
Final Answer

- (i) $P(\text{not } A) = 0.58$
 (ii) $P(\text{not } B) = 0.52$
 (iii) $P(A \text{ or } B) = 0.74$

Question 6

A and B are two mutually exclusive events of an experiment. If $P(\text{not } A) = 0.65$, $P(A \cup B) = 0.65$ and $P(B) = p$, find the value of p .

TYPE 1 Addition Theorem – Direct Application



Mutually exclusive events A and B

Given: $P(A') = 0.65$, $P(A \cup B) = 0.65$, $P(B) = p$, $A \cap B = \emptyset$

Concept / Formula Used: $P(A) = 1 - P(A')$ $P(A \cup B) = P(A) + P(B)$

Solution:

First, find $P(A)$ using the complement rule:

$$\begin{aligned} P(A) &= 1 - P(A') \\ &= 1 - 0.65 \\ &= 0.35 \end{aligned}$$

Since A and B are mutually exclusive events, $P(A \cap B) = 0$.

Applying the addition theorem for mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

Substitute the given and calculated values into the equation:

$$\begin{aligned} 0.65 &= 0.35 + p \\ p &= 0.65 - 0.35 \\ p &= 0.30 \end{aligned}$$

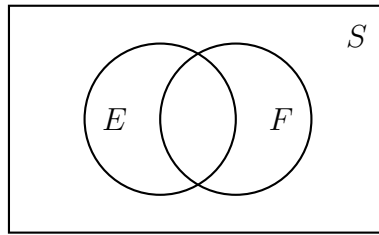
Final Answer

$$p = 0.3$$

Question 7

E and F are two events associated with a random experiment for which $P(E) = 0.60$, $P(E \text{ or } F) = 0.85$, $P(E \text{ and } F) = 0.42$. Find $P(F)$.

TYPE 1 Addition Theorem – Direct Application

Overlapping events E and F

Given: $P(E) = 0.60$, $P(E \cup F) = 0.85$, $P(E \cap F) = 0.42$

Concept / Formula Used: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

Solution:

Using the general addition theorem of probability:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Substitute the given values:

$$0.85 = 0.60 + P(F) - 0.42$$

$$0.85 = (0.60 - 0.42) + P(F)$$

$$0.85 = 0.18 + P(F)$$

Solving for $P(F)$:

$$P(F) = 0.85 - 0.18$$

$$P(F) = 0.67$$

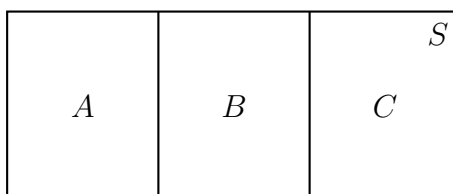
Final Answer

$$P(F) = 0.67$$

Question 8

If A, B and C are mutually exclusive and exhaustive events and it is known that $P(A \cup B) = 0.63$, calculate $P(C)$.

TYPE 4 Addition Theorem with Exhaustive Events ($P(A \cup B) = 1$)

Partition of sample space S into three mutually exclusive events A, B, C

Given: A, B, C are mutually exclusive and exhaustive events; $P(A \cup B) = 0.63$

Concept / Formula Used: $P(A) + P(B) + P(C) = 1$

Solution:

Since A, B, C are mutually exclusive and exhaustive events, their union forms the entire sample space S :

$$P(A \cup B \cup C) = 1$$

Because the events are mutually exclusive:

$$P(A \cup B \cup C) = P(A \cup B) + P(C)$$

Equating the two expressions:

$$P(A \cup B) + P(C) = 1$$

Substitute $P(A \cup B) = 0.63$:

$$0.63 + P(C) = 1$$

$$P(C) = 1 - 0.63$$

$$P(C) = 0.37$$

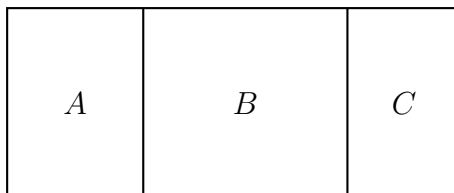
Final Answer

$$P(C) = 0.37$$

Question 9

A, B, C are three mutually exclusive and exhaustive events associated with a random experiment. Find $P(A)$, it being given that $P(B) = \frac{3}{2}P(A)$ and $P(C) = \frac{1}{2}P(B)$.

TYPE 4 Addition Theorem with Exhaustive Events ($P(A \cup B) = 1$)



Sample space partitioned into A, B, C

Given: A, B, C mutually exclusive and exhaustive; $P(B) = \frac{3}{2}P(A)$, $P(C) = \frac{1}{2}P(B)$

Concept / Formula Used: $\sum P(E_i) = 1$

Solution:

Let $P(A) = x$.

Express $P(B)$ in terms of x :

$$P(B) = \frac{3}{2}P(A) = \frac{3}{2}x$$

Express $P(C)$ in terms of x :

$$\begin{aligned}P(C) &= \frac{1}{2}P(B) \\&= \frac{1}{2} \left(\frac{3}{2}x \right) \\&= \frac{3}{4}x\end{aligned}$$

Since A, B, C are mutually exclusive and exhaustive events:

$$P(A) + P(B) + P(C) = 1$$

Substitute $P(A), P(B), P(C)$ in terms of x :

$$x + \frac{3}{2}x + \frac{3}{4}x = 1$$

Taking a common denominator of 4:

$$\begin{aligned}\left(\frac{4x + 6x + 3x}{4} \right) &= 1 \\ \frac{13x}{4} &= 1 \\ 13x &= 4 \\ x &= \frac{4}{13}\end{aligned}$$

Therefore:

$$P(A) = \frac{4}{13}$$

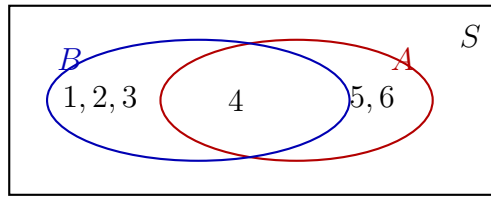
Final Answer

$$P(A) = \frac{4}{13}$$

Question 10

A die is thrown once. Let A be the event that the number obtained is greater than 3 and B be the event that the number obtained is less than 5, then find $P(A \cup B)$.

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)



Sample space $S = \{1, 2, 3, 4, 5, 6\}$ showing sets A and B

Given: Sample space $S = \{1, 2, 3, 4, 5, 6\}$; $A = \{x \in S \mid x > 3\}$; $B = \{x \in S \mid x < 5\}$

Concept / Formula Used: $P(E) = \frac{n(E)}{n(S)}$ $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Solution:

The total number of outcomes when a die is thrown once is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$n(S) = 6$$

Listing outcomes for event A (number obtained is > 3):

$$A = \{4, 5, 6\}$$

$$n(A) = 3$$

Listing outcomes for event B (number obtained is < 5):

$$B = \{1, 2, 3, 4\}$$

$$n(B) = 4$$

Now, find the union of A and B :

$$A \cup B = \{4, 5, 6\} \cup \{1, 2, 3, 4\}$$

$$= \{1, 2, 3, 4, 5, 6\}$$

The number of elements in $A \cup B$ is:

$$n(A \cup B) = 6$$

Therefore, the probability $P(A \cup B)$ is:

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)}$$

$$= \frac{6}{6}$$

$$= 1$$

Final Answer

$$P(A \cup B) = 1$$

Question 11

A and B are two candidates seeking admission in an engineering college. The probability that A is selected is 0.5 and the probability that both are selected is at most 0.3. Is it possible that the probability of B getting selected is 0.7?

TYPE 6 Validity/Consistency Check of Probability Values

★ Likely Exam Question

Given: $P(A) = 0.5$, $P(A \cap B) \leq 0.3$; candidate selection $P(B) = 0.7$

Concept / Formula Used: $P(A \cup B) \leq 1$

Solution:

Let $P(A) = 0.5$ and suppose $P(B) = 0.7$.

We know that for any events A and B , the probability of their union cannot exceed 1:

$$P(A \cup B) \leq 1$$

By the addition theorem of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substituting this into the inequality:

$$\begin{aligned} P(A) + P(B) - P(A \cap B) &\leq 1 \\ P(A) + P(B) - 1 &\leq P(A \cap B) \end{aligned}$$

Substitute the values $P(A) = 0.5$ and $P(B) = 0.7$:

$$\begin{aligned} 0.5 + 0.7 - 1 &\leq P(A \cap B) \\ 1.2 - 1 &\leq P(A \cap B) \\ 0.2 &\leq P(A \cap B) \end{aligned}$$

Thus, if $P(B) = 0.7$, we must have:

$$0.2 \leq P(A \cap B)$$

We are given that $P(A \cap B) \leq 0.3$. Combining both bounds gives:

$$0.2 \leq P(A \cap B) \leq 0.3$$

Since there exists a range of valid values for $P(A \cap B)$ (e.g., $P(A \cap B) = 0.25$), this situation contains no logical contradiction.

Hence, it is possible that the probability of B getting selected is 0.7.

Hence Proved

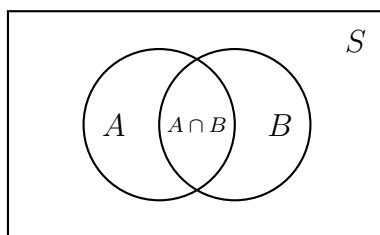
Yes, it is possible for $P(B)$ to be 0.7.

Common Mistake: Incorrectly assuming $P(A \cup B) = P(A) + P(B)$ without accounting for $P(A \cap B)$ leads to $0.5 + 0.7 = 1.2 > 1$, which wrongly makes one conclude it is impossible.

Question 12

The probability of an event A occurring is $\frac{2}{3}$ and the probability of event B not occurring is $\frac{5}{9}$. If the probability of getting success in atleast one of the two events is $\frac{4}{5}$, what is the probability of success in both the events?

TYPE 2 Addition Theorem + Complement (Find P(neither) / P(complement of union))



Venn diagram of events A and B

Given: $P(A) = \frac{2}{3}$, $P(B') = \frac{5}{9}$, $P(A \cup B) = \frac{4}{5}$

Concept / Formula Used: $P(B) = 1 - P(B')$ $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

Solution:

First, calculate $P(B)$ from $P(B')$:

$$\begin{aligned} P(B) &= 1 - P(B') \\ &= 1 - \frac{5}{9} \\ &= \frac{9 - 5}{9} \\ &= \frac{4}{9} \end{aligned}$$

The event "success in at least one of the two events" is $A \cup B$, and "success in both events" is $A \cap B$.

By the addition theorem:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Rearranging to solve for $P(A \cap B)$:

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= \frac{2}{3} + \frac{4}{9} - \frac{4}{5} \end{aligned}$$

Find the Least Common Multiple (LCM) of the denominators 3, 9, 5, which is 45:

$$\frac{2}{3} = \frac{2 \times 15}{45} = \frac{30}{45}$$

$$\frac{4}{9} = \frac{4 \times 5}{45} = \frac{20}{45}$$

$$\frac{4}{5} = \frac{4 \times 9}{45} = \frac{36}{45}$$

Substitute these fractions back into the expression:

$$\begin{aligned} P(A \cap B) &= \frac{30}{45} + \frac{20}{45} - \frac{36}{45} \\ &= \frac{30 + 20 - 36}{45} \\ &= \frac{14}{45} \end{aligned}$$

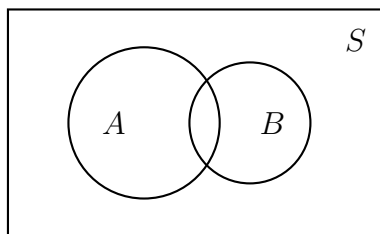
Final Answer

$$P(A \cap B) = \frac{14}{45}$$

Question 13

Tickets numbered from 1 to 20 are mixed up together and then a ticket is drawn at random. What is the probability that the ticket has a number which is multiple of 3 or 7?

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)



Venn diagram for subsets of multiples of 3 and multiples of 7 in $\{1, \dots, 20\}$

Given: Sample space $S = \{1, 2, 3, \dots, 20\}$, $n(S) = 20$

Concept / Formula Used: $P(A \cup B) = \frac{n(A \cup B)}{n(S)}$

Solution:

The total number of tickets is:

$$\begin{aligned} S &= \{1, 2, 3, \dots, 20\} \\ n(S) &= 20 \end{aligned}$$

Let A be the event of drawing a ticket with a number that is a multiple of 3:

$$A = \{3, 6, 9, 12, 15, 18\}$$

$$n(A) = 6$$

Let B be the event of drawing a ticket with a number that is a multiple of 7:

$$B = \{7, 14\}$$

$$n(B) = 2$$

Find the intersection $A \cap B$ (multiples of both 3 and 7, i.e., multiples of 21 between 1 and 20):

$$A \cap B = \emptyset$$

$$n(A \cap B) = 0$$

Since A and B have no common elements:

$$A \cup B = \{3, 6, 7, 9, 12, 14, 15, 18\}$$

$$n(A \cup B) = 6 + 2 = 8$$

The required probability is:

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)}$$

$$= \frac{8}{20}$$

$$= \frac{2}{5}$$

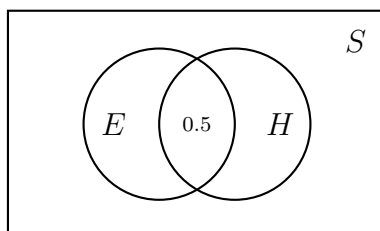
Final Answer

$$P(\text{multiple of 3 or 7}) = \frac{2}{5}$$

Question 14

The probability that a student will pass the final examination in both English and Hindi is 0.5 and the probability of passing neither is 0.1. If the probability of passing the English examination is 0.75 what is the probability of passing the Hindi examination?

TYPE 3 Addition Theorem + De Morgan's Law



Venn diagram for events E (passing English) and H (passing Hindi)

Given: $P(E \cap H) = 0.5$, $P(E' \cap H') = 0.1$, $P(E) = 0.75$

Concept / Formula Used: $P(E' \cap H') = 1 - P(E \cup H)$ $P(E \cup H) = P(E) + P(H) - P(E \cap H)$

Solution:

Let E be the event of passing English and H be the event of passing Hindi.

We are given that $P(E' \cap H') = 0.1$.

By De Morgan's Law:

$$\begin{aligned} E' \cap H' &= (E \cup H)' \\ P((E \cup H)') &= 1 - P(E \cup H) \end{aligned}$$

Substitute $P(E' \cap H') = 0.1$:

$$\begin{aligned} 0.1 &= 1 - P(E \cup H) \\ P(E \cup H) &= 1 - 0.1 \\ P(E \cup H) &= 0.9 \end{aligned}$$

Now, apply the addition theorem of probability:

$$P(E \cup H) = P(E) + P(H) - P(E \cap H)$$

Substitute the values $P(E \cup H) = 0.9$, $P(E) = 0.75$, and $P(E \cap H) = 0.5$:

$$\begin{aligned} 0.9 &= 0.75 + P(H) - 0.5 \\ 0.9 &= 0.25 + P(H) \\ P(H) &= 0.9 - 0.25 \\ P(H) &= 0.65 \end{aligned}$$

Final Answer

$$P(H) = 0.65$$

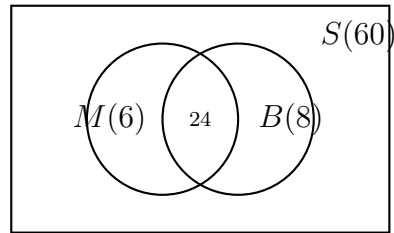
Question 15

In a class of 60 students, 30 opted for Mathematics, 32 opted for Biology and 24 opted for both Mathematics and Biology. If one of these students is selected at random, find the probability that:

- the student opted for Mathematics or Biology
- the student has opted neither Mathematics nor Biology
- the student has opted Mathematics but not Biology

TYPE 7 Venn Diagram / Two-Way Classification Counting Word Problems

★ Likely Exam Question



Venn diagram of student options in class of 60

Given: Total students $n(S) = 60$; Mathematics $n(M) = 30$; Biology $n(B) = 32$; Both $n(M \cap B) = 24$

Concept / Formula Used: $n(M \cup B) = n(M) + n(B) - n(M \cap B)$ $P(E) = \frac{n(E)}{n(S)}$

Solution:

Let M be the set of students who opted for Mathematics and B be the set of students who opted for Biology.

(i) Calculate the number of students who opted for Mathematics or Biology:

$$\begin{aligned} n(M \cup B) &= n(M) + n(B) - n(M \cap B) \\ &= 30 + 32 - 24 \\ &= 62 - 24 \\ &= 38 \end{aligned}$$

The probability $P(M \cup B)$ is:

$$\begin{aligned} P(M \cup B) &= \frac{n(M \cup B)}{n(S)} \\ &= \frac{38}{60} \\ &= \frac{19}{30} \end{aligned}$$

(ii) The number of students who opted for neither subject is:

$$\begin{aligned} n((M \cup B)') &= n(S) - n(M \cup B) \\ &= 60 - 38 \\ &= 22 \end{aligned}$$

The probability $P(M' \cap B')$ is:

$$\begin{aligned} P(M' \cap B') &= \frac{22}{60} \\ &= \frac{11}{30} \end{aligned}$$

(iii) The number of students who opted for Mathematics but not Biology is:

$$\begin{aligned} n(M \cap B') &= n(M) - n(M \cap B) \\ &= 30 - 24 \\ &= 6 \end{aligned}$$

The probability $P(M \cap B')$ is:

$$\begin{aligned} P(M \cap B') &= \frac{6}{60} \\ &= \frac{1}{10} \end{aligned}$$

Final Answer

(i) $P(\text{Maths or Biology}) = \frac{19}{30}$

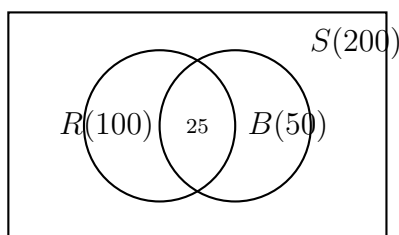
(ii) $P(\text{neither}) = \frac{11}{30}$

(iii) $P(\text{Maths but not Biology}) = \frac{1}{10}$

Question 16

A bag contains 150 nuts and 50 bolts. Half of the bolts and half of the nuts are rusted. One item is drawn at random from the bag, find the probability that it is either rusted or a bolt.

TYPE 7 Venn Diagram / Two-Way Classification Counting Word Problems



Categorization of bag contents

Given: Nuts = 150, Bolts = 50; Rusted nuts = 75, Rusted bolts = 25

Concept / Formula Used: $n(R \cup B) = n(R) + n(B) - n(R \cap B)$

Solution:

Calculate the total number of items in the sample space:

$$n(S) = 150 + 50 = 200$$

Let R be the event of drawing a rusted item, and B be the event of drawing a bolt.

Number of bolts:

$$n(B) = 50$$

Calculate the number of rusted items $n(R)$:

$$\text{Rusted nuts} = \frac{1}{2} \times 150 = 75$$

$$\text{Rusted bolts} = \frac{1}{2} \times 50 = 25$$

$$n(R) = 75 + 25 = 100$$

The number of items that are both rusted and bolts ($R \cap B$) is:

$$n(R \cap B) = 25$$

Using the addition formula for the number of elements:

$$\begin{aligned} n(R \cup B) &= n(R) + n(B) - n(R \cap B) \\ &= 100 + 50 - 25 \\ &= 125 \end{aligned}$$

The required probability is:

$$\begin{aligned} P(R \cup B) &= \frac{n(R \cup B)}{n(S)} \\ &= \frac{125}{200} \\ &= \frac{125 \div 25}{200 \div 25} \\ &= \frac{5}{8} \end{aligned}$$

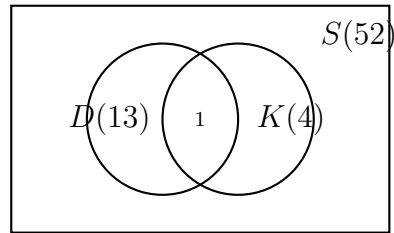
Final Answer

$$P(\text{rusted or bolt}) = \frac{5}{8}$$

Question 17

A card is drawn from a well shuffled pack of playing cards. What is the probability that it is either a diamond or a king or both?

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)



Venn diagram for Diamond cards and King cards in a standard deck

Given: Total cards $n(S) = 52$; Diamond cards $n(D) = 13$; King cards $n(K) = 4$

Concept / Formula Used: $P(D \cup K) = P(D) + P(K) - P(D \cap K)$

Solution:

Let D be the event of drawing a diamond card, and K be the event of drawing a king. The number of diamond cards in a deck is:

$$n(D) = 13 \implies P(D) = \frac{13}{52}$$

The number of kings in a deck is:

$$n(K) = 4 \implies P(K) = \frac{4}{52}$$

The card that is both a diamond and a king is the King of Diamonds:

$$n(D \cap K) = 1 \implies P(D \cap K) = \frac{1}{52}$$

Using the addition theorem of probability:

$$\begin{aligned} P(D \cup K) &= P(D) + P(K) - P(D \cap K) \\ &= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} \\ &= \frac{13 + 4 - 1}{52} \\ &= \frac{16}{52} \\ &= \frac{16 \div 4}{52 \div 4} \\ &= \frac{4}{13} \end{aligned}$$

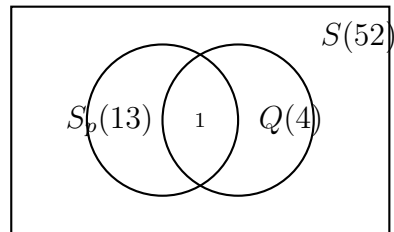
Final Answer

$$P(\text{diamond or king}) = \frac{4}{13}$$

Question 18

A card is drawn at random from a pack of 52 playing cards. What is the probability that the card drawn is neither a spade nor a queen?

TYPE 2 Addition Theorem + Complement (Find $P(\text{neither})$ / $P(\text{complement of union})$)



Venn diagram showing spade cards and queen cards

Given: Total cards $n(S) = 52$; Spades $n(S_p) = 13$; Queens $n(Q) = 4$

Concept / Formula Used: $P(S'_p \cap Q') = 1 - P(S_p \cup Q)$

Solution:

Let S_p be the event of drawing a spade and Q be the event of drawing a queen.

The count of favorable single events:

$$n(S_p) = 13$$

$$n(Q) = 4$$

The card that is both a spade and a queen is the Queen of Spades:

$$n(S_p \cap Q) = 1$$

Calculate the number of cards that are either a spade or a queen:

$$\begin{aligned} n(S_p \cup Q) &= n(S_p) + n(Q) - n(S_p \cap Q) \\ &= 13 + 4 - 1 \\ &= 16 \end{aligned}$$

Therefore, the probability of drawing a spade or a queen is:

$$P(S_p \cup Q) = \frac{16}{52} = \frac{4}{13}$$

The probability that the card drawn is neither a spade nor a queen is:

$$\begin{aligned} P(S'_p \cap Q') &= 1 - P(S_p \cup Q) \\ &= 1 - \frac{4}{13} \\ &= \frac{13 - 4}{13} \\ &= \frac{9}{13} \end{aligned}$$

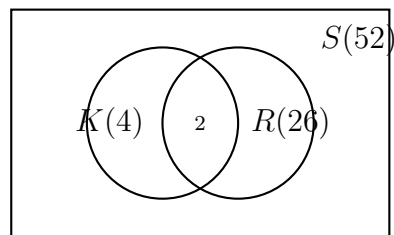
Final Answer

$$P(\text{neither spade nor queen}) = \frac{9}{13}$$

Question 19

A card is drawn at random from well shuffled pack of 52 playing cards. Find the probability that it is neither a king nor a red card.

TYPE 2 Addition Theorem + Complement (Find $P(\text{neither}) / P(\text{complement of union})$)



Venn diagram for King cards and Red cards

Given: Total cards $n(S) = 52$; Kings $n(K) = 4$; Red cards $n(R) = 26$

Concept / Formula Used: $P(K' \cap R') = 1 - P(K \cup R)$

Solution:

Let K be the event of drawing a king, and R be the event of drawing a red card.

Number of king cards:

$$n(K) = 4$$

Number of red cards (Hearts and Diamonds):

$$n(R) = 26$$

The cards that are both kings and red are the King of Hearts and King of Diamonds:

$$n(K \cap R) = 2$$

Using the counting formula for the union:

$$\begin{aligned} n(K \cup R) &= n(K) + n(R) - n(K \cap R) \\ &= 4 + 26 - 2 \\ &= 28 \end{aligned}$$

The probability of drawing a red card or a king is:

$$P(K \cup R) = \frac{28}{52} = \frac{7}{13}$$

Using De Morgan's Law, the probability of drawing neither a king nor a red card is:

$$\begin{aligned} P(K' \cap R') &= 1 - P(K \cup R) \\ &= 1 - \frac{7}{13} \\ &= \frac{13 - 7}{13} \\ &= \frac{6}{13} \end{aligned}$$

Final Answer

$$P(\text{neither king nor red card}) = \frac{6}{13}$$

Question 20

Find the probability of getting an odd number on the first die or a total of 8 in a single throw of two dice.

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

Grid representing outcomes of throwing two dice

Given: Throw of two dice: total outcomes $n(S) = 6 \times 6 = 36$

Concept / Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Solution:

The sample space for throwing two dice consists of 36 ordered pairs:

$$n(S) = 36$$

Let A be the event of getting an odd number on the first die.

The outcomes for A are those where the first die is 1, 3, or 5:

$$A = \{(1, 1), (1, 2), \dots, (1, 6), (3, 1), \dots, (3, 6), (5, 1), \dots, (5, 6)\}$$
$$n(A) = 3 \times 6 = 18$$

Let B be the event of getting a total sum of 8 on the two dice:

$$B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$
$$n(B) = 5$$

Find the intersection $A \cap B$ (odd number on first die AND sum equal to 8):

$$A \cap B = \{(3, 5), (5, 3)\}$$
$$n(A \cap B) = 2$$

Applying the addition theorem for sets:

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 18 + 5 - 2 \\ &= 21 \end{aligned}$$

The probability is:

$$\begin{aligned} P(A \cup B) &= \frac{n(A \cup B)}{n(S)} \\ &= \frac{21}{36} \\ &= \frac{21 \div 3}{36 \div 3} \\ &= \frac{7}{12} \end{aligned}$$

Final Answer

$$P(A \cup B) = \frac{7}{12}$$

Question 21

If two dice are thrown simultaneously, find the probability of getting

- (i) a sum of 7 or 11
- (ii) a doublet or a total of 6

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)

$$6 \times 6 = 36 \text{ outcomes}$$

Sample space outcomes layout for two dice

Given: Two dice thrown simultaneously: $n(S) = 36$

Concept / Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Solution:

(i) Let A be the event of getting a sum of 7 and B be the event of getting a sum of 11.

List outcomes for sum of 7:

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} \implies n(A) = 6$$

List outcomes for sum of 11:

$$B = \{(5, 6), (6, 5)\} \implies n(B) = 2$$

Since a sum cannot be both 7 and 11 simultaneously:

$$A \cap B = \emptyset \implies n(A \cap B) = 0$$

The probability is:

$$\begin{aligned} P(A \cup B) &= \frac{n(A) + n(B)}{n(S)} \\ &= \frac{6 + 2}{36} \\ &= \frac{8}{36} \\ &= \frac{2}{9} \end{aligned}$$

(ii) Let D be the event of getting a doublet and S_6 be the event of getting a total of 6.

List outcomes for doublets:

$$D = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\} \implies n(D) = 6$$

List outcomes for sum of 6:

$$S_6 = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} \implies n(S_6) = 5$$

Find the intersection $D \cap S_6$ (a doublet that adds up to 6):

$$D \cap S_6 = \{(3, 3)\} \implies n(D \cap S_6) = 1$$

Apply the addition formula:

$$\begin{aligned} n(D \cup S_6) &= n(D) + n(S_6) - n(D \cap S_6) \\ &= 6 + 5 - 1 \\ &= 10 \end{aligned}$$

The required probability is:

$$P(D \cup S_6) = \frac{10}{36} = \frac{5}{18}$$

Final Answer

(i) $P(\text{sum of 7 or 11}) = \frac{2}{9}$

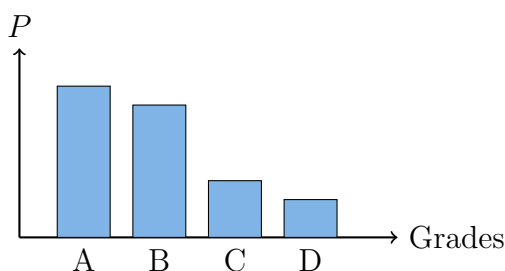
(ii) $P(\text{doublet or sum of 6}) = \frac{5}{18}$

Question 22

The probabilities that a student will get A, B, C or D grade are 0.4, 0.35, 0.15 and 0.1 respectively. Find the probability that she will get

- (i) B or C grade
- (ii) atmost C grade

TYPE 1 Addition Theorem – Direct Application



Mutually exclusive grade probability distribution

Given: $P(A) = 0.4$, $P(B) = 0.35$, $P(C) = 0.15$, $P(D) = 0.1$; events are mutually exclusive

Concept / Formula Used: $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Solution:

Since a student can obtain only one grade, the events A, B, C, D are mutually exclusive.

(i) The probability of getting B or C grade:

$$\begin{aligned} P(B \cup C) &= P(B) + P(C) \\ &= 0.35 + 0.15 \\ &= 0.50 \end{aligned}$$

(ii) "Atmost C grade" means getting grade C or a lower grade (grade D):

$$\begin{aligned} P(\text{atmost C grade}) &= P(C \cup D) \\ &= P(C) + P(D) \\ &= 0.15 + 0.10 \\ &= 0.25 \end{aligned}$$

Final Answer

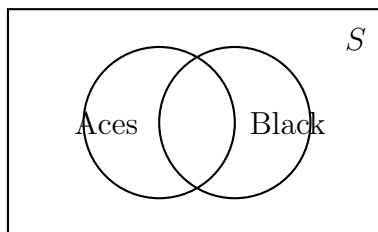
(i) $P(\text{B or C grade}) = 0.5$

(ii) $P(\text{atmost C grade}) = 0.25$

Question 23

Two cards are drawn at random from a pack of 52 cards. What is probability that the cards are either both aces or both black cards?

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)



Overlapping categories for 2-card selection

Given: Total cards = 52; Aces = 4; Black cards = 26; Black aces = 2

Concept / Formula Used: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Solution:

The total number of ways to draw 2 cards from 52 cards is:

$$n(S) = \binom{52}{2} = \frac{52 \times 51}{2 \times 1} = 1326$$

Let A be the event that both cards are aces. Since there are 4 aces:

$$n(A) = \binom{4}{2} = \frac{4 \times 3}{2 \times 1} = 6$$

Let B be the event that both cards are black cards. Since there are 26 black cards:

$$n(B) = \binom{26}{2} = \frac{26 \times 25}{2 \times 1} = 325$$

The event $A \cap B$ is that both cards are black aces. Since there are 2 black aces (Ace of Spades and Ace of Clubs):

$$n(A \cap B) = \binom{2}{2} = 1$$

Using the inclusion-exclusion principle:

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 6 + 325 - 1 \\ &= 330 \end{aligned}$$

The required probability is:

$$\begin{aligned} P(A \cup B) &= \frac{n(A \cup B)}{n(S)} \\ &= \frac{330}{1326} \end{aligned}$$

Dividing numerator and denominator by 6:

$$\begin{aligned} P(A \cup B) &= \frac{330 \div 6}{1326 \div 6} \\ &= \frac{55}{221} \end{aligned}$$

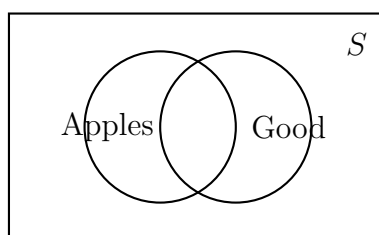
Final Answer

$$P(\text{both aces or both black}) = \frac{55}{221}$$

Question 24

A basket contains 20 apples and 10 oranges out of which 5 apples and 3 oranges are defective. If a person takes out 2 fruits at random, find the probability that either both are apples or both are good.

TYPE 5 Addition Theorem – Classical Probability Word Problems (dice/cards/counting)



Venn diagram for fruits selection

Given: Apples = 20 (5 defective, 15 good); Oranges = 10 (3 defective, 7 good); Total fruits = 30

Concept / Formula Used: $n(A \cup G) = n(A) + n(G) - n(A \cap G)$ $\binom{n}{2} = \frac{n(n-1)}{2}$

Solution:

Summarize the fruit counts:

$$\text{Total Apples} = 20 \quad (\text{Good} = 15, \text{Defective} = 5)$$

$$\text{Total Oranges} = 10 \quad (\text{Good} = 7, \text{Defective} = 3)$$

$$\text{Total Fruits} = 30$$

$$\text{Total Good Fruits} = 15 + 7 = 22$$

Total ways to pick 2 fruits from 30 fruits:

$$n(S) = \binom{30}{2} = \frac{30 \times 29}{2} = 435$$

Let A be the event that both fruits are apples:

$$n(A) = \binom{20}{2} = \frac{20 \times 19}{2} = 190$$

Let G be the event that both fruits are good:

$$n(G) = \binom{22}{2} = \frac{22 \times 21}{2} = 231$$

The intersection $A \cap G$ is the event that both fruits are good apples:

$$n(A \cap G) = \binom{15}{2} = \frac{15 \times 14}{2} = 105$$

Applying the principle of inclusion-exclusion:

$$\begin{aligned} n(A \cup G) &= n(A) + n(G) - n(A \cap G) \\ &= 190 + 231 - 105 \\ &= 421 - 105 \\ &= 316 \end{aligned}$$

The required probability is:

$$\begin{aligned} P(A \cup G) &= \frac{n(A \cup G)}{n(S)} \\ &= \frac{316}{435} \end{aligned}$$

Final Answer

$$P(\text{both apples or both good}) = \frac{316}{435}$$

13.2 – Conditional Probability

Question 1

- (i) If $P(A) = \frac{1}{2}$, $P(B) = 0$, then what can you say about $P(A|B)$?
 (ii) If A and B are events such that $P(A|B) = P(B|A)$, then show that $P(A) = P(B)$.

TYPE 8 Conditional Probability – Definition/Property Proofs

Given: For part (i): $P(A) = \frac{1}{2}$, $P(B) = 0$. For part (ii): $P(A|B) = P(B|A)$.

Concept / Formula Used: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Solution:

Part (i): By the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Since $P(B) = 0$, the denominator is zero. Division by zero is mathematically undefined. Therefore, $P(A|B)$ is not defined.

Part (ii): Given that:

$$P(A|B) = P(B|A)$$

Using the definition of conditional probability on both sides:

$$\frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{P(A)}$$

Assuming $P(A \cap B) \neq 0$, dividing both sides by $P(A \cap B)$ yields:

$$\frac{1}{P(B)} = \frac{1}{P(A)}$$

Taking reciprocals on both sides:

$$P(A) = P(B)$$

Hence Proved

$P(A|B)$ is undefined for (i), and $P(A) = P(B)$ is proved for (ii).

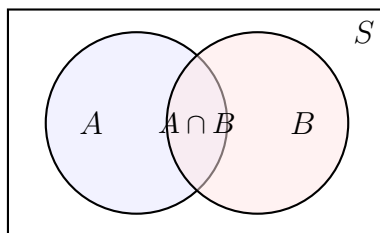
Common Mistake: Assuming $P(A|B) = 0$ when $P(B) = 0$ is a common error; division by zero makes the conditional probability undefined.

Question 2

If A and B are events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$, then find:

- (i) $P(A|B)$
- (ii) $P(B|A)$

TYPE 9 Conditional Probability – Direct Calculation from Joint & Marginal Probabilities



Venn diagram representation of events A and B

Given: $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{4}$.

Concept / Formula Used: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Solution:

Part (i): Using the formula for conditional probability $P(A|B)$:

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{\frac{1}{4}}{\frac{1}{3}} \\ &= \frac{1}{4} \times \frac{3}{1} \\ &= \frac{3}{4} \end{aligned}$$

Part (ii): Using the formula for conditional probability $P(B|A)$:

$$\begin{aligned} P(B|A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{\frac{1}{4}}{\frac{1}{2}} \\ &= \frac{1}{4} \times \frac{2}{1} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

(i) $P(A|B) = \frac{3}{4}$

(ii) $P(B|A) = \frac{1}{2}$

Question 3

If A and B are two events such that $P(A) = \frac{5}{8}$, $P(B) = \frac{1}{2}$ and $P(A|B) = \frac{3}{4}$, then find the value of $P(B|A)$.

TYPE 10 Conditional Probability – Two-Dice / Word-Problem Sample-Space Counting

Given: $P(A) = \frac{5}{8}$, $P(B) = \frac{1}{2}$, $P(A|B) = \frac{3}{4}$.

Concept / Formula Used: $P(A \cap B) = P(A|B) \cdot P(B)$

Solution:

First, we express $P(A \cap B)$ using $P(A|B)$ and $P(B)$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplying both sides by $P(B)$:

$$\begin{aligned} P(A \cap B) &= P(A|B) \cdot P(B) \\ &= \frac{3}{4} \cdot \frac{1}{2} \\ &= \frac{3}{8} \end{aligned}$$

Now, we calculate $P(B|A)$:

$$\begin{aligned} P(B|A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{\frac{3}{8}}{\frac{5}{8}} \\ &= \frac{3}{8} \times \frac{8}{5} \\ &= \frac{3}{5} \end{aligned}$$

Final Answer

$$P(B|A) = \frac{3}{5}$$

Question 4

If the events A and B are such that $P(A) = \frac{1}{4}$, $P(B|A) = \frac{2}{3}$, $P(A|B) = \frac{1}{2}$, then find $P(B)$.

TYPE 9 Conditional Probability – Direct Calculation from Joint & Marginal Probabilities

Given: $P(A) = \frac{1}{4}$, $P(B|A) = \frac{2}{3}$, $P(A|B) = \frac{1}{2}$.

Concept / Formula Used: $P(A \cap B) = P(B|A) \cdot P(A)$ $P(B) = \frac{P(A \cap B)}{P(A|B)}$

Solution:

First, we derive the probability of the intersection $P(A \cap B)$:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Rearranging to find $P(A \cap B)$:

$$\begin{aligned} P(A \cap B) &= P(B|A) \cdot P(A) \\ &= \frac{2}{3} \cdot \frac{1}{4} \\ &= \frac{2}{12} \\ &= \frac{1}{6} \end{aligned}$$

Next, using the definition of $P(A|B)$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Rearranging to solve for $P(B)$:

$$\begin{aligned} P(B) &= \frac{P(A \cap B)}{P(A|B)} \\ &= \frac{\frac{1}{6}}{\frac{1}{2}} \\ &= \frac{1}{6} \times \frac{2}{1} \\ &= \frac{2}{6} \\ &= \frac{1}{3} \end{aligned}$$

Final Answer

$$P(B) = \frac{1}{3}$$

Question 5

- (i) If $P(A) = 0.6$, $P(B) = 0.5$ and $P(B|A) = 0.4$, find $P(A \cup B)$ and $P(A|B)$.
(ii) If $P(\text{not } A) = 0.7$, $P(B) = 0.7$ and $P(B|A) = 0.5$, then find $P(A|B)$ and $P(A \cup B)$.

TYPE 11 Conditional Probability Combined with Addition Theorem

Given: Part (i): $P(A) = 0.6$, $P(B) = 0.5$, $P(B|A) = 0.4$. Part (ii): $P(\text{not } A) = 0.7$, $P(B) = 0.7$, $P(B|A) = 0.5$.

Concept / Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Solution:

Part (i): First, find $P(A \cap B)$:

$$\begin{aligned}P(A \cap B) &= P(B|A) \cdot P(A) \\&= 0.4 \times 0.6 \\&= 0.24\end{aligned}$$

Now, find $P(A \cup B)$:

$$\begin{aligned}P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\&= 0.6 + 0.5 - 0.24 \\&= 1.1 - 0.24 \\&= 0.86\end{aligned}$$

Next, calculate $P(A|B)$:

$$\begin{aligned}P(A|B) &= \frac{P(A \cap B)}{P(B)} \\&= \frac{0.24}{0.5} \\&= 0.48\end{aligned}$$

Part (ii): First, find $P(A)$ using complement rule:

$$\begin{aligned}P(A) &= 1 - P(\text{not } A) \\&= 1 - 0.7 \\&= 0.3\end{aligned}$$

Now, determine $P(A \cap B)$:

$$\begin{aligned}P(A \cap B) &= P(B|A) \cdot P(A) \\&= 0.5 \times 0.3 \\&= 0.15\end{aligned}$$

Next, calculate $P(A|B)$:

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{0.15}{0.7} \\ &= \frac{15}{70} \\ &= \frac{3}{14} \end{aligned}$$

Finally, find $P(A \cup B)$:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.3 + 0.7 - 0.15 \\ &= 1.0 - 0.15 \\ &= 0.85 \end{aligned}$$

Final Answer

(i) $P(A \cup B) = 0.86$, $P(A|B) = 0.48$

(ii) $P(A|B) = \frac{3}{14}$, $P(A \cup B) = 0.85$

Question 6

Compute $P(A|B)$ when $P(A) = \frac{1}{5}$, $P(B) = \frac{2}{5}$, $P(A \cup B) = \frac{3}{5}$.

TYPE 11 Conditional Probability Combined with Addition Theorem

Given: $P(A) = \frac{1}{5}$, $P(B) = \frac{2}{5}$, $P(A \cup B) = \frac{3}{5}$.

Concept / Formula Used: $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

Solution:

Using the Addition Law of Probability to isolate $P(A \cap B)$:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \frac{3}{5} &= \frac{1}{5} + \frac{2}{5} - P(A \cap B) \\ \frac{3}{5} &= \frac{3}{5} - P(A \cap B) \end{aligned}$$

Subtracting $\frac{3}{5}$ from both sides:

$$\begin{aligned} -P(A \cap B) &= 0 \\ P(A \cap B) &= 0 \end{aligned}$$

Now, compute $P(A|B)$:

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{0}{\frac{2}{5}} \\ &= 0 \end{aligned}$$

Final Answer

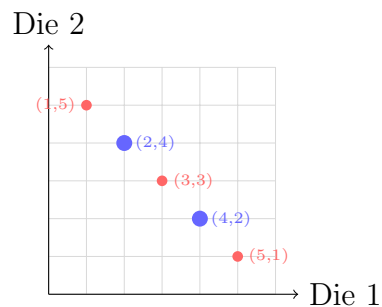
$$P(A|B) = 0$$

Question 7

A die is thrown twice and the sum of numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?

TYPE 10 Conditional Probability – Two-Dice / Word-Problem Sample-Space Counting

★ Likely Exam Question



Grid of outcomes for rolling two dice

Given: Sum of numbers on two rolls of a die is 6.

To Find: Probability that 4 has appeared at least once given the sum is 6.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Let B be the event that the sum of the numbers appearing is 6. The outcomes in B are:

$$B = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

The number of elements in event B is:

$$n(B) = 5$$

Let A be the event that the number 4 appears at least once. The outcomes common to both A and B are:

$$A \cap B = \{(2, 4), (4, 2)\}$$

The number of elements in $A \cap B$ is:

$$n(A \cap B) = 2$$

Therefore, the required conditional probability is:

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ &= \frac{2}{5} \end{aligned}$$

Final Answer

The required conditional probability is $\frac{2}{5}$.

Question 8

A pair of dice is thrown and the sum of the numbers is observed to be even. What is the probability that both dice have come up with even numbers?

TYPE 10 Conditional Probability – Two-Dice / Word-Problem Sample-Space Counting

Given: Sum of numbers on a pair of dice is even.

To Find: Probability that both dice show even numbers.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Let B be the event that the sum of the numbers is even. A sum is even if both numbers are odd or both numbers are even. Number of outcomes where both numbers are odd:

$$\text{Odd outcomes} = \{1, 3, 5\} \times \{1, 3, 5\} \implies 3 \times 3 = 9$$

Number of outcomes where both numbers are even:

$$\text{Even outcomes} = \{2, 4, 6\} \times \{2, 4, 6\} \implies 3 \times 3 = 9$$

Total outcomes in event B :

$$n(B) = 9 + 9 = 18$$

Let A be the event that both dice show even numbers. The common outcomes for A and B are all outcomes where both dice are even:

$$n(A \cap B) = 9$$

The required conditional probability is:

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ &= \frac{9}{18} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

The required probability is $\frac{1}{2}$.

Question 9

Given that the numbers appearing on rolling two dice together are different. What is the probability that the sum of numbers appearing on two dice is 6?

TYPE 10 Conditional Probability – Two-Dice / Word-Problem Sample-Space Counting

Given: Numbers on the two dice are different.

To Find: Probability that the sum of the numbers is 6.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Total number of outcomes when rolling two dice is:

$$n(S) = 6 \times 6 = 36$$

Let B be the event that the numbers appearing on the two dice are different. Outcomes where numbers are the same (doubles) are:

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\} \Rightarrow 6 \text{ outcomes}$$

Therefore, the number of outcomes in B is:

$$n(B) = 36 - 6 = 30$$

Let A be the event that the sum of the numbers is 6. Outcomes with sum 6 are $\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$. The intersection $A \cap B$ consists of outcomes with sum 6 where the numbers are different:

$$A \cap B = \{(1, 5), (2, 4), (4, 2), (5, 1)\}$$

The number of elements in $A \cap B$ is:

$$n(A \cap B) = 4$$

Thus, the required conditional probability is:

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ &= \frac{4}{30} \\ &= \frac{2}{15} \end{aligned}$$

Final Answer

The required probability is $\frac{2}{15}$.

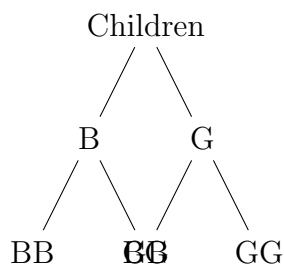
Question 10

A family has two children.

- What is the probability that both the children are boys given that at least one of them is a boy?
- What is the probability that both children are boys if it is known that the elder child is a boy?
- What is the probability that both children are girls if it is known that the elder child is a girl?

TYPE 12 Conditional Probability – Discrete Sample-Space Word Problems (parity/gender/digits)

★ Likely Exam Question



Tree diagram for gender of two children

Given: A family has two children.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Let B denote a boy and G denote a girl. The sample space for the gender order (elder, younger) of two children is:

$$\begin{aligned} S &= \{BB, BG, GB, GG\} \\ n(S) &= 4 \end{aligned}$$

Part (i): Let A = both children are boys = $\{BB\}$. Let E_1 = at least one child is a boy = $\{BB, BG, GB\}$. Here, $n(E_1) = 3$. The common outcome is $A \cap E_1 = \{BB\}$, so $n(A \cap E_1) = 1$.

$$\begin{aligned} P(A|E_1) &= \frac{n(A \cap E_1)}{n(E_1)} \\ &= \frac{1}{3} \end{aligned}$$

Part (ii): Let E_2 = elder child is a boy = $\{BB, BG\}$. Here, $n(E_2) = 2$. The common outcome is $A \cap E_2 = \{BB\}$, so $n(A \cap E_2) = 1$.

$$\begin{aligned} P(A|E_2) &= \frac{n(A \cap E_2)}{n(E_2)} \\ &= \frac{1}{2} \end{aligned}$$

Part (iii): Let C = both children are girls = $\{GG\}$. Let E_3 = elder child is a girl = $\{GB, GG\}$. Here, $n(E_3) = 2$. The common outcome is $C \cap E_3 = \{GG\}$, so $n(C \cap E_3) = 1$.

$$\begin{aligned} P(C|E_3) &= \frac{n(C \cap E_3)}{n(E_3)} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

- (i) $\frac{1}{3}$
- (ii) $\frac{1}{2}$
- (iii) $\frac{1}{2}$

Question 11

12 cards numbered 1 to 12 are placed in a box, mixed up thoroughly and then a card is drawn at random from the box. If it is known that the number on the drawn card is greater than 5, then find the probability that it is an odd number.

TYPE 13 Conditional Probability – Card Draw Sample-Space Counting

Given: Card numbers in box: $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. Drawn number > 5 .

To Find: Probability that the drawn card is an odd number.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be the sample space. Let B be the event that the number drawn is greater than 5:

$$B = \{6, 7, 8, 9, 10, 11, 12\}$$

$$n(B) = 7$$

Let A be the event that the number drawn is odd:

$$A = \{1, 3, 5, 7, 9, 11\}$$

The intersection $A \cap B$ represents numbers that are both greater than 5 and odd:

$$A \cap B = \{7, 9, 11\}$$

$$n(A \cap B) = 3$$

Therefore, the required conditional probability is:

$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

$$= \frac{3}{7}$$

Final Answer

The required probability is $\frac{3}{7}$.

Question 12

Two digits are selected at random from the digits 1 to 9. If the sum is even, then find the probability that both digits are odd.

TYPE 12 Conditional Probability – Discrete Sample-Space Word Problems (parity/gender/digits)

Given: Digits available: $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Sum of selected digits is even.

To Find: Probability that both selected digits are odd.

Concept / Formula Used: ${}^n_r = \frac{n!}{r!(n-r)!}$ $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

The given set of digits is $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, which contains:

$$\text{Odd digits} = \{1, 3, 5, 7, 9\} \quad (5 \text{ digits})$$

$$\text{Even digits} = \{2, 4, 6, 8\} \quad (4 \text{ digits})$$

Two digits are selected without order. The sum of two numbers is even if either both are odd or both are even. Let B be the event that the sum of the two digits is even:

$$\begin{aligned} n(B) &= \binom{5}{2} + \binom{4}{2} \\ &= \frac{5 \times 4}{2 \times 1} + \frac{4 \times 3}{2 \times 1} \\ &= 10 + 6 \\ &= 16 \end{aligned}$$

Let A be the event that both selected digits are odd. The number of ways to pick two odd digits is:

$$\begin{aligned} n(A \cap B) &= \binom{5}{2} \\ &= 10 \end{aligned}$$

Therefore, the conditional probability is:

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ &= \frac{10}{16} \\ &= \frac{5}{8} \end{aligned}$$

Final Answer

The required probability is $\frac{5}{8}$.

Question 13

One card is drawn from a well-shuffled pack of 52 cards. If E is the event 'the card drawn is a king or a queen' and F is the event 'the card drawn is a queen or an ace', then find the probability of conditional event $E|F$.

TYPE 13 Conditional Probability – Card Draw Sample-Space Counting

Given: Standard pack of 52 cards. Event E : King or Queen, Event F : Queen or Ace.

To Find: $P(E|F)$

Concept / Formula Used: $P(E|F) = \frac{P(E \cap F)}{P(F)}$

Solution:

Total number of cards in the deck is 52. The number of cards in event E (King or Queen):

$$n(E) = 4 \text{ Kings} + 4 \text{ Queens} = 8$$

The number of cards in event F (Queen or Ace):

$$n(F) = 4 \text{ Queens} + 4 \text{ Aces} = 8$$

Thus:

$$P(F) = \frac{8}{52}$$

The event $E \cap F$ represents drawing a card that is both (King or Queen) AND (Queen or Ace):

$$\begin{aligned} E \cap F &= \text{Queen} \\ n(E \cap F) &= 4 \text{ Queens} \end{aligned}$$

Thus:

$$P(E \cap F) = \frac{4}{52}$$

Now, calculating $P(E|F)$:

$$\begin{aligned} P(E|F) &= \frac{P(E \cap F)}{P(F)} \\ &= \frac{\frac{4}{52}}{\frac{8}{52}} \\ &= \frac{4}{8} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

$$P(E|F) = \frac{1}{2}$$

Question 14

- (i) I roll two dice and get a sum more than 9. What is the probability that the number on the first die is even?
- (ii) I roll two dice and get an even number on the first die. What is the probability that sum is more than 9?

TYPE 10 Conditional Probability – Two-Dice / Word-Problem Sample-Space Counting

Given: Two dice rolled.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Part (i): Let B be the event that the sum of the numbers is more than 9 (sum ≥ 10). Outcomes in B :

$$B = \{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$$

$$n(B) = 6$$

Let A be the event that the number on the first die is even. The common outcomes ($A \cap B$) are those in B where the first coordinate is even (4 or 6):

$$A \cap B = \{(4, 6), (6, 4), (6, 5), (6, 6)\}$$

$$n(A \cap B) = 4$$

The conditional probability is:

$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

Part (ii): Let B be the event that the number on the first die is even (2, 4, or 6). Number of outcomes in B :

$$n(B) = 3 \times 6 = 18$$

Let A be the event that the sum is more than 9. The common outcomes ($A \cap B$) are outcomes where first die is even and sum > 9 :

$$A \cap B = \{(4, 6), (6, 4), (6, 5), (6, 6)\}$$

$$n(A \cap B) = 4$$

The conditional probability is:

$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

$$= \frac{4}{18}$$

$$= \frac{2}{9}$$

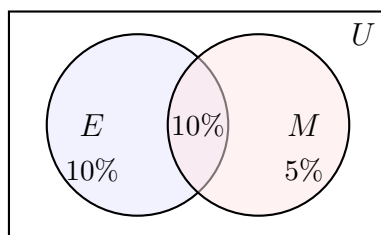
Final Answer

- (i) $\frac{2}{3}$
- (ii) $\frac{2}{9}$

Question 15

In a certain school, 20% of the students failed in English, 15% of the students failed in Mathematics and 10% of the students failed in both English and Mathematics. A student is selected at random. If he passed in English, what is the probability that he also passed in Mathematics?

TYPE 14 Conditional Probability – Word Problems with Set/Percentage Data



Venn diagram for failure in English and Mathematics

Given: $P(E) = 0.20$, $P(M) = 0.15$, $P(E \cap M) = 0.10$, where E and M denote failure in English and Mathematics respectively.

To Find: $P(M^c|E^c)$, the probability that a student passed Mathematics given that he passed English.

Concept / Formula Used: $P(E^c \cap M^c) = 1 - P(E \cup M)$

Solution:

Let E be the event that the student failed in English. Let M be the event that the student failed in Mathematics. Given values:

$$P(E) = 0.20$$

$$P(M) = 0.15$$

$$P(E \cap M) = 0.10$$

First, compute the probability that a student failed in at least one subject, $P(E \cup M)$:

$$\begin{aligned} P(E \cup M) &= P(E) + P(M) - P(E \cap M) \\ &= 0.20 + 0.15 - 0.10 \\ &= 0.25 \end{aligned}$$

The event that a student passed English is E^c .

$$\begin{aligned} P(E^c) &= 1 - P(E) \\ &= 1 - 0.20 \\ &= 0.80 \end{aligned}$$

The event that a student passed both English and Mathematics is $E^c \cap M^c = (E \cup M)^c$.

$$\begin{aligned} P(E^c \cap M^c) &= 1 - P(E \cup M) \\ &= 1 - 0.25 \\ &= 0.75 \end{aligned}$$

We require $P(M^c|E^c)$, the probability that the student passed Mathematics given that he passed English:

$$\begin{aligned} P(M^c|E^c) &= \frac{P(E^c \cap M^c)}{P(E^c)} \\ &= \frac{0.75}{0.80} \\ &= \frac{75}{80} \\ &= \frac{15}{16} \end{aligned}$$

Final Answer

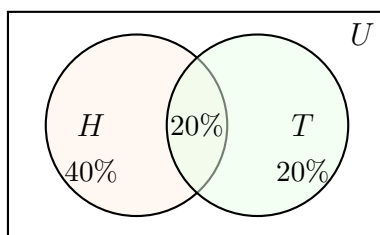
The required probability is $\frac{15}{16}$.

Question 16

60% students read Hindi newspaper, 40% students read Tamil newspaper and 20% students read both Hindi and Tamil newspaper. Find the probability that a student selected at random reads:

- Tamil newspaper given that he has already read Hindi newspaper.
- Hindi newspaper given that he has already read Tamil newspaper.
- Neither Hindi nor Tamil newspaper.

TYPE 14 Conditional Probability – Word Problems with Set/Percentage Data



Venn diagram for Hindi and Tamil newspaper readers

Given: $P(H) = 0.60$, $P(T) = 0.40$, $P(H \cap T) = 0.20$, where H and T denote reading Hindi and Tamil newspapers respectively.

Concept / Formula Used: $\frac{P(H \cap T)}{P(T)} \quad P(H^c \cap T^c) = 1 - P(H \cup T)$	$P(T H) = \frac{P(H \cap T)}{P(H)} \quad P(H T) =$
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Solution:

Given probabilities:

$$P(H) = 0.60$$

$$P(T) = 0.40$$

$$P(H \cap T) = 0.20$$

Part (i): Probability that student reads Tamil newspaper given that he reads Hindi newspaper:

$$\begin{aligned} P(T|H) &= \frac{P(H \cap T)}{P(H)} \\ &= \frac{0.20}{0.60} \\ &= \frac{2}{6} \\ &= \frac{1}{3} \end{aligned}$$

Part (ii): Probability that student reads Hindi newspaper given that he reads Tamil newspaper:

$$\begin{aligned} P(H|T) &= \frac{P(H \cap T)}{P(T)} \\ &= \frac{0.20}{0.40} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Part (iii): First, find $P(H \cup T)$:

$$\begin{aligned} P(H \cup T) &= P(H) + P(T) - P(H \cap T) \\ &= 0.60 + 0.40 - 0.20 \\ &= 0.80 \end{aligned}$$

Probability of reading neither newspaper:

$$\begin{aligned} P(H^c \cap T^c) &= 1 - P(H \cup T) \\ &= 1 - 0.80 \\ &= 0.20 \\ &= \frac{1}{5} \end{aligned}$$

Final Answer

- (i) $\frac{1}{3}$
 (ii) $\frac{1}{2}$
 (iii) $\frac{1}{5}$ (or 0.2)

Question 17

A committee of 4 students is selected from a group consisting of 7 boys and 4 girls. Find the probability that there are exactly 2 boys in the committee, given that at least one girl must be there in the committee.

TYPE 15 Conditional Probability – Combinatorial/Permutation Sample-Space Counting

Given: Total group: 7 boys, 4 girls. Committee size: 4 students.

To Find: Probability of selecting exactly 2 boys given at least 1 girl is selected.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Total number of students in the group:

$$7 \text{ boys} + 4 \text{ girls} = 11 \text{ students}$$

Total ways to form a committee of 4 students out of 11:

$$\begin{aligned} n(S) &= \binom{11}{4} \\ &= \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} \\ &= 330 \end{aligned}$$

Let B be the event that at least one girl is included in the committee. The complement event B^c is that no girls (all 4 boys) are selected:

$$\begin{aligned} n(B^c) &= \binom{7}{4} \\ &= \frac{7 \times 6 \times 5 \times 4}{4 \times 3 \times 2 \times 1} \\ &= 35 \end{aligned}$$

Therefore, the number of outcomes in event B is:

$$\begin{aligned} n(B) &= n(S) - n(B^c) \\ &= 330 - 35 \\ &= 295 \end{aligned}$$

Let A be the event that exactly 2 boys (and hence 2 girls) are in the committee. Since a committee of 2 boys and 2 girls already contains at least one girl, $A \cap B = A$. The number of ways to select 2 boys from 7 and 2 girls from 4:

$$\begin{aligned} n(A \cap B) &= \binom{7}{2} \times \binom{4}{2} \\ &= \left(\frac{7 \times 6}{2 \times 1} \right) \times \left(\frac{4 \times 3}{2 \times 1} \right) \\ &= 21 \times 6 \\ &= 126 \end{aligned}$$

Therefore, the required conditional probability is:

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ &= \frac{126}{295} \end{aligned}$$

Final Answer

The required probability is $\frac{126}{295}$.

Question 18

Three friends A, B and C got their photograph clicked. Find the probability that B is standing at the central position, given that A is standing at the left corner.

TYPE 15 Conditional Probability – Combinatorial/Permutation Sample-Space Counting

Pos 1	Pos 2	Pos 3
Left	Center	Right

Schematic representation of three standing positions

Given: Three friends A, B, C standing in a row.

To Find: Probability B is in the center given A is on the left corner.

Concept / Formula Used: $P(A|B) = \frac{n(A \cap B)}{n(B)}$

Solution:

Total arrangements of three friends A, B, C in a line:

$$n(S) = 3! = 6$$

The sample space of all arrangements is:

$$S = \{ABC, ACB, BAC, BCA, CAB, CBA\}$$

Let E_1 be the event that A is standing at the left corner (Position 1):

$$E_1 = \{ABC, ACB\}$$

$$n(E_1) = 2$$

Let E_2 be the event that B is standing at the central position (Position 2). The intersection $E_2 \cap E_1$ consists of arrangements where A is on the left corner AND B is in the central position:

$$E_2 \cap E_1 = \{ABC\}$$

$$n(E_2 \cap E_1) = 1$$

Therefore, the required conditional probability is:

$$P(E_2|E_1) = \frac{n(E_2 \cap E_1)}{n(E_1)}$$

$$= \frac{1}{2}$$

Final Answer

The required probability is $\frac{1}{2}$.

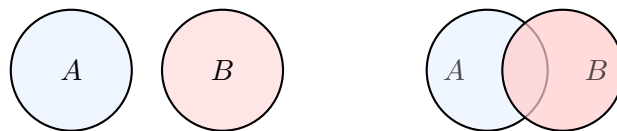
13.3 – Multiplication Theorem & Independent Events

Question 1

Fill in the blanks:

- (i) If events A and B are mutually exclusive, then $P(A \cap B)$ is _____, and if A and B are independent, then $P(A \cap B)$ is _____.
- (ii) If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$, then the statement that events A and B are independent is _____, and value of $P(A \cup B)$ is _____.
- (iii) When a die is rolled twice, the probability of getting a 2 followed by a 4 is _____.
- (iv) When two cards are drawn from a pack of cards with replacement, the probability of getting an ace followed by a king is _____, while the probability of getting two hearts is _____.
- (v) When two cards are drawn from a pack of cards without replacement, the probability of getting an ace followed by a king is _____, while the probability of getting two aces is _____.

TYPE 16 Conceptual Comparison – Mutually Exclusive vs Independent Events



Mutually Exclusive ($A \cap B = \emptyset$) Independent ($P(A \cap B) = P(A)P(B)$)

Comparison of Mutually Exclusive and Independent Events

Given: Events A and B under various probability conditions.

Concept / Formula Used: $P(A \cap B) = 0$ $P(A \cap B) = P(A) \cdot P(B)$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Solution:

- (i) For mutually exclusive events, no outcome is common, so $A \cap B = \emptyset$, giving:

$$P(A \cap B) = 0$$

For independent events, by definition of independence:

$$P(A \cap B) = P(A) \cdot P(B)$$

- (ii) Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$. Calculating the product of individual probabilities:

$$\begin{aligned} P(A) \cdot P(B) &= 0.5 \times 0.4 \\ &= 0.20 \end{aligned}$$

Since $P(A \cap B) = P(A) \cdot P(B) = 0.2$, the statement that A and B are independent is **True**.

Now, applying the addition theorem of probability:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.5 + 0.4 - 0.2 \\ &= 0.7 \end{aligned}$$

- (iii) Total outcomes when a die is rolled once is 6. The probability of getting a 2 on the first roll:

$$P(2 \text{ on 1st roll}) = \frac{1}{6}$$

The probability of getting a 4 on the second roll:

$$P(4 \text{ on 2nd roll}) = \frac{1}{6}$$

Since the two rolls are independent:

$$\begin{aligned} P(2 \text{ followed by } 4) &= \frac{1}{6} \times \frac{1}{6} \\ &= \frac{1}{36} \end{aligned}$$

- (iv) Total cards in a standard deck = 52. Number of aces = 4, number of kings = 4, number of hearts = 13. Since cards are drawn **with replacement**, events are independent.

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{King}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{Ace followed by King}) = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$$

For drawing two hearts with replacement:

$$P(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$$

$$P(\text{Two Hearts}) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

- (v) Since cards are drawn **without replacement**:

$$P(\text{1st card Ace}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{2nd card King} \mid \text{1st card Ace}) = \frac{4}{51}$$

$$P(\text{Ace followed by King}) = \frac{1}{13} \times \frac{4}{51} = \frac{4}{663}$$

For drawing two aces without replacement:

$$P(\text{1st card Ace}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{2nd card Ace} \mid \text{1st card Ace}) = \frac{3}{51} = \frac{1}{17}$$

$$P(\text{Two Aces}) = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}$$

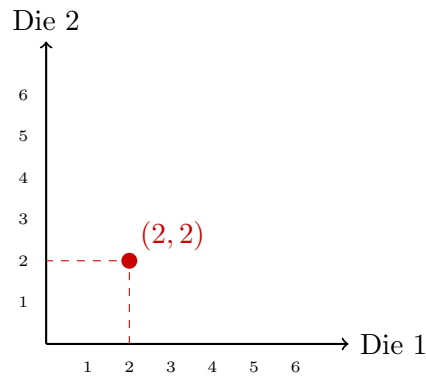
Final Answer

- (i) 0; $P(A) \cdot P(B)$
 (ii) True (or independent); 0.7
 (iii) $\frac{1}{36}$
 (iv) $\frac{1}{169}$; $\frac{1}{16}$
 (v) $\frac{4}{663}$; $\frac{1}{221}$

Question 2

What is the probability of obtaining an even prime number on each die when a pair of dice is rolled?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)



Sample Space Grid for Two Dice

Given: A pair of fair six-sided dice is rolled.

Concept / Formula Used: $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

Solution:

The sample space for a single fair die is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

The only prime number that is even in S is 2.

Let E_1 be the event of obtaining an even prime number on the first die.

$$\begin{aligned} E_1 &= \{2\} \\ P(E_1) &= \frac{1}{6} \end{aligned}$$

Let E_2 be the event of obtaining an even prime number on the second die.

$$\begin{aligned} E_2 &= \{2\} \\ P(E_2) &= \frac{1}{6} \end{aligned}$$

Since the outcome on the second die is independent of the outcome on the first die:

$$\begin{aligned} P(E_1 \cap E_2) &= P(E_1) \cdot P(E_2) \\ &= \frac{1}{6} \times \frac{1}{6} \\ &= \frac{1}{36} \end{aligned}$$

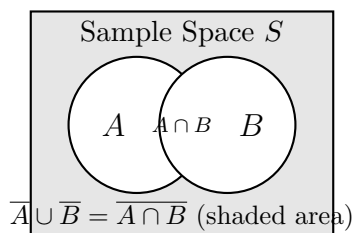
Final Answer

The probability of obtaining an even prime number on each die is $\frac{1}{36}$.

Question 3

- (i) Events A and B are such that $P(A) = \frac{1}{2}$, $P(B) = \frac{7}{12}$ and $P(\overline{A} \cup \overline{B}) = \frac{1}{4}$. State whether A and B are independent.
- (ii) Let E and F be events with $P(E) = \frac{3}{5}$, $P(F) = \frac{3}{10}$ and $P(E \cap F) = \frac{1}{5}$. Are E and F independent?

TYPE 18 Independence Verification (Check $P(A \cap B) = P(A)P(B)$)



Venn Diagram for De Morgan's Law

Given: (i) $P(A) = \frac{1}{2}$, $P(B) = \frac{7}{12}$, $P(\overline{A} \cup \overline{B}) = \frac{1}{4}$
(ii) $P(E) = \frac{3}{5}$, $P(F) = \frac{3}{10}$, $P(E \cap F) = \frac{1}{5}$

Concept / Formula Used: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ $P(A \cap B) = 1 - P(\overline{A \cap B})$ $P(X \cap Y) = P(X) \cdot P(Y)$

Solution:

(i) Using De Morgan's Law: $\overline{A \cap B} = \overline{A} \cup \overline{B}$ Taking probabilities on both sides:

$$P(\overline{A \cap B}) = P(\overline{A \cap B}) = \frac{1}{4} \quad \text{Using the complement probability identity:} \quad P(A \cap B) = 1 - P(\overline{A \cap B})$$

$$= 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

Now calculate the product of individual probabilities:

$$P(A) \cdot P(B) = \frac{1}{2} \times \frac{7}{12} = \frac{7}{24}$$

Ex- pressing $P(A \cap B)$ with common denominator 24: $P(A \cap B) = \frac{3}{4} = \frac{18}{24}$ Since $\frac{18}{24} \neq \frac{7}{24}$, we have: $P(A \cap B) \neq P(A) \cdot P(B)$ Therefore, events A and B are **not independent**.

(ii) Calculate the product of individual probabilities of events E and F :

$$P(E) \cdot P(F) = \frac{3}{5} \times \frac{3}{10} = \frac{9}{50}$$

Given probability of intersection: $P(E \cap F) = \frac{1}{5} = \frac{10}{50}$ Comparing the two values:

$$\frac{10}{50} \neq \frac{9}{50} \quad \text{Therefore, events } E \text{ and } F \text{ are not independent.}$$

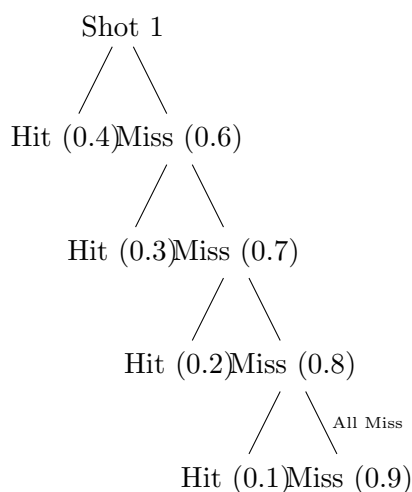
$$P(E \cap F) \neq P(E) \cdot P(F)$$

Final Answer

- (i) Events A and B are **not independent**.
 (ii) Events E and F are **not independent**.

Question 4

- (i) If $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$, then find $P(A \cap B)$ if A and B are independent events.
- (ii) Given two independent events A and B such that $P(A) = 0.3$ and $P(B) = 0.6$, find $P(A' \cap B')$.
- (iii) If events A and B are independent such that $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{3}$, find $P(A \cup B)$.
- (iv) If A and B are two independent events such that $P(\text{not } B) = 0.65$ and $P(A \cup B) = 0.85$, then find $P(A)$.
- (v) The probability of event A occurring is $\frac{1}{3}$ and of event B occurring is $\frac{1}{2}$. If A and B are independent events, then find the probability of neither A nor B occurring.
- (vi) A and B appeared for an interview for two vacancies. The probability of A 's selection is $\frac{4}{5}$ and B 's selection is $\frac{1}{3}$. Find the probability that none of them will be selected.
- (vii) Teena is practising for an upcoming Rifle Shooting tournament. The probability of her shooting the target in the 1st, 2nd, 3rd and 4th shots are 0.4, 0.3, 0.2 and 0.1 respectively. Find the probability of at least one shot of Teena hitting the target.

TYPE 19 Solving for Unknown Parameters Using Independence

Tree Diagram for Independent Shots

Given: Various probability values for independent events across subparts (i) to (vii).

Concept / Formula Used: $P(A \cap B) = P(A) \cdot P(B)$ $P(A' \cap B') = P(A') \cdot P(B')$
 $P(\text{At least one}) = 1 - P(\text{None})$

Solution:

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\begin{aligned} \text{(i) Since } A \text{ and } B \text{ are independent:} &= \frac{3}{5} \times \frac{1}{5} \\ &= \frac{3}{25} \end{aligned}$$

(ii) Since A and B are independent, their complements A' and B' are also independent.

Calculate individual complement probabilities: $P(A') = 1 - P(A) = 1 - 0.3 = 0.7$
 $P(B') = 1 - P(B) = 1 - 0.6 = 0.4$

$$P(A' \cap B') = P(A') \cdot P(B')$$

$$\begin{aligned} \text{Now calculate } P(A' \cap B'): &= 0.7 \times 0.4 \\ &= 0.28 \end{aligned}$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\begin{aligned} \text{(iii) Since } A \text{ and } B \text{ are independent:} &= \frac{3}{5} \times \frac{2}{3} && \text{Applying the addition} \\ &= \frac{2}{5} \end{aligned}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned} &= \frac{3}{5} + \frac{2}{3} - \frac{2}{5} \\ &= \left(\frac{3}{5} - \frac{2}{5} \right) + \frac{2}{3} \\ \text{formula:} &= \frac{1}{5} + \frac{2}{3} \\ &= \frac{3+10}{15} \\ &= \frac{13}{15} \end{aligned}$$

(iv) Given $P(\text{not } B) = P(B') = 0.65$. $P(B) = 1 - P(B') = 1 - 0.65 = 0.35$ Since A and B are independent, $P(A \cap B) = P(A) \cdot P(B) = 0.35P(A)$. Substitute into

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.85 = P(A) + 0.35 - 0.35P(A)$$

$$\text{addition formula: } 0.85 - 0.35 = P(A)(1 - 0.35)$$

$$0.50 = 0.65P(A)$$

$$P(A) = \frac{0.50}{0.65} = \frac{50}{65} = \frac{10}{13}$$

$$(v) \quad P(\text{neither } A \text{ nor } B) = P(A' \cap B').$$

$$P(A') = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(B') = 1 - P(B) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(A' \cap B') = P(A') \cdot P(B')$$

$$\begin{aligned} \text{are independent:} \quad &= \frac{2}{3} \times \frac{1}{2} \\ &= \frac{1}{3} \end{aligned}$$

(vi) Let A be the event of A 's selection and B be the event of B 's selection. Selections

$$P(A') = 1 - P(A) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$P(B') = 1 - P(B) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(A' \cap B') = P(A') \cdot P(B')$$

$$\begin{aligned} \text{selected:} \quad &= \frac{1}{5} \times \frac{2}{3} \\ &= \frac{2}{15} \end{aligned}$$

(vii) Let T_1, T_2, T_3, T_4 be the events of hitting the target in 1st, 2nd, 3rd, 4th shots. Given $P(T_1) = 0.4$, $P(T_2) = 0.3$, $P(T_3) = 0.2$, $P(T_4) = 0.1$. Calculate probabilities

$$P(T'_1) = 1 - 0.4 = 0.6$$

$$P(T'_2) = 1 - 0.3 = 0.7$$

$$P(T'_3) = 1 - 0.2 = 0.8$$

$$P(T'_4) = 1 - 0.1 = 0.9$$

$$P(\text{all miss}) = P(T'_1) \cdot P(T'_2) \cdot P(T'_3) \cdot P(T'_4)$$

$$= 0.6 \times 0.7 \times 0.8 \times 0.9$$

$$= 0.42 \times 0.72$$

$$= 0.3024$$

$$P(\text{at least one hit}) = 1 - P(\text{all miss})$$

$$\text{the target:} \quad = 1 - 0.3024$$

$$= 0.6976$$

Final Answer

(i) $\frac{3}{25}$

(ii) 0.28

(iii) $\frac{13}{15}$

(iv) $\frac{10}{13}$

(v) $\frac{1}{3}$

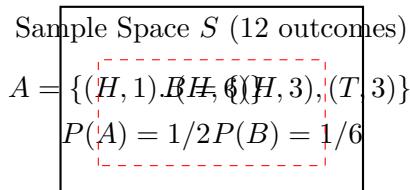
(vi) $\frac{2}{15}$

(vii) 0.6976

Question 5

- (i) A fair coin and an unbiased die are tossed. Let A be the event 'head appears on the coin' and B be the event '3 on the die'. Check whether A and B are independent events or not.
- (ii) A die is thrown. If E is the event 'the number appearing is a multiple of 3' and F be the event 'the number appearing is even', then find whether E and F are independent.

TYPE 18 Independence Verification (Check $P(A \cap B) = P(A)P(B)$)



Sample Space Diagram for Coin and Die

Given: (i) Toss of a fair coin and an unbiased die. A = 'head', B = '3'.
 (ii) Throw of a fair die. E = 'multiple of 3', F = 'even number'.

Concept / Formula Used: $P(A \cap B) = P(A) \cdot P(B)$

Solution:

- (i) Total outcomes in sample space S : $S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$
- $n(S) = 12$
- $A = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6)\} \implies n(A) = 6$
- Events A , B , and $A \cap B$: $B = \{(H, 3), (T, 3)\} \implies n(B) = 2$
- $A \cap B = \{(H, 3)\} \implies n(A \cap B) = 1$
- $P(A) = \frac{6}{12} = \frac{1}{2}$
- Calculating probabilities: $P(B) = \frac{2}{12} = \frac{1}{6}$ Check independence condition: $P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$
- $P(A \cap B) = \frac{1}{12}$
- Since $P(A \cap B) = P(A) \cdot P(B)$, A and B are **independent events**.
- (ii) Sample space for throwing a die: $S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$ Events E , F ,
- $E = \{3, 6\} \implies n(E) = 2$ $P(E) = \frac{2}{6} = \frac{1}{3}$
- and $E \cap F$: $F = \{2, 4, 6\} \implies n(F) = 3$ Calculating probabilities: $P(F) = \frac{3}{6} = \frac{1}{2}$
- $E \cap F = \{6\} \implies n(E \cap F) = 1$ $P(E \cap F) = \frac{1}{6}$

Check independence condition: $P(E) \cdot P(F) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$ Since $P(E \cap F) = P(E) \cdot P(F)$, events E and F are **independent**.

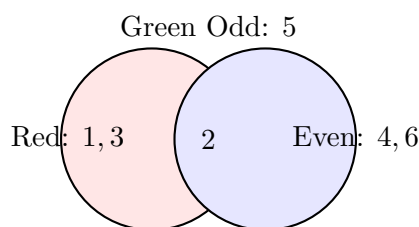
Final Answer

- (i) A and B are **independent events**.
 (ii) E and F are **independent events**.

Question 6

- (i) A die marked 1, 2, 3 in red and 4, 5, 6 in green is tossed. Let A be the event 'number is even' and B be the event 'number is red'. Are A and B independent?
 (ii) If $P(A) = 0.2$, $P(B) = p$, $P(A \cup B) = 0.6$ and A and B are given to be independent events, find the value of p .

TYPE 18 Independence Verification (Check $P(A \cap B) = P(A)P(B)$)



Venn Diagram of Die Outcomes

- Given:** (i) Die faces: 1, 2, 3 (red); 4, 5, 6 (green). A = 'even', B = 'red'.
 (ii) $P(A) = 0.2$, $P(B) = p$, $P(A \cup B) = 0.6$, A and B are independent.

Concept / Formula Used: $P(A \cap B) = P(A) \cdot P(B)$

Solution:

- (i) Sample space $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$. Events defined as: $A = \{2, 4, 6\} \implies n(A) = 3$
 $B = \{1, 2, 3\} \implies n(B) = 3$
 $A \cap B = \{2\} \implies n(A \cap B) = 1$

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

Calculating probabilities: $P(B) = \frac{3}{6} = \frac{1}{2}$ Calculating product of individual prob-

$$P(A \cap B) = \frac{1}{6}$$

abilities: $P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ Comparing $P(A \cap B)$ and $P(A) \cdot P(B)$: $\frac{1}{6} \neq \frac{1}{4}$
 $P(A \cap B) \neq P(A) \cdot P(B)$

Hence, A and B are **not independent**.

$$P(A \cap B) = P(A) \cdot P(B)$$

(ii) Since A and B are independent events: $= 0.2 \times p$ Apply addition
 $= 0.2p$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.6 = 0.2 + p - 0.2p$$

$$0.6 - 0.2 = p(1 - 0.2)$$

theorem of probability: $0.4 = 0.8p$

$$p = \frac{0.4}{0.8}$$

$$p = 0.5 = \frac{1}{2}$$

Final Answer

(i) A and B are **not independent**.

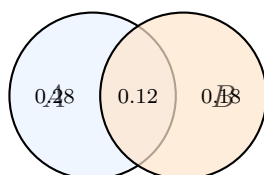
(ii) $p = 0.5$ (or $\frac{1}{2}$)

Question 7

Bhuvan is preparing chapati and curry for a dinner party at his home. Let event A = Bhuvan prepares chapati well. Let event B = Bhuvan cooks curry well. Given $P(A) = 0.40$, $P(B) = 0.30$ and $P(A \cap B) = 0.12$. Which one of the following is true about the events A and B ?

- (a) They are mutually exclusive but not independent.
- (b) They are independent but not mutually exclusive.

TYPE 16 Conceptual Comparison – Mutually Exclusive vs Independent Events



Venn Diagram for Chapati and Curry Preparation

Given: $P(A) = 0.40$, $P(B) = 0.30$, $P(A \cap B) = 0.12$

Concept / Formula Used: $P(A \cap B) = 0$ $P(A \cap B) = P(A) \cdot P(B)$

Solution:

First, check for mutual exclusivity: $P(A \cap B) = 0.12 \neq 0$ Since $P(A \cap B) \neq 0$, the events A and B are **not mutually exclusive**.

Second, check for independence: $P(A) \cdot P(B) = 0.40 \times 0.30$ Since $P(A \cap B) = P(A) \cdot P(B)$
 $= 0.12$
 $P(B) = 0.12$, the events A and B are **independent**.

Therefore, the events are independent but not mutually exclusive, matching option (b).

Final Answer

(b) They are independent but not mutually exclusive.

Question 8

A coin is tossed thrice and all the eight outcomes are assumed to be equally likely. In which of the cases are the events E_1 and E_2 independent? In which cases are they mutually exclusive?

- (i) E_1 : the first throw results in head
 E_2 : the last throw results in tail
- (ii) E_1 : the number of heads is two
 E_2 : the last throw results in head
- (iii) E_1 : the number of heads is odd
 E_2 : the number of tails is odd.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

E_1 outcomes

E_2 outcomes

Sample space outcomes for three coin tosses

Given: Sample space $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ with total number of outcomes $n(S) = 8$.

Concept / Formula Used: $P(A \cap B) = P(A) \cdot P(B)$ $A \cap B = \emptyset$ $P(A \cap B) = 0$

Solution:

$$E_1 = \{HHH, HHT, HTH, HTT\} \implies n(E_1) = 4$$

$$P(E_1) = \frac{4}{8} = \frac{1}{2}$$

$$E_2 = \{HHT, HTT, THT, TTT\} \implies n(E_2) = 4$$

(i) For part (i):

$$P(E_2) = \frac{4}{8} = \frac{1}{2}$$

Now check-

$$E_1 \cap E_2 = \{HHT, HTT\} \implies n(E_1 \cap E_2) = 2$$

$$P(E_1 \cap E_2) = \frac{2}{8} = \frac{1}{4}$$

ing independence: $P(E_1) \cdot P(E_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ Since $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) = \frac{1}{4}$, the events E_1 and E_2 are **independent**.

Since $E_1 \cap E_2 = \{HHT, HTT\} \neq \emptyset$, $P(E_1 \cap E_2) \neq 0$, so they are **not mutually exclusive**.

$$E_1 = \{HHT, HTH, THH\} \Rightarrow n(E_1) = 3$$

$$P(E_1) = \frac{3}{8}$$

$$E_2 = \{HHH, HTH, THH, TTH\} \Rightarrow n(E_2) = 4$$

(ii) For part (ii):

$$P(E_2) = \frac{4}{8} = \frac{1}{2}$$

Checking

$$E_1 \cap E_2 = \{HTH, THH\} \Rightarrow n(E_1 \cap E_2) = 2$$

$$P(E_1 \cap E_2) = \frac{2}{8} = \frac{1}{4}$$

independence: $P(E_1) \cdot P(E_2) = \frac{3}{8} \times \frac{1}{2} = \frac{3}{16}$ Since $P(E_1 \cap E_2) = \frac{1}{4} \neq \frac{3}{16}$, the events E_1 and E_2 are **not independent**.

Since $E_1 \cap E_2 \neq \emptyset$, they are **not mutually exclusive**.

$$E_1 = \{HHH, HTT, THT, TTH\} \Rightarrow n(E_1) = 4$$

$$P(E_1) = \frac{4}{8} = \frac{1}{2}$$

(iii) For part (iii):

$$E_2 = \{TTT, THH, HTH, HHT\} \Rightarrow n(E_2) = 4$$

Since any out-

$$P(E_2) = \frac{4}{8} = \frac{1}{2}$$

come with an odd number of heads in 3 tosses has an even number of tails: $E_1 \cap E_2 = \emptyset \Rightarrow P(E_1 \cap E_2) = 0$. Since $E_1 \cap E_2 = \emptyset$, the events E_1 and E_2 are **mutually exclusive**.

Checking independence: $P(E_1) \cdot P(E_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ Since $P(E_1 \cap E_2) = 0 \neq \frac{1}{4}$, they are **not independent**.

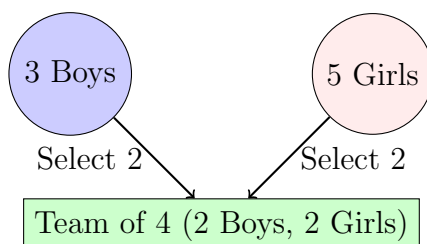
Final Answer

- (i) Independent, but not mutually exclusive.
- (ii) Neither independent nor mutually exclusive.
- (iii) Mutually exclusive, but not independent.

Question 9

- (i) Out of 8 outstanding students of a school, in which there are 3 boys and 5 girls, a team of 4 students is to be selected for a quiz competition. Find the probability that 2 boys and 2 girls are selected.
- (ii) Two cards are drawn at random and one by one without replacement from a well-shuffled pack of 52 playing cards. Find the probability that one card is red and the other is black.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)



Selection schema for boys and girls

Given:

- (i) Total students $N = 8$ (Boys $B = 3$, Girls $G = 5$). Number of students to select $k = 4$.
 (ii) Total cards $N = 52$ (Red cards $R = 26$, Black cards $B = 26$). Number of cards drawn $k = 2$ without replacement.

Concept / Formula Used: $P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}} = \frac{\binom{N_1}{r_1} \times \binom{N_2}{r_2}}{\binom{N}{k}}$

Solution:

(i) Total number of ways to select 4 students out of 8: $n(S) = \binom{8}{4} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$
 $n(E) = \binom{3}{2} \times \binom{5}{2}$

Number of ways to select 2 boys from 3 and 2 girls from 5: $\binom{3}{2} = 3$
 $\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$
 $n(E) = 3 \times 10 = 30$

Therefore, the required probability is: $P(E) = \frac{n(E)}{n(S)} = \frac{30}{70} = \frac{3}{7}$

(ii) Total number of ways to draw 2 cards out of 52: $n(S) = \binom{52}{2} = \frac{52 \times 51}{2 \times 1} = 1326$

Number of ways to select 1 red card from 26 and 1 black card from 26: $n(E) = \binom{26}{1} \times \binom{26}{1} = 26$

Therefore, the required probability is: $P(E) = \frac{676}{1326}$ Dividing numerator and denominator by 26: $P(E) = \frac{26}{51}$

Final Answer

- (i) Probability = $\frac{3}{7}$
 (ii) Probability = $\frac{26}{51}$

Question 10

A bag contains 19 tickets, numbered from 1 to 19. Two tickets are drawn randomly in succession with replacement. Find the probability that both the tickets drawn are even numbered.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

★ Likely Exam Question

Given: Total tickets in bag = 19 (numbered 1 to 19). Two draws made with replacement.

Concept / Formula Used: $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

Solution:

The even-numbered tickets in the range 1 to 19 are: {2, 4, 6, 8, 10, 12, 14, 16, 18} Counting the even-numbered tickets: Number of even tickets = 9

Total tickets = 19 Let E_1 be the event that the first ticket drawn is even, and E_2 be the event that the second ticket drawn is even. $P(E_1) = \frac{9}{19}$ Since the draw is done with replacement, the total number of tickets and

the number of even tickets remain unchanged for the second draw: $P(E_2) = \frac{9}{19}$ Since

$$P(\text{both tickets are even}) = P(E_1) \cdot P(E_2)$$

the draws are independent:

$$\begin{aligned} &= \frac{9}{19} \times \frac{9}{19} \\ &= \frac{81}{361} \end{aligned}$$

Final Answer

$$\text{Required Probability} = \frac{81}{361}$$

Question 11

A company has estimated that the probabilities of success for three products introduced in the market are $\frac{1}{3}$, $\frac{2}{5}$ and $\frac{2}{3}$ respectively. Assuming independence, find the probability that at least one product is successful.

TYPE 20 Probability of At-Least-One / All-Fail Using Complement & Independence

★ Likely Exam Question

Given: Events A, B, C are independent with $P(A) = \frac{1}{3}$, $P(B) = \frac{2}{5}$, $P(C) = \frac{2}{3}$.

Concept / Formula Used: $P(\text{at least one successful}) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) = 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$

Solution:

Let A, B, C denote the events that the first, second, and third products are successful re-

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$$

spectively. The probabilities of failure for each product are: $P(\bar{B}) = 1 - P(B) = 1 - \frac{2}{5} = \frac{3}{5}$

$$P(\bar{C}) = 1 - P(C) = 1 - \frac{2}{3} = \frac{1}{3}$$

Since the products succeed or fail independently, the complement events $\bar{A}, \bar{B}, \bar{C}$ are also

$$P(\text{none of the products is successful}) = P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

independent:

$$= \frac{2}{3} \times \frac{3}{5} \times \frac{1}{3} \quad \text{Using the com-}$$

$$= \frac{6}{45} = \frac{2}{15}$$

$$P(\text{at least one product is successful}) = 1 - P(\text{none is successful})$$

plementary probability formula:

$$= 1 - \frac{2}{15}$$

$$= \frac{13}{15}$$

Final Answer

$$\text{Required Probability} = \frac{13}{15}$$

Question 12

A bag contains 4 white, 3 red and 2 blue balls. Four balls are drawn one by one. Find the probability that all balls are white when the drawing is done:

- without replacement.
- with replacement.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Total = 9 balls

Bag: 4 W, 3 R, 2 B

Bag contents representation

Given: White balls $W = 4$, Red balls $R = 3$, Blue balls $B = 2$. Total balls = $4 + 3 + 2 = 9$. Number of balls drawn = 4.

Concept / Formula Used: $P(W_1 \cap W_2 \cap W_3 \cap W_4) = P(W_1) \cdot P(W_2|W_1) \cdot P(W_3|W_1W_2) \cdot P(W_4|W_1W_2W_3)$ $P(W_1 \cap W_2 \cap W_3 \cap W_4) = [P(W)]^4$

Solution:

$$\begin{aligned}
 P(\text{1st ball is white}) &= \frac{4}{9} \\
 P(\text{2nd ball is white}) &= \frac{3}{8} \\
 P(\text{3rd ball is white}) &= \frac{2}{7} \\
 P(\text{4th ball is white}) &= \frac{1}{6}
 \end{aligned}$$

(i) Without replacement: Multiplying the conditional probabilities:

$$\begin{aligned}
 P(\text{all 4 white}) &= \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{1}{6} \\
 &= \frac{24}{3024} \\
 &= \frac{1}{126}
 \end{aligned}$$

(ii) With replacement: In each draw, the probability of drawing a white ball remains

constant: $P(W) = \frac{4}{9}$ Since each draw is independent:

$$\begin{aligned}
 P(\text{all 4 white}) &= \left(\frac{4}{9}\right)^4 \\
 &= \frac{256}{6561}
 \end{aligned}$$

Final Answer

(i) Without replacement: $\frac{1}{126}$

(ii) With replacement: $\frac{256}{6561}$

Question 13

A bag contains 4 white, 3 red and 2 blue balls. Four balls are drawn one by one. Find the probability that no ball is white when the drawing is done:

- without replacement.
- with replacement.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: White balls $W = 4$, Non-white balls (Red + Blue) $= 3 + 2 = 5$. Total balls $= 9$. Number of balls drawn $= 4$.

Concept / Formula Used: $P(E) = \frac{\binom{N_{\text{non-white}}}{k}}{\binom{N_{\text{total}}}{k}} \quad P(E) = [P(\text{non-white})]^k$

Solution:

(i) Without replacement: Drawing 4 balls with no white ball means all 4 balls must

be drawn from the 5 non-white balls.

$$P(\text{1st non-white}) = \frac{5}{9}$$

$$P(\text{2nd non-white}) = \frac{4}{8}$$

$$P(\text{3rd non-white}) = \frac{3}{7}$$

$$P(\text{4th non-white}) = \frac{2}{6}$$

Multiplying these

$$P(\text{no white}) = \frac{5}{9} \times \frac{4}{8} \times \frac{3}{7} \times \frac{2}{6}$$

$$\begin{aligned} \text{probabilities:} &= \frac{120}{3024} \\ &= \frac{5}{126} \end{aligned}$$

(ii) With replacement: The probability of drawing a non-white ball in any single draw

$$\begin{aligned} \text{is: } P(\text{non-white}) &= \frac{5}{9} \quad \text{Since each draw is independent:} & P(\text{no white}) &= \left(\frac{5}{9}\right)^4 \\ & & &= \frac{625}{6561} \end{aligned}$$

Final Answer

(i) Without replacement: $\frac{5}{126}$

(ii) With replacement: $\frac{625}{6561}$

Question 14

- (i) A bag contains 5 red, 6 white and 7 black balls. Two balls are drawn at random. What is the probability that both balls are red or both are black?
- (ii) There are 2 red and 3 black balls in a bag. Three balls are taken out at random from the bag. Find the probability of getting 2 red and 1 black ball or 1 red and 2 black balls.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given:

- (i) Red $R = 5$, White $W = 6$, Black $B = 7$. Total = 18. Draw $k = 2$.
- (ii) Red $R = 2$, Black $B = 3$. Total = 5. Draw $k = 3$.

Concept / Formula Used: $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Solution:

- (i) Total ways to draw 2 balls from 18 balls: $n(S) = \binom{18}{2} = \frac{18 \times 17}{2 \times 1} = 153$ Let E_1 be the event of drawing 2 red balls, and E_2 be the event of drawing 2 black

$$\begin{aligned}
 \text{balls.} \quad n(E_1) &= \binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10 \implies P(E_1) = \frac{10}{153} \\
 n(E_2) &= \binom{7}{2} = \frac{7 \times 6}{2 \times 1} = 21 \implies P(E_2) = \frac{21}{153} \quad \text{Since } E_1 \text{ and } E_2 \text{ are mutually} \\
 P(\text{both red or both black}) &= P(E_1) + P(E_2) \\
 \text{exclusive:} \quad &= \frac{10}{153} + \frac{21}{153} \\
 &= \frac{31}{153}
 \end{aligned}$$

(ii) Total ways to draw 3 balls from 5 balls: $n(S) = \binom{5}{3} = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$ Let A be the event of getting 2 red and 1 black ball, and B be the event of getting 1 red and 2

$$\begin{aligned}
 \text{black balls.} \quad n(A) &= \binom{2}{2} \times \binom{3}{1} = 1 \times 3 = 3 \\
 n(B) &= \binom{2}{1} \times \binom{3}{2} = 2 \times 3 = 6 \quad \text{Since } A \text{ and } B \text{ are mutually exclusive:} \\
 P(A \cup B) &= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} \\
 &= \frac{3}{10} + \frac{6}{10} \\
 &= \frac{9}{10}
 \end{aligned}$$

Final Answer

$$\begin{aligned}
 \text{(i) Probability} &= \frac{31}{153} \\
 \text{(ii) Probability} &= \frac{9}{10}
 \end{aligned}$$

Question 15

A bag contains 5 white, 7 red and 4 black balls. Four balls are drawn one by one with replacement. What is the probability that none is white?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: White $W = 5$, Red $R = 7$, Black $B = 4$. Total = $5 + 7 + 4 = 16$. Non-white balls = $7 + 4 = 11$. Number of draws = 4 with replacement.

Concept / Formula Used: $P(\text{none white}) = [P(\text{non-white})]^4$

Solution:

The total number of balls in the bag is: Total balls = $5 + 7 + 4 = 16$ The number of non-white balls (red or black) is: Non-white balls = $7 + 4 = 11$ In any single draw, the probability of drawing a non-white ball is: $P(\text{non-white}) = \frac{11}{16}$ Since the four draws are

$$P(\text{none is white}) = \left(\frac{11}{16}\right)^4$$

done with replacement, each draw is independent:

$$= \frac{11^4}{16^4} = \frac{14641}{65536}$$

Final Answer

$$\text{Required Probability} = \frac{14641}{65536}$$

Question 16

A bag contains 1 white, 5 red and 4 black balls. If three balls are drawn one by one without replacement, then find the probability of drawing a white ball followed by two red balls.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: White $W = 1$, Red $R = 5$, Black $B = 4$. Total = 10. Draws = 3 without replacement. Sequence requested: $W \rightarrow R \rightarrow R$.

Concept / Formula Used: $P(W_1 \cap R_2 \cap R_3) = P(W_1) \cdot P(R_2|W_1) \cdot P(R_3|W_1 \cap R_2)$

Solution:

The total number of balls initially present is: $N = 1 + 5 + 4 = 10$. We need to find the probability of the sequence (W, R, R) : $P(W_1)$ = Probability of 1st ball being white = $\frac{1}{10}$

After drawing 1 white ball, 9 balls remain in the bag (5 red and 4 black): $P(R_2|W_1)$ = Probability of 2nd

After drawing 1 red ball, 8 balls remain in the bag (4 red and 4 black): $P(R_3|W_1 \cap R_2)$ = Probability of 3rd

$$P(W_1 \cap R_2 \cap R_3) = \frac{1}{10} \times \frac{5}{9} \times \frac{4}{8}$$

Multiplying the sequential probabilities:

$$= \frac{20}{720} = \frac{1}{36}$$

Final Answer

$$\text{Required Probability} = \frac{1}{36}$$

Question 17

From a bag containing 20 tickets, numbered from 1 to 20, two tickets are drawn at random. Find the probability that

- both the tickets have prime numbers on them
- on one there is a prime number and on the other there is a multiple of 4.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Tickets numbered 1 to 20. $N = 20$. Draw $k = 2$ tickets at random.

Concept / Formula Used: $P(E) = \frac{n(E)}{n(S)}$ $n(S) = {}^{(20)}_2$

Solution:

Total number of ways to draw 2 tickets out of 20: $n(S) = {}^{(20)}_2 = \frac{20 \times 19}{2 \times 1} = 190$ Identify the subsets of numbers from 1 to 20:

Prime numbers = $\{2, 3, 5, 7, 11, 13, 17, 19\} \implies \text{Count} = 8$

Multiples of 4 = $\{4, 8, 12, 16, 20\} \implies \text{Count} = 5$

Note that the sets of prime numbers and multiples of 4 are disjoint.

(i) Both tickets have prime numbers: Number of ways to select 2 prime numbers out

$$\text{of 8: } n(E_1) = {}^{(8)}_2 = \frac{8 \times 7}{2 \times 1} = 28$$

$$P(E_1) = \frac{28}{190} = \frac{14}{95}$$

(ii) One prime number and one multiple of 4: Number of ways to select 1 prime number

$$\text{from 8 and 1 multiple of 4 from 5: } n(E_2) = {}^{(8)}_1 \times {}^{(5)}_1 = 8 \times 5 = 40$$

$$P(E_2) = \frac{40}{190} = \frac{4}{19}$$

Final Answer

$$(i) \text{ Probability} = \frac{14}{95}$$

$$(ii) \text{ Probability} = \frac{4}{19}$$

Question 18

An urn contains 4 red and 7 blue balls. Two balls are drawn at random with replacement. Find the probability of getting

- (i) 2 red balls
- (ii) 2 blue balls
- (iii) one red and one blue ball.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Red balls $R = 4$, Blue balls $B = 7$. Total balls = 11. Number of draws = 2 with replacement.

Concept / Formula Used: $P(R) = \frac{4}{11}$ $P(B) = \frac{7}{11}$

Solution:

In any single draw: $P(R) = \frac{4}{11}$ Since the draws are with replacement, each draw is independent.
 $P(B) = \frac{7}{11}$

$$(i) \text{ Probability of getting 2 red balls (RR): } P(RR) = P(R) \cdot P(R) = \frac{4}{11} \times \frac{4}{11} = \frac{16}{121}$$

$$(ii) \text{ Probability of getting 2 blue balls (BB): } P(BB) = P(B) \cdot P(B) = \frac{7}{11} \times \frac{7}{11} = \frac{49}{121}$$

(iii) Probability of getting one red and one blue ball: This can happen in two orderings:

$$P(RB) = P(R) \cdot P(B) = \frac{4}{11} \times \frac{7}{11} = \frac{28}{121}$$

$$P(BR) = P(B) \cdot P(R) = \frac{7}{11} \times \frac{4}{11} = \frac{28}{121}$$

(R, B) or (B, R) .

$$P(\text{one red, one blue}) = P(RB) + P(BR) = \frac{28}{121} + \frac{28}{121} = \frac{56}{121}$$

Final Answer

$$(i) P(2 \text{ red}) = \frac{16}{121}$$

$$(ii) P(2 \text{ blue}) = \frac{49}{121}$$

$$(iii) P(1 \text{ red and 1 blue}) = \frac{56}{121}$$

Question 19

Three balls are drawn one by one without replacement from an urn containing 3 white, 3 red and 3 black balls. What is the probability that at least one ball is red?

TYPE 20 Probability of At-Least-One / All-Fail Using Complement & Independence

Given: White $W = 3$, Red $R = 3$, Black $B = 3$. Total = 9. Non-red balls = $3+3 = 6$. Draw 3 balls without replacement.

Concept / Formula Used: $P(\text{at least one red}) = 1 - P(\text{no red}) = 1 - \frac{\binom{N_{\text{non-red}}}{3}}{\binom{N_{\text{total}}}{3}}$

Solution:

Total number of balls in the urn: $N = 3 + 3 + 3 = 9$ Total ways to draw 3 balls out of 9

without replacement: $n(S) = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$ Non-red balls present in the urn =

3 white + 3 black = 6. Number of ways to choose 3 non-red balls: $n(\text{no red}) = \binom{6}{3} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$

Probability of drawing no red ball: $P(\text{no red}) = \frac{n(\text{no red})}{n(S)} = \frac{20}{84} = \frac{5}{21}$ Using the complementary probability principle:

$$P(\text{at least one red}) = 1 - P(\text{no red}) = 1 - \frac{5}{21} = \frac{16}{21}$$

Final Answer

$$\text{Required Probability} = \frac{16}{21}$$

Question 20

Three distinct numbers are chosen randomly from the first 50 natural numbers. Find the probability that all the three numbers are divisible by 2 and 3 both.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Natural numbers $N = \{1, 2, 3, \dots, 50\}$. Total count = 50. Draw $k = 3$ distinct numbers.

Concept / Formula Used: $\text{lcm}(2, 3) = 6$

Solution:

A number is divisible by both 2 and 3 if it is a multiple of their least common multiple: $\text{lcm}(2, 3) = 6$. The multiples of 6 among the first 50 natural numbers are: $\{6, 12, 18, 24, 30, 36, 42, 48\}$. Counting these multiples: Number of favorable numbers = 8. Total number of ways to

choose 3 distinct numbers from 50: $n(S) = \binom{50}{3} = \frac{50 \times 49 \times 48}{3 \times 2 \times 1} = 19600$. Number of ways to choose 3 numbers from the 8 multiples of 6: $n(E) = \binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$.

Therefore, the required probability is: $P(E) = \frac{n(E)}{n(S)} = \frac{56}{19600}$. Simplifying the fraction by dividing numerator and denominator by 56: $P(E) = \frac{1}{350}$.

Final Answer

$$\text{Required Probability} = \frac{1}{350}$$

Question 21

The probability of finding a green signal on a busy crossing X is 30%. What is the probability of finding a green signal on two consecutive days out of three?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Probability of green signal $p = 30\% = \frac{3}{10} = 0.3$. Probability of non-green signal $q = 1 - 0.3 = 0.7 = \frac{7}{10}$. Total days = 3.

Concept / Formula Used: $P(\text{consecutive green on 2 days out of 3}) = P(GGN) + P(NGG) + P(GGG)$

Solution:

Let G denote finding a green signal and N denote not finding a green signal on any given day. The probabilities for a single day are: $P(G) = p = 0.3$ Finding a green signal
 $P(N) = q = 1 - p = 0.7$
 on two consecutive days out of three means that either Day 1 & Day 2 are green, or Day 2 & Day 3 are green (or all three days are green). The mutually exclusive outcomes that satisfy this condition are:

1. GGN : Green on Day 1, Green on Day 2, Not Green on Day 3
2. NGG : Not Green on Day 1, Green on Day 2, Green on Day 3
3. GGG : Green on all 3 days

$$P(GGN) = 0.3 \times 0.3 \times 0.7 = 0.063$$

Calculating the probabilities for each outcome: $P(NGG) = 0.7 \times 0.3 \times 0.3 = 0.063$ Sum-

$$P(GGG) = 0.3 \times 0.3 \times 0.3 = 0.027$$

ming these mutually exclusive probabilities: $P(\text{consecutive green signals}) = P(GGN) + P(NGG) + P(GGG)$
 $= 0.063 + 0.063 + 0.027$
 $= 0.153$

In fractional form: $P = \frac{153}{1000}$

Final Answer

Required Probability = $0.153 = \frac{153}{1000}$

Question 22

- (i) A box B_1 contains 1 white ball and 3 red balls. Another box B_2 contains 2 white balls and 3 red balls. If one ball is drawn at random from each of the boxes B_1 and B_2 , then find the probability that the two balls drawn are of the same colour.
- (ii) One bag contains 3 white balls, 7 red balls and 15 black balls. Another bag contains 10 white balls, 6 red balls and 9 black balls. One ball is taken from each bag. Find the probability that both balls will be of the same colour.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given:

- (i) Box B_1 : 1 W, 3 R (Total 4). Box B_2 : 2 W, 3 R (Total 5).
- (ii) Bag 1: 3 W, 7 R, 15 B (Total 25). Bag 2: 10 W, 6 R, 9 B (Total 25).

Concept / Formula Used: $P(\text{same colour}) = \sum_c P(\text{Colour } c \text{ from Container 1}) \times P(\text{Colour } c \text{ from Container 2})$

Solution:

(i) For Box B_1 and Box B_2 :
 In B_1 : $P(W_1) = \frac{1}{4}$, $P(R_1) = \frac{3}{4}$
 In B_2 : $P(W_2) = \frac{2}{5}$, $P(R_2) = \frac{3}{5}$ The two balls are of the

same colour if both are White or both are Red:
 $P(\text{both White}) = P(W_1) \cdot P(W_2) = \frac{1}{4} \times \frac{2}{5} = \frac{2}{20}$

$$P(\text{both Red}) = P(R_1) \cdot P(R_2) = \frac{3}{4} \times \frac{3}{5} = \frac{9}{20}$$

Summing these mutually exclusive probabilities: $P(\text{same colour}) = \frac{2}{20} + \frac{9}{20} = \frac{11}{20}$

(ii) For Bag 1 and Bag 2:
 In Bag 1 (Total 25) : $P(W_1) = \frac{3}{25}$, $P(R_1) = \frac{7}{25}$, $P(B_1) = \frac{15}{25}$
 In Bag 2 (Total 25) : $P(W_2) = \frac{10}{25}$, $P(R_2) = \frac{6}{25}$, $P(B_2) = \frac{9}{25}$

$$P(\text{both White}) = \frac{3}{25} \times \frac{10}{25} = \frac{30}{625}$$

Calculating probabilities for each matching colour: $P(\text{both Red}) = \frac{7}{25} \times \frac{6}{25} = \frac{42}{625}$

$$P(\text{both Black}) = \frac{15}{25} \times \frac{9}{25} = \frac{135}{625}$$

Summing these mutually exclusive probabilities:
 $P(\text{same colour}) = \frac{30}{625} + \frac{42}{625} + \frac{135}{625}$
 $= \frac{207}{625}$

Final Answer

(i) Probability = $\frac{11}{20}$

(ii) Probability = $\frac{207}{625}$

Question 23

An urn contains 7 white, 5 black and 3 red balls. Two balls are drawn at random. Find the probability that

- (i) both balls are red
- (ii) one ball is red, the other is black
- (iii) one ball is white.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: White $W = 7$, Black $B = 5$, Red $R = 3$. Total balls = 15. Number of balls drawn = 2.

Concept / Formula Used: $n(S) = \binom{15}{2} = \frac{15 \times 14}{2} = 105$

Solution:

Total number of ways to draw 2 balls out of 15: $n(S) = \binom{15}{2} = \frac{15 \times 14}{2 \times 1} = 105$

(i) Both balls are red: Number of ways to select 2 red balls from 3 red balls: $n(E_1) = \binom{3}{2} = 3$
 $P(E_1) = \frac{3}{105} = \frac{1}{35}$

(ii) One ball is red and the other is black: Number of ways to select 1 red from 3 and 1 black from 5:
 $n(E_2) = \binom{3}{1} \times \binom{5}{1} = 3 \times 5 = 15$
 $P(E_2) = \frac{15}{105} = \frac{1}{7}$

(iii) Exactly one ball is white (1 white and 1 non-white): Non-white balls = 5 black + 3 red = 8. Number of ways to select 1 white from 7 and 1 non-white from 8:
 $n(E_3) = \binom{7}{1} \times \binom{8}{1} = 7 \times 8 = 56$
 $P(E_3) = \frac{56}{105}$
 Simplifying by dividing numerator and denominator by 7: $P(E_3) = \frac{8}{15}$

Final Answer

- (i) $P(\text{both red}) = \frac{1}{35}$
 (ii) $P(1 \text{ red, } 1 \text{ black}) = \frac{1}{7}$
 (iii) $P(\text{exactly 1 white}) = \frac{8}{15}$

Question 24

In a group of students, there are 3 boys and 3 girls. Four students are to be selected at random from the group. Find the probability that either 3 boys and 1 girl, or 3 girls and 1 boy are selected.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Boys $B = 3$, Girls $G = 3$. Total students = 6. Select $k = 4$ students.

Concept / Formula Used: $P(E) = \frac{\binom{3}{3}\binom{3}{1} + \binom{3}{1}\binom{3}{3}}{\binom{6}{4}}$

Solution:

Total number of ways to select 4 students out of 6: $n(S) = \binom{6}{4} = \binom{6}{2} = \frac{6 \times 5}{2 \times 1} = 15$

Let E_1 be the event of selecting 3 boys and 1 girl: $n(E_1) = \binom{3}{3} \times \binom{3}{1} = 1 \times 3 = 3$ Let

E_2 be the event of selecting 3 girls and 1 boy: $n(E_2) = \binom{3}{1} \times \binom{3}{3} = 3 \times 1 = 3$ Since E_1

and E_2 are mutually exclusive events: $n(E_1 \cup E_2) = n(E_1) + n(E_2) = 3 + 3 = 6$ Therefore, the required probability is: $P(E_1 \cup E_2) = \frac{6}{15} = \frac{2}{5}$

Final Answer

Required Probability = $\frac{2}{5}$

Question 25

Three cards are drawn from a pack of well-shuffled 52 cards. Find the probability that

- (i) all the three cards are of the same suit
- (ii) one is a king, the other is a queen and the third is a jack.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Total cards $N = 52$. Number of cards drawn $k = 3$.

Concept / Formula Used: $n(S) = \binom{52}{3} = \frac{52 \times 51 \times 50}{6} = 22100$ $4 \times \binom{13}{3}$ $\binom{4}{1} \times \binom{4}{1} \times \binom{4}{1}$

Solution:

Total number of ways to draw 3 cards from 52 cards: $n(S) = \binom{52}{3} = \frac{52 \times 51 \times 50}{3 \times 2 \times 1} = 22100$

- (i) All three cards are of the same suit: There are 4 suits (Spades, Hearts, Diamonds, Clubs), each containing 13 cards. Number of ways to select 3 cards from one particular suit = $\binom{13}{3}$: $\binom{13}{3} = \frac{13 \times 12 \times 11}{3 \times 2 \times 1} = 286$ Since there are 4 suits, total favorable

$$n(E_1) = 4 \times 286 = 1144$$

ways: $P(E_1) = \frac{1144}{22100}$ Simplifying by dividing numerator and denominator by 22

by 52: $P(E_1) = \frac{4}{425}$

- (ii) One is a king, one is a queen, and one is a jack: In a pack of 52 cards, there are 4 kings, 4 queens, and 4 jacks.

$$n(E_2) = \binom{4}{1} \times \binom{4}{1} \times \binom{4}{1} = 4 \times 4 \times 4 = 64$$

$$P(E_2) = \frac{64}{22100}$$

Simplifying by dividing numerator and denominator by 4: $P(E_2) = \frac{16}{5525}$

Final Answer

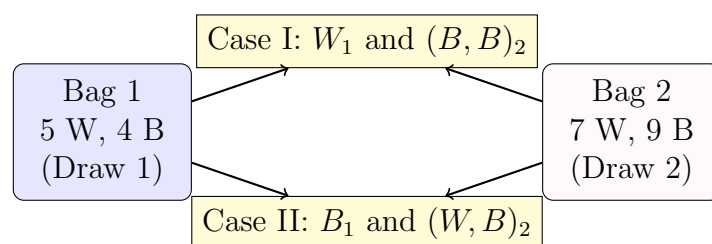
$$(i) P(\text{same suit}) = \frac{22}{425}$$

$$(ii) P(1 \text{ King, } 1 \text{ Queen, } 1 \text{ Jack}) = \frac{16}{5525}$$

Question 26

A bag contains 5 white and 4 black balls and another bag contains 7 white and 9 black balls. A ball is drawn from the first bag and two balls drawn from the second bag. What is the probability of drawing one white and two black balls?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)



Drawings from Bag 1 and Bag 2

Given: Bag 1: 5 W, 4 B (Total 9). Draw 1 ball.
 Bag 2: 7 W, 9 B (Total 16). Draw 2 balls.
 Total target count: 1 White ball and 2 Black balls.

Concept / Formula Used: $P(\text{Target}) = P(\text{Case I}) + P(\text{Case II})$

Solution:

To get a total of 1 White and 2 Black balls across the two draws, there are two mutually exclusive cases:

Case I: 1 White ball from Bag 1 AND 2 Black balls from Bag 2.

$$\begin{aligned}
 P(\text{White from Bag 1}) &= \frac{5}{9} \\
 P(2 \text{ Black from Bag 2}) &= \frac{\binom{9}{2}}{\binom{16}{2}} \\
 \binom{9}{2} &= \frac{9 \times 8}{2} = 36 \\
 \binom{16}{2} &= \frac{16 \times 15}{2} = 120 \\
 P(2 \text{ Black from Bag 2}) &= \frac{36}{120} = \frac{3}{10} \\
 P(\text{Case I}) &= \frac{5}{9} \times \frac{3}{10} = \frac{15}{90}
 \end{aligned}$$

$P(\text{Black from B})$
 $P(1 \text{ White and } 1 \text{ Black from B})$

Case II: 1 Black ball from Bag 1 AND 1 White, 1 Black ball from Bag 2.

 $P(\text{Case II})$

$$P(\text{Total}) = P(\text{Case I}) + P(\text{Case II})$$

$$= \frac{1}{6} + \frac{7}{30}$$

$$= \frac{5}{30} + \frac{7}{30}$$

$$= \frac{12}{30} = \frac{2}{5}$$

Summing the probabilities of these two mutually exclusive cases:

Final Answer

$$\text{Required Probability} = \frac{2}{5}$$

Question 27

Two dice are rolled once. Find the probability that

- the numbers on two dice are different
- the total of numbers on the two dice is at least 4.
- the product of the numbers on two dice is odd.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Two standard 6-sided dice are rolled once. Total outcome sample space $n(S) = 6 \times 6 = 36$.

Concept / Formula Used: $P(E) = \frac{n(E)}{36}$

Solution:

Sample space size $n(S) = 36$.

- (i) Numbers on the two dice are different: The outcomes where the numbers are the same (doubles) are: $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\} \implies \text{Count} = 6$ Num-

$$n(E_1) = 36 - 6 = 30$$

ber of outcomes where numbers are different:

$$P(E_1) = \frac{30}{36} = \frac{5}{6}$$

- (ii) Total of numbers on the two dice is at least 4: It is easier to calculate the complemen-

$$\text{Sum} = 2 : (1, 1) \implies 1 \text{ outcome}$$

$$\text{Sum} = 3 : (1, 2), (2, 1) \implies 2 \text{ outcomes}$$

tary event where the sum is less than 4 (i.e., sum = 2 or sum = 3): $n(\text{Sum} < 4) = 1 + 2 = 3$

$$n(\text{Sum} \geq 4) = 36 - 3 = 33$$

$$P(E_2) = \frac{33}{36} = \frac{11}{12}$$

(iii) Product of the numbers on the two dice is odd: The product of two integers is odd if and only if both integers are odd. Odd numbers on a die are $\{1, 3, 5\}$ (3 choices

$$n(E_3) = 3 \times 3 = 9$$

for each die): $P(E_3) = \frac{9}{36} = \frac{1}{4}$

Final Answer

$$(i) P(\text{different numbers}) = \frac{5}{6}$$

$$(ii) P(\text{sum} \geq 4) = \frac{11}{12}$$

$$(iii) P(\text{product is odd}) = \frac{1}{4}$$

Question 28

In a single throw of three dice, find the probability of getting

- (i) same number on the three dice
- (ii) a total of 5
- (iii) a total of at most 5
- (iv) a total of at least 5.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Three standard 6-sided dice are thrown. Sample space size $n(S) = 6^3 = 216$.

Concept / Formula Used: $P(E) = \frac{n(E)}{216}$

Solution:

Total number of possible outcomes: $n(S) = 6 \times 6 \times 6 = 216$

(i) Same number on all three dice (triplets): The favorable outcomes are $\{(1, 1, 1), (2, 2, 2), (3, 3, 3), (4, 4, 4), (5, 5, 5), (6, 6, 6)\}$

$$n(E_1) = 6$$

$$P(E_1) = \frac{6}{216} = \frac{1}{36}$$

- (ii) A total of 5: Possible partitions of 5 into 3 positive integers:
- $$(1, 1, 3) \Rightarrow \frac{3!}{2!} = 3 \text{ arrangements: } (1, 1, 3), (1, 3, 1), (3, 1, 1)$$
- $$(1, 2, 2) \Rightarrow \frac{3!}{2!} = 3 \text{ arrangements: } (1, 2, 2), (2, 1, 2), (2, 2, 1)$$
- $$n(E_2) = 3 + 3 = 6$$
- $$P(E_2) = \frac{6}{216} = \frac{1}{36}$$
- Sum = 3 : (1, 1, 1) $\Rightarrow$ 1 outcome
 Sum = 4 : (1, 1, 2), (1, 2, 1), (2, 1, 1) $\Rightarrow$ 3 outcomes
 Sum = 5 : 6 outcomes (from part ii)
- (iii) A total of at most 5: Possible sums ≤ 5 are 3, 4, and 5:
- $$n(\text{Sum} \leq 5) = 1 + 3 + 6 = 10$$
- $$P(E_3) = \frac{10}{216} = \frac{5}{108}$$
- $$n(\text{Sum} < 5) = n(\text{Sum} \leq 4) = 4$$
- $$P(\text{Sum} < 5) = \frac{4}{216}$$
- (iv) A total of at least 5: Using the complementary probability principle:
- $$P(\text{Sum} \geq 5) = 1 - P(\text{Sum} < 5)$$
- $$= 1 - \frac{4}{216}$$
- $$= \frac{212}{216} = \frac{53}{54}$$

Final Answer

- (i) $P(\text{same number}) = \frac{1}{36}$
 (ii) $P(\text{total } 5) = \frac{1}{36}$
 (iii) $P(\text{total } \leq 5) = \frac{5}{108}$
 (iv) $P(\text{total } \geq 5) = \frac{53}{54}$

Question 29

The probabilities of two students A and B coming to school in time are $\frac{2}{7}$ and $\frac{4}{7}$ respectively. Assuming that the events, 'A coming in time' and 'B coming in time' are independent, find the probability of only one of them coming to school in time.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Events A and B are independent with $P(A) = \frac{2}{7}$ and $P(B) = \frac{4}{7}$.

Concept / Formula Used: $P(\text{exactly one happens}) = P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B)$

Solution:

Let A be the event that student A comes in time, and B be the event that student B comes

in time. The probabilities of their complements are:

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{2}{7} = \frac{5}{7}$$

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{4}{7} = \frac{3}{7}$$

The event "only one of them comes to school in time" is represented by $(A \cap \bar{B}) \cup (\bar{A} \cap B)$. Since A and B are independent, A and $\bar{B}$ are independent, as are $\bar{A}$ and B :

$$P(A \cap \bar{B}) = P(A) \cdot P(\bar{B}) = \frac{2}{7} \times \frac{3}{7} = \frac{6}{49}$$

$$P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B) = \frac{5}{7} \times \frac{4}{7} = \frac{20}{49}$$

$$P(\text{only one in time}) = P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= \frac{6}{49} + \frac{20}{49}$$

$$= \frac{26}{49}$$

Final Answer

$$\text{Required Probability} = \frac{26}{49}$$

Question 30

Two horses are considered for race. The probability of selection of first horse is $\frac{1}{5}$ and that of second is $\frac{2}{3}$. Find the probability that:

- both will be selected.
- only one of them will be selected.
- none of them will be selected.
- at least one of them will be selected.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

★ Likely Exam Question

Given: Independent events H_1 and H_2 with $P(H_1) = \frac{1}{5}$ and $P(H_2) = \frac{2}{3}$.

Concept / Formula Used: $P(H_1 \cap H_2) = P(H_1) \cdot P(H_2)$ $P(\text{only one}) = P(H_1)P(\bar{H}_2) + P(\bar{H}_1)P(H_2)$ $P(\text{none}) = P(\bar{H}_1) \cdot P(\bar{H}_2)$ $P(\text{at least one}) = 1 - P(\text{none})$

Solution:

Let H_1 be the event that the first horse is selected, and H_2 be the event that the second horse is selected. The given probabilities are:

$$P(H_1) = \frac{1}{5} \implies P(\bar{H}_1) = 1 - \frac{1}{5} = \frac{4}{5}$$

$$P(H_2) = \frac{2}{3} \implies P(\bar{H}_2) = 1 - \frac{2}{3} = \frac{1}{3}$$

the selection of each horse is an independent event:

- $$P(H_1 \cap H_2) = P(H_1) \cdot P(H_2)$$
- (i) Probability that both will be selected: $= \frac{1}{5} \times \frac{2}{3} = \frac{2}{15}$
- $$P(\text{only one}) = P(H_1) \cdot P(\bar{H}_2) + P(\bar{H}_1) \cdot P(H_2)$$
- (ii) Probability that only one of them will be selected: $= \left(\frac{1}{5} \times \frac{1}{3}\right) + \left(\frac{4}{5} \times \frac{2}{3}\right)$
 $= \frac{1}{15} + \frac{8}{15} = \frac{9}{15} = \frac{3}{5}$
- $$P(\bar{H}_1 \cap \bar{H}_2) = P(\bar{H}_1) \cdot P(\bar{H}_2)$$
- (iii) Probability that none of them will be selected: $= \frac{4}{5} \times \frac{1}{3} = \frac{4}{15}$
- $$P(\text{at least one}) = 1 - P(\text{none})$$
- (iv) Probability that at least one of them will be selected: $= 1 - \frac{4}{15} = \frac{11}{15}$

Final Answer

- (i) $P(\text{both selected}) = \frac{2}{15}$
- (ii) $P(\text{only one selected}) = \frac{3}{5}$
- (iii) $P(\text{none selected}) = \frac{4}{15}$
- (iv) $P(\text{at least one selected}) = \frac{11}{15}$

Question 31

A husband and a wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{7}$ and that of wife's selection is $\frac{1}{5}$. What is the probability that

- (i) both of them will be selected?
- (ii) none of them will be selected?
- (iii) only one of them will be selected?
- (iv) at least one of them will be selected?

TYPE 21 Probability of At-Least-One / All-Fail Using Complement & Independence

Given: Let H be the event that the husband is selected and W be the event that the wife is selected. $P(H) = \frac{1}{7}$ and $P(W) = \frac{1}{5}$. The selection of the husband and the selection of the wife are independent events.

Concept / Formula Used: $P(H \cap W) = P(H) \cdot P(W)$ $P(\text{at least one}) = 1 - P(\text{none})$

Solution:

Since the husband's and wife's selections are independent events, we first find the complementary (non-selection) probabilities:

$$P(H') = 1 - P(H) = 1 - \frac{1}{7} = \frac{6}{7}$$

plementary (non-selection) probabilities:

$$P(W') = 1 - P(W) = 1 - \frac{1}{5} = \frac{4}{5}$$

$$P(H \cap W) = P(H) \cdot P(W)$$

(i) **Both of them selected:** Since H and W are independent,

$$\begin{aligned} &= \frac{1}{7} \times \frac{1}{5} \\ &= \frac{1}{35} \end{aligned}$$

(ii) **None of them selected:** Since H' and W' are also independent,

$$\begin{aligned} P(H' \cap W') &= P(H') \cdot P(W') \\ &= \frac{6}{7} \times \frac{4}{5} \\ &= \frac{24}{35} \end{aligned}$$

(iii) **Only one of them selected:** This means either the husband is selected and the wife is not, or the wife is selected and the husband is not – these two cases are mutually exclusive, so their probabilities add:

$$P(\text{only one}) = P(H \cap W') + P(H' \cap W)$$

$$= \left(\frac{1}{7} \times \frac{4}{5} \right) + \left(\frac{6}{7} \times \frac{1}{5} \right)$$

$$= \frac{4}{35} + \frac{6}{35}$$

$$= \frac{10}{35}$$

$$= \frac{2}{7}$$

(iv) **At least one of them selected:** Using the complement of “none selected” from

$$P(\text{at least one}) = 1 - P(H' \cap W')$$

part (ii):

$$= 1 - \frac{24}{35}$$

$$= \frac{11}{35}$$

Final Answer

$$(i) P(\text{both selected}) = \frac{1}{35}$$

$$(ii) P(\text{none selected}) = \frac{24}{35}$$

$$(iii) P(\text{only one selected}) = \frac{2}{7}$$

$$(iv) P(\text{at least one selected}) = \frac{11}{35}$$

Question 32

- (i) A problem is given to three students whose chances of solving it are $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{3}$ respectively. Find the probability that the problem is solved.
- (ii) A problem in Mathematics is given to three students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that exactly one of them solves the problem.
- (iii) A problem in Mathematics is given to three students A, B and C. Their chances of solving the problem are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that
- exactly two students will solve the problem.
 - atleast two students will solve the problem.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

★ Likely Exam Question

Given: Let A, B, C be the events that the three students solve the problem respectively.

Concept / Formula Used: $P(A \cup B \cup C) = 1 - P(A')P(B')P(C')$
 $P(\text{exactly one solves}) = P(A)P(B')P(C') + P(A')P(B)P(C') + P(A')P(B')P(C)$

Solution:

$$P(A) = \frac{1}{4} \implies P(A') = 1 - \frac{1}{4} = \frac{3}{4}$$

Part (i): Let the three events be A, B, C with probabilities: $P(B) = \frac{1}{5} \implies P(B') = 1 - \frac{1}{5} = \frac{4}{5}$

$$P(C) = \frac{1}{3} \implies P(C') = 1 - \frac{1}{3} = \frac{2}{3}$$

The problem is solved if at least one student solves it. Using the complement rule:
 $P(\text{problem is solved}) = 1 - P(\text{none solves})$

$$= 1 - P(A') \cdot P(B') \cdot P(C')$$

$$= 1 - \left(\frac{3}{4} \times \frac{4}{5} \times \frac{2}{3} \right)$$

$$= 1 - \frac{2}{5}$$

$$= \frac{3}{5}$$

$$P(A) = \frac{1}{2} \implies P(A') = 1 - \frac{1}{2} = \frac{1}{2}$$

Part (ii): Here, the probabilities of solving the problem are: $P(B) = \frac{1}{3} \implies P(B') = 1 - \frac{1}{3} = \frac{2}{3}$

$$P(C) = \frac{1}{4} \implies P(C') = 1 - \frac{1}{4} = \frac{3}{4}$$

The probability that exactly one student solves the problem is:

$$\begin{aligned}
 P(\text{exactly one solves}) &= P(A \cap B' \cap C') + P(A' \cap B \cap C') + P(A' \cap B' \cap C) \\
 &= P(A)P(B')P(C') + P(A')P(B)P(C') + P(A')P(B')P(C) \\
 &= \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4}\right) + \left(\frac{1}{2} \times \frac{2}{3} \times \frac{1}{4}\right) + \left(\frac{1}{2} \times \frac{1}{3} \times \frac{3}{4}\right) \\
 &= \frac{6}{24} + \frac{2}{24} + \frac{3}{24} \\
 &= \frac{11}{24}
 \end{aligned}$$

Part (iii): Given $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(C) = \frac{1}{4}$.

Hence $P(A') = \frac{1}{2}$, $P(B') = \frac{2}{3}$, $P(C') = \frac{3}{4}$.

(a) Probability that exactly two students solve the problem:

$$\begin{aligned}
 P(\text{exactly two}) &= P(A \cap B \cap C') + P(A \cap B' \cap C) + P(A' \cap B \cap C) \\
 &= P(A)P(B)P(C') + P(A)P(B')P(C) + P(A')P(B)P(C) \\
 &= \left(\frac{1}{2} \times \frac{1}{3} \times \frac{3}{4}\right) + \left(\frac{1}{2} \times \frac{2}{3} \times \frac{1}{4}\right) + \left(\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}\right) \\
 &= \frac{3}{24} + \frac{2}{24} + \frac{1}{24} \\
 &= \frac{6}{24} = \frac{1}{4}
 \end{aligned}$$

(b) Probability that at least two students solve the problem:

$$\begin{aligned}
 P(\text{atleast two}) &= P(\text{exactly two}) + P(\text{all three}) \\
 &= \frac{1}{4} + P(A)P(B)P(C) \\
 &= \frac{6}{24} + \left(\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}\right) \\
 &= \frac{6}{24} + \frac{1}{24} \\
 &= \frac{7}{24}
 \end{aligned}$$

Final Answer

(i) $P(\text{problem solved}) = \frac{3}{5}$

(ii) $P(\text{exactly one solves}) = \frac{11}{24}$

(iii) (a) $P(\text{exactly two solve}) = \frac{1}{4}$, (b) $P(\text{atleast two solve}) = \frac{7}{24}$

Question 33

Three persons A, B and C shoot to hit a target. Their probabilities of hitting the target are $\frac{5}{6}$, $\frac{4}{5}$ and $\frac{3}{4}$ respectively. Find the probability that

- exactly two persons hit the target.
- atleast one person hits the target.

★ Likely Exam Question

Given: $P(A) = \frac{5}{6}$, $P(B) = \frac{4}{5}$, $P(C) = \frac{3}{4}$

Concept / Formula Used: $P(A') = 1 - P(A)$ $P(\text{atleast one hits}) = 1 - P(A')P(B')P(C')$

Solution:

$$P(A') = 1 - \frac{5}{6} = \frac{1}{6}$$

First, calculate the complement probabilities of missing the target: $P(B') = 1 - \frac{4}{5} = \frac{1}{5}$

$$P(C') = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\text{exactly two hit}) = P(A)P(B)P(C')$$

$$= \left(\frac{5}{6} \times \frac{4}{5} \times \frac{1}{4}\right)$$

Part (i): Probability that exactly two persons hit the target

$$= \frac{20}{120} + \frac{15}{120} + \frac{47}{120}$$

$$P(\text{atleast one hits}) = 1 - P(\text{none hit})$$

$$= 1 - P(A')P(B')P(C')$$

Part (ii): Probability that at least one person hits the target

$$= 1 - \left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right)$$

$$= 1 - \frac{1}{120}$$

$$= \frac{119}{120}$$

Final Answer

$$(i) P(\text{exactly two hit}) = \frac{47}{120}$$

$$(ii) P(\text{atleast one hits}) = \frac{119}{120}$$

Question 34

A machine operates if all of its three components function. The probability that the first component fails during the year is 0.14, the probability that the second component fails is 0.10 and the probability that the third component fails is 0.05. What is the probability that the machine will fail during the year?

TYPE 20 Probability of At-Least-One / All-Fail Using Complement & Independence

Given: $P(F_1) = 0.14$, $P(F_2) = 0.10$, $P(F_3) = 0.05$ where F_i represents the event of component i failing.

Concept / Formula Used: $P(\text{Machine Fails}) = 1 - P(\text{Machine Operates}) = 1 - P(F'_1)P(F'_2)P(F'_3)$

Solution:

First, find the probability of each component functioning (not failing) during the year:

$$P(F'_1) = 1 - 0.14 = 0.86$$

$$P(F'_2) = 1 - 0.10 = 0.90 \text{ Since the machine operates only if all three components function}$$

$$P(F'_3) = 1 - 0.05 = 0.95$$

simultaneously, and component operations are independent:

$$P(\text{Machine operates}) = P(F'_1) \times P(F'_2) \times P(F'_3)$$

$$= 0.86 \times 0.90 \times 0.95$$

$$= 0.7353$$

$$P(\text{Machine fails}) = 1 - P(\text{Machine operates})$$

Therefore, the probability that the machine fails during the year is:

$$= 1 - 0.7353$$

$$= 0.2647$$

Final Answer

$$P(\text{Machine fails}) = 0.2647$$

Question 35

A speaks truth in 60% of the cases, while B in 40% of the cases. In what percent of cases are they likely to contradict each other in stating the same fact?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

★ Likely Exam Question

Given: $P(A) = 60\% = 0.6$, $P(B) = 40\% = 0.4$

Concept / Formula Used: $P(\text{Contradiction}) = P(A)P(B') + P(A')P(B)$

Solution:

Let A be the event that A speaks the truth, and B be the event that B speaks the truth.

$$P(A) = \frac{60}{100} = 0.6 \implies P(A') = 1 - 0.6 = 0.4$$

A and B contradict each other when one

$$P(B) = \frac{40}{100} = 0.4 \implies P(B') = 1 - 0.4 = 0.6$$

speaks the truth and the other tells a lie.

$$\begin{aligned}
 P(\text{Contradiction}) &= P(A \cap B') + P(A' \cap B) \\
 &= P(A)P(B') + P(A')P(B) \\
 &= (0.6 \times 0.6) + (0.4 \times 0.4) \\
 &= 0.36 + 0.16 \\
 &= 0.52
 \end{aligned}$$

To express as a percentage: Percentage of cases = $0.52 \times 100\% = 52\%$

Final Answer

They are likely to contradict each other in 52% of the cases.

Question 36

Ten Tickets are numbered 1 to 10. Two tickets are drawn one after the other at random. Find the probability that the number on one of the tickets is a multiple of 5 and the other a multiple of 4 when the tickets are drawn

- (i) without replacement
- (ii) with replacement.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ ($n(S) = 10$).

Multiples of 5: $M_5 = \{5, 10\}$ ($n(M_5) = 2$).

Multiples of 4: $M_4 = \{4, 8\}$ ($n(M_4) = 2$).

Concept / Formula Used: $P(E) = P(M_5 \text{ then } M_4) + P(M_4 \text{ then } M_5)$

Solution:

Part (i): Without replacement For the first ticket to be M_5 and second to be M_4 :

$$P(M_5 \text{ 1st}) = \frac{2}{10}$$

$$P(M_4 \text{ 2nd} \mid M_5 \text{ 1st}) = \frac{2}{9}$$

Similarly, for the first ticket to be M_4 and second to

$$P(M_5 \text{ then } M_4) = \frac{2}{10} \times \frac{2}{9} = \frac{4}{90}$$

$$P(M_4 \text{ 1st}) = \frac{2}{10}$$

$$\text{be } M_5: P(M_5 \text{ 2nd} \mid M_4 \text{ 1st}) = \frac{2}{9}$$

Since these two orders are mutually exclusive:

$$P(M_4 \text{ then } M_5) = \frac{2}{10} \times \frac{2}{9} = \frac{4}{90}$$

$$\begin{aligned}
 P(\text{one } M_5 \text{ and one } M_4) &= \frac{4}{90} + \frac{4}{90} \\
 &= \frac{8}{90} = \frac{4}{45}
 \end{aligned}$$

Part (ii): With replacement When tickets are drawn with replacement, total out-

comes for each draw remain 10:

$$P(M_5 \text{ then } M_4) = \frac{2}{10} \times \frac{2}{10} = \frac{4}{100}$$

$$P(M_4 \text{ then } M_5) = \frac{2}{10} \times \frac{2}{10} = \frac{4}{100}$$

Total probability:

$$P(\text{one } M_5 \text{ and one } M_4) = \frac{4}{100} + \frac{4}{100}$$

$$= \frac{8}{100} = \frac{2}{25}$$

Final Answer

(i) Without replacement: $\frac{4}{45}$

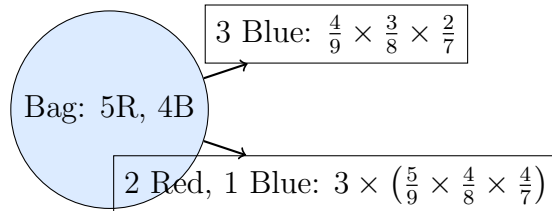
(ii) With replacement: $\frac{2}{25}$

Question 37

A bag contains 5 red discs and 4 blue discs. If 3 discs are drawn from the bag without replacement, then find the probability that

- (i) all three are blue
- (ii) there are 2 red and 1 blue
- (iii) the second disc is red.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)



Probability tree for drawing 3 discs

Given: Bag composition: 5 Red (R), 4 Blue (B). Total discs = $5 + 4 = 9$.
3 discs drawn without replacement.

Concept / Formula Used: $P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$

Solution:

Part (i): All three are blue The probability of drawing three blue discs consecutively

$$P(B_1 \cap B_2 \cap B_3) = P(B_1) \cdot P(B_2 | B_1) \cdot P(B_3 | B_1 \cap B_2)$$

$$= \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7}$$

without replacement:

$$= \frac{24}{504}$$

$$= \frac{1}{21}$$

Part (ii): 2 red and 1 blue The 3 possible sequences of drawing 2 red and 1 blue disc are RRB , RBR , and BRR . For sequence RRB : $P(RRB) = \frac{5}{9} \times \frac{4}{8} \times \frac{4}{7} = \frac{80}{504}$ For sequence RBR : $P(RBR) = \frac{5}{9} \times \frac{4}{8} \times \frac{4}{7} = \frac{80}{504}$ For sequence BRR : $P(BRR) = \frac{4}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{80}{504}$

$$P(2 \text{ Red and } 1 \text{ Blue}) = \frac{80}{504} + \frac{80}{504} + \frac{80}{504}$$

$$= \frac{240}{504} = \frac{10}{21}$$

Adding these mutually exclusive sequences:

Part (iii): The second disc is red The second disc is red in two cases: R_1R_2 or B_1R_2 .

$$P(R_1 \cap R_2) = \frac{5}{9} \times \frac{4}{8} = \frac{20}{72}$$

$$P(B_1 \cap R_2) = \frac{4}{9} \times \frac{5}{8} = \frac{20}{72}$$

$$P(2\text{nd disc is Red}) = \frac{20}{72} + \frac{20}{72}$$

$$= \frac{40}{72} = \frac{5}{9}$$

Final Answer

(i) $P(\text{all three blue}) = \frac{1}{21}$

(ii) $P(2 \text{ red and } 1 \text{ blue}) = \frac{10}{21}$

(iii) $P(2\text{nd disc is red}) = \frac{5}{9}$

Question 38

A bag contains 5 white and 3 black balls. Four balls are successively drawn out without replacement. What is the probability that they are alternately of different colours? What would be the probability of this event if balls are drawn with replacement?

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Bag composition: 5 White (W), 3 Black (B). Total balls = 8.
Number of draws = 4.

Concept / Formula Used: $P(\text{Alternating}) = P(WBWB) + P(BWBW)$

Solution:

Case 1: Without replacement The two sequences with alternating colours are $WBWB$ and $BWBW$.

Probability of sequence $WBWB$:

$$P(WBWB) = \frac{5}{8} \times \frac{3}{7} \times \frac{4}{6} \times \frac{2}{5}$$

$$= \frac{120}{1680} = \frac{1}{14}$$

$$\begin{aligned}
 P(BWBW) &= \frac{3}{8} \times \frac{5}{7} \times \frac{2}{6} \times \frac{4}{5} \\
 \text{Probability of sequence } BWBW: &= \frac{120}{1680} = \frac{1}{14}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{Alternating without replacement}) &= \frac{1}{14} + \frac{1}{14} \\
 \text{Total probability without replacement:} &= \frac{2}{14} = \frac{1}{7}
 \end{aligned}$$

Case 2: With replacement Here $P(W) = \frac{5}{8}$ and $P(B) = \frac{3}{8}$ remain constant for every draw.

$$\begin{aligned}
 P(WBWB) &= \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} \\
 \text{Probability of sequence } WBWB: &= \frac{225}{4096}
 \end{aligned}$$

$$\begin{aligned}
 P(BWBW) &= \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8} \\
 \text{Probability of sequence } BWBW: &= \frac{225}{4096}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{Alternating with replacement}) &= \frac{225}{4096} + \frac{225}{4096} \\
 \text{Total probability with replacement:} &= \frac{450}{4096} = \frac{225}{2048}
 \end{aligned}$$

Final Answer

$$\begin{aligned}
 \text{Without replacement: } &\frac{1}{7} \\
 \text{With replacement: } &\frac{225}{2048}
 \end{aligned}$$

Question 39

A bag contains one black and two white balls. A drawing from the bag consists of taking a ball from the bag and keeping it out if it is white but putting it back if it is black. Calculate the probabilities that

- the first drawing is a white ball
- the second drawing is a white ball
- the third drawing is a white ball.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Initial bag state: 1 Black (B), 2 White (W). Total = 3 balls.
Rule: If W is drawn, keep it out. If B is drawn, put it back.

Concept / Formula Used: $P(W_k) = \sum P(\text{path to } W_k)$

Solution:

Part (i): First drawing is a white ball Initially there are 2 white balls out of 3 balls:

$$P(W_1) = \frac{2}{3}$$

Part (ii): Second drawing is a white ball Consider all paths to the 2nd drawing:

1. Path $W_1 \rightarrow W_2$: After W_1 , bag has 1B and 1W (total 2). $P(W_1 \cap W_2) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$

2. Path $B_1 \rightarrow W_2$: After B_1 , black is replaced, bag still has 1B and 2W (total 3).

$$P(B_1 \cap W_2) = \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$$

$$P(W_2) = \frac{1}{3} + \frac{2}{9}$$

Summing the probabilities:

$$= \frac{3}{9} + \frac{2}{9} = \frac{5}{9}$$

Part (iii): Third drawing is a white ball Possible paths to draw a white ball on 3rd draw:

1. Path $W_1W_2W_3$: After W_1 and W_2 , both white balls are drawn, leaving 0 white balls in the bag. $P(W_1W_2W_3) = \frac{2}{3} \times \frac{1}{2} \times 0 = 0$

2. Path $W_1B_2W_3$: W_1 leaves 1B, 1W. B_2 is replaced, so bag still has 1B, 1W. $P(W_1B_2W_3) = \frac{2}{3} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{6}$

3. Path $B_1W_2W_3$: B_1 leaves 1B, 2W. W_2 leaves 1B, 1W. $P(B_1W_2W_3) = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{2} = \frac{1}{9}$

4. Path $B_1B_2W_3$: B_1 and B_2 are both replaced, bag remains 1B, 2W. $P(B_1B_2W_3) = \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{27}$

$$P(W_3) = 0 + \frac{1}{6} + \frac{1}{9} + \frac{2}{27}$$

Summing all path probabilities:

$$= \frac{9 + 6 + 4}{54} = \frac{19}{54}$$

Final Answer

$$(i) P(W_1) = \frac{2}{3}$$

$$(ii) P(W_2) = \frac{5}{9}$$

$$(iii) P(W_3) = \frac{19}{54}$$

Question 40

A bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. If one ball is drawn from each bag, then find the probability that

- both are white
- both are black
- one is white and one is black.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)**Given:** Bag 1: 4 White (W_1), 2 Black (B_1) $\Rightarrow$ Total = 6 balls.Bag 2: 3 White (W_2), 5 Black (B_2) $\Rightarrow$ Total = 8 balls.**Concept / Formula Used:** $P(W_1) = \frac{4}{6} = \frac{2}{3}$ $P(B_1) = \frac{2}{6} = \frac{1}{3}$ $P(W_2) = \frac{3}{8}$ $P(B_2) = \frac{5}{8}$ **Solution:**

$$P(\text{both white}) = P(W_1) \times P(W_2)$$

$$\begin{aligned} \text{Part (i): Both are white} \quad &= \frac{4}{6} \times \frac{3}{8} \\ &= \frac{12}{48} = \frac{1}{4} \end{aligned}$$

$$P(\text{both black}) = P(B_1) \times P(B_2)$$

$$\begin{aligned} \text{Part (ii): Both are black} \quad &= \frac{2}{6} \times \frac{5}{8} \\ &= \frac{10}{48} = \frac{5}{24} \end{aligned}$$

Part (iii): One is white and one is black This can occur in two mutually exclusive

$$P(\text{one white, one black}) = P(W_1)P(B_2) + P(B_1)P(W_2)$$

$$\begin{aligned} &= \left(\frac{4}{6} \times \frac{5}{8}\right) + \left(\frac{2}{6} \times \frac{3}{8}\right) \\ \text{ways: } (W_1 \cap B_2) \text{ or } (B_1 \cap W_2). \quad &= \frac{20}{48} + \frac{6}{48} \\ &= \frac{26}{48} = \frac{13}{24} \end{aligned}$$

Final Answer

$$(i) P(\text{both white}) = \frac{1}{4}$$

$$(ii) P(\text{both black}) = \frac{5}{24}$$

$$(iii) P(\text{one white, one black}) = \frac{13}{24}$$

Question 41

A bag contains 2 black, 4 white and 3 red balls. One ball is drawn at random and kept aside. Then another ball is drawn and kept aside. This process is continued till all the balls are drawn. Find the probability that the ball drawn are in the sequence 2 black, 4 white and 3 red.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: Bag contents: 2 Black (B), 4 White (W), 3 Red (R). Total = 9 balls.
Sequence required: $B, B, W, W, W, W, R, R, R$.

Concept / Formula Used: $P(\text{sequence}) = \frac{n_B! \cdot n_W! \cdot n_R!}{N!}$

Solution:

The total number of balls is $N = 2 + 4 + 3 = 9$. We need to draw the exact se-

$$P(B_1) = \frac{2}{9}$$

$$P(B_2) = \frac{1}{8}$$

$$P(W_1) = \frac{4}{7}$$

$$P(W_2) = \frac{3}{6}$$

quence $B_1 B_2 W_1 W_2 W_3 W_4 R_1 R_2 R_3$ without replacement: $P(W_3) = \frac{2}{5}$ Multiplying all

$$P(W_4) = \frac{1}{4}$$

$$P(R_1) = \frac{3}{3} = 1$$

$$P(R_2) = \frac{2}{2} = 1$$

$$P(R_3) = \frac{1}{1} = 1$$

$$P(\text{sequence}) = \frac{2}{9} \times \frac{1}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} \times \frac{3}{3} \times \frac{2}{2} \times \frac{1}{1}$$

$$= \frac{2! \times 4! \times 3!}{9!}$$

$$\text{step probabilities: } = \frac{2 \times 24 \times 6}{362880}$$

$$= \frac{288}{362880}$$

$$= \frac{1}{1260}$$

Final Answer

$$P(\text{sequence}) = \frac{1}{1260}$$

Question 42

A policeman fires four bullets on a dacoit. The probability that the dacoit will be killed by one bullet is 0.6. What is the probability that the dacoit is still alive?

TYPE 20 Probability of At-Least-One / All-Fail Using Complement & Independence

Given: Number of bullets $n = 4$.

Probability of dacoit being killed by one bullet $p = 0.6$.

Concept / Formula Used: $P(\text{Alive}) = P(\text{misses with all 4 bullets}) = (1 - p)^n$

Solution:

Probability that a single bullet fails to kill the dacoit: $q = 1 - p = 1 - 0.6 = 0.4$ The da-
 $P(\text{Dacoit is still alive}) = q^4$
 coit remains alive if and only if all 4 bullets fail to kill him: $= (0.4)^4$
 $= 0.0256$

Final Answer

$P(\text{Dacoit is still alive}) = 0.0256$

Question 43

A and B toss a coin alternately till one of them gets a head and wins the game. If A starts first, then find the probability that B will win the game.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

★ Likely Exam Question

Given: A fair coin is tossed alternately by A and B. A tosses first.

$$P(H) = p = \frac{1}{2}, P(T) = q = \frac{1}{2}.$$

Concept / Formula Used: $S_{\infty} = \frac{a}{1-r}$

Solution:

For B to win, the game must progress such that B throws Head on B's 1st, 2nd, 3rd,
 1st turn win for B : TA then $HB \Rightarrow q \cdot p = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

...turn: 2nd turn win for B : $TA, TB, TA, HB \Rightarrow q^3 \cdot p = \left(\frac{1}{2}\right)^3 \times \frac{1}{2} = \frac{1}{16}$ Summing

$$\text{3rd turn win for B : } q^5 \cdot p = \left(\frac{1}{2}\right)^5 \times \frac{1}{2} = \frac{1}{64}$$

all probabilities: $P(\text{B wins}) = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$ This is an infinite Geometric Progress-

sion with first term $a = \frac{1}{4}$ and common ratio $r = \frac{1}{4}$.

$$\begin{aligned} P(\text{B wins}) &= \frac{a}{1-r} \\ &= \frac{\frac{1}{4}}{1-\frac{1}{4}} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} \\ &= \frac{1}{3} \end{aligned}$$

Final Answer

$$P(\text{B wins}) = \frac{1}{3}$$

Question 44

Two aeroplanes I and II bomb a target in succession. The probabilities of aeroplanes I and II scoring a hit correctly are 0.3 and 0.2 respectively. The second plane will bomb only if the first plane misses the target. Find the probability that the target is hit by the second plane.

TYPE 17 Multiplication Theorem – Successive/Independent Draw Word Problems (With/Without Replacement)

Given: $P(H_1) = 0.3 \implies P(M_1) = 1 - 0.3 = 0.7$
 $P(H_2) = 0.2$

Concept / Formula Used: $P(\text{Hit by 2nd plane}) = P(M_1 \cap H_2) = P(M_1) \times P(H_2)$

Solution:

The second aeroplane bombs the target if and only if the first aeroplane misses the target. $P(\text{First plane misses}) = P(M_1) = 1 - 0.3 = 0.7$ Since the bombing attempts are

$$\begin{aligned} P(\text{Target hit by second plane}) &= P(M_1) \times P(H_2) \\ \text{independent:} &= 0.7 \times 0.2 \\ &= 0.14 \end{aligned}$$

Final Answer

$$P(\text{Target hit by second plane}) = 0.14$$

Question 45

Aman and Bhuvan throw a pair of dice alternately. In order to win, they have to get a sum of 8. Find their respective probabilities of winning if Aman starts the game.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Winning condition: sum of numbers on pair of dice equals 8.
 Aman starts first.

Concept / Formula Used: $P(\text{sum } 8) = p = \frac{5}{36}$ $q = 1 - p = \frac{31}{36}$
 $P(\text{First player wins}) = \frac{p}{1-q^2}$ $P(\text{Second player wins}) = \frac{qp}{1-q^2}$

Solution:

Outcomes giving a sum of 8 from throwing two dice:

$$\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$

Total number of favourable outcomes = 5 out of 36 sample space outcomes.

$$p = P(\text{getting sum } 8) =$$

$$q = P(\text{not getting sum } 8) =$$

$$P(\text{Aman wins}) = p + q^2p + q^4p + \dots$$

$$= \frac{p}{1 - q^2}$$

$$= \frac{\frac{5}{36}}{1 - \left(\frac{31}{36}\right)^2}$$

Aman wins if he gets sum 8 on 1st, 3rd, 5th, ... throw:

$$= \frac{\frac{5}{36}}{\frac{1296 - 961}{1296}}$$

$$= \frac{5}{36} \times \frac{1296}{335}$$

$$= \frac{5 \times 36}{335}$$

$$= \frac{180}{335} = \frac{36}{67}$$

$$P(\text{Bhuvan wins}) = 1 - P(\text{Aman wins})$$

Bhuvan's winning probability:

$$= 1 - \frac{36}{67}$$

$$= \frac{31}{67}$$

Final Answer

Probability of Aman winning = $\frac{36}{67}$

Probability of Bhuvan winning = $\frac{31}{67}$

Question 46

A and B throw a pair of dice alternately till one of them gets the sum of numbers as multiples of 6 and wins the game. If A starts first, find the probability of B winning the game.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Winning sum: multiples of 6 (i.e. sum of 6 or 12).

A starts first.

Concept / Formula Used: $p = P(\text{sum 6 or 12}) = \frac{6}{36} = \frac{1}{6}$ $q = \frac{5}{6}$
 $P(\text{B wins}) = \frac{qp}{1-q^2}$

Solution:

Favourable outcomes for sum of 6: $\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$ (5 outcomes).

Favourable outcomes for sum of 12: $\{(6, 6)\}$ (1 outcome).

Total favourable outcomes = $5 + 1 = 6$. $p = \frac{6}{36} = \frac{1}{6}$ B wins on 2nd, 4th, 6th, ... throw
 $q = 1 - \frac{1}{6} = \frac{5}{6}$

$$P(\text{B wins}) = qp + q^3p + q^5p + \dots$$

$$= \frac{qp}{1 - q^2}$$

$$= \frac{\frac{5}{6} \times \frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2}$$

of the game:

$$= \frac{\frac{5}{36}}{1 - \frac{25}{36}}$$

$$= \frac{\frac{5}{36}}{\frac{11}{36}}$$

$$= \frac{5}{11}$$

Final Answer

$$P(\text{B wins}) = \frac{5}{11}$$

Question 47

A and B cut a pack of 52 playing cards alternately and every time the card is put back. The pack is well-shuffled after each cut. A starts the game and the one who cuts a heart first wins. What are their respective probabilities of winning?

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Pack of 52 cards, total hearts = 13.

Card is replaced and shuffled after each cut.

A starts first.

Concept / Formula Used: $p = \frac{13}{52} = \frac{1}{4}$ $q = 1 - p = \frac{3}{4}$

Solution:

Probability of cutting a heart on any turn: $p = \frac{13}{52} = \frac{1}{4}$ A wins if A cuts a heart on
 $q = 1 - \frac{1}{4} = \frac{3}{4}$

$$P(\text{A wins}) = p + q^2p + q^4p + \dots$$

$$= \frac{p}{1 - q^2}$$

$$= \frac{\frac{1}{4}}{1 - \left(\frac{3}{4}\right)^2}$$

1st, 3rd, 5th, ... turn:

$$= \frac{\frac{1}{4}}{1 - \frac{9}{16}}$$

$$= \frac{\frac{1}{4}}{\frac{7}{16}}$$

$$= \frac{1}{4} \times \frac{16}{7} = \frac{4}{7}$$

$$P(\text{B wins}) = qp + q^3p + q^5p + \dots$$

$$= q \times P(\text{A wins})$$

4th, 6th, ... turn:

$$= \frac{3}{4} \times \frac{4}{7} = \frac{3}{7}$$

B wins if B cuts a heart on 2nd,

Final Answer

$$\text{Probability of A winning} = \frac{4}{7}$$

$$\text{Probability of B winning} = \frac{3}{7}$$

Question 48

A and B throw a pair of dice alternately. A wins the game if he gets a total of 7 and B wins the game if he gets a total of 10. If A starts the game, find the probability that B wins the game.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: A wins if sum = 7. B wins if sum = 10.

A starts the game.

Concept / Formula Used: $P(\text{B wins}) = \frac{q_A p_B}{1 - q_A q_B}$

Solution:

For A (sum = 7): outcomes are $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ (6 outcomes).

$$p_A = \frac{6}{36} = \frac{1}{6} \implies q_A = 1 - \frac{1}{6} = \frac{5}{6}$$

For B (sum = 10): outcomes are $\{(4, 6), (5, 5), (6, 4)\}$ (3 outcomes).

$$p_B = \frac{3}{36} = \frac{1}{12} \implies q_B = 1 - \frac{1}{12} = \frac{11}{12}$$

B wins on B's 1st turn (A misses,

B hits), 2nd turn (A misses, B misses, A misses, B hits), etc.:

$$\begin{aligned}
 P(\text{B wins}) &= q_A p_B + (q_A q_B) q_A p_B + (q_A q_B)^2 q_A p_B + \dots \\
 &= \frac{q_A p_B}{1 - q_A q_B} \\
 &= \frac{\frac{5}{6} \times \frac{1}{12}}{1 - \left(\frac{5}{6} \times \frac{11}{12}\right)} \\
 &= \frac{\frac{5}{72}}{1 - \frac{55}{72}} \\
 &= \frac{\frac{5}{72}}{\frac{17}{72}} = \frac{5}{17}
 \end{aligned}$$

Final Answer

$$P(\text{B wins}) = \frac{5}{17}$$

Question 49

A, B and C in order toss a fair coin. The first one to throw a head wins. What are their respective chances of winning? Assume that the game may continue indefinitely.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Three players A, B, C toss in order A, B, C, A, B, C, ...

Fair coin: $p = P(H) = \frac{1}{2}$, $q = P(T) = \frac{1}{2}$.

Concept	/	Formula	Used:	$P(\text{A wins}) = \frac{p}{1-q^3}$	$P(\text{B wins}) = \frac{qp}{1-q^3}$	$P(\text{C wins}) = \frac{q^2p}{1-q^3}$
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Solution:

The game repeats every 3 tosses with common ratio $r = q^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$. $1 - r = 1 - \frac{1}{8} = \frac{7}{8}$

$$P(A) = p + q^3p + q^6p + \dots$$

A's chance of winning: A gets Head on 1st, 4th, 7th turn:

$$= \frac{p}{1-r} = \frac{\frac{1}{2}}{\frac{7}{8}} = \frac{1}{2} \times \frac{8}{7} = \frac{4}{7}$$

$$P(B) = qp + q^4p + q^7p + \dots$$

B's chance of winning: B gets Head on 2nd, 5th, 8th turn:

$$= q \cdot P(A) = \frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$$

$$P(C) = q^2p + q^5p + q^8p + \dots$$

C's chance of winning: C gets Head on 3rd, 6th, 9th turn:

$$= q^2 \cdot P(A) = \frac{1}{4} \times \frac{4}{7} = \frac{1}{7}$$

Final Answer

Chances of winning:

$$P(A) = \frac{4}{7}, \quad P(B) = \frac{2}{7}, \quad P(C) = \frac{1}{7}$$

Question 50

A, B and C throw a die one after the other in the same order till one of them gets a '6' and wins the game. Find their respective probability of winning, if A starts the game.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Three players throw die in order A, B, C.

Winning condition: throw a '6'.

$$p = \frac{1}{6}, \quad q = \frac{5}{6}.$$

Concept / Formula Used: $r = q^3 = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$

Solution:

$$\begin{aligned} \text{First calculate common ratio factor:} \quad r &= \left(\frac{5}{6}\right)^3 = \frac{125}{216} \\ 1 - r &= 1 - \frac{125}{216} = \frac{91}{216} \end{aligned}$$

$$\begin{aligned} P(A) &= \frac{p}{1 - r} \\ \text{Probability of A winning:} \quad &= \frac{\frac{1}{6}}{\frac{91}{216}} \\ &= \frac{1}{6} \times \frac{216}{91} = \frac{36}{91} \end{aligned}$$

$$\begin{aligned} P(B) &= q \times P(A) \\ \text{Probability of B winning:} \quad &= \frac{5}{6} \times \frac{36}{91} = \frac{30}{91} \end{aligned}$$

$$\begin{aligned} P(C) &= q^2 \times P(A) \\ \text{Probability of C winning:} \quad &= \left(\frac{5}{6}\right)^2 \times \frac{36}{91} \\ &= \frac{25}{36} \times \frac{36}{91} = \frac{25}{91} \end{aligned}$$

Final Answer

$$P(A \text{ wins}) = \frac{36}{91}, \quad P(B \text{ wins}) = \frac{30}{91}, \quad P(C \text{ wins}) = \frac{25}{91}$$

Question 51

In a hockey match, both teams A and B scored same number of goals up to the end of the game, so to decide the winner, the referee asked both the captains to throw a die alternately and decided that the team, whose captain gets a six first, will be declared the winner. If the captain of team A was asked to start, find their respective probabilities of winning the match and state whether the decision of the referee was fair or not.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: Die thrown alternately by captains A and B. Captain A starts.
Winning condition: throw a six ($p = 1/6, q = 5/6$).

Concept / Formula Used: $P(\text{A wins}) = \frac{p}{1-q^2}$ $P(\text{B wins}) = \frac{qp}{1-q^2}$

Solution:

Probability of getting six on a single throw: $p = \frac{1}{6} \implies q = 1 - \frac{1}{6} = \frac{5}{6}$ Probability of

$$\begin{aligned}
 P(\text{A wins}) &= \frac{p}{1-q^2} \\
 &= \frac{\frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2} \\
 \text{Team A winning:} \quad &= \frac{\frac{1}{6}}{1 - \frac{25}{36}} \\
 &= \frac{\frac{1}{6}}{\frac{11}{36}} = \frac{6}{11}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{B wins}) &= 1 - P(\text{A wins}) \\
 &= 1 - \frac{6}{11} = \frac{5}{11}
 \end{aligned}$$

Probability of Team B winning:

Comparing the probabilities:

$$P(\text{A wins}) = \frac{6}{11} \neq \frac{5}{11} = P(\text{B wins})$$

Since Team A has a higher probability of winning solely because captain A throws first, the referee's decision is not fair.

Final Answer

$$P(\text{Team A wins}) = \frac{6}{11}, \quad P(\text{Team B wins}) = \frac{5}{11}$$

The decision of the referee was **not fair**.

Question 52

A and B throw a die alternately till one of them gets a '6' and wins the game. Find their respective probabilities of winning, if A starts the game first.

TYPE 21 Infinite Alternating-Turns Game (Geometric Series Probability)

Given: A and B throw die alternately. A starts.

$$p = P(6) = \frac{1}{6}, q = \frac{5}{6}.$$

Concept / Formula Used: $P(\text{A wins}) = \frac{p}{1-q^2}$, $P(\text{B wins}) = \frac{qp}{1-q^2}$

Solution:

$$p = \frac{1}{6} \implies q = \frac{5}{6}$$

$$1 - q^2 = 1 - \left(\frac{5}{6}\right)^2 = 1 - \frac{25}{36} = \frac{11}{36}$$

Winning probability for A: $P(A) = \frac{\frac{1}{6}}{\frac{11}{36}} = \frac{6}{11}$ Win-

ning probability for B: $P(B) = \frac{\frac{5}{6} \times \frac{1}{6}}{\frac{11}{36}} = \frac{5}{11}$

Final Answer

$$P(\text{A wins}) = \frac{6}{11}, \quad P(\text{B wins}) = \frac{5}{11}$$

Question 53

A and B are two independent events such that $P(A \cap \bar{B}) = \frac{1}{4}$ and $P(\bar{A} \cap B) = \frac{1}{6}$. Find $P(A)$ and $P(B)$.

TYPE 19 Solving for Unknown Parameters Using Independence

★ Likely Exam Question

Given: $P(A \cap \bar{B}) = \frac{1}{4}$, $P(\bar{A} \cap B) = \frac{1}{6}$, A and B are independent.

Concept / Formula Used: $P(A \cap \bar{B}) = P(A)P(\bar{B}) = P(A)(1 - P(B))$

Solution:

Let $P(A) = x$ and $P(B) = y$. Using independence of A and B:

$$P(A \cap \bar{B}) = P(A)(1 - P(B)) = x(1 - y)$$

$$P(\bar{A} \cap B) = (1 - P(A))P(B) = (1 - x)y$$

$$(x - xy) - (y - xy) = \frac{1}{4} - \frac{1}{6}$$

Subtracting equation (2) from equation (1):

$$x - y = \frac{3 - 2}{12} = \frac{1}{12} \quad \text{Substi-}$$

$$x = y + \frac{1}{12}$$

$$y - \left(y + \frac{1}{12}\right)y = \frac{1}{6}$$

Substitute $x = y + \frac{1}{12}$ into equation (2): $y - y^2 - \frac{y}{12} = \frac{1}{6}$ Multiply the entire equation by 12:

$$\frac{11}{12}y - y^2 = \frac{1}{6}$$

$$11y - 12y^2 = 2$$

$$12y^2 - 11y + 2 = 0$$

Factorize the quadratic equation: $3y(4y - 1) - 2(4y - 1) = 0$

$$(3y - 2)(4y - 1) = 0$$

This gives two possible values for y :

$$y = \frac{1}{4} \quad \text{or} \quad y = \frac{2}{3}$$

Case 1: If $y = P(B) = \frac{1}{4}$: $x = P(A) = \frac{1}{4} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$

Case 2: If $y = P(B) = \frac{2}{3}$: $x = P(A) = \frac{2}{3} + \frac{1}{12} = \frac{9}{12} = \frac{3}{4}$

Final Answer

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{4} \quad \text{OR} \quad P(A) = \frac{3}{4}, P(B) = \frac{2}{3}$$

Question 54

X and Y are the two events such that $P(X | Y) = 0.2$ and $P(Y) = 0.5$. Find the value of $P(X' \cap Y)$. Show your steps.

TYPE 22 Conditional Probability Calculation within an Independent-Events Context

Given: $P(X | Y) = 0.2, P(Y) = 0.5$

Concept / Formula Used: $P(X | Y) = \frac{P(X \cap Y)}{P(Y)}$ $P(X' \cap Y) = P(Y) - P(X \cap Y)$

Solution:

From the definition of conditional probability: $P(X | Y) = \frac{P(X \cap Y)}{P(Y)}$ Substitute the

$$0.2 = \frac{P(X \cap Y)}{0.5}$$

given values into the formula: $P(X \cap Y) = 0.2 \times 0.5$ Now, to find $P(X' \cap Y)$:

$$P(X \cap Y) = 0.1$$

$$P(X' \cap Y) = P(Y) - P(X \cap Y)$$

$$= 0.5 - 0.1$$

$$= 0.4$$

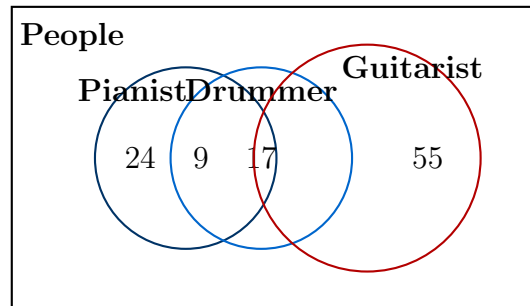
Final Answer

$$P(X' \cap Y) = 0.4$$

Question 55

Shown below is a Venn diagram representing the number of people in a music class who play different instruments. Find the probability that a guitarist selected at random is also a pianist.

TYPE 22 Conditional Probability Calculation within an Independent-Events Context



Venn Diagram of Music Class

Given: From the Venn Diagram:

Number of Pianists only = 24

Number of Pianists who are also Guitarists = 9

Number of Drummer-Guitarists = 17

Number of Guitarists only = 55

Concept / Formula Used: $P(\text{Pianist} \mid \text{Guitarist}) = \frac{n(\text{Pianist} \cap \text{Guitarist})}{n(\text{Guitarist})}$

Solution:

$$n(\text{Guitarist}) = (\text{Pianist} \cap \text{Guitarist}) + (\text{Drummer} \cap \text{Guitarist})$$

$$\begin{aligned} \text{First, calculate the total number of guitarists } n(G): \\ &= 9 + 17 + 55 \\ &= 81 \end{aligned}$$

The number of guitarists who are also pianists is: $n(\text{Pianist} \cap \text{Guitarist}) = 9$ Therefore,

$$P(\text{Pianist} \mid \text{Guitarist}) = \frac{n(\text{Pianist} \cap \text{Guitarist})}{n(\text{Guitarist})}$$

$$\begin{aligned} \text{the probability that a guitarist selected at random is also a pianist is:} \\ &= \frac{9}{81} \\ &= \frac{1}{9} \end{aligned}$$

Final Answer

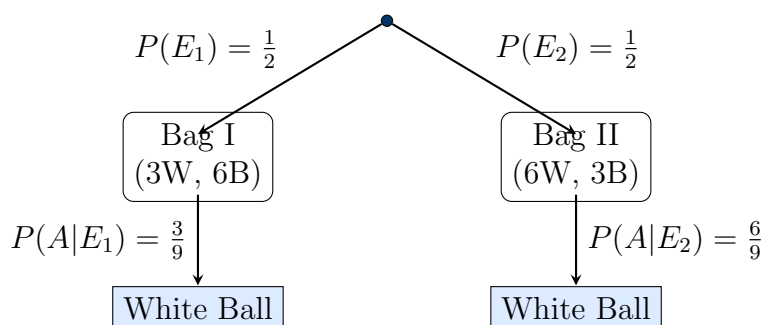
$$P(\text{Pianist} \mid \text{Guitarist}) = \frac{1}{9}$$

13.4 – Law of Total Probability

Question 1

A bag contains 3 white and 6 black balls while another bag contains 6 white and 3 black balls. A bag is selected at random and a ball is drawn. Find the probability that the ball drawn is of white colour.

TYPE 23 Law of Total Probability – Direct Application



Probability tree for selecting a bag and drawing a white ball

Given: Let E_1 be the event of selecting Bag I, E_2 be the event of selecting Bag II, and A be the event that the ball drawn is white. $P(E_1) = \frac{1}{2}$, $P(E_2) = \frac{1}{2}$.

Concept / Formula Used: $P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$

Solution:

Bag I contains 3 white balls and 6 black balls, giving a total of: Total balls in Bag I = $3 + 6 = 9$

The probability of drawing a white ball from Bag I is: $P(A|E_1) = \frac{3}{9} = \frac{1}{3}$ Bag II contains 6 white balls and 3 black balls, giving a total of: Total balls in Bag II = $6 + 3 = 9$ The probability of drawing a white ball from Bag II is: $P(A|E_2) = \frac{6}{9} = \frac{2}{3}$ Using the Law of

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$$

$$= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{2}{3}$$

Total Probability: $= \frac{1}{6} + \frac{2}{6}$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

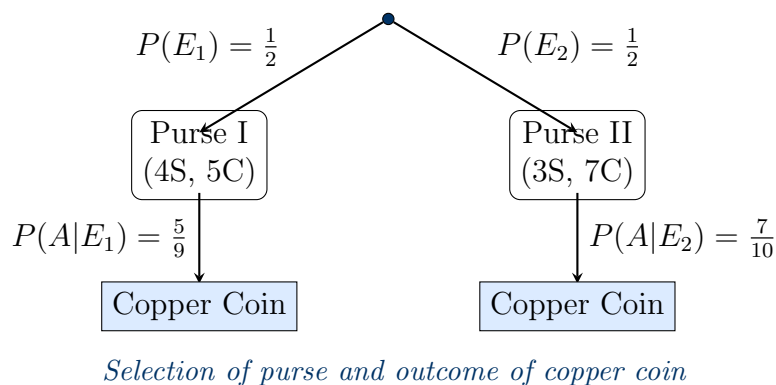
Final Answer

The probability that the ball drawn is of white colour is $\frac{1}{2}$.

Question 2

A purse contains 4 silver and 5 copper coins. A second purse contains 3 silver and 7 copper coins. If a coin is taken out at random from one of the purses, what is the probability that it is a copper coin?

TYPE 23 Law of Total Probability – Direct Application



Given: Let E_1 be the event of selecting Purse I, E_2 be the event of selecting Purse II, and A be the event that the coin drawn is copper. $P(E_1) = \frac{1}{2}$, $P(E_2) = \frac{1}{2}$.

Concept / Formula Used: $P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$

Solution:

Purse I contains 4 silver coins and 5 copper coins: Total coins in Purse I = $4 + 5 = 9$

The probability of drawing a copper coin from Purse I is: $P(A|E_1) = \frac{5}{9}$ Purse II contains

3 silver coins and 7 copper coins: Total coins in Purse II = $3 + 7 = 10$ The probability of drawing a copper coin from Purse II is: $P(A|E_2) = \frac{7}{10}$ Applying the Law of Total

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$$

$$\text{Probability: } = \frac{1}{2} \times \frac{5}{9} + \frac{1}{2} \times \frac{7}{10}$$

Taking the least common multiple

$$= \frac{5}{18} + \frac{7}{20}$$

$$P(A) = \frac{5 \times 10 + 7 \times 9}{180}$$

of 18 and 20, which is 180:

$$= \frac{50 + 63}{180}$$

$$= \frac{113}{180}$$

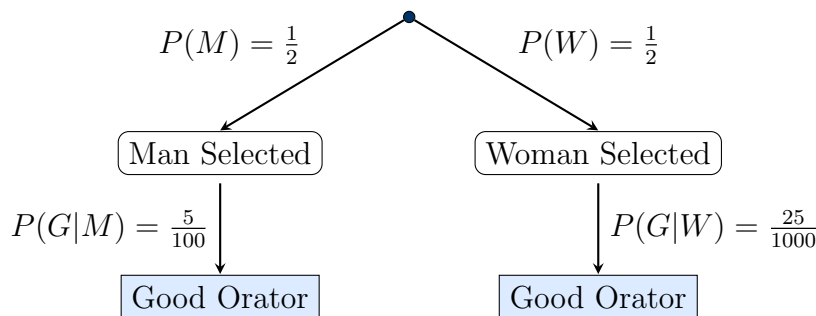
Final Answer

The probability that the drawn coin is a copper coin is $\frac{113}{180}$.

Question 3

Suppose that 5 men out of 100 and 25 women out of 1000 are good orators. Assuming that there are equal number of men and women, find the probability of choosing a good orator.

TYPE 23 Law of Total Probability – Direct Application



Event diagram for choosing a good orator

Given: Let M be the event of choosing a man, W be the event of choosing a woman, and G be the event of choosing a good orator. $P(M) = \frac{1}{2}$, $P(W) = \frac{1}{2}$.

Concept / Formula Used: $P(G) = P(M)P(G|M) + P(W)P(G|W)$

Solution:

Given that 5 men out of 100 are good orators: $P(G|M) = \frac{5}{100} = \frac{1}{20}$ Given that 25 women out of 1000 are good orators: $P(G|W) = \frac{25}{1000} = \frac{1}{40}$ By the Law of Total Prob-

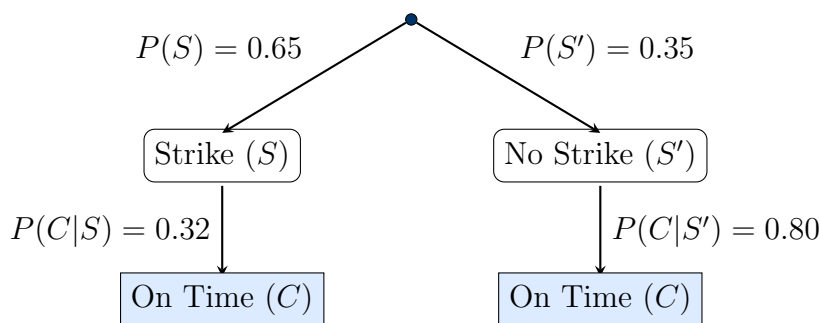
$$\begin{aligned}
 P(G) &= P(M)P(G|M) + P(W)P(G|W) \\
 \text{ability: } &= \frac{1}{2} \times \frac{1}{20} + \frac{1}{2} \times \frac{1}{40} && \text{Finding a common denominator of 80:} \\
 &= \frac{1}{40} + \frac{1}{80} \\
 P(G) &= \frac{2+1}{80} \\
 &= \frac{3}{80}
 \end{aligned}$$

Final Answer

The probability of choosing a good orator is $\frac{3}{80}$.

Question 4

A person has undertaken a construction job. The probabilities are 0.65 that there will be strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

TYPE 23 Law of Total Probability – Direct Application*Event tree for construction completion on time*

Given: Let S be the event that there is a strike, S' be the event of no strike, and C be the event that the job is completed on time. $P(S) = 0.65$, $P(C|S') = 0.80$, $P(C|S) = 0.32$.

Concept / Formula Used: $P(C) = P(S)P(C|S) + P(S')P(C|S')$

Solution:

$$P(S') = 1 - P(S)$$

First, calculate the probability of no strike: $= 1 - 0.65$ Now, apply the Law of $= 0.35$

Total Probability to find $P(C)$: $P(C) = P(S)P(C|S) + P(S')P(C|S')$ Calculating each $= (0.65) \times (0.32) + (0.35) \times (0.80)$

product separately: $0.65 \times 0.32 = 0.208$ Adding the two terms: $P(C) = 0.208 + 0.280$
 $0.35 \times 0.80 = 0.280$ $= 0.488$

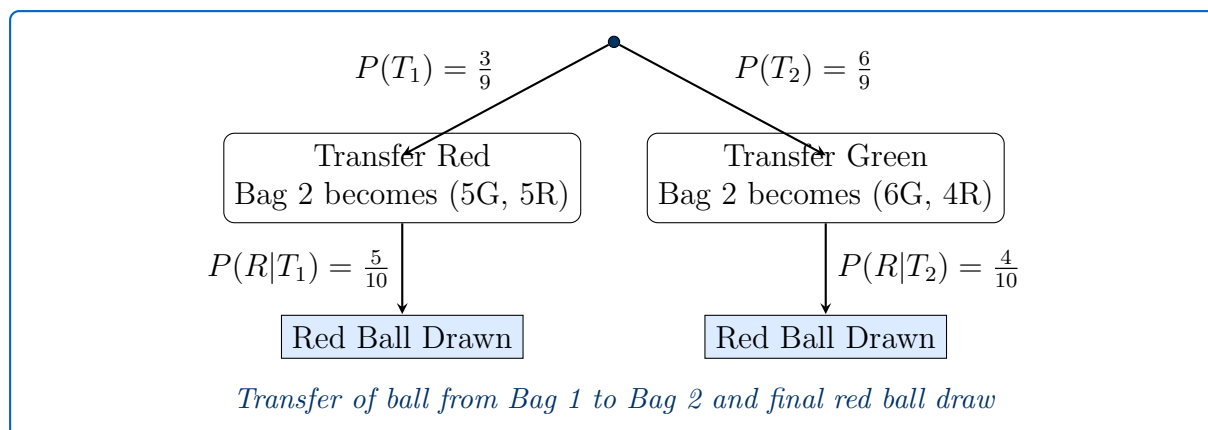
Final Answer

The probability that the construction job will be completed on time is 0.488.

Question 5

There are two bags. One bag contains six green and three red balls. The second bag contains five green and four red balls. One ball is transferred from the first bag to the second bag. Then one ball is drawn from the second bag. Find the probability that it is a red ball.

TYPE 23a Law of Total Probability – With Ball Transfer Between Bags (Polya-type)



Given: Let T_1 be the event of transferring a red ball from Bag 1, T_2 be the event of transferring a green ball from Bag 1, and R be the event of drawing a red ball from Bag 2.

Concept / Formula Used: $P(R) = P(T_1)P(R|T_1) + P(T_2)P(R|T_2)$

Solution:

Bag 1 contains 6 green balls and 3 red balls (total 9 balls). $P(T_1) = \frac{3}{9} = \frac{1}{3}$ **Case 1:** A red ball is transferred (T_1). Bag 2 originally contained 5 green and 4 red balls. After receiving a red ball, it contains 5 green and 5 red balls (total 10 balls). $P(R|T_1) = \frac{5}{10} = \frac{1}{2}$

Case 2: A green ball is transferred (T_2). After receiving a green ball, Bag 2 contains 6 green and 4 red balls (total 10 balls). $P(R|T_2) = \frac{4}{10} = \frac{2}{5}$ Using the Law of Total

$$\begin{aligned}
 P(R) &= P(T_1)P(R|T_1) + P(T_2)P(R|T_2) \\
 \text{Probability: } &= \frac{1}{3} \times \frac{1}{2} + \frac{2}{3} \times \frac{2}{5} \quad \text{Finding a common denominator of} \\
 &= \frac{1}{6} + \frac{4}{15} \\
 30: \quad P(R) &= \frac{5+8}{30} \\
 &= \frac{13}{30}
 \end{aligned}$$

Final Answer

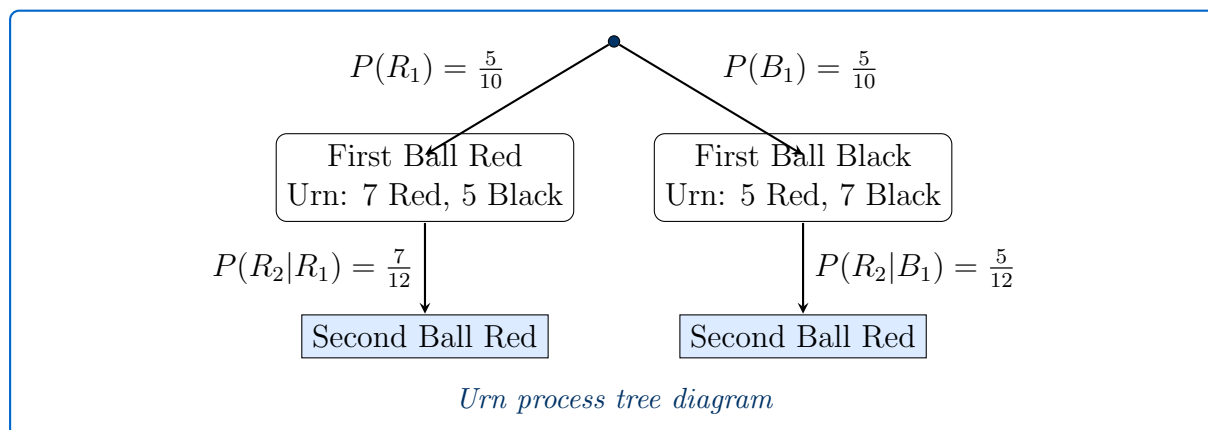
The probability that the ball drawn from the second bag is red is $\frac{13}{30}$.

Question 6

An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and is then returned to the urn. Moreover, 2 additional balls of the same colour are put in the urn and then a ball is drawn at random. What is the probability that the second ball is red?

TYPE 23a Law of Total Probability – With Ball Transfer Between Bags (Polya-type)

★ Likely Exam Question



Given: Initial contents of urn: 5 red, 5 black (total 10). Let R_1 be the event that the first ball drawn is red, B_1 be the event that the first ball drawn is black, and R_2 be the event that the second ball drawn is red.

Concept / Formula Used: $P(R_2) = P(R_1)P(R_2|R_1) + P(B_1)P(R_2|B_1)$

Solution:

The probabilities for the first draw are: $P(R_1) = \frac{5}{10} = \frac{1}{2}$ **Case 1:** The first ball drawn is Red (R_1). Two red balls are added to the urn. The urn now contains $5 + 2 = 7$ red balls and 5 black balls (total 12 balls). $P(R_2|R_1) = \frac{7}{12}$ **Case 2:** The first ball drawn is Black (B_1). Two black balls are added to the urn. The urn now contains 5 red balls and $5 + 2 = 7$ black balls (total 12 balls). $P(R_2|B_1) = \frac{5}{12}$ Now apply the Law of Total

$$P(R_2) = P(R_1)P(R_2|R_1) + P(B_1)P(R_2|B_1)$$

$$\begin{aligned}
 &= \frac{1}{2} \times \frac{7}{12} + \frac{1}{2} \times \frac{5}{12} \\
 \text{Probability: } &= \frac{7}{24} + \frac{5}{24} \\
 &= \frac{12}{24} \\
 &= \frac{1}{2}
 \end{aligned}$$

Final Answer

The probability that the second ball is red is $\frac{1}{2}$.

Question 7

There are two bags, one of which contains 3 black and 4 white balls while the other contains 4 black and 3 white balls. A die is thrown. If it shows up 1 or 3, a ball is

taken from the first bag; but if it shows up any other number, a ball is chosen from the second bag. Find the probability of choosing a black ball.

TYPE 23 Law of Total Probability – Direct Application

Given: Let the bags be denoted as: Bag₁ containing 3 black balls and 4 white balls (total 7 balls), and Bag₂ containing 4 black balls and 3 white balls (total 7 balls). Let E_1 be the event that the die shows up 1 or 3, E_2 be the event that the die shows up any other number, and A be the event of choosing a black ball.

To Find: The probability of choosing a black ball, $P(A)$.

Concept / Formula Used: Law of Total Probability: $P(A) = P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)$

Solution:

First, we calculate the probabilities of selecting each bag based on the roll of a fair die. The sample space of throwing a single fair die is:

$$S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$$

For event E_1 (die shows 1 or 3):

$$E_1 = \{1, 3\} \implies n(E_1) = 2$$

$$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

For event E_2 (die shows 2, 4, 5, or 6):

$$E_2 = \{2, 4, 5, 6\} \implies n(E_2) = 4$$

$$P(E_2) = \frac{n(E_2)}{n(S)} = \frac{4}{6} = \frac{2}{3}$$

Next, we find the conditional probabilities of drawing a black ball from each bag:

- If E_1 occurs, the ball is drawn from Bag₁, which contains 3 black and 4 white balls:

$$P(A|E_1) = \frac{\text{Number of black balls in Bag}_1}{\text{Total balls in Bag}_1} = \frac{3}{7}$$

- If E_2 occurs, the ball is drawn from Bag₂, which contains 4 black and 3 white balls:

$$P(A|E_2) = \frac{\text{Number of black balls in Bag}_2}{\text{Total balls in Bag}_2} = \frac{4}{7}$$

$$\begin{aligned}
 P(A) &= P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) \\
 &= \left(\frac{1}{3}\right) \cdot \left(\frac{3}{7}\right) + \left(\frac{2}{3}\right) \cdot \left(\frac{4}{7}\right) \\
 &= \frac{3}{21} + \frac{8}{21} \\
 &= \frac{11}{21}
 \end{aligned}$$

By applying the Law of Total Probability, we obtain:

Final Answer

The probability of choosing a black ball is $\frac{11}{21}$.

Question 8

Urn I has 2 white and 3 black balls, urn II has 4 white and 1 black ball and urn III has 3 white and 2 black balls. An urn is selected at random and a ball is drawn at random. What is the probability of drawing a white ball?

TYPE 23 Law of Total Probability – Direct Application

To Find: The probability of drawing a white ball, $P(A)$.

Concept / Formula Used: Law of Total Probability: If $E_1, E_2, \dots, E_n$ are mutually exclusive and exhaustive events, then for any event A :

$$P(A) = \sum_{i=1}^n P(E_i)P(A|E_i)$$

Solution:

Let us define the events as follows:

- E_1 : Urn I is selected.
- E_2 : Urn II is selected.
- E_3 : Urn III is selected.
- A : A white ball is drawn.

Since one of the three urns is selected at random, the probabilities of selecting each urn

$$P(E_1) = \frac{1}{3}$$

are equal: $P(E_2) = \frac{1}{3}$

$$P(E_3) = \frac{1}{3}$$

Next, we determine the conditional probabilities of drawing a white ball from each specific urn:

- Urn I contains 2 white and 3 black balls (Total = 5 balls):

$$P(A|E_1) = \frac{2}{5}$$

- Urn II contains 4 white and 1 black ball (Total = 5 balls):

$$P(A|E_2) = \frac{4}{5}$$

- Urn III contains 3 white and 2 black balls (Total = 5 balls):

$$P(A|E_3) = \frac{3}{5}$$

By the Law of Total Probability, the overall probability of drawing a white ball is:
 $P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3)$

$$\begin{aligned} &= \left(\frac{1}{3} \times \frac{2}{5}\right) + \left(\frac{1}{3} \times \frac{4}{5}\right) + \left(\frac{1}{3} \times \frac{3}{5}\right) \\ &= \frac{1}{3} \left(\frac{2}{5} + \frac{4}{5} + \frac{3}{5}\right) \\ &= \frac{1}{3} \times \frac{9}{5} \\ &= \frac{3}{5} \end{aligned}$$

Final Answer

The probability of drawing a white ball is $\frac{3}{5}$.

Question 9

In a factory, a product is manufactured by any of three machines A, B and C. They produce respectively 25%, 35% and 40% of the total products. A product is selected at random. Find the probability that it is defective. Assume that machines A, B and C produce respectively 5%, 4% and 2% defective items.

TYPE 23 Law of Total Probability – Direct Application

To Find: The probability that a randomly selected product is defective, $P(D)$.

Concept / Formula Used: Law of Total Probability: For mutually exclusive and exhaustive events $E_1, E_2, \dots, E_n$, the probability of an event D is given by:

$$P(D) = \sum_{i=1}^n P(E_i) \cdot P(D|E_i)$$

Solution:

Let us define the following events:

- E_1 : The product is manufactured by machine A.
- E_2 : The product is manufactured by machine B.
- E_3 : The product is manufactured by machine C.
- D : The selected product is defective.

From the given information, the probabilities of selecting products from machines A, B,

$$P(E_1) = 25\% = \frac{25}{100}$$

and C are: $P(E_2) = 35\% = \frac{35}{100}$

$$P(E_3) = 40\% = \frac{40}{100}$$

The conditional probabilities of producing a defective product by each machine are:

$$P(D|E_1) = 5\% = \frac{5}{100}$$

$$P(D|E_2) = 4\% = \frac{4}{100}$$

$$P(D|E_3) = 2\% = \frac{2}{100}$$

By the Law of Total Probability, the probability that a randomly selected product is defective is: $P(D) = P(E_1) \cdot P(D|E_1) + P(E_2) \cdot P(D|E_2) + P(E_3) \cdot P(D|E_3)$

$$\begin{aligned} P(D) &= \left(\frac{25}{100} \times \frac{5}{100} \right) + \left(\frac{35}{100} \times \frac{4}{100} \right) + \left(\frac{40}{100} \times \frac{2}{100} \right) \\ &= \frac{125}{10000} + \frac{140}{10000} + \frac{80}{10000} \\ &= \frac{125 + 140 + 80}{10000} \\ &= \frac{345}{10000} \end{aligned}$$

Substituting the values into the formula:

Simplifying the fraction by dividing both the numerator and the denominator by 5:

$$\begin{aligned} P(D) &= \frac{345 \div 5}{10000 \div 5} \\ &= \frac{69}{2000} \end{aligned}$$

Final Answer

The probability that the selected product is defective is $\frac{69}{2000}$.

Question 10

Bag A contains 1 white, 2 blue and 3 red balls. Bag B contains 3 white, 3 blue and 2 red balls. Bag C contains 2 white, 3 blue and 4 red balls. One bag is selected at random, and then two balls are drawn from the selected bag. Find the probability that the balls drawn are white and red.

TYPE 23 Law of Total Probability – Direct Application

Concept / Formula Used: Law of Total Probability: If $E_1, E_2, \dots, E_n$ are mutually exclusive and exhaustive events, then for any event A :

$$P(A) = \sum_{i=1}^n P(E_i)P(A | E_i)$$

Given: Let the bags be A , B , and C . Bag A contains 1 white, 2 blue, and 3 red balls (total 6 balls). Bag B contains 3 white, 3 blue, and 2 red balls (total 8 balls). Bag C contains 2 white, 3 blue, and 4 red balls (total 9 balls). One bag is selected at random, and two balls are drawn.

To Find: The probability that the two drawn balls are one white and one red.

Solution:

Let E_1 , E_2 , and E_3 be the events of selecting Bag A , Bag B , and Bag C respectively.

$$P(E_1) = \frac{1}{3}$$

Since one of the three bags is chosen at random: $P(E_2) = \frac{1}{3}$

$$P(E_3) = \frac{1}{3}$$

Let A be the event that the two balls drawn from the selected bag are one white and one red. We calculate the conditional probabilities of drawing one white and one red ball from each bag:

Case 1: Drawing from Bag A (1 white, 2 blue, 3 red; total 6 balls) The total number of ways to draw 2 balls from 6 balls is $\binom{6}{2}$. The number of ways to draw 1 white

$$\begin{aligned} P(A | E_1) &= \frac{\binom{1}{1} \times \binom{3}{1}}{\binom{6}{2}} \\ &= \frac{1 \times 3}{\frac{6 \times 5}{2 \times 1}} \\ &= \frac{3}{15} \\ &= \frac{1}{5} \end{aligned}$$

ball and 1 red ball is $\binom{1}{1} \times \binom{3}{1}$.

Case 2: Drawing from Bag B (3 white, 3 blue, 2 red; total 8 balls) The total number of ways to draw 2 balls from 8 balls is $\binom{8}{2}$. The number of ways to draw 1 white

$$\begin{aligned} P(A | E_2) &= \frac{\binom{3}{1} \times \binom{2}{1}}{\binom{8}{2}} \\ &= \frac{3 \times 2}{\frac{8 \times 7}{2 \times 1}} \\ &= \frac{6}{28} \\ &= \frac{3}{14} \end{aligned}$$

ball and 1 red ball is $\binom{3}{1} \times \binom{2}{1}$.

Case 3: Drawing from Bag C (2 white, 3 blue, 4 red; total 9 balls) The total number of ways to draw 2 balls from 9 balls is $\binom{9}{2}$. The number of ways to draw 1 white

$$\begin{aligned}
 P(A | E_3) &= \frac{\binom{2}{1} \times \binom{4}{1}}{\binom{9}{2}} \\
 &= \frac{2 \times 4}{\frac{9 \times 8}{2 \times 1}} \\
 &= \frac{8}{36} \\
 &= \frac{2}{9}
 \end{aligned}$$

ball and 1 red ball is $\binom{2}{1} \times \binom{4}{1}$.

By the Law of Total Probability, the probability of drawing one white and one red ball

$$\begin{aligned}
 P(A) &= P(E_1)P(A | E_1) + P(E_2)P(A | E_2) + P(E_3)P(A | E_3) \\
 \text{is: } &= \frac{1}{3} \left(\frac{1}{5} \right) + \frac{1}{3} \left(\frac{3}{14} \right) + \frac{1}{3} \left(\frac{2}{9} \right) \\
 &= \frac{1}{3} \left(\frac{1}{5} + \frac{3}{14} + \frac{2}{9} \right)
 \end{aligned}$$

To find the sum, we take the least common multiple of the denominators 5, 14, and 9,

$$\begin{aligned}
 P(A) &= \frac{1}{3} \left(\frac{1 \times 126}{630} + \frac{3 \times 45}{630} + \frac{2 \times 70}{630} \right) \\
 &= \frac{1}{3} \left(\frac{126 + 135 + 140}{630} \right) \\
 &= \frac{1}{3} \left(\frac{401}{630} \right) \\
 &= \frac{401}{1890}
 \end{aligned}$$

which is $5 \times 14 \times 9 = 630$:

Final Answer

The probability that the balls drawn are white and red is $\frac{401}{1890}$.

Common Mistake: A common mistake is to assume that the order of drawing matters and calculate the probability of drawing a white ball first and then a red ball without considering the reverse order (red first, then white). Using combinations $\binom{n}{r}$ automatically handles the unordered selection of the two balls.

Question 11

Bag A contains 4 white balls and 3 black balls, while bag B contains 3 white balls and 5 black balls. Two balls are drawn from bag A and placed in bag B. Then, what is the probability of drawing a white ball from bag B? (ISC 2019)

TYPE 23a Law of Total Probability – With Ball Transfer Between Bags (Polya-type)

Given: Bag A contains 4 white balls and 3 black balls (total 7 balls). Bag B contains 3 white balls and 5 black balls (total 8 balls). Two balls are drawn from Bag A and transferred to Bag B.

To Find: The probability of drawing a white ball from Bag B after the transfer.

Concept / Formula Used: Law of Total Probability:

$$P(A) = \sum_{i=1}^n P(E_i)P(A|E_i)$$

where $E_1, E_2, \dots, E_n$ are mutually exclusive and exhaustive events.

Solution:

Let E_1 , E_2 , and E_3 be the events representing the transfer of balls from Bag A to Bag B:

- E_1 : Both transferred balls are white.
- E_2 : One transferred ball is white and one is black.
- E_3 : Both transferred balls are black.

Let W be the event of drawing a white ball from Bag B after the transfer.

First, we calculate the probabilities of the events E_1 , E_2 , and E_3 . The total number of balls in Bag A is $4 + 3 = 7$. The number of ways to choose 2 balls out of 7 is:

$$\binom{7}{2} = \frac{7 \times 6}{2 \times 1} = 21$$

$$P(E_1) = \frac{\binom{4}{2}}{\binom{7}{2}} = \frac{6}{21}$$

The probabilities of the transfer events are: $P(E_2) = \frac{\binom{4}{1} \times \binom{3}{1}}{\binom{7}{2}} = \frac{4 \times 3}{21} = \frac{12}{21}$

$$P(E_3) = \frac{\binom{3}{2}}{\binom{7}{2}} = \frac{3}{21}$$

After transferring 2 balls from Bag A to Bag B, the total number of balls in Bag B becomes $8 + 2 = 10$.

Now, we find the conditional probabilities of drawing a white ball from Bag B under each case:

- **Case 1:** If E_1 occurs (2 white balls are transferred), Bag B now contains $3 + 2 = 5$ white balls and 5 black balls. $P(W|E_1) = \frac{5}{10}$
- **Case 2:** If E_2 occurs (1 white and 1 black ball are transferred), Bag B now contains $3 + 1 = 4$ white balls and $5 + 1 = 6$ black balls. $P(W|E_2) = \frac{4}{10}$

- **Case 3:** If E_3 occurs (2 black balls are transferred), Bag B now contains 3 white balls and $5 + 2 = 7$ black balls. $P(W|E_3) = \frac{3}{10}$

By the Law of Total Probability, the probability of drawing a white ball from Bag B is:
 $P(W) = P(E_1)P(W|E_1) + P(E_2)P(W|E_2) + P(E_3)P(W|E_3)$

$$\begin{aligned}
 &= \left(\frac{6}{21} \times \frac{5}{10}\right) + \left(\frac{12}{21} \times \frac{4}{10}\right) + \left(\frac{3}{21} \times \frac{3}{10}\right) \\
 &= \frac{30}{210} + \frac{48}{210} + \frac{9}{210} \\
 &= \frac{30 + 48 + 9}{210} \\
 &= \frac{87}{210}
 \end{aligned}$$

Dividing both the numerator and the denominator by their greatest common divisor, which is 3: $P(W) = \frac{29}{70}$

Final Answer

The probability of drawing a white ball from Bag B is $\frac{29}{70}$.

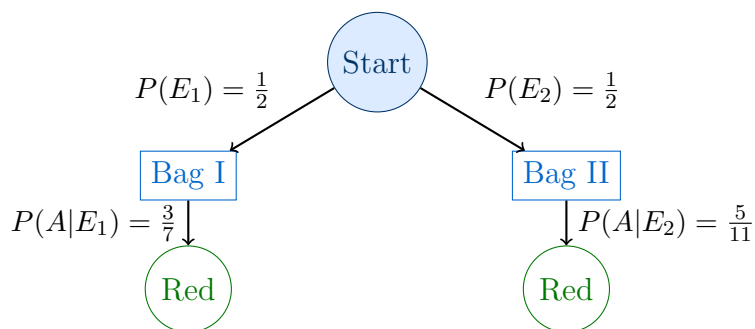
13.5 – Bayes' Theorem

Question 1

Bag I contains 3 red and 4 black balls while another bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from bag II.

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Probability tree for bag selection and drawing a red ball

Given: Let E_1 be the event of selecting Bag I, and E_2 be the event of selecting Bag II.

Let A be the event of drawing a red ball.

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

Bag I composition: 3 red, 4 black (Total = 7 balls)

Bag II composition: 5 red, 6 black (Total = 11 balls)

$$P(A|E_1) = \frac{3}{7}, P(A|E_2) = \frac{5}{11}$$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

By applying Bayes' Theorem to find the probability that the red ball was drawn from

$$P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$$

$$= \frac{\frac{1}{2} \cdot \frac{5}{11}}{\frac{1}{2} \cdot \frac{3}{7} + \frac{1}{2} \cdot \frac{5}{11}}$$

$$= \frac{\frac{5}{22}}{\frac{1}{2} \left(\frac{3}{7} + \frac{5}{11} \right)}$$

$$= \frac{\frac{5}{22}}{\frac{1}{2} \left(\frac{33 + 35}{77} \right)}$$

Bag II:

$$= \frac{\frac{5}{22}}{\frac{1}{2} \cdot \frac{68}{77}}$$

$$= \frac{\frac{5}{22}}{\frac{34}{77}}$$

$$= \frac{5}{22} \times \frac{77}{34}$$

$$= \frac{5 \times 7}{2 \times 34}$$

$$= \frac{35}{68}$$

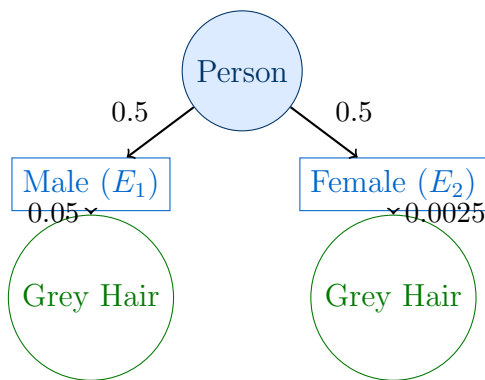
Final Answer

The probability that the drawn red ball came from Bag II is $\frac{35}{68}$.

Common Mistake: Forgetting to divide by the total probability of selecting a red ball across both bags is a common error.

Question 2

Suppose 5% of men and 0.25% of women have grey hairs. A grey-haired person is selected at random. What is the probability of this person being male? Assume that there are equal number of males and females.

TYPE 24 Bayes' Theorem – Direct Application

Tree diagram for male/female selection and grey hair

Given: Let E_1 be the event that the person selected is male, and E_2 be the event that the person is female.

Let A be the event that the person has grey hair.

$$P(E_1) = 0.5, P(E_2) = 0.5$$

$$P(A|E_1) = 5\% = 0.05$$

$$P(A|E_2) = 0.25\% = 0.0025$$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned}
 P(E_1|A) &= \frac{0.5 \times 0.05}{0.5 \times 0.05 + 0.5 \times 0.0025} \\
 &= \frac{0.5 \times 0.05}{0.5 \times (0.05 + 0.0025)} \\
 &= \frac{0.05}{0.0525} \\
 &= \frac{500}{525} \\
 &= \frac{20}{21}
 \end{aligned}$$

Substitute the given probabilities into Bayes' Theorem:

Final Answer

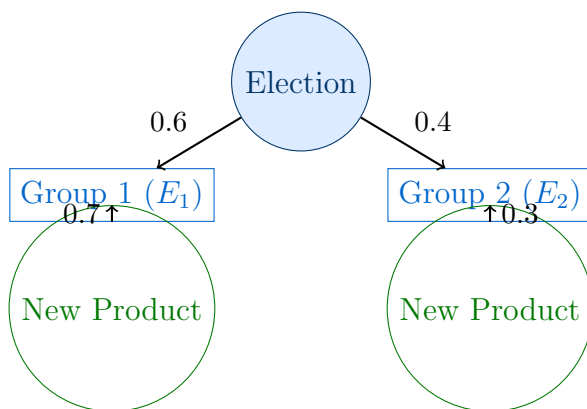
The probability that the selected grey-haired person is male is $\frac{20}{21}$.

Common Mistake: Converting 0.25% as 0.25 instead of 0.0025 is a frequent percentage conversion mistake.

Question 3

Two groups are competing for the positions on Board of Directors of a corporation. The probabilities that the first group and the second group will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product is 0.7 and the corresponding probability is 0.3 if the second group wins. Find the probability that the new product introduced was by second group.

TYPE 24 Bayes' Theorem – Direct Application



Probability tree for corporate board outcome

Given: Let E_1 be the event that the first group wins, and E_2 be the event that the second group wins.

Let A be the event that a new product is introduced.

$$P(E_1) = 0.6, P(E_2) = 0.4$$

$$P(A|E_1) = 0.7, P(A|E_2) = 0.3$$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned} P(E_2|A) &= \frac{0.4 \times 0.3}{(0.6 \times 0.7) + (0.4 \times 0.3)} \\ &= \frac{0.12}{0.42 + 0.12} \\ &= \frac{0.12}{0.54} \\ &= \frac{12}{54} \\ &= \frac{2}{9} \end{aligned}$$

Substitute the given values into Bayes' formula:

Final Answer

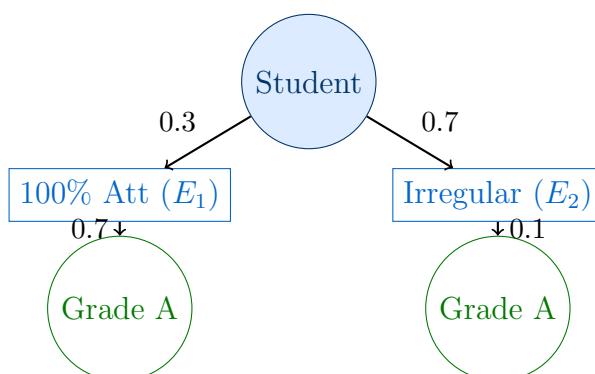
The probability that the new product was introduced by the second group is $\frac{2}{9}$.

Common Mistake: Mixing up $P(E_1)$ and $P(E_2)$ in the numerator when solving for the second group.

Question 4

Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous years result report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade. At the end of the year, one student is chosen at random and he was found to have an A grade. What is the probability that the student has 100% attendance?

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for attendance and grades

Given: Let E_1 be the event that a student has 100% attendance, and E_2 be the event that a student is irregular.

Let A be the event that a student attains grade A.

$$P(E_1) = 30\% = 0.3, P(E_2) = 70\% = 0.7$$

$$P(A|E_1) = 70\% = 0.7, P(A|E_2) = 10\% = 0.1$$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$P(E_1|A) = \frac{0.3 \times 0.7}{(0.3 \times 0.7) + (0.7 \times 0.1)}$$

$$= \frac{0.21}{0.21 + 0.07}$$

Substitute the given probabilities:

$$= \frac{0.21}{0.28}$$

$$= \frac{21}{28}$$

$$= \frac{3}{4}$$

Final Answer

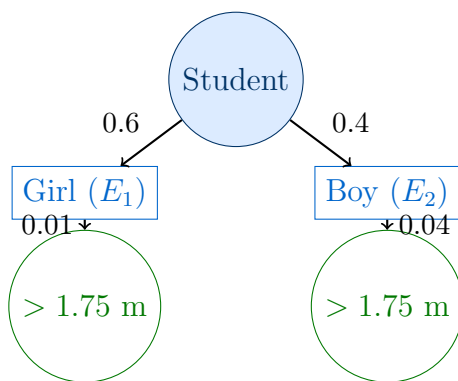
The probability that the student has 100% attendance is $\frac{3}{4}$.

Common Mistake: Confusing conditional probability $P(A|E_1)$ with joint probability $P(A \cap E_1)$.

Question 5

In a certain college, 4% of boys and 1% of girls are taller than 1.75 metres. Furthermore, 60% of the students in the college are girls. A student is selected at random from the college and is found to be taller than 1.75 metres. Find the probability that the selected student is a girl.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for height conditional probability

Given: Let E_1 be the event that the student is a girl, and E_2 be the event that the student is a boy.

Let A be the event that the student is taller than 1.75 m.

$$P(E_1) = 60\% = 0.6$$

Since E_1 and E_2 form a partition, $P(E_2) = 1 - 0.6 = 0.4$

$$P(A|E_1) = 1\% = 0.01, P(A|E_2) = 4\% = 0.04$$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$P(E_1|A) = \frac{0.6 \times 0.01}{(0.6 \times 0.01) + (0.4 \times 0.04)}$$

$$= \frac{0.006}{0.006 + 0.016}$$

$$\text{Calculating step-by-step: } = \frac{0.006}{0.022}$$

$$= \frac{6}{22}$$

$$= \frac{3}{11}$$

Final Answer

The probability that the selected tall student is a girl is $\frac{3}{11}$.

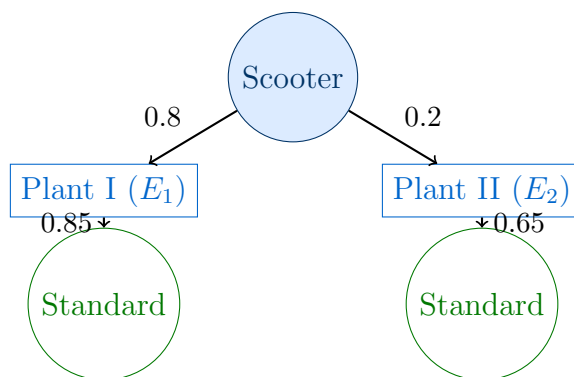
Common Mistake: Assuming equal probability for boys and girls when the problem explicitly specifies 60% girls.

Question 6

A company has two plants to manufacture scooters. Plant I manufactures 80% of the scooters and plant II manufactures 20%. At plant I, 85 out of 100 scooters are rated

as of standard quality and at plant II, 65 out of 100 scooters are rated as of standard quality. A scooter is chosen at random and is found to be of standard quality. What is the probability that it has come from plant I?

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for plant production quality

Given: Let E_1 be the event that the scooter is from Plant I, and E_2 be the event that it is from Plant II.

Let A be the event that the chosen scooter is of standard quality.

$$P(E_1) = 80\% = 0.8, P(E_2) = 20\% = 0.2$$

$$P(A|E_1) = \frac{85}{100} = 0.85, P(A|E_2) = \frac{65}{100} = 0.65$$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned} P(E_1|A) &= \frac{0.8 \times 0.85}{(0.8 \times 0.85) + (0.2 \times 0.65)} \\ &= \frac{0.68}{0.68 + 0.13} \\ &= \frac{0.68}{0.81} \\ &= \frac{68}{81} \end{aligned}$$

Substituting the given values into the Bayes' formula:

Final Answer

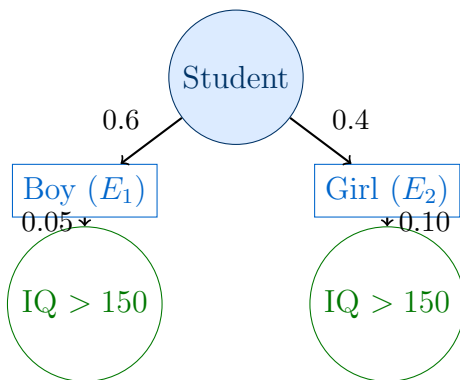
The probability that the scooter came from Plant I is $\frac{68}{81}$.

Common Mistake: Incorrect decimal placement during multiplication of percentages.

Question 7

In a class, having 60% boys, 5% of the boys and 10% of the girls have an I.Q. of more than 150. A student is selected at random and found to have an I.Q. of more than 150. Find the probability that the selected student is a boy.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for student gender and IQ

Given: Let E_1 be the event that the selected student is a boy, and E_2 be the event that the student is a girl.

Let A be the event that the student has an I.Q. > 150 .

$P(E_1) = 60\% = 0.6$, $P(E_2) = 40\% = 0.4$

$P(A|E_1) = 5\% = 0.05$, $P(A|E_2) = 10\% = 0.10$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned}
 P(E_1|A) &= \frac{0.6 \times 0.05}{(0.6 \times 0.05) + (0.4 \times 0.10)} \\
 &= \frac{0.03}{0.03 + 0.04} \\
 &= \frac{0.03}{0.07} \\
 &= \frac{3}{7}
 \end{aligned}$$

Substitute all values into Bayes' formula:

Final Answer

The probability that the selected high-IQ student is a boy is $\frac{3}{7}$.

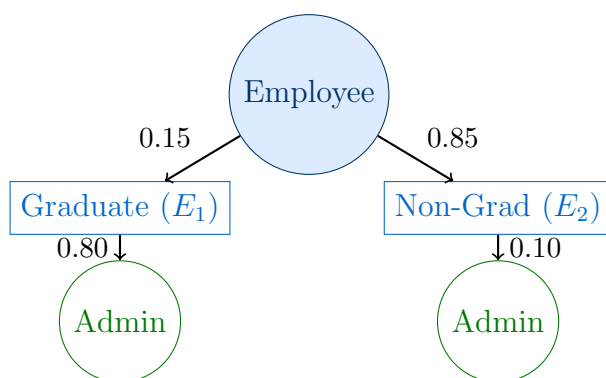
Common Mistake: Taking $P(E_2) = 0.6$ instead of calculating $P(E_2) = 1 - 0.6 = 0.4$.

Question 8

In a company, 15% of the employees are graduates and 85% of the employees are non-graduates. As per the annual report of the company, 80% of the graduates and 10% of the non-graduates are in the administrative positions. Find the probability that an employee selected at random from those working in administrative positions will be a graduate.

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Tree diagram for employee qualifications

Given: Let E_1 be the event that the selected employee is a graduate, and E_2 be the event that the employee is a non-graduate.

Let A be the event that the employee holds an administrative position.

$$P(E_1) = 15\% = 0.15, P(E_2) = 85\% = 0.85$$

$$P(A|E_1) = 80\% = 0.80, P(A|E_2) = 10\% = 0.10$$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$P(E_1|A) = \frac{0.15 \times 0.80}{(0.15 \times 0.80) + (0.85 \times 0.10)}$$

$$= \frac{0.120}{0.120 + 0.085}$$

$$= \frac{0.120}{0.205}$$

$$= \frac{120}{205}$$

$$= \frac{24}{41}$$

Substitute values into the Bayes' formula:

Final Answer

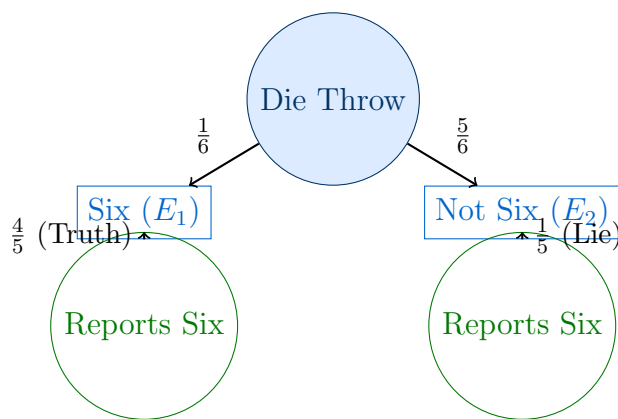
The probability that the selected administrative employee is a graduate is $\frac{24}{41}$.

Common Mistake: Dividing by 0.120 instead of the total administrative probability 0.205.

Question 9

A man is known to speak truth 4 out of 5 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

TYPE 24 Bayes' Theorem – Direct Application

★ **Likely Exam Question**

Tree diagram for die throw and truthfulness

Given: Let E_1 be the event that a six actually occurs on the die.

Let E_2 be the event that a six does not occur on the die.

Let A be the event that the man reports a six.

$$P(E_1) = \frac{1}{6}, P(E_2) = 1 - \frac{1}{6} = \frac{5}{6}$$

Probability the man speaks the truth: $P(A|E_1) = \frac{4}{5}$

Probability the man lies (reports a six when it is not a six): $P(A|E_2) = 1 - \frac{4}{5} = \frac{1}{5}$

Concept / Formula Used: $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned}
 P(E_1|A) &= \frac{\frac{1}{6} \cdot \frac{4}{5}}{\left(\frac{1}{6} \cdot \frac{4}{5}\right) + \left(\frac{5}{6} \cdot \frac{1}{5}\right)} \\
 &= \frac{\frac{4}{30}}{\frac{4}{30} + \frac{5}{30}} \\
 &= \frac{\frac{4}{30}}{\frac{9}{30}} \\
 &= \frac{4}{9}
 \end{aligned}$$

Substitute the respective probabilities into Bayes' Theorem:

Final Answer

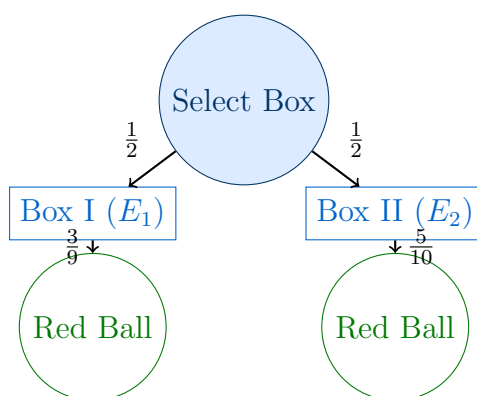
The probability that it is actually a six is $\frac{4}{9}$.

Common Mistake: Assuming the probability of lying is $\frac{5}{6}$ instead of $1 - \frac{4}{5} = \frac{1}{5}$.

Question 10

There are two boxes, namely box-I and box-II. Box-I contains 3 red and 6 black balls. Box-II contains 5 red and 5 black balls. One of the two boxes is selected at random and a ball is drawn at random. The ball drawn is found to be red. Find the probability that this red ball comes from box-II.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for box selection

Given: Let E_1 be the event of choosing Box-I, and E_2 be the event of choosing Box-II. Let A be the event of drawing a red ball.

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

$$\text{Box-I: 3 Red, 6 Black (Total = 9 balls)} \implies P(A|E_1) = \frac{3}{9} = \frac{1}{3}$$

$$\text{Box-II: 5 Red, 5 Black (Total = 10 balls)} \implies P(A|E_2) = \frac{5}{10} = \frac{1}{2}$$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$P(E_2|A) = \frac{\frac{1}{2} \cdot \frac{1}{2}}{\left(\frac{1}{2} \cdot \frac{1}{3}\right) + \left(\frac{1}{2} \cdot \frac{1}{2}\right)}$$

$$= \frac{\frac{1}{4}}{\frac{1}{2} \left(\frac{1}{3} + \frac{1}{2}\right)}$$

Substituting values:

$$= \frac{\frac{1}{4}}{\frac{1}{2} \cdot \frac{5}{6}}$$

$$= \frac{\frac{1}{4}}{\frac{5}{12}}$$

$$= \frac{1}{4} \times \frac{12}{5}$$

$$= \frac{3}{5}$$

Final Answer

The probability that the red ball comes from Box-II is $\frac{3}{5}$.

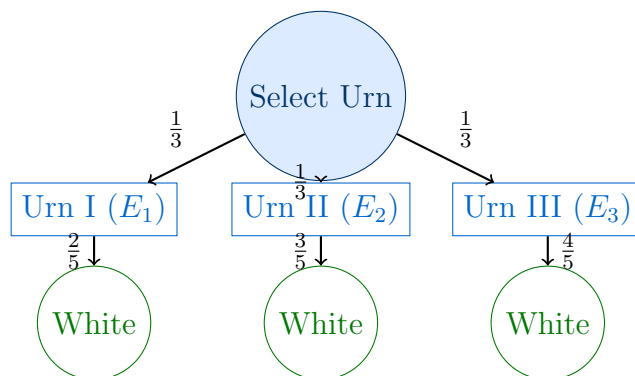
Common Mistake: Calculating total balls as denominators incorrectly without checking individual box counts.

Question 11 (i)

There are three urns containing 2 white and 3 black balls, 3 white and 2 black balls, and 4 white and 1 black ball, respectively. There is an equal opportunity of each urn being chosen. A ball is drawn at random from the chosen urn and it is found to be white. Find the probability that ball drawn was from the second urn.

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Tree diagram for three urns

Given: Let E_1, E_2, E_3 be the events of choosing Urn I, Urn II, and Urn III respectively. Since choices are equally likely, $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$.

Let A be the event of drawing a white ball.

Urn I: 2W, 3B (Total = 5) $\implies P(A|E_1) = \frac{2}{5}$

Urn II: 3W, 2B (Total = 5) $\implies P(A|E_2) = \frac{3}{5}$

Urn III: 4W, 1B (Total = 5) $\implies P(A|E_3) = \frac{4}{5}$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)}$

Solution:

$$\begin{aligned}
 P(E_2|A) &= \frac{\frac{1}{3} \cdot \frac{3}{5}}{\left(\frac{1}{3} \cdot \frac{2}{5}\right) + \left(\frac{1}{3} \cdot \frac{3}{5}\right) + \left(\frac{1}{3} \cdot \frac{4}{5}\right)} \\
 &= \frac{\frac{1}{3} \cdot \frac{3}{5}}{\frac{1}{3} \left(\frac{2}{5} + \frac{3}{5} + \frac{4}{5}\right)} \\
 &= \frac{\frac{3}{5}}{2 + 3 + 4} \\
 &= \frac{3}{9} \\
 &= \frac{1}{3}
 \end{aligned}$$

Substitute all given probabilities:

Final Answer

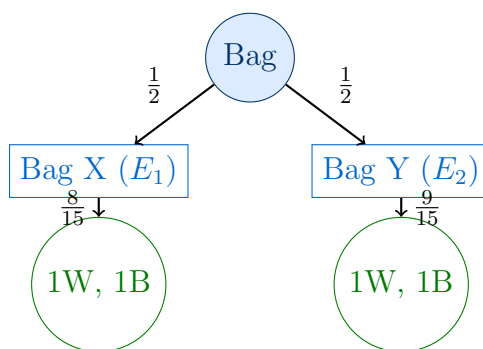
The probability that the white ball was drawn from the second urn is $\frac{1}{3}$.

Common Mistake: Cancelling out $\frac{1}{3}$ improperly across terms in complex fractions.

Question 11 (ii)

A bag X contains 4 white balls and 2 black balls, while another bag Y contains 3 white balls and 3 black balls. Two balls are drawn (without replacement) at random from one of the bags and were found to be one white and one black. Find the probability that the balls were drawn from bag Y.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for selecting two balls without replacement

Given: Let E_1 be the event of selecting Bag X, and E_2 be the event of selecting Bag Y.

Let A be the event of drawing 1 white and 1 black ball without replacement.

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

Bag X: 4W, 2B (Total = 6 balls). Number of ways to choose 2 balls = $\binom{6}{2} = 15$.

$$P(A|E_1) = \frac{\binom{4}{1} \cdot \binom{2}{1}}{\binom{6}{2}} = \frac{4 \times 2}{15} = \frac{8}{15}$$

Bag Y: 3W, 3B (Total = 6 balls). Number of ways to choose 2 balls = $\binom{6}{2} = 15$.

$$P(A|E_2) = \frac{\binom{3}{1} \cdot \binom{3}{1}}{\binom{6}{2}} = \frac{3 \times 3}{15} = \frac{9}{15}$$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned} P(E_2|A) &= \frac{\frac{1}{2} \cdot \frac{9}{15}}{\left(\frac{1}{2} \cdot \frac{8}{15}\right) + \left(\frac{1}{2} \cdot \frac{9}{15}\right)} \\ &= \frac{\frac{1}{2} \cdot \frac{9}{15}}{\frac{1}{2} \left(\frac{8}{15} + \frac{9}{15}\right)} \\ &= \frac{9}{\frac{15}{17}} \\ &= \frac{9}{17} \end{aligned}$$

Substitute the computed conditional probabilities into Bayes' Theorem:

Final Answer

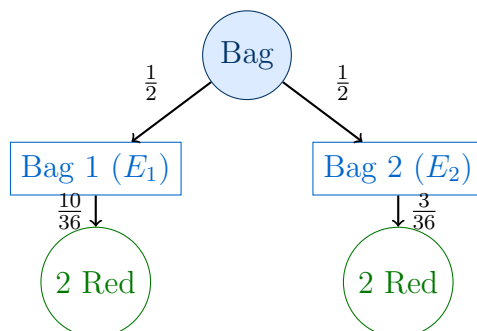
The probability that the balls were drawn from Bag Y is $\frac{9}{17}$.

Common Mistake: Treating the draw as with-replacement draw instead of using combinations for without-replacement.

Question 12

A bag contains 5 red and 4 black balls, a second bag contains 3 red and 6 black balls. One of the two bags is selected at random and two balls are drawn at random (without replacement) both of which are found to be red. Find the probability that the balls are drawn from the second bag.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for selecting two red balls

Given: Let E_1 be the event of selecting Bag 1, and E_2 be the event of selecting Bag 2.

Let A be the event of drawing 2 red balls without replacement.

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

$$\text{Bag 1: 5R, 4B (Total = 9). } P(A|E_1) = \frac{\binom{5}{2}}{\binom{9}{2}} = \frac{10}{36}$$

$$\text{Bag 2: 3R, 6B (Total = 9). } P(A|E_2) = \frac{\binom{3}{2}}{\binom{9}{2}} = \frac{3}{36}$$

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2)}$

Solution:

$$\begin{aligned} P(E_2|A) &= \frac{\frac{1}{2} \cdot \frac{3}{36}}{\left(\frac{1}{2} \cdot \frac{10}{36}\right) + \left(\frac{1}{2} \cdot \frac{3}{36}\right)} \\ &= \frac{\frac{3}{72}}{\frac{10}{72} + \frac{3}{72}} \\ &= \frac{\frac{3}{72}}{\frac{13}{72}} \\ &= \frac{3}{13} \end{aligned}$$

Substitute the terms into the formula:

Final Answer

The probability that the two red balls were drawn from the second bag is $\frac{3}{13}$.

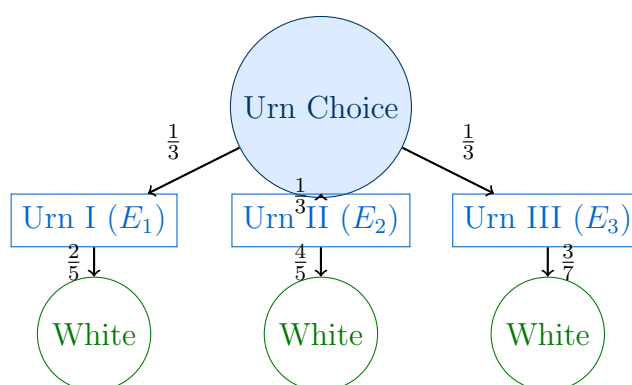
Common Mistake: Calculating combination $\binom{9}{2}$ incorrectly as $9 \times 8 = 72$ instead of $\frac{9 \times 8}{2} = 36$.

Question 13

Urn I has 2 white and 3 black balls, urn II has 4 white and 1 black ball and urn III has 3 white and 4 black balls. An urn is selected at random and a ball is drawn at random.

- What is the probability of drawing a white ball?
- If the ball drawn is white, then what is the probability that urn I was selected?

TYPE 24a Bayes' Theorem Combined Explicitly with Total Probability Theorem



Tree diagram for three urns with white/black outcomes

Given: Let E_1, E_2, E_3 be the events of selecting Urn I, Urn II, and Urn III respectively.

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Let A be the event of drawing a white ball.

$$\text{Urn I: } 2W, 3B \text{ (Total = 5)} \implies P(A|E_1) = \frac{2}{5}$$

$$\text{Urn II: } 4W, 1B \text{ (Total = 5)} \implies P(A|E_2) = \frac{4}{5}$$

$$\text{Urn III: } 3W, 4B \text{ (Total = 7)} \implies P(A|E_3) = \frac{3}{7}$$

Concept / Formula Used: $P(A) = \sum_{i=1}^3 P(E_i) \cdot P(A|E_i)$ $P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(A)}$

Solution:

$$P(A) = P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) +$$

$$= \frac{1}{3} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{4}{5} + \frac{1}{3} \cdot \frac{3}{7}$$

$$= \frac{1}{3} \left(\frac{2}{5} + \frac{4}{5} + \frac{3}{7} \right)$$

Part (i): Probability of drawing a white ball $P(A)$:

$$= \frac{1}{3} \left(\frac{6}{5} + \frac{3}{7} \right)$$

$$= \frac{1}{3} \left(\frac{42 + 15}{35} \right)$$

$$= \frac{1}{3} \cdot \frac{57}{35}$$

$$= \frac{19}{35}$$

Part (ii): Probability that Urn I was selected given ball is white $P(E_1|A)$:

$$P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(A)}$$

$$= \frac{\frac{1}{3} \cdot \frac{2}{5}}{\frac{19}{35}}$$

$$= \frac{\frac{2}{15}}{\frac{19}{35}}$$

$$= \frac{2}{15} \times \frac{35}{19}$$

$$= \frac{2 \times 7}{3 \times 19}$$

$$= \frac{14}{57}$$

Final Answer

$$(i) P(\text{White ball}) = \frac{19}{35}$$

$$(ii) P(\text{Urn I}|\text{White}) = \frac{14}{57}$$

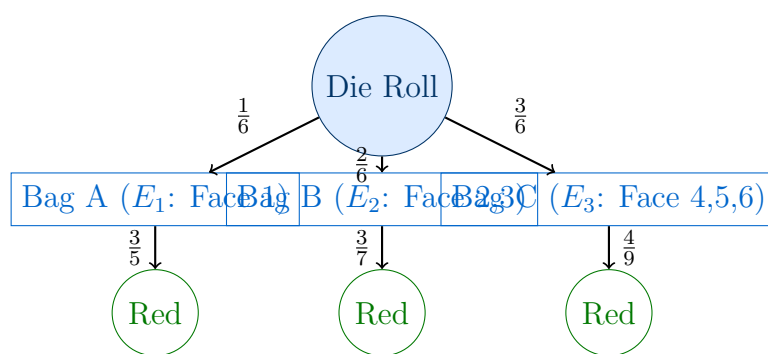
Common Mistake: Taking common denominator incorrectly for $\frac{6}{5} + \frac{3}{7}$.

Question 14

A fair die is rolled. If face 1 turns up, a ball is drawn from bag A. If face 2 or 3 turn up, a ball is drawn from bag B. If face 4 or 5 or 6 turn up, a ball is drawn from bag C. Bag A contains 3 red and 2 white balls, bag B contains 3 red and 4 white balls and bag C contains 4 red and 5 white balls. The die is rolled, a bag is picked up and a ball is drawn. If the ball drawn is red, what is the probability that it is drawn from bag B?

TYPE 24b Bayes' Theorem – Multi-Stage/Compound Prior (Ball Transfer)

★ Likely Exam Question



Tree diagram for die face selection to bag

Given: Let E_1, E_2, E_3 be the events of picking Bag A, Bag B, and Bag C respectively.

$$P(E_1) = P(\text{Face 1}) = \frac{1}{6}$$

$$P(E_2) = P(\text{Face 2 or 3}) = \frac{2}{6} = \frac{1}{3}$$

$$P(E_3) = P(\text{Face 4, 5, or 6}) = \frac{3}{6} = \frac{1}{2}$$

Let R be the event of drawing a red ball.

$$\text{Bag A: 3R, 2W (Total = 5)} \implies P(R|E_1) = \frac{3}{5}$$

$$\text{Bag B: 3R, 4W (Total = 7)} \implies P(R|E_2) = \frac{3}{7}$$

$$\text{Bag C: 4R, 5W (Total = 9)} \implies P(R|E_3) = \frac{4}{9}$$

Concept / Formula Used: $P(E_2|R) = \frac{P(E_2) \cdot P(R|E_2)}{P(E_1) \cdot P(R|E_1) + P(E_2) \cdot P(R|E_2) + P(E_3) \cdot P(R|E_3)}$

Solution:

Calculate the total probability of drawing a Red ball $P(R)$:

$$\begin{aligned}
 P(R) &= P(E_1) \cdot P(R|E_1) + P(E_2) \cdot P(R|E_2) \\
 &= \left(\frac{1}{6} \cdot \frac{3}{5}\right) + \left(\frac{1}{3} \cdot \frac{3}{7}\right) + \left(\frac{1}{2} \cdot \frac{4}{9}\right) \\
 &= \frac{3}{30} + \frac{3}{21} + \frac{4}{18} \\
 &= \frac{1}{10} + \frac{1}{7} + \frac{2}{9} \\
 P(R) &= \frac{63 + 90 + 140}{630}
 \end{aligned}$$

Taking the LCM of denominators 10, 7, 9 which is 630:

Now, cal-

$$\begin{aligned}
 P(E_2|R) &= \frac{P(E_2) \cdot P(R|E_2)}{P(R)} \\
 &= \frac{\frac{1}{3} \cdot \frac{3}{7}}{\frac{293}{630}}
 \end{aligned}$$

culate the posterior probability $P(E_2|R)$:

$$\begin{aligned}
 &= \frac{\frac{1}{3} \cdot \frac{3}{7}}{\frac{293}{630}} \\
 &= \frac{1}{7} \times \frac{630}{293} \\
 &= \frac{90}{293}
 \end{aligned}$$

Final Answer

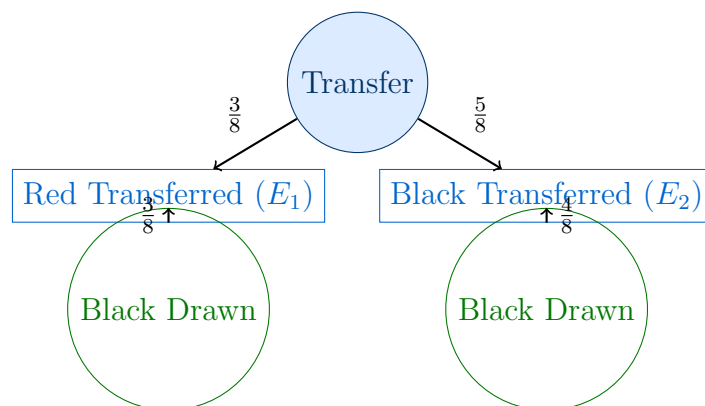
The probability that the red ball was drawn from Bag B is $\frac{90}{293}$.

Common Mistake: Assuming equal probabilities $1/3$ for all three bags, ignoring the die face probabilities given in the question.

Question 15

There are two bags, I and II. Bag I contains 3 red and 5 black balls and bag II contains 4 red and 3 black balls. One ball is transferred randomly from bag I to bag II and then a ball is drawn randomly from bag II. If the ball so drawn is found to be black in colour, then find the probability that the transferred ball is also black.

TYPE 24b Bayes' Theorem – Multi-Stage/Compound Prior (Ball Transfer)



Tree diagram for transfer from Bag I to Bag II

Given: Bag I: 3 Red, 5 Black (Total = 8 balls)

Bag II: 4 Red, 3 Black (Total = 7 balls)

Let E_1 be the event that a red ball is transferred from Bag I to Bag II.

Let E_2 be the event that a black ball is transferred from Bag I to Bag II.

$$P(E_1) = \frac{3}{8}, P(E_2) = \frac{5}{8}$$

Let B be the event that a black ball is drawn from Bag II.

If E_1 occurs, Bag II contains $4 + 1 = 5$ Red and 3 Black balls (Total = 8) $\Rightarrow$

$$P(B|E_1) = \frac{3}{8}$$

If E_2 occurs, Bag II contains 4 Red and $3 + 1 = 4$ Black balls (Total = 8) $\Rightarrow$

$$P(B|E_2) = \frac{4}{8} = \frac{1}{2}$$

Concept / Formula Used: $P(E_2|B) = \frac{P(E_2) \cdot P(B|E_2)}{P(E_1) \cdot P(B|E_1) + P(E_2) \cdot P(B|E_2)}$

Solution:

$$P(E_2|B) = \frac{\frac{5}{8} \cdot \frac{4}{8}}{\left(\frac{3}{8} \cdot \frac{3}{8}\right) + \left(\frac{5}{8} \cdot \frac{4}{8}\right)}$$

$$= \frac{\frac{20}{64}}{\frac{9}{64} + \frac{20}{64}}$$

Substitute values step-by-step:

$$= \frac{\frac{20}{64}}{\frac{29}{64}}$$

$$= \frac{20}{29}$$

Final Answer

The probability that the transferred ball was black is $\frac{20}{29}$.

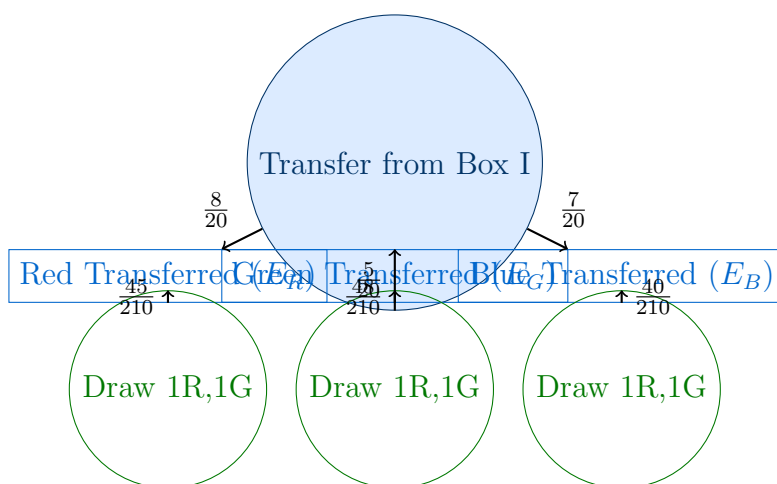
Common Mistake: Forgetting to increase the total count of balls in Bag II from 7 to 8 after the transfer.

Question 16

There are two boxes, box I and box II. Each box has 8 red balls, 5 green balls and 7 blue balls. Sahil takes out one ball from box I and places it in box II. He then draws two balls from box II simultaneously.

- What is the probability of Sahil selecting a green ball from box I?
- After Sahil transfers a ball from box I to box II, what is the probability of drawing one red ball and one green ball simultaneously from box II?
- Given that the balls drawn from box II are red and green, what is the probability that ball transferred from box I to box II was green?

TYPE 24b Bayes' Theorem – Multi-Stage/Compound Prior (Ball Transfer)



Transfer of one ball and selection of two balls

Given: Box I composition: 8 Red, 5 Green, 7 Blue (Total = 20 balls)

Box II composition before transfer: 8 Red, 5 Green, 7 Blue (Total = 20 balls)

Let E_R, E_G, E_B be the events of transferring a Red, Green, and Blue ball from Box I respectively.

$$P(E_R) = \frac{8}{20}, P(E_G) = \frac{5}{20}, P(E_B) = \frac{7}{20}$$

Let A be the event of drawing 1 Red and 1 Green ball simultaneously from Box II (which now has 21 balls).

Total combinations of selecting 2 balls from 21: $\binom{21}{2} = \frac{21 \times 20}{2} = 210$.

Concept / Formula Used: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Solution:

Part (i): Probability of selecting a green ball from Box I:

$$P(\text{Green from Box I}) = P(E_G) = \frac{5}{20} = \frac{1}{4}$$

Part (ii): Total probability of drawing 1 Red and 1 Green ball from Box II:

Conditional probabilities based on transferred ball: - If E_R occurs, Box II has 9R, 5G, 7B:

$$P(A|E_R) = \frac{\binom{9}{1} \cdot \binom{5}{1}}{210} = \frac{9 \times 5}{210} = \frac{45}{210}$$

- If E_G occurs, Box II has 8R, 6G, 7B:

$$P(A|E_G) = \frac{\binom{8}{1} \cdot \binom{6}{1}}{210} = \frac{8 \times 6}{210} = \frac{48}{210}$$

- If E_B occurs, Box II has 8R, 5G, 8B:

$$P(A|E_B) = \frac{\binom{8}{1} \cdot \binom{5}{1}}{210} = \frac{8 \times 5}{210} = \frac{40}{210}$$

$$P(A) = P(E_R) \cdot P(A|E_R) + P(E_G) \cdot P(A|E_G) + P(E_B) \cdot P(A|E_B)$$

$$= \left(\frac{8}{20} \cdot \frac{45}{210} \right) + \left(\frac{5}{20} \cdot \frac{48}{210} \right) + \left(\frac{7}{20} \cdot \frac{40}{210} \right)$$

By Law of Total Probability:
$$= \frac{360 + 240 + 280}{4200}$$

$$= \frac{880}{4200}$$

$$= \frac{88}{420} = \frac{22}{105}$$

Part (iii): Probability that transferred ball was green given drawn balls are

$$P(E_G|A) = \frac{P(E_G) \cdot P(A|E_G)}{P(A)}$$

$$= \frac{\frac{5}{20} \cdot \frac{48}{210}}{\frac{880}{4200}}$$

Red and Green:

$$= \frac{\frac{240}{4200}}{\frac{880}{4200}}$$

$$= \frac{240}{880}$$

$$= \frac{3}{11}$$

Final Answer

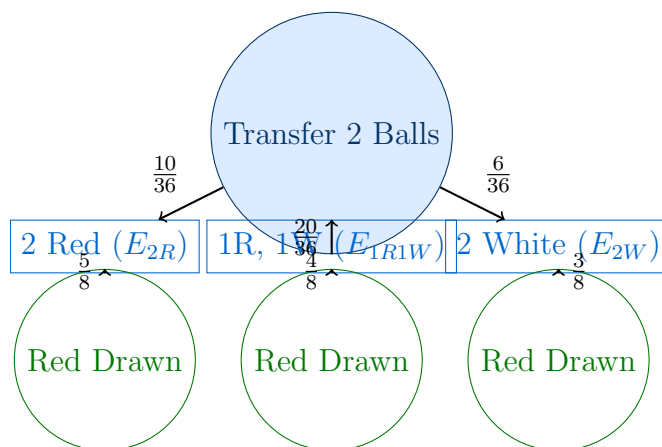
- (i) $P(\text{Green from Box I}) = \frac{1}{4}$
- (ii) $P(1R, 1G \text{ from Box II}) = \frac{22}{105}$
- (iii) $P(\text{Transferred ball was Green} | 1R, 1G) = \frac{3}{11}$

Common Mistake: Forgetting to update total count to 21 balls in Box II after transfer for all case combinations.

Question 17

Bag I contains 5 red and 4 white balls and Bag II contains 3 red and 3 white balls. Two balls are transferred from Bag I to Bag II and then one ball is drawn from the Bag II. If the ball drawn from the Bag II is red, then find the probability that one red ball and one white ball are transferred from Bag I to Bag II.

TYPE 24b Bayes' Theorem – Multi-Stage/Compound Prior (Ball Transfer)



Transfer of 2 balls tree structure

Given: Bag I: 5 Red, 4 White (Total = 9 balls). Number of ways to choose 2 balls: $\binom{9}{2} = 36$.

Bag II: 3 Red, 3 White (Total = 6 balls).

Let events for transferred balls from Bag I be:

$$E_{2R}: 2 \text{ Red balls transferred} \implies P(E_{2R}) = \frac{\binom{5}{2}}{36} = \frac{10}{36}$$

$$E_{1R1W}: 1 \text{ Red and 1 White ball transferred} \implies P(E_{1R1W}) = \frac{\binom{5}{1} \cdot \binom{4}{1}}{36} = \frac{20}{36}$$

$$E_{2W}: 2 \text{ White balls transferred} \implies P(E_{2W}) = \frac{\binom{4}{2}}{36} = \frac{6}{36}$$

Let A be the event of drawing a Red ball from Bag II (which now has 8 balls).

$$P(A|E_{2R}) = \frac{3+2}{8} = \frac{5}{8}$$

$$P(A|E_{1R1W}) = \frac{3+1}{8} = \frac{4}{8}$$

$$P(A|E_{2W}) = \frac{3+0}{8} = \frac{3}{8}$$

Concept	/	Formula	Used:	$P(E_{1R1W} A)$	=
$\frac{P(E_{1R1W}) \cdot P(A E_{1R1W})}{P(E_{2R}) \cdot P(A E_{2R}) + P(E_{1R1W}) \cdot P(A E_{1R1W}) + P(E_{2W}) \cdot P(A E_{2W})}$					

Solution:

$$\begin{aligned}
 P(E_{1R1W}|A) &= \frac{\frac{20}{36} \cdot \frac{4}{8}}{\left(\frac{10}{36} \cdot \frac{5}{8}\right) + \left(\frac{20}{36} \cdot \frac{4}{8}\right) + \left(\frac{6}{36} \cdot \frac{3}{8}\right)} \\
 &= \frac{\frac{80}{288}}{\frac{50}{288} + \frac{80}{288} + \frac{18}{288}} \\
 &= \frac{\frac{80}{288}}{\frac{148}{288}} \\
 &= \frac{80}{148} \\
 &= \frac{20}{37}
 \end{aligned}$$

Substitute all values into Bayes' formula:

Final Answer

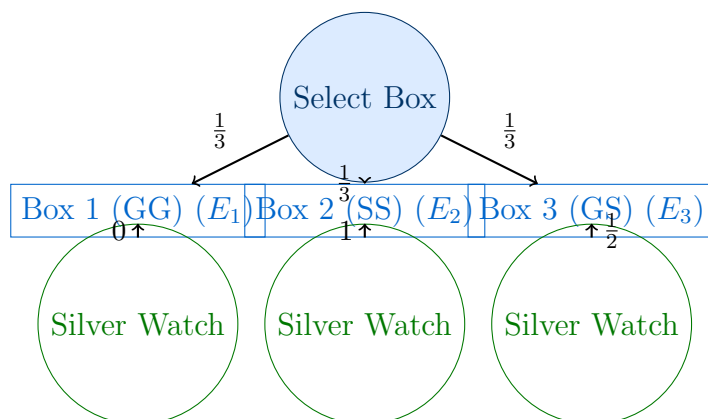
The probability that 1 red and 1 white ball were transferred is $\frac{20}{37}$.

Common Mistake: Forgetting to multiply combinations $\binom{5}{1} \times \binom{4}{1} = 20$ for mixed transfer case.

Question 18

Each of three identical jewellery boxes has two drawers. In each drawer of the first box there is a gold watch. In each drawer of the second box there is a silver watch. In one drawer of the third box there is a gold watch while in the other there is a silver watch. If we select a box at random, open one of the drawers and find it to contain a silver watch, then what is the probability that the other drawer has the gold watch?

TYPE 24c Bayes' Theorem – Bertrand's Box Paradox



Tree diagram for jewellery box drawers

Given: Let E_1, E_2, E_3 be the events of selecting Box 1, Box 2, and Box 3 respectively.

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Contents: Box 1: (G, G), Box 2: (S, S), Box 3: (G, S).

Let A be the event that the opened drawer contains a silver watch.

$$P(A|E_1) = 0, P(A|E_2) = 1, P(A|E_3) = \frac{1}{2}$$

The other drawer has a gold watch if and only if Box 3 was selected.

Concept / Formula Used: $P(E_3|A) = \frac{P(E_3) \cdot P(A|E_3)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)}$

Solution:

$$\begin{aligned}
 P(E_3|A) &= \frac{\frac{1}{3} \cdot \frac{1}{2}}{\left(\frac{1}{3} \cdot 0\right) + \left(\frac{1}{3} \cdot 1\right) + \left(\frac{1}{3} \cdot \frac{1}{2}\right)} \\
 &= \frac{\frac{1}{6}}{0 + \frac{1}{3} + \frac{1}{6}} \\
 &= \frac{\frac{1}{6}}{\frac{1}{2}} \\
 &= \frac{1}{3}
 \end{aligned}$$

Substitute values into Bayes' formula:

Final Answer

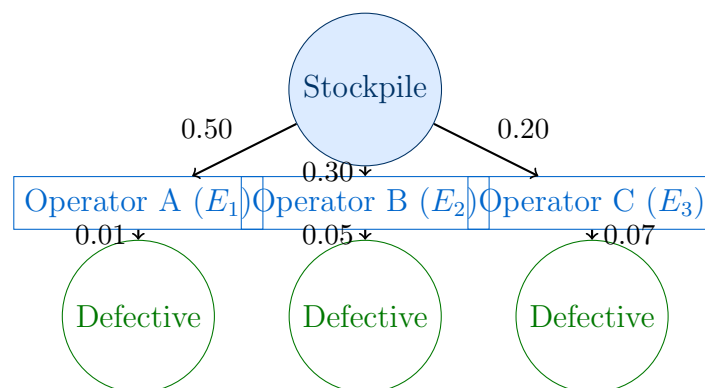
The probability that the other drawer has a gold watch is $\frac{1}{3}$.

Common Mistake: Assuming the probability is $\frac{1}{2}$ by mistakenly focusing only on the remaining watch type.

Question 19

A manufacturer has three machine operators A, B and C. The first operator A produces 1% of defective items, where as the other two operators B and C produce 5% and 7% defective items respectively. A is on the job for 50% of time, B is on the job 30% of the time and C is on the job for 20% of the time. All the items are put into one stockpile and then one item is chosen at random from this and is found to be defective. What is the probability that it was produced by A?

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for machine operator outputs

Given: Let E_1, E_2, E_3 be the events that item is produced by operators A, B, and C respectively.

$$P(E_1) = 50\% = 0.5, P(E_2) = 30\% = 0.3, P(E_3) = 20\% = 0.2$$

Let D be the event of selecting a defective item.

$$P(D|E_1) = 1\% = 0.01, P(D|E_2) = 5\% = 0.05, P(D|E_3) = 7\% = 0.07$$

Concept / Formula Used: $P(E_1|D) = \frac{P(E_1) \cdot P(D|E_1)}{P(E_1) \cdot P(D|E_1) + P(E_2) \cdot P(D|E_2) + P(E_3) \cdot P(D|E_3)}$

Solution:

$$P(E_1|D) = \frac{0.5 \times 0.01}{(0.5 \times 0.01) + (0.3 \times 0.05) + (0.2 \times 0.07)}$$

Substitute all given percentages:

$$= \frac{0.005}{0.005 + 0.015 + 0.014}$$

$$= \frac{0.005}{0.034}$$

$$= \frac{5}{34}$$

Final Answer

The probability that the defective item was produced by operator A is $\frac{5}{34}$.

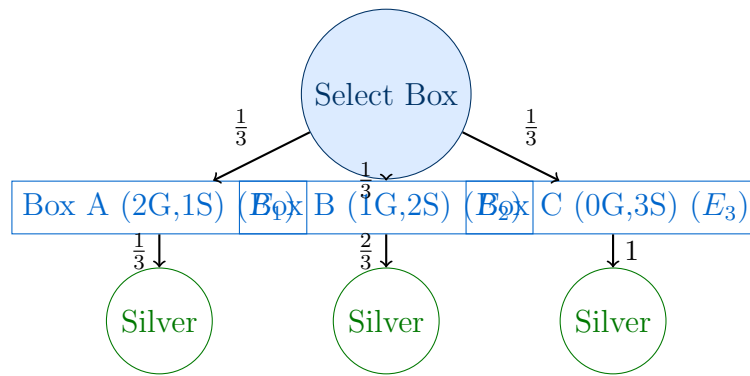
Common Mistake: Misplacing decimals when computing $0.5 \times 0.01 = 0.005$.

Question 20

Given three identical boxes A, B and C, box A contains 2 gold and 1 silver coin, box B contains 1 gold and 2 silver coins and box C contains 3 silver coins. A person chooses a box at random and takes out a coin. If the coin drawn is of silver, find the probability that it has been drawn from the box which has the remaining two coins also of silver.

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Tree diagram for three coin boxes

Given: Let E_1, E_2, E_3 be the events of choosing Box A, Box B, and Box C respectively.

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Let S be the event of drawing a Silver coin.

$$\text{Box A (2G, 1S): } P(S|E_1) = \frac{1}{3}$$

$$\text{Box B (1G, 2S): } P(S|E_2) = \frac{2}{3}$$

$$\text{Box C (0G, 3S): } P(S|E_3) = \frac{3}{3} = 1$$

The box having remaining two coins as silver is Box C.

Concept / Formula Used: $P(E_3|S) = \frac{P(E_3) \cdot P(S|E_3)}{P(E_1) \cdot P(S|E_1) + P(E_2) \cdot P(S|E_2) + P(E_3) \cdot P(S|E_3)}$

Solution:

$$P(E_3|S) = \frac{\frac{1}{3} \cdot 1}{\left(\frac{1}{3} \cdot \frac{1}{3}\right) + \left(\frac{1}{3} \cdot \frac{2}{3}\right) + \left(\frac{1}{3} \cdot 1\right)}$$

$$= \frac{\frac{1}{3}}{\frac{1}{3} \left(\frac{1}{3} + \frac{2}{3} + 1\right)}$$

$$= \frac{1}{\frac{1}{3} + \frac{2}{3} + 1}$$

$$= \frac{1}{2}$$

Substitute all parameters into Bayes' formula:

Final Answer

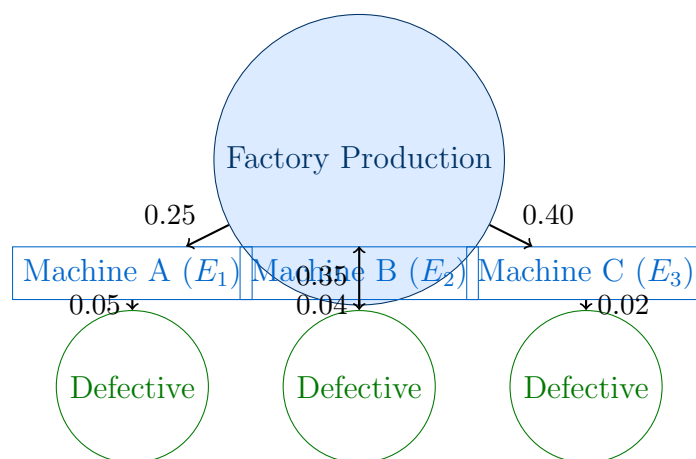
The probability that the silver coin was drawn from Box C is $\frac{1}{2}$.

Common Mistake: Identifying the required box incorrectly as Box B instead of Box C.

Question 21 (i)

In a bolt factory, three machines A, B and C manufacture 25%, 35% and 40% of the total production respectively. Of their respective outputs, 5%, 4% and 2% are defective. A bolt is drawn at random from the total production and it is found to be defective. Find the probability that it was manufactured by machine C.

TYPE 24 Bayes' Theorem – Direct Application



Tree diagram for bolt factory production

Given: Let E_1, E_2, E_3 be the events that a bolt is manufactured by Machines A, B, and C respectively.

$$P(E_1) = 25\% = 0.25, P(E_2) = 35\% = 0.35, P(E_3) = 40\% = 0.40$$

Let D be the event that the bolt is defective.

$$P(D|E_1) = 5\% = 0.05, P(D|E_2) = 4\% = 0.04, P(D|E_3) = 2\% = 0.02$$

Concept / Formula Used: $P(E_3|D) = \frac{P(E_3) \cdot P(D|E_3)}{P(E_1) \cdot P(D|E_1) + P(E_2) \cdot P(D|E_2) + P(E_3) \cdot P(D|E_3)}$

Solution:

$$\begin{aligned} P(E_3|D) &= \frac{0.40 \times 0.02}{(0.25 \times 0.05) + (0.35 \times 0.04) + (0.40 \times 0.02)} \\ &= \frac{0.0080}{0.0125 + 0.0140 + 0.0080} \\ &= \frac{0.0080}{0.0345} \\ &= \frac{80}{345} \\ &= \frac{16}{69} \end{aligned}$$

Substitute all values step-by-step:

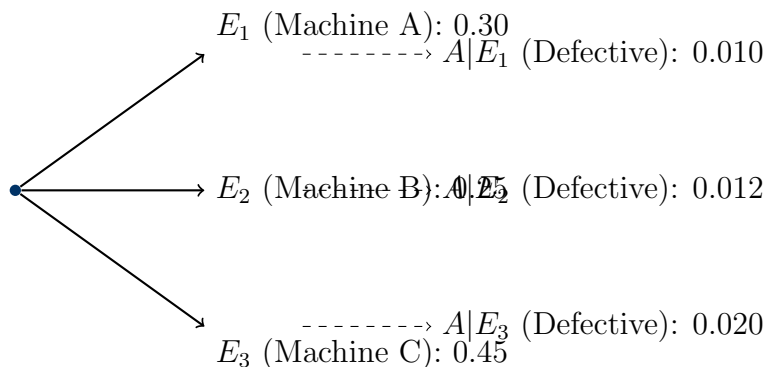
Final Answer

The probability that the defective bolt was manufactured by Machine C is $\frac{16}{69}$.

Common Mistake: Dividing by Machine C production probability 0.40 instead of total defective probability 0.0345.

Question 21 (ii)

In a factory, machine A produces 30% of total output, machine B produces 25% and machine C produces the remaining output. The defective items produced by machines A, B and C are 1%, 1.2% and 2% respectively. An item is picked at random from a day's output and found to be defective. Find the probability that it was produced by machine B.

TYPE 24 Bayes' Theorem – Direct Application

Probability tree for Machine A, B, C outputs and defect rates

Given: Let E_1, E_2, E_3 be the events that an item is produced by machines A, B, C respectively, and A be the event that the chosen item is defective. $P(E_1) = 0.30$, $P(E_2) = 0.25$, $P(E_3) = 1 - (0.30 + 0.25) = 0.45$, $P(A|E_1) = 0.01$, $P(A|E_2) = 0.012$, $P(A|E_3) = 0.02$.

To Find: The conditional probability $P(E_2|A)$ that the item was produced by machine B given that it is defective.

Concept / Formula Used: $P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)}$

Solution:

First, calculate the prior probabilities for each machine:

$$P(E_1) = \frac{30}{100} = 0.30$$

$$P(E_2) = \frac{25}{100} = 0.25$$

$$P(E_3) = 1 - (0.30 + 0.25) = 0.45$$

$$P(A|E_1) = \frac{1}{100} = 0.01$$

Next, write down the given conditional probabilities of producing a defective item: $P(A|E_2) = \frac{1.2}{100} = 0.012$

$$P(A|E_3) = \frac{2}{100} = 0.02$$

Using the Law of Total Probability, calculate the total probability $P(A)$ of picking a

$$P(A) = P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)$$

$$\begin{aligned} &= (0.30 \times 0.010) + (0.25 \times 0.012) + (0.45 \times 0.020) \\ \text{defective item:} \quad &= 0.0030 + 0.0030 + 0.0090 \\ &= 0.0150 \end{aligned}$$

$$\begin{aligned} P(E_2|A) &= \frac{P(E_2) \cdot P(A|E_2)}{P(A)} \\ &= \frac{0.0030}{0.0150} \\ &= \frac{3}{15} \\ &= \frac{1}{5} = 0.2 \end{aligned}$$

Now, apply Bayes' Theorem to determine $P(E_2|A)$:

Final Answer

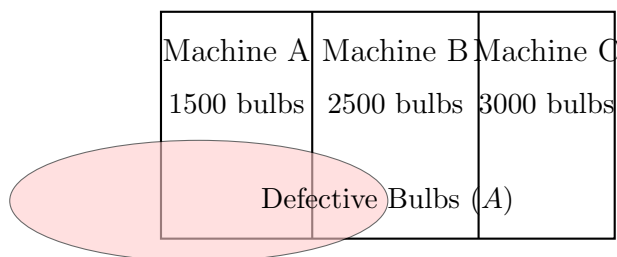
$$P(E_2|A) = \frac{1}{5} = 0.2$$

Common Mistake: A common mistake is forgetting to calculate $P(E_3) = 1 - (0.30 + 0.25) = 0.45$ and assuming all three machines have equal production output.

Question 22

A factory has three machines A, B and C producing 1500, 2500 and 3000 bulbs per day respectively. Machine A produces 1.5% defective bulbs, machine B produces 2% defective bulbs and machine C produces 2.5% defective bulbs. At the end of the day, a bulb is drawn at random and it is found to be defective. What is the probability that this defective bulb has been produced by machine B?

TYPE 24 Bayes' Theorem – Direct Application



Event partition diagram for total daily bulb output

Given: Let E_1, E_2, E_3 be the events of picking a bulb produced by machines A, B, C respectively. Total daily production = $1500 + 2500 + 3000 = 7000$ bulbs. Defective rates: $P(D|E_1) = 1.5\%$, $P(D|E_2) = 2\%$, $P(D|E_3) = 2.5\%$.

To Find: The conditional probability $P(E_2|D)$ that the defective bulb was produced by machine B.

Concept / Formula Used: $P(E_2|D) = \frac{P(E_2) \cdot P(D|E_2)}{\sum_{i=1}^3 P(E_i) \cdot P(D|E_i)}$

Solution:

$$P(E_1) = \frac{1500}{7000} = \frac{15}{70} = \frac{3}{14}$$

First, compute the prior probabilities of selection for each machine: $P(E_2) = \frac{2500}{7000} = \frac{25}{70} = \frac{5}{14}$

$$P(E_3) = \frac{3000}{7000} = \frac{30}{70} = \frac{6}{14}$$

$$P(D|E_1) = \frac{1.5}{100} = \frac{3}{200}$$

Express the defective rates as common fractions: $P(D|E_2) = \frac{2}{100} = \frac{4}{200}$

$$P(D|E_3) = \frac{2.5}{100} = \frac{5}{200}$$

$$P(E_1 \cap D) = \frac{3}{14} \times \frac{3}{200} = \frac{9}{2800}$$

Compute the joint probabilities $P(E_i \cap D) = P(E_i) \cdot P(D|E_i)$: $P(E_2 \cap D) = \frac{5}{14} \times \frac{4}{200} = \frac{20}{2800}$

$$P(E_3 \cap D) = \frac{6}{14} \times \frac{5}{200} = \frac{30}{2800}$$

$$P(D) = \frac{9}{2800} + \frac{20}{2800} + \frac{30}{2800}$$

Sum to find the total probability of selecting a defective bulb $P(D)$:

$$= \frac{9 + 20 + 30}{2800} = \frac{59}{2800}$$

$$P(E_2|D) = \frac{P(E_2 \cap D)}{P(D)}$$

Apply Bayes' Theorem to find $P(E_2|D)$:

$$= \frac{\frac{20}{2800}}{\frac{59}{2800}} = \frac{20}{59}$$

Final Answer

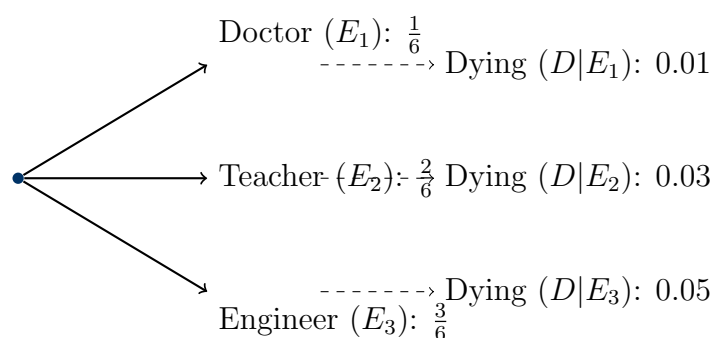
$$P(E_2|D) = \frac{20}{59}$$

Common Mistake: Avoid simplifying fractions to different denominators early in the step; keeping a common denominator of 2800 allows direct cancellation of denominators in Bayes' formula.

Question 23

An insurance company insured 4000 doctors, 8000 teachers and 12000 engineers. The probability of a doctor, a teacher and an engineer dying before the age of 58 years are 0.01, 0.03 and 0.05 respectively. If one of the insured person dies before the age of 58 years, find the probability that he is a doctor.

TYPE 24 Bayes' Theorem – Direct Application



Probability paths for insured professions and mortality risk

Given: Let E_1, E_2, E_3 be the events that the insured person is a doctor, teacher, engineer respectively. Number of insured persons: Doctors = 4000, Teachers = 8000, Engineers = 12000. Total = 24000. Let D be the event that the person dies before age 58. $P(D|E_1) = 0.01$, $P(D|E_2) = 0.03$, $P(D|E_3) = 0.05$.

To Find: The conditional probability $P(E_1|D)$ that the deceased person was a doctor.

Concept / Formula Used: $P(E_1|D) = \frac{P(E_1) \cdot P(D|E_1)}{P(E_1) \cdot P(D|E_1) + P(E_2) \cdot P(D|E_2) + P(E_3) \cdot P(D|E_3)}$

Solution:

$$P(E_1) = \frac{4000}{24000} = \frac{1}{6}$$

Calculate the prior probabilities of selecting a person from each profession: $P(E_2) = \frac{8000}{24000} = \frac{2}{6}$

$$P(E_3) = \frac{12000}{24000} = \frac{3}{6}$$

$$P(E_1 \cap D) = \frac{1}{6} \times 0.01 = \frac{0.01}{6} = \frac{1}{600}$$

Calculate the joint probability $P(E_i \cap D)$ for each profession: $P(E_2 \cap D) = \frac{2}{6} \times 0.03 = \frac{0.06}{6} = \frac{6}{600}$

$$P(E_3 \cap D) = \frac{3}{6} \times 0.05 = \frac{0.15}{6} = \frac{15}{600}$$

Calculate the total probability of death before age 58, $P(D)$:

$$\begin{aligned} P(D) &= P(E_1 \cap D) + P(E_2 \cap D) + P(E_3 \cap D) \\ &= \frac{1}{600} + \frac{6}{600} + \frac{15}{600} \\ &= \frac{1 + 6 + 15}{600} \\ &= \frac{22}{600} \end{aligned}$$

Apply Bayes' Theorem to find $P(E_1|D)$:

$$\begin{aligned} P(E_1|D) &= \frac{P(E_1 \cap D)}{P(D)} \\ &= \frac{\frac{1}{600}}{\frac{22}{600}} \\ &= \frac{1}{22} \end{aligned}$$

Final Answer

$$P(E_1|D) = \frac{1}{22}$$

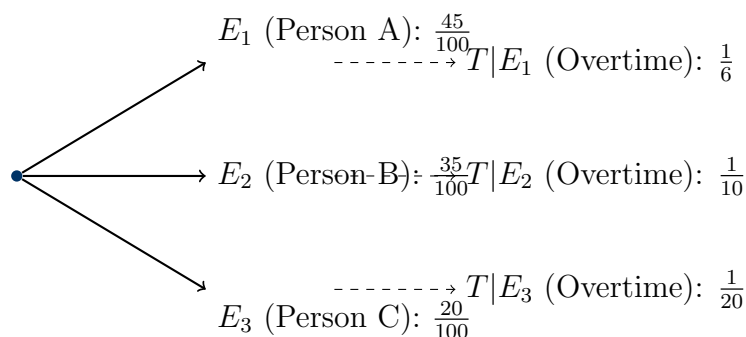
Common Mistake: Do not confuse the count of people with probabilities directly; always divide by the total population (24000) first to obtain $P(E_i)$.

Question 24

In an automobile factory, certain parts are to be fixed into the chassis in a section before it moves to the next section. On a given day, one of the three persons A, B and C carries out the task. A has 45% chance, B has 35% chance and C has 20% chance of doing the task. The probability that A, B and C will take more time than the allotted time is $\frac{1}{6}$, $\frac{1}{10}$ and $\frac{1}{20}$ respectively. If it is found that the time taken is more than the allotted time, what is the probability that A has done the task?

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Tree diagram for worker assignment and time overrun

Given: Let E_1, E_2, E_3 be the events that person A, B, C carries out the task respectively. Let T be the event that time taken is more than allotted time.
 $P(E_1) = \frac{45}{100} = \frac{9}{20}$, $P(E_2) = \frac{35}{100} = \frac{7}{20}$, $P(E_3) = \frac{20}{100} = \frac{4}{20}$. $P(T|E_1) = \frac{1}{6}$, $P(T|E_2) = \frac{1}{10}$, $P(T|E_3) = \frac{1}{20}$.

To Find: The conditional probability $P(E_1|T)$ that person A did the task given that it took more than allotted time.

Concept / Formula Used: $P(E_1|T) = \frac{P(E_1) \cdot P(T|E_1)}{\sum_{i=1}^3 P(E_i) \cdot P(T|E_i)}$

Solution:

$$P(E_1 \cap T) = \frac{9}{20} \times \frac{1}{6} = \frac{9}{120} = \frac{3}{40}$$

Calculate individual joint probabilities $P(E_i \cap T)$: $P(E_2 \cap T) = \frac{7}{20} \times \frac{1}{10} = \frac{7}{200}$

$$P(E_3 \cap T) = \frac{4}{20} \times \frac{1}{20} = \frac{4}{400} = \frac{1}{100}$$

$$P(E_1 \cap T) = \frac{3 \times 15}{40 \times 15} = \frac{45}{600}$$

Convert all fractions to a common denominator of 600: $P(E_2 \cap T) = \frac{7 \times 3}{200 \times 3} = \frac{21}{600}$

$$P(E_3 \cap T) = \frac{1 \times 6}{100 \times 6} = \frac{6}{600}$$

$$P(T) = \frac{45}{600} + \frac{21}{600} + \frac{6}{600}$$

Compute total probability $P(T)$: $= \frac{45 + 21 + 6}{600}$

$$= \frac{72}{600}$$

$$P(E_1|T) = \frac{P(E_1 \cap T)}{P(T)}$$

Apply Bayes' Theorem to obtain $P(E_1|T)$: $= \frac{\frac{45}{600}}{\frac{72}{600}}$

$$= \frac{45}{72}$$

Simplifying by dividing numerator and denominator by 9: $P(E_1|T) = \frac{45 \div 9}{72 \div 9} = \frac{5}{8}$

Final Answer

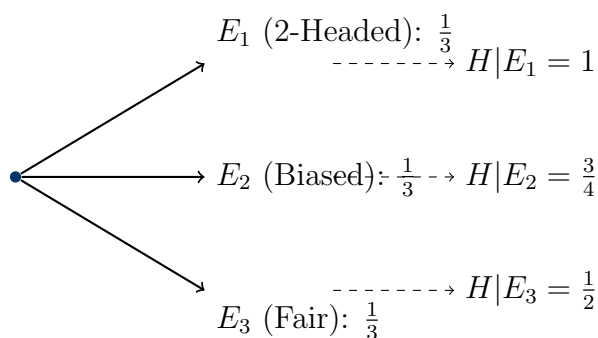
$$P(E_1|T) = \frac{5}{8}$$

Common Mistake: Ensure percentages are converted correctly to fractions before multiplying with conditional probabilities.

Question 25 (i)

There are three coins. One is a two-headed coin, another is a biased coin that comes up heads 75% of the time and third is an unbiased coin. One of the three coins is chosen at random and tossed. If it shows heads, then what is the probability that it was the two-headed coin?

TYPE 24 Bayes' Theorem – Direct Application



Probability tree for selecting coin type and tossing Heads

Given: Let E_1 be event of picking two-headed coin, E_2 be event of picking biased coin, E_3 be event of picking unbiased coin. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$. Let H be the event of getting a head. $P(H|E_1) = 1$, $P(H|E_2) = 75\% = \frac{3}{4}$, $P(H|E_3) = \frac{1}{2}$.

To Find: The conditional probability $P(E_1|H)$.

Concept / Formula Used: $P(E_1|H) = \frac{P(E_1) \cdot P(H|E_1)}{P(E_1) \cdot P(H|E_1) + P(E_2) \cdot P(H|E_2) + P(E_3) \cdot P(H|E_3)}$

Solution:

$$P(E_1 \cap H) = \frac{1}{3} \times 1 = \frac{1}{3}$$

Calculate individual joint probabilities $P(E_i \cap H)$: $P(E_2 \cap H) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$

$$P(E_3 \cap H) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

Find the total probability of getting a head, $P(H)$: $P(H) = \frac{1}{3} + \frac{1}{4} + \frac{1}{6}$

$$\begin{aligned}
 P(H) &= \frac{4 + 3 + 2}{12} \\
 \text{Take LCM of denominators } (3, 4, 6) &= 12: \\
 &= \frac{9}{12} = \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 P(E_1|H) &= \frac{P(E_1 \cap H)}{P(H)} \\
 &= \frac{\frac{1}{3}}{\frac{4}{3}} \\
 &= \frac{1}{3} \times \frac{3}{4} \\
 &= \frac{1}{4}
 \end{aligned}$$

Apply Bayes' Theorem to find $P(E_1|H)$:

Final Answer

$$P(E_1|H) = \frac{1}{4}$$

Common Mistake: For a two-headed coin, $P(H|E_1) = 1$, not $\frac{1}{2}$.

Question 25 (ii)

There are three coins. One is a coin having tail on both faces, another is a biased coin that comes up tails 70% of the time and the third is an unbiased coin. One of the coin is chosen at random and tossed, it shows tail. Find the probability that it was a coin with tail on both faces.

TYPE 24 Bayes' Theorem – Direct Application

Given: Let E_1, E_2, E_3 be selecting two-tailed coin, biased coin, unbiased coin respectively. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$. Let T be event of getting a tail. $P(T|E_1) = 1$, $P(T|E_2) = 70\% = \frac{7}{10}$, $P(T|E_3) = \frac{1}{2} = \frac{5}{10}$.

To Find: The conditional probability $P(E_1|T)$.

Concept / Formula Used: $P(E_1|T) = \frac{P(E_1) \cdot P(T|E_1)}{\sum_{i=1}^3 P(E_i) \cdot P(T|E_i)}$

Solution:

$$P(E_1 \cap T) = \frac{1}{3} \times 1 = \frac{1}{3} = \frac{10}{30}$$

$$\text{Calculate joint probabilities: } P(E_2 \cap T) = \frac{1}{3} \times \frac{7}{10} = \frac{7}{30}$$

$$P(E_3 \cap T) = \frac{1}{3} \times \frac{5}{10} = \frac{5}{30}$$

$$P(T) = \frac{10}{30} + \frac{7}{30} + \frac{5}{30}$$

Calculate total probability of observing a tail $P(T)$:

$$= \frac{10 + 7 + 5}{30}$$

$$= \frac{22}{30}$$

$$\begin{aligned}
 P(E_1|T) &= \frac{P(E_1 \cap T)}{P(T)} \\
 &= \frac{\frac{10}{30}}{\frac{22}{30}} \\
 &= \frac{10}{22} \\
 &= \frac{5}{11}
 \end{aligned}$$

Apply Bayes' Theorem:

Final Answer

$$P(E_1|T) = \frac{5}{11}$$

Common Mistake: Do not confuse tail probability of 70% with head probability (30%) since the observed outcome is a tail.

Question 25 (iii)

There are two coins. One of them is a biased coin such that $P(\text{head}) : P(\text{tail})$ is 1 : 3 and other coin is a fair coin. A coin is selected at random and tossed once. If the coin showed head, then find the probability that it is a biased coin.

TYPE 24 Bayes' Theorem – Direct Application

Given: Let E_1 be event of selecting biased coin, E_2 be event of selecting fair coin. $P(E_1) = P(E_2) = \frac{1}{2}$. For biased coin, $P(H|E_1) : P(T|E_1) = 1 : 3 \implies P(H|E_1) = \frac{1}{1+3} = \frac{1}{4}$. For fair coin, $P(H|E_2) = \frac{1}{2}$.

To Find: The conditional probability $P(E_1|H)$.

Concept / Formula Used: $P(E_1|H) = \frac{P(E_1) \cdot P(H|E_1)}{P(E_1) \cdot P(H|E_1) + P(E_2) \cdot P(H|E_2)}$

Solution:

Determine the conditional head probability for the biased coin: $P(H|E_1) = \frac{1}{1+3} = \frac{1}{4}$

Determine the joint probabilities:

$$\begin{aligned}
 P(E_1 \cap H) &= \frac{1}{2} \times \frac{1}{4} = \frac{1}{8} \\
 P(E_2 \cap H) &= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \frac{2}{8}
 \end{aligned}$$

Calculate total probability of showing head $P(H)$: $P(H) = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$

Apply Bayes' Theorem: $P(E_1|H) = \frac{\frac{1}{8}}{\frac{3}{8}} = \frac{1}{3}$

Final Answer

$$P(E_1|H) = \frac{1}{3}$$

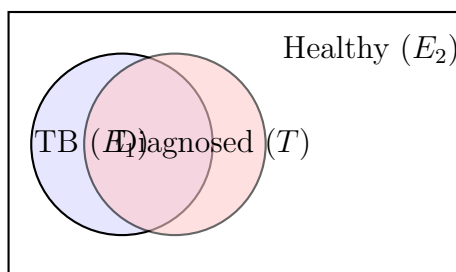
Common Mistake: Remember 1 : 3 ratio implies total parts = 4, so probability of head is $\frac{1}{4}$ and not $\frac{1}{3}$.

Question 26

By examining the chest X-ray, the probability that TB is detected when a person is actually suffering from it is 0.99. The probability of a healthy person diagnosed to have TB is 0.001. In a certain city, 1 in 1000 people suffers from TB. A person is selected at random and is diagnosed to have TB. What is the probability that he/she actually has TB?

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question



Venn Diagram for Population TB Status and Test Diagnosis

Given: Let E_1 be event person has TB, E_2 be event person is healthy. $P(E_1) = \frac{1}{1000} = 0.001$, $P(E_2) = 1 - 0.001 = 0.999$. Let T be event person is diagnosed to have TB. $P(T|E_1) = 0.99$, $P(T|E_2) = 0.001$.

To Find: The conditional probability $P(E_1|T)$ that a person diagnosed with TB actually has TB.

Concept / Formula Used: $P(E_1|T) = \frac{P(E_1) \cdot P(T|E_1)}{P(E_1) \cdot P(T|E_1) + P(E_2) \cdot P(T|E_2)}$

Solution:

$$P(E_1 \cap T) = P(E_1) \cdot P(T|E_1)$$

$$\begin{aligned} \text{Calculate the joint probabilities:} \quad &= 0.001 \times 0.99 \\ &= 0.00099 \end{aligned}$$

$$\begin{aligned} P(E_2 \cap T) &= P(E_2) \cdot P(T|E_2) \\ &= 0.999 \times 0.001 \\ &= 0.000999 \end{aligned}$$

$$\begin{aligned} \text{Sum to find total probability of positive diagnosis } P(T): \quad &P(T) = P(E_1 \cap T) + P(E_2 \cap T) \\ &= 0.00099 + 0.000999 \\ &= 0.001989 \end{aligned}$$

$$P(E_1|T) = \frac{P(E_1 \cap T)}{P(T)}$$

Apply Bayes' Theorem to find $P(E_1|T)$:

$$= \frac{0.00099}{0.001989}$$

$$= \frac{990}{1989}$$

Dividing numerator and denominator by 9:

$$P(E_1|T) = \frac{990 \div 9}{1989 \div 9}$$

$$= \frac{110}{221} \approx 0.4977$$

Final Answer

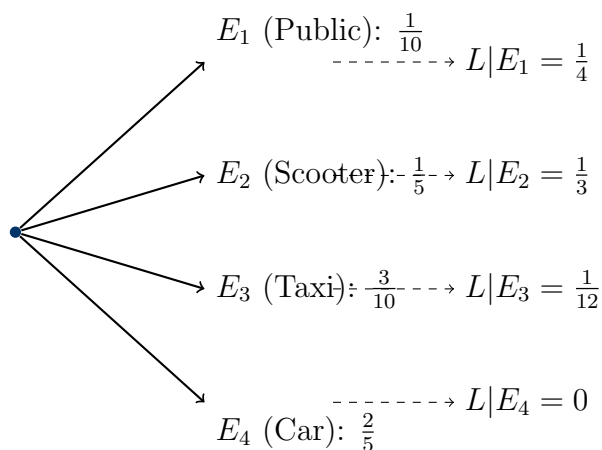
$$P(E_1|T) = \frac{110}{221}$$

Common Mistake: Even with a high test sensitivity of 0.99, low disease prevalence (0.001) means positive test results have roughly 50% probability of being false positives.

Question 27

A doctor is to visit a patient. From the past experience, it is known that the probability that he will come by public transport, scooter, taxi or personal car are respectively $\frac{1}{10}$, $\frac{1}{5}$, $\frac{3}{10}$ and $\frac{2}{5}$. The probabilities that he will be late are $\frac{1}{4}$, $\frac{1}{3}$ or $\frac{1}{12}$ if he comes by public transport, scooter or taxi respectively, but if he comes by personal car he will not be late. When he arrives, he is late. What is the probability that he came by scooter?

TYPE 24 Bayes' Theorem – Direct Application



Probability diagram for modes of transport and lateness

Given: Let E_1, E_2, E_3, E_4 be events doctor arrives by public transport, scooter, taxi, personal car. $P(E_1) = \frac{1}{10}$, $P(E_2) = \frac{1}{5}$, $P(E_3) = \frac{3}{10}$, $P(E_4) = \frac{2}{5}$. Let L be event that doctor is late. $P(L|E_1) = \frac{1}{4}$, $P(L|E_2) = \frac{1}{3}$, $P(L|E_3) = \frac{1}{12}$, $P(L|E_4) = 0$.

To Find: The conditional probability $P(E_2|L)$.

Concept / Formula Used: $P(E_2|L) = \frac{P(E_2) \cdot P(L|E_2)}{\sum_{i=1}^4 P(E_i) \cdot P(L|E_i)}$

Solution:

$$P(E_1 \cap L) = \frac{1}{10} \times \frac{1}{4} = \frac{1}{40}$$

$$P(E_2 \cap L) = \frac{1}{5} \times \frac{1}{3} = \frac{1}{15}$$

Calculate joint probabilities $P(E_i \cap L)$:

$$P(E_3 \cap L) = \frac{3}{10} \times \frac{1}{12} = \frac{3}{120} = \frac{1}{40}$$

$$P(E_4 \cap L) = \frac{2}{5} \times 0 = 0$$

$$P(E_1 \cap L) = \frac{3}{120}$$

$$P(E_2 \cap L) = \frac{8}{120}$$

$$P(E_3 \cap L) = \frac{3}{120}$$

$$P(E_4 \cap L) = 0$$

Convert all non-zero terms to a common denominator of 120:

$$P(L) = \frac{3}{120} + \frac{8}{120} + \frac{3}{120} + 0$$

Calculate total probability of being late $P(L)$:

$$= \frac{3+8+3}{120}$$

$$= \frac{14}{120}$$

$$P(E_2|L) = \frac{P(E_2 \cap L)}{P(L)}$$

Apply Bayes' Theorem to find $P(E_2|L)$:

$$= \frac{\frac{8}{120}}{\frac{14}{120}}$$

$$= \frac{8}{14} = \frac{4}{7}$$

Final Answer

$$P(E_2|L) = \frac{4}{7}$$

Common Mistake: Do not omit E_4 from the setup, even though $P(L|E_4) = 0$; showing it explicitly ensures total probability law completeness.

Question 28

Bag A contains 2 white, 1 black and 3 red balls; Bag B contains 3 white, 2 black and 4 red balls; Bag C contains 4 white, 3 black and 2 red balls. One bag is chosen at random and 2 balls are drawn at random from that Bag. If the randomly drawn balls happened to be red and black, what is the probability that both balls come from Bag

B?

TYPE 24 Bayes' Theorem – Direct Application

Bag A	Bag B	Bag C
2W, 1B, 3R	3W, 2B, 4R	4W, 3B, 2R

Structure of 3 Bags with different ball combinations

Given: Let E_1, E_2, E_3 be selecting Bag A, Bag B, Bag C respectively. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$. Let D be event of drawing 1 Red and 1 Black ball without replacement.

To Find: The conditional probability $P(E_2|D)$.

Concept / Formula Used: $P(1R, 1B) = \frac{\binom{\text{Red}}{1} \times \binom{\text{Black}}{1}}{\binom{\text{Total}}{2}}$

Solution:

Calculate total balls in each bag: - Bag A: $2 + 1 + 3 = 6$ balls. - Bag B: $3 + 2 + 4 = 9$ balls. - Bag C: $4 + 3 + 2 = 9$ balls.

Calculate conditional probabilities $P(D|E_i)$ of drawing 1 Red and 1 Black ball:

$$\text{For Bag A: } P(D|E_1) = \frac{\binom{3}{1} \times \binom{1}{1}}{\binom{6}{2}} = \frac{3 \times 1}{\frac{6 \times 5}{2}} = \frac{3}{15} = \frac{1}{5}$$

$$\text{For Bag B: } P(D|E_2) = \frac{\binom{4}{1} \times \binom{2}{1}}{\binom{9}{2}} = \frac{4 \times 2}{\frac{9 \times 8}{2}} = \frac{8}{36} = \frac{2}{9}$$

$$\text{For Bag C: } P(D|E_3) = \frac{\binom{2}{1} \times \binom{3}{1}}{\binom{9}{2}} = \frac{2 \times 3}{36} = \frac{6}{36} = \frac{1}{6}$$

$$P(E_1 \cap D) = \frac{1}{3} \times \frac{1}{5} = \frac{1}{15}$$

$$\text{Calculate joint probabilities } P(E_i \cap D) = \frac{1}{3}P(D|E_i): P(E_2 \cap D) = \frac{1}{3} \times \frac{2}{9} = \frac{2}{27}$$

$$P(E_3 \cap D) = \frac{1}{3} \times \frac{1}{6} = \frac{1}{18}$$

$$P(E_1 \cap D) = \frac{18}{270}$$

$$\text{Find total probability } P(D) \text{ with common denominator LCM}(15, 27, 18) = 270: P(E_2 \cap D) = \frac{20}{270}$$

$$P(E_3 \cap D) = \frac{15}{270}$$

$$P(D) = \frac{18 + 20 + 15}{270} = \frac{53}{270}$$

Apply Bayes' Theorem for $P(E_2|D)$: $P(E_2|D) = \frac{\frac{20}{270}}{\frac{53}{270}} = \frac{20}{53}$

Final Answer

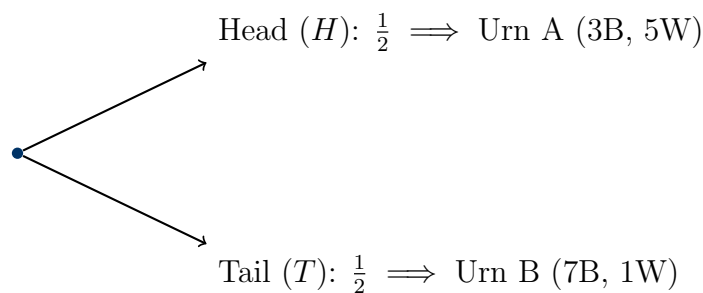
$$P(E_2|D) = \frac{20}{53}$$

Common Mistake: Be careful when calculating combinations $\binom{n}{2}$; ensure total balls in Bag A is 6 while in Bag B and C it is 9.

Question 29

A coin is tossed. If it turns up heads, 2 balls will be drawn from urn A, otherwise 2 balls will be drawn from urn B. Urn A contains 3 black and 5 white balls. Urn B contains 7 black and 1 white ball. In both cases, selections are to be made with replacement. What is the probability that urn A is used given that both the balls drawn are black?

TYPE 24 Bayes' Theorem – Direct Application



Coin toss decision leading to Urn A or Urn B

Given: Coin is fair: $P(H) = \frac{1}{2}$, $P(T) = \frac{1}{2}$. Urn A: 3 black, 5 white (total 8). Urn B: 7 black, 1 white (total 8). Selections are WITH REPLACEMENT. Let $2B$ be event of drawing 2 black balls.

To Find: The conditional probability $P(H|2B)$.

Concept / Formula Used: $P(2B|H) = (P(B|A))^2$, $P(2B|T) = (P(B|B))^2$

Solution:

Calculate single-ball black drawing probability for each urn:

$$P(B|A) = \frac{3}{8}$$

$$P(B|B) = \frac{7}{8}$$

Since draws are made with replacement, the two draws are independent:

$$P(2B|H) = \left(\frac{3}{8}\right)^2 = \frac{9}{64}$$

$$P(2B|T) = \left(\frac{7}{8}\right)^2 = \frac{49}{64}$$

Calculate joint probabilities:

$$P(H \cap 2B) = \frac{1}{2} \times \frac{9}{64} = \frac{9}{128}$$

$$P(T \cap 2B) = \frac{1}{2} \times \frac{49}{64} = \frac{49}{128}$$

Find total probability $P(2B)$: $P(2B) = \frac{9}{128} + \frac{49}{128} = \frac{58}{128}$

Apply Bayes' Theorem: $P(H|2B) = \frac{\frac{9}{128}}{\frac{58}{128}} = \frac{9}{58}$

Final Answer

$$P(H|2B) = \frac{9}{58}$$

Common Mistake: The question states "with replacement", so do NOT use combinations without replacement $\binom{3}{2}/\binom{8}{2}$; instead square the single draw probability $(3/8)^2$.

Question 30

Redo the above problem assuming that the coin is biased and turns up heads 2 times out of 3.

TYPE 24 Bayes' Theorem – Direct Application

Given: Biased coin probabilities: $P(H) = \frac{2}{3}, P(T) = 1 - \frac{2}{3} = \frac{1}{3}$. Conditional probabilities for 2 black balls with replacement: $P(2B|H) = \frac{9}{64}, P(2B|T) = \frac{49}{64}$.

To Find: The conditional probability $P(H|2B)$.

Concept / Formula Used: $P(H|2B) = \frac{P(H) \cdot P(2B|H)}{P(H) \cdot P(2B|H) + P(T) \cdot P(2B|T)}$

Solution:

Compute joint probabilities with the updated prior coin probabilities:

$$P(H \cap 2B) = \frac{2}{3} \times \frac{9}{64} = \frac{18}{192}$$

$$P(T \cap 2B) = \frac{1}{3} \times \frac{49}{64} = \frac{49}{192}$$

Calculate total probability of getting 2 black balls $P(2B)$:

$$\begin{aligned} P(2B) &= P(H \cap 2B) + P(T \cap 2B) \\ &= \frac{18}{192} + \frac{49}{192} \\ &= \frac{18 + 49}{192} \\ &= \frac{67}{192} \end{aligned}$$

Apply Bayes' Theorem:

$$\begin{aligned} P(H|2B) &= \frac{P(H \cap 2B)}{P(2B)} \\ &= \frac{\frac{18}{192}}{\frac{67}{192}} \\ &= \frac{18}{67} \end{aligned}$$

Final Answer

$$P(H|2B) = \frac{18}{67}$$

Common Mistake: Check that $P(H) + P(T) = \frac{2}{3} + \frac{1}{3} = 1$; ensure numerator uses $P(H) = \frac{2}{3}$ instead of $\frac{1}{2}$.

Question 31

There are three urns. Urn I contains 1 white, 2 black and 3 red balls. Urn II contains 2 white, 1 black and 1 red balls. Urn III contains 4 white, 5 black and 3 red balls. One urn is chosen at random and 2 balls drawn without replacement. They happen to be white and red. What is the probability that they are from urn I, II or III?

TYPE 24 Bayes' Theorem – Direct Application

Urn I	Urn II	Urn III
1W, 2B, 3R	2W, 1B, 1R	4W, 5B, 3R
Total = 6	Total = 4	Total = 12

Urn contents and sample space selection

Given: Let E_1, E_2, E_3 be selection of Urn I, II, III respectively. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$. Let A be event of drawing 1 White and 1 Red ball without replacement.

To Find: The conditional probabilities $P(E_1|A)$, $P(E_2|A)$, and $P(E_3|A)$.

Concept / Formula Used: $P(1W, 1R) = \frac{\binom{\text{White}}{1} \times \binom{\text{Red}}{1}}{\binom{\text{Total}}{2}}$

Solution:

Calculate conditional probabilities $P(A|E_i)$:

For Urn I (1W, 2B, 3R; Total = 6): $P(A|E_1) = \frac{\binom{1}{1} \times \binom{3}{1}}{\binom{6}{2}} = \frac{1 \times 3}{15} = \frac{3}{15} = \frac{1}{5}$

For Urn II (2W, 1B, 1R; Total = 4): $P(A|E_2) = \frac{\binom{2}{1} \times \binom{1}{1}}{\binom{4}{2}} = \frac{2 \times 1}{6} = \frac{2}{6} = \frac{1}{3}$

For Urn III (4W, 5B, 3R; Total = 12): $P(A|E_3) = \frac{\binom{4}{1} \times \binom{3}{1}}{\binom{12}{2}} = \frac{4 \times 3}{66} = \frac{12}{66} = \frac{2}{11}$

$$P(E_1 \cap A) = \frac{1}{3} \times \frac{1}{5} = \frac{1}{15} = \frac{33}{495}$$

Compute joint probabilities $P(E_i \cap A) = \frac{1}{3}P(A|E_i)$: $P(E_2 \cap A) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9} = \frac{55}{495}$

$$P(E_3 \cap A) = \frac{1}{3} \times \frac{2}{11} = \frac{2}{33} = \frac{30}{495}$$

Calculate total probability $P(A)$: $P(A) = \frac{33 + 55 + 30}{495} = \frac{118}{495}$

Now calculate posterior probabilities for each urn:

For Urn I: $P(E_1|A) = \frac{\frac{33}{495}}{\frac{118}{495}} = \frac{33}{118}$

For Urn II: $P(E_2|A) = \frac{\frac{55}{495}}{\frac{118}{495}} = \frac{55}{118}$

For Urn III: $P(E_3|A) = \frac{\frac{30}{495}}{\frac{118}{495}} = \frac{30}{118}$

Final Answer

$$P(\text{Urn I}|A) = \frac{33}{118}$$

$$P(\text{Urn II}|A) = \frac{55}{118}$$

$$P(\text{Urn III}|A) = \frac{30}{118}$$

Common Mistake: Verify that sum of posterior probabilities $\frac{33}{118} + \frac{55}{118} + \frac{30}{118} = \frac{118}{118} = 1$ to check correctness.

Question 32

A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn randomly one-by-one without replacement and are found to be both aces. Find the probability of the lost card being an ace.

TYPE 24 Bayes' Theorem – Direct Application

Given: Let E_1 be event that the lost card is an Ace, E_2 be event that the lost card is not an Ace. $P(E_1) = \frac{4}{52} = \frac{1}{13}$, $P(E_2) = \frac{48}{52} = \frac{12}{13}$. Let A be event of drawing 2 Aces from remaining 51 cards.

To Find: The conditional probability $P(E_1|A)$.

Concept / Formula Used: $P(A|E_1) = \frac{\binom{\text{Remaining Aces}}{2}}{\binom{51}{2}}$

Solution:

If E_1 occurs (lost card was an Ace), 3 Aces and 48 non-Aces remain in the deck: $P(A|E_1) = \frac{\binom{3}{2}}{\binom{51}{2}} = \frac{3}{\frac{51 \times 50}{2}}$

If E_2 occurs (lost card was NOT an Ace), 4 Aces and 47 non-Aces remain in the deck:

$$P(A|E_2) = \frac{\binom{4}{2}}{\binom{51}{2}} = \frac{6}{1275} = \frac{2}{425}$$

Compute joint probabilities:

$$P(E_1 \cap A) = \frac{1}{13} \times \frac{1}{425} = \frac{1}{5525}$$

$$P(E_2 \cap A) = \frac{12}{13} \times \frac{2}{425} = \frac{24}{5525}$$

Compute total probability $P(A)$: $P(A) = \frac{1}{5525} + \frac{24}{5525} = \frac{25}{5525}$

$$P(E_1|A) = \frac{P(E_1 \cap A)}{P(A)}$$

Apply Bayes' Theorem to find $P(E_1|A)$:

$$= \frac{\frac{1}{5525}}{\frac{25}{5525}} = \frac{1}{25}$$

Final Answer

$$P(E_1|A) = \frac{1}{25}$$

Common Mistake: Remember that after losing 1 card, the sample space size for drawing 2 cards becomes $\binom{51}{2} = 1275$, not $\binom{52}{2}$.

Question 33

A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be hearts. Find the probability of the missing card to be a heart.

TYPE 24 Bayes' Theorem – Direct Application

Given: Let E_1 be event that missing card is a Heart, E_2 be event that missing card is not a Heart. Total hearts = 13, non-hearts = 39. $P(E_1) = \frac{13}{52} = \frac{1}{4}$, $P(E_2) = \frac{39}{52} = \frac{3}{4}$. Let H be event that 2 cards drawn from remaining 51 are Hearts.

To Find: The conditional probability $P(E_1|H)$.

Concept / Formula Used: $P(E_1|H) = \frac{P(E_1) \cdot P(H|E_1)}{P(E_1) \cdot P(H|E_1) + P(E_2) \cdot P(H|E_2)}$

Solution:

If E_1 occurs (lost card is a Heart), 12 Hearts remain out of 51 cards: $P(H|E_1) = \frac{\binom{12}{2}}{\binom{51}{2}} = \frac{\frac{12 \times 11}{2}}{\frac{51 \times 50}{2}} = \frac{132}{2550}$

If E_2 occurs (lost card is not a Heart), 13 Hearts remain out of 51 cards: $P(H|E_2) = \frac{\binom{13}{2}}{\binom{51}{2}} = \frac{\frac{13 \times 12}{2}}{\frac{51 \times 50}{2}} = \frac{156}{2550}$

Compute joint probabilities:

$$P(E_1 \cap H) = \frac{1}{4} \times \frac{132}{2550} = \frac{33}{2550}$$

$$P(E_2 \cap H) = \frac{3}{4} \times \frac{156}{2550} = \frac{468}{4 \times 2550} = \frac{117}{2550}$$

Compute total probability $P(H)$: $P(H) = \frac{33 + 117}{2550} = \frac{150}{2550}$

$$P(E_1|H) = \frac{\frac{33}{2550}}{\frac{150}{2550}}$$

$$= \frac{33}{150}$$

$$= \frac{11}{50}$$

Apply Bayes' Theorem to find $P(E_1|H)$:

Final Answer

$$P(E_1|H) = \frac{11}{50}$$

Common Mistake: Be careful when simplifying $\frac{33}{150}$; dividing numerator and denominator by 3 gives $\frac{11}{50}$.

Question 34

A jewellery seller has precious gems in white and red colour which he has put in three boxes. The distribution of these gems is shown in the table given below:

Box	Number of Gems	
	White	Red
I	1	2
II	2	3
III	3	1

He wants to gift two gems to his mother. So, he asks her to select one box at random and pick out any two gems one after the other without replacement from the selected box. The mother selects one white and one red gem. Calculate the probability that the gems drawn are from Box II.

TYPE 24 Bayes' Theorem – Direct Application

★ Likely Exam Question

Box I	Box II	Box III
1 White, 2 Red	2 White, 3 Red	3 White, 1 Red
Total = 3	Total = 5	Total = 4

Gems distribution in three boxes

Given: Let E_1, E_2, E_3 be selecting Box I, II, III respectively. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$. Let G be event of drawing 1 White and 1 Red gem without replacement.

To Find: The conditional probability $P(E_2|G)$.

Concept / Formula Used: $P(1W, 1R) = \frac{\binom{\text{White}}{1} \times \binom{\text{Red}}{1}}{\binom{\text{Total}}{2}}$

Solution:

Calculate conditional probabilities $P(G|E_i)$ for each box:

$$\text{For Box I (1 White, 2 Red; Total = 3): } P(G|E_1) = \frac{\binom{1}{1} \times \binom{2}{1}}{\binom{3}{2}} = \frac{1 \times 2}{3} = \frac{2}{3}$$

$$\text{For Box II (2 White, 3 Red; Total = 5): } P(G|E_2) = \frac{\binom{2}{1} \times \binom{3}{1}}{\binom{5}{2}} = \frac{2 \times 3}{10} = \frac{6}{10} = \frac{3}{5}$$

$$\text{For Box III (3 White, 1 Red; Total = 4): } P(G|E_3) = \frac{\binom{3}{1} \times \binom{1}{1}}{\binom{4}{2}} = \frac{3 \times 1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$P(E_1 \cap G) = \frac{1}{3} \times \frac{2}{3} = \frac{2}{9} = \frac{20}{90}$$

$$\text{Calculate joint probabilities } P(E_i \cap G) = \frac{1}{3}P(G|E_i): P(E_2 \cap G) = \frac{1}{3} \times \frac{3}{5} = \frac{1}{5} = \frac{18}{90}$$

$$P(E_3 \cap G) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} = \frac{15}{90}$$

$$\text{Compute total probability } P(G): P(G) = \frac{20}{90} + \frac{18}{90} + \frac{15}{90} = \frac{53}{90}$$

$$\text{Apply Bayes' Theorem to find } P(E_2|G): P(E_2|G) = \frac{P(E_2 \cap G)}{P(G)} = \frac{\frac{18}{90}}{\frac{53}{90}} = \frac{18}{53}$$

Final Answer

$$P(E_2|G) = \frac{18}{53}$$

Common Mistake: Ensure that $\binom{3}{2} = 3$, $\binom{5}{2} = 10$, and $\binom{4}{2} = 6$ are computed correctly for each box.

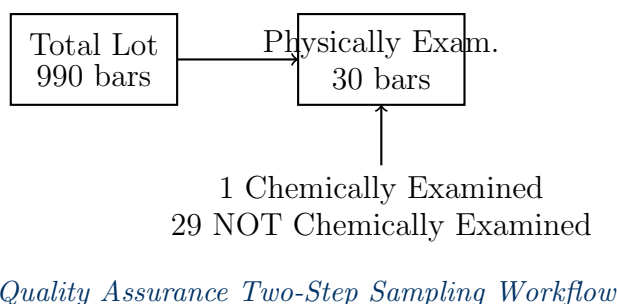
Question 35

Dobleron, a chocolate is known for its distinctive shape and its taste. The quality assurance team of the brand undertakes quality test in two steps:

- **Step 1:** 30 out of every 990 chocolate bars are randomly picked for physical examination, where the shape of the bar is examined.
- **Step 2:** For every 30 chocolate bars whose shape is examined, 1 chocolate bar is randomly picked for chemical examination, where the composition and taste of the chocolate is examined.

If one chocolate bar is randomly picked out of 990 bars, what is the probability that it was not chemically examined, provided it was physically examined?

TYPE 24 Bayes' Theorem – Direct Application



Given: Out of 30 physically examined bars, exactly 1 bar is chemically examined. Let P be event that a bar is physically examined, and C be event that a bar is chemically examined. Given $P(C|P) = \frac{1}{30}$.

To Find: The conditional probability $P(C'|P)$ that the bar was NOT chemically examined, given that it was physically examined.

Concept / Formula Used: $P(C'|P) = 1 - P(C|P)$

Solution:

From Step 2 of the quality assurance process, among the group of 30 physically examined bars:

Number of chemically examined bars = 1

Number of NOT chemically examined bars = $30 - 1 = 29$

Therefore, the conditional probability that a bar is chemically examined given it was physically examined is: $P(C|P) = \frac{1}{30}$

$$\begin{aligned}
 P(C'|P) &= 1 - P(C|P) \\
 &= 1 - \frac{1}{30} \\
 &= \frac{30 - 1}{30} \\
 &= \frac{29}{30}
 \end{aligned}$$

Using the complement rule for conditional probability:

Final Answer

$$P(\text{Not Chemically Examined} \mid \text{Physically Examined}) = \frac{29}{30}$$

Common Mistake: Notice that the question asks for conditional probability *given* it was physically examined; thus the total universe for this conditional step is the 30 physically examined bars, not the 990 total bars.

Question 36

Three companies viz. Amber, Bonzi and Comet are manufacturing cars which run on petrol and on battery as well. Their production share in the market is 60% (printed as 80% in some editions), 30% and 10% respectively. Of their respective production capacities, 20%, 10% and 5% cars respectively are electric (or battery operated).

Based on the above, answer the following:

- (i) (a) What is the probability that a randomly selected car is an electric car?

OR

- (b) What is the probability that a randomly selected car is a petrol car?
- (ii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Comet?
- (iii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Amber or Bonzi?

TYPE 24a Bayes' Theorem Combined Explicitly with Total Probability Theorem

Given: Let the events of selecting a car manufactured by Amber, Bonzi, and Comet be A , B , and C respectively. Let E be the event that a randomly selected car is electric (battery operated), and P be the event that it is a petrol car.

$$P(A) = 60\% = 0.60 \quad (\text{Note: The textbook contains a typo printing } 0.80)$$

The market production shares are: $P(B) = 30\% = 0.30$

$$P(C) = 10\% = 0.10$$

Note that $P(A) + P(B) + P(C) = 0.60 + 0.30 + 0.10 = 1.00$.

The conditional probabilities of a car being electric given its manufacturer are:

$$P(E|A) = 20\% = 0.20$$

$$P(E|B) = 10\% = 0.10$$

$$P(E|C) = 5\% = 0.05$$

Concept / Formula Used: Total Probability Theorem: $P(E) = \sum P(A_i)P(E|A_i)$
 Bayes' Theorem: $P(A_i|E) = \frac{P(A_i)P(E|A_i)}{P(E)}$

Solution:**Part (i)**

(a) Probability that a randomly selected car is electric:

By the Theorem of Total Probability, the probability $P(E)$ that a randomly selected car

$$\begin{aligned}
 P(E) &= P(A) \cdot P(E|A) + P(B) \cdot P(E|B) + P(C) \cdot P(E|C) \\
 &= 0.60 \times 0.20 + 0.30 \times 0.10 + 0.10 \times 0.05 \\
 \text{is electric is given by:} \quad &= 0.120 + 0.030 + 0.005 \\
 &= 0.155
 \end{aligned}$$

(b) Probability that a randomly selected car is a petrol car:

Since a car is either electric or petrol-operated, the probability $P(P)$ of selecting a petrol

$$\begin{aligned}
 P(P) &= 1 - P(E) \\
 \text{car is:} \quad &= 1 - 0.155 \\
 &= 0.845
 \end{aligned}$$

Part (ii) Probability that the electric car was manufactured by Comet:

$$\begin{aligned}
 P(C|E) &= \frac{P(C) \cdot P(E|C)}{P(E)} \\
 &= \frac{0.10 \times 0.05}{0.155} \\
 &= \frac{0.005}{0.155} \\
 &= \frac{5}{155} \\
 &= \frac{1}{31}
 \end{aligned}$$

We need to find the posterior probability $P(C|E)$. By Bayes' Theorem:

Part (iii) Probability that the electric car was manufactured by Amber or Bonzi:

We need to find $P(A \cup B|E)$. Since the events A and B are mutually exclusive: $P(A \cup B|E) = P(A|E) + P(B|E)$. Alternatively, since a car must be manufactured by Amber, Bonzi, or Comet, we have:

$$\begin{aligned}
 P(A \cup B|E) &= 1 - P(C|E) \\
 &= 1 - \frac{1}{31} \\
 &= \frac{30}{31}
 \end{aligned}$$

Final Answer

(i) (a) Probability of an electric car: 0.155

OR

(b) Probability of a petrol car: 0.845

(ii) Probability that the electric car is from Comet: $\frac{1}{31}$

(iii) Probability that the electric car is from Amber or Bonzi: $\frac{30}{31}$

Common Mistake: The textbook contains a typographical error in the question statement, listing Amber's production share as 80% instead of 60%. If 80% is used, the sum of the production shares exceeds 100% ($80\% + 30\% + 10\% = 120\%$), and the calculated probabilities will not match the textbook's official answer key. Always verify that the sum of prior probabilities equals 1 in such partition problems.

13.6 – Probability Distribution (Random Variables)

Question 1

A random variable X has the following probability distribution:

X	0	1	2	3	4	5	6	7
$P(X)$	a	$4a$	$3a$	$7a$	$8a$	$10a$	$6a$	$9a$

- Determine the value of a .
- Find $P(X < 3)$, $P(X \geq 4)$, and $P(0 < X < 5)$.

TYPE 25 Probability Distribution – Verify/Determine PMF (Sums to 1)

Given: Probability distribution of random variable X expressed in terms of constant a .

Concept / Formula Used: $\sum_i P(X_i) = 1$

Solution:

- Since the total probability for a random variable must equal 1:

$$P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7) = 1$$

$$a + 4a + 3a + 7a + 8a + 10a + 6a + 9a = 1$$

$$(1 + 4 + 3 + 7 + 8 + 10 + 6 + 9)a = 1$$

$$58a = 1$$

$$a = \frac{1}{58}$$

- Now evaluating the required probabilities using $a = \frac{1}{58}$:

$$\begin{aligned}
 P(X < 3) &= P(X = 0) + P(X = 1) + P(X = 2) \\
 &= a + 4a + 3a \\
 &= 8a
 \end{aligned}$$

First, for $P(X < 3)$:

$$\begin{aligned}
 &= 8 \times \frac{1}{48} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 P(X \geq 4) &= P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) \\
 &= 8a + 10a + 6a + 9a \\
 &= 33a
 \end{aligned}$$

Next, for $P(X \geq 4)$:

$$\begin{aligned}
 &= 33 \times \frac{1}{48} \\
 &= \frac{11}{16}
 \end{aligned}$$

$$\begin{aligned}
 P(0 < X < 5) &= P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\
 &= 4a + 3a + 7a + 8a \\
 &= 22a
 \end{aligned}$$

Finally, for $P(0 < X < 5)$:

$$\begin{aligned}
 &= 22 \times \frac{1}{48} \\
 &= \frac{11}{24}
 \end{aligned}$$

Final Answer

(i) $a = \frac{1}{48}$

(ii) $P(X < 3) = \frac{1}{6}, \quad P(X \geq 4) = \frac{11}{16}, \quad P(0 < X < 5) = \frac{11}{24}$

Common Mistake: Including $X = 3$ in $P(X < 3)$ or including $X = 0$ and $X = 5$ in the strict inequality $P(0 < X < 5)$ are common boundary errors.

Question 2

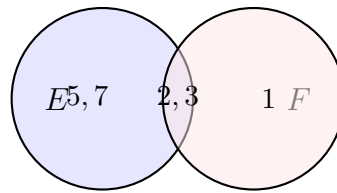
A random variable X has the probability distribution:

X	1	2	3	4	5	6	7	8
$P(X)$	0.15	0.23	0.12	0.1	0.2	0.08	0.07	0.05

If events $E = \{X \text{ is a prime number}\}$ and $F = \{X < 4\}$, then find $P(E \cup F)$.

TYPE 25b Probability Distribution – Construct PMF from Experiment

Sample Space $\{1, 2, 3, 4, 5, 6, 7, 8\}$



Venn Diagram of Events E and F

Given: Distribution of X over values $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and event definitions $E = \{X \text{ is prime}\}$, $F = \{X < 4\}$.

Concept / Formula Used: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ $P(E \cup F) = \sum_{x \in E \cup F} P(X = x)$

Solution:

First, identify the outcomes corresponding to each event from the sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$:

The prime numbers in S are 2, 3, 5, 7. Therefore:

$$E = \{2, 3, 5, 7\}$$

The values of $X < 4$ in S are 1, 2, 3. Therefore:

$$F = \{1, 2, 3\}$$

Taking the union of sets E and F : $E \cup F = \{2, 3, 5, 7\} \cup \{1, 2, 3\}$
 $= \{1, 2, 3, 5, 7\}$

Now, calculate $P(E \cup F)$ by summing the individual probabilities:

$$P(E \cup F) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 5) + P(X = 7)$$

$$= 0.15 + 0.23 + 0.12 + 0.20 + 0.07$$

$$= 0.77$$

Final Answer

$$P(E \cup F) = 0.77$$

Common Mistake: Misclassifying 1 as a prime number is a standard error. Remember that 1 is neither prime nor composite.

Question 3

A random variable X has the following probability distribution where k is some num-

ber:

$$P(x) = \begin{cases} k, & \text{if } x = 0 \\ 2k, & \text{if } x = 1 \\ 3k, & \text{if } x = 2 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Determine the value of k .
 (ii) Find $P(X < 2)$, $P(X \leq 2)$, and $P(X \geq 2)$.

TYPE 25 Probability Distribution – Verify/Determine PMF (Sums to 1)

Given: Probability mass function $P(X = x)$ defined piecewise in terms of parameter k .

Concept / Formula Used: $\sum_x P(X = x) = 1$

Solution:

$$\sum P(X = x) = 1$$

$$P(X = 0) + P(X = 1) + P(X = 2) = 1$$

- (i) The non-zero probabilities are for $x = 0, 1, 2$:

$$k + 2k + 3k = 1$$

$$6k = 1$$

$$k = \frac{1}{6}$$

- (ii) Using $k = \frac{1}{6}$:

$$P(X < 2) = P(X = 0) + P(X = 1)$$

$$= k + 2k$$

$$= 3k$$

For $P(X < 2)$:

$$= 3 \times \frac{1}{6}$$

$$= \frac{1}{2}$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= k + 2k + 3k$$

$$= 6k$$

For $P(X \leq 2)$:

$$= 6 \times \frac{1}{6}$$

$$= 1$$

$$P(X \geq 2) = P(X = 2) + P(X > 2) \\ = 3k + 0$$

$$\text{For } P(X \geq 2): \quad = 3 \times \frac{1}{6} \\ = \frac{1}{2}$$

Final Answer

$$(i) k = \frac{1}{6}$$

$$(ii) P(X < 2) = \frac{1}{2}, \quad P(X \leq 2) = 1, \quad P(X \geq 2) = \frac{1}{2}$$

Question 4

Let X denote the number of colleges where you will apply after your results and $P(X = x)$ denote your probability of getting admission in x number of colleges. It is given that

$$P(X = x) = \begin{cases} kx, & \text{if } x = 0 \text{ or } 1 \\ 2kx, & \text{if } x = 2 \\ k(5 - x), & \text{if } x = 3 \text{ or } 4 \\ 0, & \text{if } x > 4 \end{cases}$$

where k is a positive constant. Find the value of k . Also find the probability that you will get admission in

- (i) exactly one college
- (ii) at most two colleges
- (iii) at least 2 colleges

TYPE 25 Probability Distribution – Verify/Determine PMF (Sums to 1)

Given: Random variable $X \in \{0, 1, 2, 3, 4\}$ with given piecewise PMF expression.

Concept / Formula Used: $\sum_{x=0}^4 P(X = x) = 1$

Solution:

$$P(X = 0) = k(0) = 0$$

$$P(X = 1) = k(1) = k$$

First, evaluate the individual probabilities in terms of k : $P(X = 2) = 2k(2) = 4k$

$$P(X = 3) = k(5 - 3) = 2k$$

$$P(X = 4) = k(5 - 4) = k$$

$$P(0) + P(1) + P(2) + P(3) + P(4) = 1$$

$$0 + k + 4k + 2k + k = 1$$

Summing all probabilities to 1:

$$8k = 1$$

$$k = \frac{1}{8}$$

Now, evaluate the required probabilities using $k = \frac{1}{8}$:

$$\begin{aligned} P(X = 1) &= k \\ \text{(i) Probability of admission in exactly one college } (X = 1): &= \frac{1}{8} \end{aligned}$$

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= 0 + k + 4k \\ &= 5k \end{aligned}$$

$$\begin{aligned} \text{(ii) Probability of admission in at most two colleges } (X \leq 2): &= 5 \times \frac{1}{8} \\ &= \frac{5}{8} \end{aligned}$$

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) + P(X = 4) \\ &= 4k + 2k + k \\ &= 7k \end{aligned}$$

$$\begin{aligned} \text{(iii) Probability of admission in at least two colleges } (X \geq 2): &= 7 \times \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$

Final Answer

$$k = \frac{1}{8}$$

$$\text{(i) } P(\text{exactly 1 college}) = \frac{1}{8}$$

$$\text{(ii) } P(\text{at most 2 colleges}) = \frac{5}{8}$$

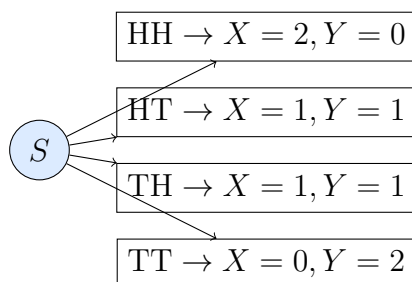
$$\text{(iii) } P(\text{at least 2 colleges}) = \frac{7}{8}$$

Question 5

A coin is tossed twice (or two coins are tossed simultaneously).

- Find the probability distribution of X , the number of heads.
- If a random variable Y is defined on this sample space as the number of tails, then find the probability distribution of Y .

TYPE 25b Probability Distribution – Construct PMF from Experiment



Sample Space Diagram for Two Coin Tosses

Given: Experiment of tossing two coins. Random variables: X = number of heads, Y = number of tails.

Concept / Formula Used: $S = \{HH, HT, TH, TT\}$

Solution:

$$P(X = 0) = P(\{TT\}) = \frac{1}{4}$$

(i) X can take values 0, 1, 2: $P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}$ The probability dis-

$$P(X = 2) = P(\{HH\}) = \frac{1}{4}$$

tribution table of X is:

X	0	1	2
$P(X)$	1/4	1/2	1/4

$$P(Y = 0) = P(\{HH\}) = \frac{1}{4}$$

(ii) Y can take values 0, 1, 2: $P(Y = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}$ The probability dis-

$$P(Y = 2) = P(\{TT\}) = \frac{1}{4}$$

tribution table of Y is:

Y	0	1	2
$P(Y)$	1/4	1/2	1/4

Final Answer

Probability distribution of X (Heads):

X	0	1	2
$P(X)$	1/4	1/2	1/4

Probability distribution of Y (Tails):

Y	0	1	2
$P(Y)$	1/4	1/2	1/4

Question 6

A die is rolled. If a random variable X is defined as the number on the upper face, then find its probability distribution.

TYPE 25b Probability Distribution – Construct PMF from Experiment

Given: Rolling a fair six-sided die, where X is the number on the upper face.

Concept / Formula Used: $S = \{1, 2, 3, 4, 5, 6\}$ $P(X = x) = \frac{1}{6}$

Solution:

The possible values that the random variable X can take are 1, 2, 3, 4, 5, 6.

$$P(X = 1) = \frac{1}{6}$$

$$P(X = 2) = \frac{1}{6}$$

$$P(X = 3) = \frac{1}{6}$$

$$P(X = 4) = \frac{1}{6}$$

$$P(X = 5) = \frac{1}{6}$$

$$P(X = 6) = \frac{1}{6}$$

Since the die is fair, each face is equally likely to appear:

Thus, the probability distribution table is:

X	1	2	3	4	5	6
$P(X)$	1/6	1/6	1/6	1/6	1/6	1/6

Final Answer

The probability distribution of X is given by:

X	1	2	3	4	5	6
$P(X)$	1/6	1/6	1/6	1/6	1/6	1/6

Question 7

Two balls are drawn at random from a bag containing 2 red balls and 3 blue balls, without replacement. Let the variable X denote the number of red balls. Find the probability distribution of X .

TYPE 25a Probability Distribution – Hypergeometric (Sampling Without Replacement)

Given: Bag composition: 2 red, 3 blue balls. Total balls = 5. Draw size = 2 without replacement. X = number of red balls.

Concept / Formula Used: $P(X = r) = \frac{{}^R C_r \times {}^B C_{2-r}}{{}^N C_2}$

Solution:

Total number of balls = 2+3 = 5. Number of ways to draw 2 balls out of 5: ${}^5 C_2 = \frac{5 \times 4}{2 \times 1} = 10$

The random variable X (number of red balls drawn) can take values 0, 1, 2:

$$P(X = 0) = \frac{{}^2 C_0 \times {}^3 C_2}{{}^5 C_2}$$

For $X = 0$ (drawing 0 red and 2 blue balls):

$$= \frac{1 \times 3}{10}$$

$$= \frac{3}{10}$$

$$P(X = 1) = \frac{{}^2 C_1 \times {}^3 C_1}{{}^5 C_2}$$

For $X = 1$ (drawing 1 red and 1 blue ball):

$$= \frac{2 \times 3}{10}$$

$$= \frac{6}{10} = \frac{3}{5}$$

$$P(X = 2) = \frac{{}^2 C_2 \times {}^3 C_0}{{}^5 C_2}$$

For $X = 2$ (drawing 2 red and 0 blue balls):

$$= \frac{1 \times 1}{10}$$

$$= \frac{1}{10}$$

Checking the sum of probabilities:

$$\frac{3}{10} + \frac{6}{10} + \frac{1}{10} = \frac{10}{10} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2
$P(X)$	3/10	3/5	1/10

Question 8

An urn contains 5 white and 3 black balls. If two balls are drawn at random without replacement and X denotes the number of white balls drawn, then find its probability distribution.

TYPE 25a Probability Distribution – Hypergeometric (Sampling Without Replacement)

Given: Urn contains 5 white balls and 3 black balls. Total balls = 8. Number of balls drawn = 2. X = number of white balls.

Concept / Formula Used: $P(X = x) = \frac{{}^5C_x \times {}^3C_{2-x}}{{}^8C_2}$

Solution:

Total balls = $5 + 3 = 8$. Total number of possible outcomes when drawing 2 balls:

$${}^8C_2 = \frac{8 \times 7}{2 \times 1} = 28$$

X can take values 0, 1, 2:

$$P(X = 0) = \frac{{}^5C_0 \times {}^3C_2}{{}^8C_2}$$

$$\begin{aligned} \text{Case 1: } X = 0 \text{ (0 white and 2 black balls)} &= \frac{1 \times 3}{28} \\ &= \frac{3}{28} \end{aligned}$$

$$P(X = 1) = \frac{{}^5C_1 \times {}^3C_1}{{}^8C_2}$$

$$\begin{aligned} \text{Case 2: } X = 1 \text{ (1 white and 1 black ball)} &= \frac{5 \times 3}{28} \\ &= \frac{15}{28} \end{aligned}$$

$$P(X = 2) = \frac{{}^5C_2 \times {}^3C_0}{{}^8C_2}$$

$$\begin{aligned} \text{Case 3: } X = 2 \text{ (2 white and 0 black balls)} &= \frac{10 \times 1}{28} \\ &= \frac{10}{28} = \frac{5}{14} \end{aligned}$$

Verification:

$$\frac{3}{28} + \frac{15}{28} + \frac{10}{28} = \frac{28}{28} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2
$P(X)$	3/28	15/28	5/14

Question 9

Two cards are drawn simultaneously from a well-shuffled pack of 52 cards. Find the probability distribution of the number of jacks.

TYPE 25b Probability Distribution – Construct PMF from Experiment

Given: Total cards = 52. Number of Jacks = 4. Non-Jack cards = 48. Draw size = 2. Let X = number of Jacks.

Concept / Formula Used: ${}^{52}C_2 = \frac{52 \times 51}{2 \times 1} = 1326$ $P(X = x) = \frac{{}^4C_x \times {}^{48}C_{2-x}}{{}^{52}C_2}$

Solution:

The random variable X can take values 0, 1, 2.

$$P(X = 0) = \frac{{}^4C_0 \times {}^{48}C_2}{{}^{52}C_2}$$

$$= \frac{1 \times \frac{48 \times 47}{2}}{1326}$$

For $X = 0$ (No Jacks):

$$= \frac{1128}{1326}$$

$$= \frac{188}{221}$$

$$= \frac{188}{221}$$

$$= \frac{188}{221}$$

$$P(X = 1) = \frac{{}^4C_1 \times {}^{48}C_1}{{}^{52}C_2}$$

$$= \frac{4 \times 48}{1326}$$

For $X = 1$ (One Jack):

$$= \frac{192}{1326}$$

$$= \frac{32}{221}$$

$$= \frac{32}{221}$$

$$= \frac{32}{221}$$

$$P(X = 2) = \frac{{}^4C_2 \times {}^{48}C_0}{{}^{52}C_2}$$

$$= \frac{6 \times 1}{1326}$$

For $X = 2$ (Two Jacks):

$$= \frac{6}{1326}$$

$$= \frac{1}{221}$$

$$= \frac{1}{221}$$

$$= \frac{1}{221}$$

Verification:

$$\frac{188}{221} + \frac{32}{221} + \frac{1}{221} = \frac{221}{221} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2
$P(X)$	188/221	32/221	1/221

Question 10

Four bad oranges are mixed accidentally with 16 good oranges. Find the probability distribution of the number of bad oranges in a draw of two oranges.

TYPE 25b Probability Distribution – Construct PMF from Experiment

Given: Bad oranges = 4, Good oranges = 16. Total oranges = 20. Two oranges are drawn. X = number of bad oranges.

Concept / Formula Used: ${}^{20}C_2 = \frac{20 \times 19}{2 \times 1} = 190$

Solution:

X can take values 0, 1, 2.

$$P(X = 0) = \frac{{}^4C_0 \times {}^{16}C_2}{{}^{20}C_2}$$

For $X = 0$ (No bad oranges, 2 good oranges):

$$\begin{aligned} &= \frac{1 \times \frac{16 \times 15}{2}}{190} \\ &= \frac{120}{190} \\ &= \frac{12}{19} \end{aligned}$$

$$P(X = 1) = \frac{{}^4C_1 \times {}^{16}C_1}{{}^{20}C_2}$$

For $X = 1$ (1 bad orange, 1 good orange):

$$\begin{aligned} &= \frac{4 \times 16}{190} \\ &= \frac{64}{190} \\ &= \frac{32}{95} \end{aligned}$$

$$P(X = 2) = \frac{{}^4C_2 \times {}^{16}C_0}{{}^{20}C_2}$$

For $X = 2$ (2 bad oranges, 0 good oranges):

$$\begin{aligned} &= \frac{6 \times 1}{190} \\ &= \frac{6}{190} \\ &= \frac{3}{95} \end{aligned}$$

Sum check:

$$\frac{120}{190} + \frac{64}{190} + \frac{6}{190} = \frac{190}{190} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2
$P(X)$	12/19	32/95	3/95

Question 11

Out of a group of 60 architects, 40 are qualified and co-operative while the remaining are qualified but remain reserved. Two architects are selected from the group at random. Find the probability distribution of the number of architects who are qualified

and co-operative.

TYPE 25a Probability Distribution – Hypergeometric (Sampling Without Replacement)

Given: Total architects = 60. Qualified and co-operative = 40. Qualified and reserved = $60 - 40 = 20$. Draw size = 2. X = number of qualified and co-operative architects.

Concept / Formula Used: ${}^{60}C_2 = \frac{60 \times 59}{2 \times 1} = 1770$

Solution:

X takes values 0, 1, 2.

$$\begin{aligned}
 P(X = 0) &= \frac{{}^{40}C_0 \times {}^{20}C_2}{{}^{60}C_2} \\
 &= \frac{1 \times \frac{20 \times 19}{2}}{1770} \\
 &= \frac{190}{1770} \\
 &= \frac{19}{177}
 \end{aligned}$$

For $X = 0$ (0 co-operative, 2 reserved):

$$\begin{aligned}
 P(X = 1) &= \frac{{}^{40}C_1 \times {}^{20}C_1}{{}^{60}C_2} \\
 &= \frac{40 \times 20}{1770} \\
 &= \frac{800}{1770} \\
 &= \frac{80}{177}
 \end{aligned}$$

For $X = 1$ (1 co-operative, 1 reserved):

$$\begin{aligned}
 P(X = 2) &= \frac{{}^{40}C_2 \times {}^{20}C_0}{{}^{60}C_2} \\
 &= \frac{\frac{40 \times 39}{2} \times 1}{1770} \\
 &= \frac{780}{1770} \\
 &= \frac{78}{177} = \frac{26}{59}
 \end{aligned}$$

For $X = 2$ (2 co-operative, 0 reserved):

Verification:

$$\frac{19}{177} + \frac{80}{177} + \frac{78}{177} = \frac{177}{177} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2
$P(X)$	$19/177$	$80/177$	$26/59$

Question 12

Three balls are drawn one by one without replacement from a bag containing 5 white and 4 red balls. Find the probability distribution of the number of white balls drawn.

TYPE 25a Probability Distribution – Hypergeometric (Sampling Without Replacement)

Given: Bag has 5 white and 4 red balls. Total = 9. Draw size = 3 without replacement. Let X = number of white balls drawn.

Concept / Formula Used: ${}^9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$

Solution:

The random variable X can take values 0, 1, 2, 3.

$$P(X = 0) = \frac{{}^5C_0 \times {}^4C_3}{{}^9C_3}$$

$$\begin{aligned} \text{For } X = 0 \text{ (0 white, 3 red balls):} &= \frac{1 \times 4}{84} \\ &= \frac{4}{84} = \frac{1}{21} \end{aligned}$$

$$P(X = 1) = \frac{{}^5C_1 \times {}^4C_2}{{}^9C_3}$$

$$\begin{aligned} \text{For } X = 1 \text{ (1 white, 2 red balls):} &= \frac{5 \times 6}{84} \\ &= \frac{30}{84} = \frac{5}{14} \end{aligned}$$

$$P(X = 2) = \frac{{}^5C_2 \times {}^4C_1}{{}^9C_3}$$

$$\begin{aligned} \text{For } X = 2 \text{ (2 white, 1 red ball):} &= \frac{10 \times 4}{84} \\ &= \frac{40}{84} = \frac{10}{21} \end{aligned}$$

$$P(X = 3) = \frac{{}^5C_3 \times {}^4C_0}{{}^9C_3}$$

$$\begin{aligned} \text{For } X = 3 \text{ (3 white, 0 red balls):} &= \frac{10 \times 1}{84} \\ &= \frac{10}{84} = \frac{5}{42} \end{aligned}$$

Verification:

$$\frac{4}{84} + \frac{30}{84} + \frac{40}{84} + \frac{10}{84} = \frac{84}{84} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2	3
$P(X)$	1/21	5/14	10/21	5/42

Question 13

A box contains 12 bulbs of which 3 are defective. A sample of 3 bulbs is selected from the box. Let X denote the number of defective bulbs in the sample. Find the probability distribution of X .

TYPE 25a Probability Distribution – Hypergeometric (Sampling Without Replacement)

Given: Total bulbs = 12. Defective bulbs = 3. Non-defective bulbs = 9. Sample size = 3. X = number of defective bulbs.

Concept / Formula Used: ${}^{12}C_3 = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$

Solution:

The possible values for X are 0, 1, 2, 3.

$$\begin{aligned}
 P(X = 0) &= \frac{{}^3C_0 \times {}^9C_3}{{}^{12}C_3} \\
 &= \frac{1 \times \frac{9 \times 8 \times 7}{6}}{220} \\
 &= \frac{84}{220} \\
 &= \frac{21}{55}
 \end{aligned}$$

For $X = 0$ (0 defective, 3 non-defective):

$$\begin{aligned}
 P(X = 1) &= \frac{{}^3C_1 \times {}^9C_2}{{}^{12}C_3} \\
 &= \frac{3 \times 36}{220} \\
 &= \frac{108}{220} \\
 &= \frac{27}{55}
 \end{aligned}$$

For $X = 1$ (1 defective, 2 non-defective):

$$\begin{aligned}
 P(X = 2) &= \frac{{}^3C_2 \times {}^9C_1}{{}^{12}C_3} \\
 &= \frac{3 \times 9}{220} \\
 &= \frac{27}{220}
 \end{aligned}$$

For $X = 2$ (2 defective, 1 non-defective):

$$P(X = 3) = \frac{{}^3C_3 \times {}^9C_0}{{}^{12}C_3}$$

For $X = 3$ (3 defective, 0 non-defective):

$$= \frac{1 \times 1}{220}$$

$$= \frac{1}{220}$$

Verification:

$$\frac{84}{220} + \frac{108}{220} + \frac{27}{220} + \frac{1}{220} = \frac{220}{220} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2	3
$P(X)$	21/55	27/55	27/220	1/220

Question 14

Four bad eggs are mixed with 10 good ones. If 3 eggs are drawn one by one without replacement, then find the probability distribution of the number of bad eggs drawn.

TYPE 25b Probability Distribution – Construct PMF from Experiment

Given: Bad eggs = 4, Good eggs = 10. Total eggs = 14. Draw size = 3 without replacement. Let X = number of bad eggs drawn.

Concept / Formula Used: ${}^{14}C_3 = \frac{14 \times 13 \times 12}{3 \times 2 \times 1} = 364$

Solution:

X can take values 0, 1, 2, 3.

$$P(X = 0) = \frac{{}^4C_0 \times {}^{10}C_3}{{}^{14}C_3}$$

For $X = 0$ (0 bad, 3 good eggs):

$$= \frac{1 \times \frac{10 \times 9 \times 8}{6}}{364}$$

$$= \frac{120}{364}$$

$$= \frac{30}{91}$$

$$P(X = 1) = \frac{{}^4C_1 \times {}^{10}C_2}{{}^{14}C_3}$$

For $X = 1$ (1 bad, 2 good eggs):

$$= \frac{4 \times 45}{364}$$

$$= \frac{180}{364}$$

$$= \frac{45}{91}$$

$$\begin{aligned}
 P(X = 2) &= \frac{{}^4C_2 \times {}^{10}C_1}{{}^{14}C_3} \\
 &= \frac{6 \times 10}{364} \\
 &= \frac{60}{364} \\
 &= \frac{15}{91}
 \end{aligned}$$

For $X = 2$ (2 bad, 1 good egg):

$$\begin{aligned}
 P(X = 3) &= \frac{{}^4C_3 \times {}^{10}C_0}{{}^{14}C_3} \\
 &= \frac{4 \times 1}{364} \\
 &= \frac{4}{364} \\
 &= \frac{1}{91}
 \end{aligned}$$

For $X = 3$ (3 bad, 0 good eggs):

Verification:

$$\frac{30}{91} + \frac{45}{91} + \frac{15}{91} + \frac{1}{91} = \frac{91}{91} = 1$$

Final Answer

The probability distribution of X is:

X	0	1	2	3
$P(X)$	30/91	45/91	15/91	1/91

Question 15

A pair of dice is thrown simultaneously. If X denotes the absolute difference of numbers obtained on the pair of dice, then find the probability distribution of X .

TYPE 25b Probability Distribution – Construct PMF from Experiment

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

Matrix of Absolute Differences for Pair of Dice

Given: Experiment of rolling two dice. Random variable $X = |d_1 - d_2|$. Total outcomes = 36.

Concept / Formula Used: $X = |d_1 - d_2|$ $d_1, d_2 \in \{1, 2, 3, 4, 5, 6\}$

Solution:

Total number of possible outcomes in sample space $S = 6 \times 6 = 36$.

Let us determine the favorable outcomes for each value of X :

1. For $X = 0$ (identical numbers):

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

Number of outcomes = 6.

$$P(X = 0) = \frac{6}{36}$$

2. For $X = 1$ (difference of 1):

$$\{(1, 2), (2, 1), (2, 3), (3, 2), (3, 4), (4, 3), (4, 5), (5, 4), (5, 6), (6, 5)\}$$

Number of outcomes = 10.

$$P(X = 1) = \frac{10}{36}$$

3. For $X = 2$ (difference of 2):

$$\{(1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4)\}$$

Number of outcomes = 8.

$$P(X = 2) = \frac{8}{36}$$

4. For $X = 3$ (difference of 3):

$$\{(1, 4), (4, 1), (2, 5), (5, 2), (3, 6), (6, 3)\}$$

Number of outcomes = 6.

$$P(X = 3) = \frac{6}{36}$$

5. For $X = 4$ (difference of 4):

$$\{(1, 5), (5, 1), (2, 6), (6, 2)\}$$

Number of outcomes = 4.

$$P(X = 4) = \frac{4}{36}$$

6. For $X = 5$ (difference of 5):

$$\{(1, 6), (6, 1)\}$$

Number of outcomes = 2.

$$P(X = 5) = \frac{2}{36}$$

Verification of total probability:

$$\frac{6}{36} + \frac{10}{36} + \frac{8}{36} + \frac{6}{36} + \frac{4}{36} + \frac{2}{36} = \frac{36}{36} = 1$$

$$P(X = 0) = \frac{1}{6}$$

$$P(X = 1) = \frac{5}{18}$$

$$P(X = 2) = \frac{2}{9}$$

Simplifying each fraction:

$$P(X = 3) = \frac{1}{6}$$

$$P(X = 4) = \frac{1}{9}$$

$$P(X = 5) = \frac{1}{18}$$

Final Answer

The probability distribution of X is:

X	0	1	2	3	4	5
$P(X)$	1/6	5/18	2/9	1/6	1/9	1/18

Question 16

A group of class XI students designed a game for class XII students to be played at the farewell programme. They kept two sets of cards on the stage. Each set contained four cards numbered 1, 2, 3 and 4. Each student of class XII had to draw one card from each set simultaneously.

The score a student received was defined as the difference between the numbers shown on the cards. The possible outcome of the score is given in the table below:

		Numbers on card 1			
		1	2	3	4
Numbers on card 2	1	0	p	2	3
	2	1	0	p	2
	3	q	1	0	p
	4	3	q	1	0

- Calculate the values of p and q in the above table.
- Reshma plays the game once. Let the random variable ' S ' represents Reshma's score. Construct the probability distribution for ' S '.
- Find the probability that Reshma's score is at most 2.
- Find Reshma's expected score.

TYPE 25 Probability Distribution – Verify/Determine PMF (Sums to 1)

★ Likely Exam Question

	1	2	3	4	Card 1 (c_1)
1	0	1	2	3	
2	1	0	1	2	
3	2	1	0	1	
4	3	2	1	0	
Card 2 (c_2)					

Grid of outcome scores $S = |c_1 - c_2|$ for drawing two cards

Given: Two sets of cards numbered 1, 2, 3, 4. Score $S = |c_1 - c_2|$, where $c_1, c_2 \in \{1, 2, 3, 4\}$ are the drawn numbers.

Concept / Formula Used: $E(S) = \sum s \cdot P(S = s)$ $\sum P(S = s) = 1$

Solution:

- (i) Let c_1 be the number drawn from card set 1 and c_2 be the number drawn from card set 2.

The score S is defined as the absolute difference between the numbers shown on the cards:

$$S = |c_1 - c_2|$$

$$\begin{array}{ll}
 p = \text{Score when } c_1 = 2 \text{ and } c_2 = 1 & q = \text{Score when } c_1 = 1 \text{ and } c_2 = 3 \\
 \text{From the given table: } & = |1 - 3| \\
 = |2 - 1| & = |-2| \\
 = 1 & = 2
 \end{array}$$

Thus, $p = 1$ and $q = 2$.

- (ii) Total number of outcomes when one card is drawn from each set simultaneously:

$$n(S) = 4 \times 4 = 16$$

The possible values for the score S are 0, 1, 2, 3.

Counting the occurrences for each score:

- For $S = 0$: outcomes are (1, 1), (2, 2), (3, 3), (4, 4) $\Rightarrow$ 4 outcomes.

$$P(S = 0) = \frac{4}{16}$$

- For $S = 1$: outcomes are (1, 2), (2, 1), (2, 3), (3, 2), (3, 4), (4, 3) $\Rightarrow$ 6 outcomes.

$$P(S = 1) = \frac{6}{16}$$

- For $S = 2$: outcomes are $(1, 3), (3, 1), (2, 4), (4, 2) \implies 4$ outcomes.

$$P(S = 2) = \frac{4}{16}$$

- For $S = 3$: outcomes are $(1, 4), (4, 1) \implies 2$ outcomes.

$$P(S = 3) = \frac{2}{16}$$

The probability distribution for S is:

s	0	1	2	3
$P(S = s)$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{2}{16}$

- (iii) The probability that Reshma's score is at most 2 is given by $P(S \leq 2)$:

$$\begin{aligned}
 P(S \leq 2) &= P(S = 0) + P(S = 1) + P(S = 2) \\
 &= \frac{4}{16} + \frac{6}{16} + \frac{4}{16} \\
 &= \frac{4 + 6 + 4}{16} \\
 &= \frac{14}{16} \\
 &= \frac{7}{8}
 \end{aligned}$$

- (iv) Using the formula for expectation $E(S) = \sum s \cdot P(S = s)$:

$$\begin{aligned}
 E(S) &= 0 \cdot P(S = 0) + 1 \cdot P(S = 1) + 2 \cdot P(S = 2) + 3 \cdot P(S = 3) \\
 &= 0 \cdot \left(\frac{4}{16}\right) + 1 \cdot \left(\frac{6}{16}\right) + 2 \cdot \left(\frac{4}{16}\right) + 3 \cdot \left(\frac{2}{16}\right) \\
 &= 0 + \frac{6}{16} + \frac{8}{16} + \frac{6}{16} \\
 &= \frac{6 + 8 + 6}{16} \\
 &= \frac{20}{16} \\
 &= \frac{5}{4} \\
 &= 1.25
 \end{aligned}$$

Final Answer

(i) $p = 1, q = 2$

(ii) Probability Distribution:

s	0	1	2	3
$P(S = s)$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{2}{16}$

(iii) $P(S \leq 2) = \frac{7}{8}$

(iv) Expected score $E(S) = 1.25$

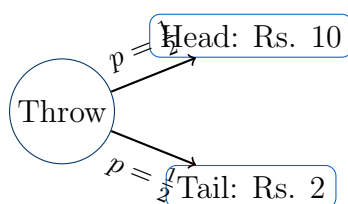
Common Mistake: When forming the pairs for score calculation, remember that order matters because cards are drawn from two separate sets; for instance, $(1, 2)$ and $(2, 1)$ are two distinct outcomes giving score 1.

13.7 – Mean and Variance of a Random Variable

Question 1

A boy throws a coin. He is to receive two rupees for getting a tail and ten rupees for getting a head. Find his expectation (mean receipt).

TYPE 26 Expectation/Mean – Direct Calculation from PMF



Outcome Tree for Single Coin Throw

Given: Random variable X denotes the receipt in rupees: $X = 10$ for Head, $X = 2$ for Tail. Probabilities are $P(H) = \frac{1}{2}$ and $T(T) = \frac{1}{2}$.

Concept / Formula Used: Expected Value $E(X) = \sum x_i \cdot P(x_i)$

Solution:

Let X be the random variable representing the amount received by the boy in rupees.

The possible outcomes of tossing a fair coin and their corresponding probabilities are:

$$P(X = 10) = P(\text{Head}) = \frac{1}{2}$$

$$P(X = 2) = P(\text{Tail}) = \frac{1}{2}$$

The probability distribution table for X is given by:

x_i	10	2
$P(x_i)$	$\frac{1}{2}$	$\frac{1}{2}$

$$\begin{aligned}
 E(X) &= \sum x_i \cdot P(x_i) \\
 &= 10 \cdot P(X = 10) + 2 \cdot P(X = 2) \\
 &= 10 \cdot \left(\frac{1}{2}\right) + 2 \cdot \left(\frac{1}{2}\right) \\
 &= 5 + 1 \\
 &= 6
 \end{aligned}$$

Using the formula for expected value:

Final Answer

The expected receipt of the boy is Rs. 6.

Question 2

A game at a school fair costs Rs. 10, with a prize of Rs. 50. If the probability of winning is $\frac{1}{10}$, is it a fair game?

TYPE 26a Expectation – Fair Game Analysis (Expected Gain = 0)

Given: Cost of playing = Rs. 10, Prize amount = Rs. 50, $P(\text{Winning}) = \frac{1}{10}$, $P(\text{Losing}) = 1 - \frac{1}{10} = \frac{9}{10}$.

Concept / Formula Used: $E(X) = 0$

Solution:

Let X denote the net gain of a player in rupees.

If the player wins:

$$\text{Net gain} = 50 - 10 = 40$$

The probability of winning is:

$$P(X = 40) = \frac{1}{10}$$

If the player loses:

$$\text{Net gain} = 0 - 10 = -10$$

The probability of losing is:

$$P(X = -10) = 1 - \frac{1}{10} = \frac{9}{10}$$

$$\begin{aligned} E(X) &= \sum x_i \cdot P(x_i) \\ &= 40 \cdot \left(\frac{1}{10}\right) + (-10) \cdot \left(\frac{9}{10}\right) \\ &= 4 - 9 \\ &= -5 \end{aligned}$$

Since $E(X) = -5 \neq 0$, the expected net gain is negative, meaning the player loses Rs. 5 on average per game.

Final Answer

No, it is not a fair game (the expected net gain is –Rs. 5).

Question 3

In a certain village, families are strictly limited to two children. The probability distribution of the number of children of any given individual is as follows:

Number of children	0	1	2
Probability	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{2}{5}$

Find the mean number of children.

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Random variable $X \in \{0, 1, 2\}$ with probabilities $P(0) = \frac{1}{10}$, $P(1) = \frac{1}{2}$, $P(2) = \frac{2}{5}$.

Concept / Formula Used: Mean $\mu = E(X) = \sum x_i \cdot P(x_i)$

Solution:

Let X be the random variable representing the number of children.

$$\begin{aligned}\sum P(x_i) &= \frac{1}{10} + \frac{1}{2} + \frac{2}{5} \\ &= \frac{1 + 5 + 4}{10}\end{aligned}$$

We check that the sum of probabilities equals 1:

$$= \frac{10}{10} = 1$$

$$\begin{aligned}\mu = E(X) &= \sum x_i \cdot P(x_i) \\ &= 0 \cdot \left(\frac{1}{10}\right) + 1 \cdot \left(\frac{1}{2}\right) + 2 \cdot \left(\frac{2}{5}\right) \\ &= 0 + \frac{1}{2} + \frac{4}{5} \\ &= \frac{5 + 8}{10} \\ &= \frac{13}{10} \\ &= 1.3\end{aligned}$$

Now, calculating the mean number of children μ :

Final Answer

The mean number of children is 1.3.

Question 4

Find the mean for the following probability distribution:

X	0	1	2	3	4
$P(X)$	0.1	0.25	0.3	0.2	0.15

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Values of X : 0, 1, 2, 3, 4 with respective probabilities 0.1, 0.25, 0.3, 0.2, 0.15.

Concept / Formula Used: Mean $\mu = E(X) = \sum x_i \cdot P(X = x_i)$

Solution:

We compute the expected value by summing the products of each value of X and its

$$E(X) = (0 \cdot 0.1) + (1 \cdot 0.25) + (2 \cdot 0.3) + (3 \cdot 0.2) + (4 \cdot 0.15)$$

$$\begin{aligned} \text{probability:} &= 0 + 0.25 + 0.60 + 0.60 + 0.60 \\ &= 2.05 \end{aligned}$$

Final Answer

The mean of the given probability distribution is 2.05.

Question 5

Find the mean μ for the following probability distribution:

X	0	1	2	3
$P(X)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Random variable $X \in \{0, 1, 2, 3\}$ with probabilities $P(0) = \frac{1}{6}$, $P(1) = \frac{1}{2}$, $P(2) = \frac{3}{10}$, $P(3) = \frac{1}{30}$.

Concept / Formula Used: Mean $\mu = \sum x_i \cdot P(x_i)$

Solution:

$$\begin{aligned} \sum P(X) &= \frac{1}{6} + \frac{1}{2} + \frac{3}{10} + \frac{1}{30} \\ &= \frac{5 + 15 + 9 + 1}{30} \end{aligned}$$

We verify the total probability first:

$$= \frac{30}{30} = 1$$

$$\mu = 0 \cdot \left(\frac{1}{6}\right) + 1 \cdot \left(\frac{1}{2}\right) + 2 \cdot \left(\frac{3}{10}\right) + 3 \cdot \left(\frac{1}{30}\right)$$

$$= 0 + \frac{1}{2} + \frac{6}{10} + \frac{3}{30}$$

$$\begin{aligned} \text{Now, computing the mean } \mu: &= \frac{1}{2} + \frac{3}{5} + \frac{1}{10} \\ &= \frac{5 + 6 + 1}{10} \end{aligned}$$

$$\begin{aligned} &= \frac{12}{10} \\ &= 1.2 \end{aligned}$$

Final Answer

$$\mu = 1.2 \text{ (or } \frac{6}{5} \text{)}$$

Question 6

A random variable X has the following probability distribution:

x_i	-2	-1	0	1	2	3
p_i	0.1	k	0.2	$2k$	0.3	k

- (i) Find the value of k .
 (ii) Calculate the mean of the distribution.

TYPE 26e Probability Distribution – Unknown Parameter Determination

Given: Random variable X with values $-2, -1, 0, 1, 2, 3$ and probabilities $0.1, k, 0.2, 2k, 0.3, k$.

Concept / Formula Used: $\sum p_i = 1$ Mean $E(X) = \sum x_i p_i$

Solution:

(i) Finding the value of k :

$$\begin{aligned} & 0.1 + k + 0.2 + 2k + 0.3 + \\ & (0.1 + 0.2 + 0.3) + (k + 2k + \\ & 0.6 + \end{aligned}$$

Since the sum of all probabilities in a probability distribution must equal 1:

(ii) Calculating the mean of the distribution:

$$p(-2) = 0.1$$

$$p(-1) = 0.1$$

$$p(0) = 0.2$$

Using $k = 0.1$, the probabilities become:

$$p(1) = 2(0.1) = 0.2$$

$$p(2) = 0.3$$

$$p(3) = 0.1$$

$$E(X) = \sum x_i p_i$$

Now, calculate $E(X)$:

$$\begin{aligned} & = (-2)(0.1) + (-1)(0.1) + (0)(0.2) + (1)(0.2) + (2)(0.3) + (3)(0.1) \\ & = -0.2 - 0.1 + 0 + 0.2 + 0.6 + 0.3 \\ & = 0.8 \end{aligned}$$

Final Answer

- (i) $k = 0.1$
 (ii) Mean $E(X) = 0.8$

Question 7

The probability distribution of a random variable X is given as below:

X	0.5	1	1.5	2
$P(X)$	k	k^2	$2k^2$	k

- Find the value of k .
- Determine the mean of the distribution.

TYPE 26e Probability Distribution – Unknown Parameter Determination

Given: Random variable $X \in \{0.5, 1, 1.5, 2\}$ with probabilities $k, k^2, 2k^2, k$.

Concept / Formula Used: $\sum P(X) = 1$ $E(X) = \sum x_i \cdot P(x_i)$

Solution:

(i) Finding the value of k :

$$\sum P(X) = 1$$

The total probability must equal 1: $k + k^2 + 2k^2 + k = 1$

$$3k^2 + 2k - 1 = 0$$

$$3k^2 + 3k - k - 1 = 0$$

Solving this quadratic equation for k : $3k(k + 1) - 1(k + 1) = 0$

$$(3k - 1)(k + 1) = 0$$

This gives $k = \frac{1}{3}$ or $k = -1$. Since probabilities must be non-negative, $k > 0$, hence:

$$k = \frac{1}{3}$$

(ii) Determining the mean of the distribution:

$$\begin{aligned} E(X) &= \sum x_i \cdot P(x_i) \\ &= 0.5(k) + 1(k^2) + 1.5(2k^2) + 2(k) \\ &= (0.5 + 2)k + (1 + 3)k^2 \\ &= 2.5k + 4k^2 \end{aligned}$$

Substitute $k = \frac{1}{3}$ into the expected value expression:

$$\begin{aligned} &= \frac{5}{2} \left(\frac{1}{3} \right) + 4 \left(\frac{1}{3} \right)^2 \\ &= \frac{5}{6} + \frac{4}{9} \\ &= \frac{15 + 8}{18} \\ &= \frac{23}{18} \end{aligned}$$

Final Answer

(i) $k = \frac{1}{3}$

(ii) Mean $E(X) = \frac{23}{18}$

Question 8

A discrete random variable X has the following probability distribution:

X	0	1	2	3	4	5
$P(X)$	$4C^2$	$3C^2$	$2C^2$	C^2	C	$2C$

(i) Find the value of C .

(ii) Find the mean of the distribution.

TYPE 26e Probability Distribution – Unknown Parameter Determination

Given: Random variable $X \in \{0, 1, 2, 3, 4, 5\}$ with probabilities $4C^2, 3C^2, 2C^2, C^2, C, 2C$.

Concept / Formula Used: $\sum P(X) = 1$ $E(X) = \sum x_i P(x_i)$

Solution:

(i) Finding the value of C :

$$\text{Using the condition } \sum P(X) = 1: \quad \begin{aligned} 4C^2 + 3C^2 + 2C^2 + C^2 + C + 2C &= 1 \\ 10C^2 + 3C - 1 &= 0 \end{aligned}$$

$$10C^2 + 5C - 2C - 1 = 0$$

$$\text{Factoring the quadratic equation: } 5C(2C + 1) - 1(2C + 1) = 0$$

$$(5C - 1)(2C + 1) = 0$$

Thus, $C = \frac{1}{5}$ or $C = -\frac{1}{2}$. Since probability cannot be negative, $C > 0$, so:

$$C = \frac{1}{5}$$

(ii) Finding the mean of the distribution:

$$\begin{aligned} E(X) &= \sum x_i \cdot P(x_i) \\ &= 0(4C^2) + 1(3C^2) + 2(2C^2) + 3(C^2) + 4(C) + 5(2C) \\ &= 3C^2 + 4C^2 + 3C^2 + 4C + 10C \\ &= 10C^2 + 14C \end{aligned}$$

$$\begin{aligned} E(X) &= 10 \left(\frac{1}{5} \right)^2 + 14 \left(\frac{1}{5} \right) \\ &= 10 \left(\frac{1}{25} \right) + \frac{14}{5} \\ \text{Substituting } C = \frac{1}{5}: \quad &= \frac{2}{5} + \frac{14}{5} \\ &= \frac{16}{5} = 3.2 \end{aligned}$$

Final Answer

(i) $C = \frac{1}{5}$

(ii) Mean $E(X) = \frac{16}{5} = 3.2$

Question 9

The probability distribution of a random variable X is given as below:

$$P(X = x) = \begin{cases} k(x + 1) & , \text{ for } x = 1, 2, 3, 4 \\ 2kx & , \text{ for } x = 5, 6, 7 \\ 0 & , \text{ otherwise} \end{cases}$$

where k is a constant. Calculate: (i) the value of k .

(ii) $E(X)$

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Random variable X defined piecewise for $x \in \{1, 2, 3, 4, 5, 6, 7\}$.

Concept / Formula Used: $\sum_x P(X = x) = 1$ $E(X) = \sum x \cdot P(X = x)$

Solution:

(i) Calculating the value of k :

$$P(1) = k(1 + 1) = 2k$$

$$P(2) = k(2 + 1) = 3k$$

$$P(3) = k(3 + 1) = 4k$$

First, evaluate individual probabilities for each value of x : $P(4) = k(4 + 1) = 5k$

$$P(5) = 2k(5) = 10k$$

$$P(6) = 2k(6) = 12k$$

$$P(7) = 2k(7) = 14k$$

$$\begin{aligned} \text{Sum of all probabilities: } \sum_{x=1}^7 P(x) &= 2k + 3k + 4k + 5k + 10k + 12k + 14k \\ &= 50k \end{aligned}$$

$$\begin{aligned} 50k &= 1 \\ \text{Since } \sum P(x) &= 1: \quad k = \frac{1}{50} \end{aligned}$$

(ii) Calculating $E(X)$:

$$\begin{aligned}
 E(X) &= \sum_{x=1}^7 x \cdot P(x) \\
 &= 1(2k) + 2(3k) + 3(4k) + 4(5k) + 5(10k) + 6(12k) + 7(14k) \\
 &= 2k + 6k + 12k + 20k + 50k + 72k + 98k \\
 &= 260k
 \end{aligned}$$

$$E(X) = 260 \cdot \left(\frac{1}{50}\right)$$

$$\begin{aligned}
 \text{Substitute } k = \frac{1}{50}: \quad &= \frac{26}{5} \\
 &= 5.2
 \end{aligned}$$

Final Answer

- (i) $k = \frac{1}{50}$
 (ii) $E(X) = \frac{26}{5} = 5.2$

Question 10

The probability distribution of a random variable X is given below:

X	1	2	3
$P(X)$	$\frac{k}{2}$	$\frac{k}{3}$	$\frac{k}{6}$

- (i) Find the value of k .
 (ii) Find $P(1 \leq x < 3)$.
 (iii) Find $E(X)$, the mean of X .

TYPE 26e Probability Distribution – Unknown Parameter Determination

Given: $X \in \{1, 2, 3\}$ with probabilities $\frac{k}{2}, \frac{k}{3}, \frac{k}{6}$.

Concept / Formula Used: $\sum P(X) = 1$ $E(X) = \sum x_i P(x_i)$

Solution:

(i) Finding the value of k :

$$\begin{aligned}
 \sum P(X) &= 1 \\
 \frac{k}{2} + \frac{k}{3} + \frac{k}{6} &= 1 \\
 k \left(\frac{3+2+1}{6} \right) &= 1 \\
 k \left(\frac{6}{6} \right) &= 1 \\
 k &= 1
 \end{aligned}$$

(ii) Finding $P(1 \leq x < 3)$:

The values of x satisfying $1 \leq x < 3$ are $x = 1$ and $x = 2$:

$$P(1 \leq x < 3) = P(X = 1) + P(X = 2) = \frac{k}{2} + \frac{k}{3}$$

$$\begin{aligned} P(1 \leq x < 3) &= \frac{1}{2} + \frac{1}{3} \\ &= \frac{3+2}{6} \\ &= \frac{5}{6} \end{aligned}$$

(iii) Finding $E(X)$:

$$\begin{aligned} E(X) &= \sum x_i P(x_i) \\ &= 1 \cdot \left(\frac{k}{2}\right) + 2 \cdot \left(\frac{k}{3}\right) + 3 \cdot \left(\frac{k}{6}\right) \\ &= \frac{k}{2} + \frac{2k}{3} + \frac{k}{2} \\ &= k + \frac{2k}{3} \\ &= \frac{5k}{3} \end{aligned}$$

Substituting $k = 1$:

$$E(X) = \frac{5}{3}$$

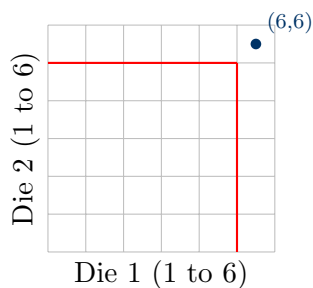
Final Answer

- (i) $k = 1$
- (ii) $P(1 \leq x < 3) = \frac{5}{6}$
- (iii) $E(X) = \frac{5}{3}$

Question 11

Two fair dice are thrown simultaneously. If X denotes the number of sixes, find the mean of X .

TYPE 26c Expectation – Binomial-Type Distribution



Sample Space Grid for Number of Sixes

Given: Two fair dice thrown. X = number of sixes. Single die: $P(\text{six}) = \frac{1}{6}$, $P(\text{not a six}) = \frac{5}{6}$.

Concept / Formula Used: Mean $E(X) = np$

Solution:

Let X denote the number of sixes obtained in a simultaneous throw of two fair dice. The possible values for X are 0, 1, 2.

Probability of getting a six on one die, $p = \frac{1}{6}$. Probability of not getting a six on one die, $q = 1 - \frac{1}{6} = \frac{5}{6}$.

$$P(X = 0) = P(\text{no six}) = q \times q = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$$

Calculating probabilities for each value of X : $P(X = 1) = P(\text{one six}) = 2 \times p \times q = 2 \times \frac{1}{6} \times \frac{5}{6} = \frac{10}{36}$

$$P(X = 2) = P(\text{two sixes}) = p \times p = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$

$$\begin{aligned}
 E(X) &= \sum x_i P(x_i) \\
 &= 0 \cdot \left(\frac{25}{36}\right) + 1 \cdot \left(\frac{10}{36}\right) + 2 \cdot \left(\frac{1}{36}\right) \\
 &= \frac{0 + 10 + 2}{36} \\
 &= \frac{12}{36} \\
 &= \frac{1}{3}
 \end{aligned}$$

Now, calculating mean $E(X)$:

Alternatively, using $E(X) = np$ with $n = 2, p = \frac{1}{6}$:

$$E(X) = 2 \times \frac{1}{6} = \frac{1}{3}$$

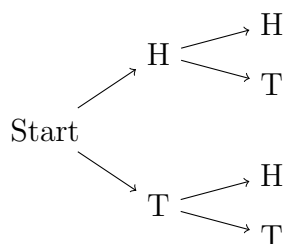
Final Answer

The mean of X is $\frac{1}{3}$.

Question 12

Find the mean of number of heads in three tosses of a fair coin.

TYPE 26c Expectation – Binomial-Type Distribution

*Probability Tree for 3 Coin Tosses*

Given: Three tosses of a fair coin. X = number of heads. $p = P(H) = \frac{1}{2}$, $q = P(T) = \frac{1}{2}$.

Concept / Formula Used: $P(X = r) = \binom{n}{r} p^r q^{n-r}$ $E(X) = np$

Solution:

Let X denote the number of heads in three tosses of a fair coin. The possible values of X are 0, 1, 2, 3.

Total number of outcomes = $2^3 = 8$.

$$P(X = 0) = P(TTT) = \binom{3}{0} \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P(X = 1) = P(HTT, THT, TTH) = \binom{3}{1} \left(\frac{1}{2}\right)^3 = \frac{3}{8}$$

Calculating probabilities for each value of X :

$$P(X = 2) = P(HHT, HTH, THH) = \binom{3}{2} \left(\frac{1}{2}\right)^3 = \frac{3}{8}$$

$$P(X = 3) = P(HHH) = \binom{3}{3} \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$\begin{aligned} E(X) &= \sum x_i P(x_i) \\ &= 0 \cdot \left(\frac{1}{8}\right) + 1 \cdot \left(\frac{3}{8}\right) + 2 \cdot \left(\frac{3}{8}\right) + 3 \cdot \left(\frac{1}{8}\right) \\ &= \frac{0 + 3 + 6 + 3}{8} \\ &= \frac{12}{8} \\ &= \frac{3}{2} = 1.5 \end{aligned}$$

Calculating the mean $E(X)$:

Final Answer

The mean number of heads is 1.5 (or $\frac{3}{2}$).

Question 13

Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the mean of the number of kings.

TYPE 26b Expectation – Hypergeometric Distribution

Given: Pack of 52 cards containing 4 Kings and 48 non-Kings. 2 cards drawn without replacement. X = number of Kings.

Concept / Formula Used: $P(X = r) = \frac{\binom{K}{r} \binom{N-K}{n-r}}{\binom{N}{n}}$

Solution:

Let X denote the number of Kings drawn. The possible values of X are 0, 1, 2.

Total ways to draw 2 cards from 52 cards without replacement:

$$\binom{52}{2} = \frac{52 \times 51}{2} = 1326$$

$$P(X = 0) = \frac{\binom{4}{0} \binom{48}{2}}{\binom{52}{2}} = \frac{1 \times \frac{48 \times 47}{2}}{1326} = \frac{1128}{1326} = \frac{188}{221}$$

Calculating probabilities: $P(X = 1) = \frac{\binom{4}{1} \binom{48}{1}}{\binom{52}{2}} = \frac{4 \times 48}{1326} = \frac{192}{1326} = \frac{32}{221}$

$$P(X = 2) = \frac{\binom{4}{2} \binom{48}{0}}{\binom{52}{2}} = \frac{6 \times 1}{1326} = \frac{6}{1326} = \frac{1}{221}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

$$\begin{aligned} E(X) &= \sum x_i P(x_i) \\ &= 0 \cdot \left(\frac{188}{221}\right) + 1 \cdot \left(\frac{32}{221}\right) + 2 \cdot \left(\frac{1}{221}\right) \end{aligned}$$

Calculating the mean $E(X)$:

$$\begin{aligned} &= \frac{32 + 2}{221} \\ &= \frac{34}{221} \end{aligned}$$

Dividing numerator and denominator by 17:

$$E(X) = \frac{34 \div 17}{221 \div 17} = \frac{2}{13}$$

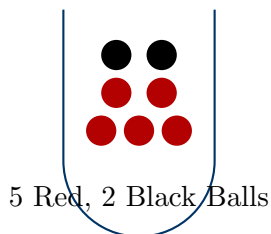
Final Answer

The mean number of kings is $\frac{2}{13}$.

Question 14

An urn contains 5 red and 2 black balls. Two balls are randomly drawn without replacement. Find the probability distribution of the black balls drawn. Also, find the mean of the black balls drawn.

TYPE 26b Expectation – Hypergeometric Distribution



5 Red, 2 Black Balls

Urn Containing Red and Black Balls

Given: Total balls = $5 + 2 = 7$. Number of black balls = 2, red balls = 5. Draw size = 2 without replacement. X = number of black balls drawn.

Concept / Formula Used: $P(X = r) = \frac{\binom{2}{r} \binom{5}{2-r}}{\binom{7}{2}}$ $E(X) = \sum x_i P(x_i)$

Solution:

Let X denote the number of black balls drawn. The possible values of X are 0, 1, 2.

Total ways to draw 2 balls from 7:

$$\binom{7}{2} = \frac{7 \times 6}{2} = 21$$

$$P(X = 0) = \frac{\binom{2}{0} \binom{5}{2}}{\binom{7}{2}} = \frac{1 \times 10}{21} = \frac{10}{21}$$

Calculating probabilities: $P(X = 1) = \frac{\binom{2}{1} \binom{5}{1}}{\binom{7}{2}} = \frac{2 \times 5}{21} = \frac{10}{21}$

$$P(X = 2) = \frac{\binom{2}{2} \binom{5}{0}}{\binom{7}{2}} = \frac{1 \times 1}{21} = \frac{1}{21}$$

The probability distribution table is:

X	0	1	2
$P(X)$	$\frac{10}{21}$	$\frac{10}{21}$	$\frac{1}{21}$

$$\begin{aligned}
 E(X) &= 0 \cdot \left(\frac{10}{21}\right) + 1 \cdot \left(\frac{10}{21}\right) + 2 \cdot \left(\frac{1}{21}\right) \\
 &= \frac{10 + 2}{21} \\
 &= \frac{12}{21} \\
 &= \frac{4}{7}
 \end{aligned}$$

Now, computing the mean $E(X)$:

Final Answer

Probability Distribution: $P(X = 0) = \frac{10}{21}$, $P(X = 1) = \frac{10}{21}$, $P(X = 2) = \frac{1}{21}$

Mean number of black balls = $\frac{4}{7}$

Question 15

3 defective bulbs are mixed with 7 good ones. Find the probability distribution of the number of defective bulbs, if 3 bulbs are drawn at random. What is the average number of defective bulbs drawn?

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Defective bulbs = 3, Good bulbs = 7, Total bulbs = 10. Sample size = 3 drawn at random. X = number of defective bulbs.

Concept / Formula Used: $P(X = r) = \frac{{}^3_r({}^{10}_{3-r})}{{}^{10}_3}$ Average $E(X) = \sum x_i P(x_i)$

Solution:

Let X denote the number of defective bulbs in a sample of 3. Possible values of X are 0, 1, 2, 3.

Total ways to select 3 bulbs from 10:

$${}^{10}_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$

$$P(X = 0) = \frac{{}^3_0({}^7_3)}{120} = \frac{1 \times 35}{120} = \frac{35}{120}$$

$$P(X = 1) = \frac{{}^3_1({}^7_2)}{120} = \frac{3 \times 21}{120} = \frac{63}{120}$$

Calculating probabilities:

$$P(X = 2) = \frac{{}^3_2({}^7_1)}{120} = \frac{3 \times 7}{120} = \frac{21}{120}$$

$$P(X = 3) = \frac{{}^3_3({}^7_0)}{120} = \frac{1 \times 1}{120} = \frac{1}{120}$$

Probability distribution table:

X	0	1	2	3
$P(X)$	$\frac{35}{120}$	$\frac{63}{120}$	$\frac{21}{120}$	$\frac{1}{120}$

$$\begin{aligned}
 E(X) &= 0 \cdot \left(\frac{35}{120}\right) + 1 \cdot \left(\frac{63}{120}\right) + 2 \cdot \left(\frac{21}{120}\right) \\
 &= \frac{63 + 42 + 3}{120} \\
 &= \frac{108}{120} \\
 &= \frac{9}{10} = 0.9
 \end{aligned}$$

Calculating the average (mean) number of defective bulbs:

Final Answer

Probability Distribution: $P(0) = \frac{35}{120}$, $P(1) = \frac{63}{120}$, $P(2) = \frac{21}{120}$, $P(3) = \frac{1}{120}$

Average number of defective bulbs = 0.9

Question 16

2 bad eggs are accidentally mixed with 10 good ones. If three eggs are drawn at random, find the probability distribution of bad eggs drawn. Also find the mean of the distribution.

TYPE 26b Expectation – Hypergeometric Distribution

Given: Bad eggs = 2, Good eggs = 10, Total eggs = 12. Draw size = 3. X = number of bad eggs drawn.

Concept / Formula Used: $P(X = r) = \frac{\binom{2}{r} \binom{10}{3-r}}{\binom{12}{3}}$ $E(X) = \sum x_i P(x_i)$

Solution:

Let X denote the number of bad eggs drawn. Since there are only 2 bad eggs in total, X can take values 0, 1, 2.

Total combinations of choosing 3 eggs out of 12:

$$\binom{12}{3} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$$

$$P(X = 0) = \frac{\binom{2}{0} \binom{10}{3}}{220} = \frac{1 \times 120}{220} = \frac{120}{220} = \frac{6}{11}$$

Calculating probabilities: $P(X = 1) = \frac{\binom{2}{1} \binom{10}{2}}{220} = \frac{2 \times 45}{220} = \frac{90}{220} = \frac{9}{22}$

$$P(X = 2) = \frac{\binom{2}{2} \binom{10}{1}}{220} = \frac{1 \times 10}{220} = \frac{10}{220} = \frac{1}{22}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

$$\begin{aligned}
 E(X) &= 0 \cdot \left(\frac{6}{11}\right) + 1 \cdot \left(\frac{9}{22}\right) + 2 \cdot \left(\frac{1}{22}\right) \\
 &= \frac{9+2}{22} \\
 &= \frac{11}{22} \\
 &= \frac{1}{2} = 0.5
 \end{aligned}$$

Calculating mean $E(X)$:

Final Answer

Probability Distribution: $P(0) = \frac{6}{11}, P(1) = \frac{9}{22}, P(2) = \frac{1}{22}$

Mean = 0.5 (or $\frac{1}{2}$)

Question 17

A box contains 4 red and 5 black marbles. Find the probability distribution of the red marbles in a random draw of three marbles. Also find the mean of the distribution.

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Red marbles = 4, Black marbles = 5, Total marbles = 9. Draw size = 3.
 X = number of red marbles.

Concept / Formula Used: $P(X = r) = \frac{{}^4C_r {}^5C_{3-r}}{{}^9C_3}$ $E(X) = \sum x_i P(x_i)$

Solution:

Let X denote the number of red marbles drawn. The possible values of X are 0, 1, 2, 3.

Total ways to select 3 marbles from 9:

$${}^9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

$$P(X = 0) = \frac{{}^4C_0 {}^5C_3}{84} = \frac{1 \times 10}{84} = \frac{10}{84} = \frac{5}{42}$$

$$P(X = 1) = \frac{{}^4C_1 {}^5C_2}{84} = \frac{4 \times 10}{84} = \frac{40}{84} = \frac{10}{21}$$

Calculating probabilities:

$$P(X = 2) = \frac{{}^4C_2 {}^5C_1}{84} = \frac{6 \times 5}{84} = \frac{30}{84} = \frac{5}{14}$$

$$P(X = 3) = \frac{{}^4C_3 {}^5C_0}{84} = \frac{4 \times 1}{84} = \frac{4}{84} = \frac{1}{21}$$

Probability distribution table:

X	0	1	2	3
$P(X)$	$\frac{10}{84}$	$\frac{40}{84}$	$\frac{30}{84}$	$\frac{4}{84}$

$$E(X) = 0 \cdot \left(\frac{10}{84}\right) + 1 \cdot \left(\frac{40}{84}\right) + 2 \cdot \left(\frac{30}{84}\right) + 3 \cdot \left(\frac{4}{84}\right)$$

Calculating the mean $E(X)$:

$$= \frac{40 + 60 + 12}{84}$$

$$= \frac{112}{84}$$

Dividing numerator and denominator by 28:

$$E(X) = \frac{112 \div 28}{84 \div 28} = \frac{4}{3}$$

Final Answer

Probability Distribution: $P(0) = \frac{10}{84}, P(1) = \frac{40}{84}, P(2) = \frac{30}{84}, P(3) = \frac{4}{84}$

Mean number of red marbles = $\frac{4}{3}$

Question 18

From a lot of 6 items containing 2 defective items, a sample of 4 items is drawn at random (without replacement). If the random variable X denotes the number of defective items in the sample, find: (i) the probability distribution of X .
(ii) the mean of the distribution.

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Total items = 6, Defective = 2, Non-defective = 4. Sample size = 4 without replacement. X = number of defective items.

Concept / Formula Used: $P(X = r) = \frac{{}^2r {}^{4-r}_{4-r}}{{}^6_4}$ $E(X) = \sum x_i P(x_i)$

Solution:

Let X denote the number of defective items in the sample of 4. Since there are 2 defective items in the lot, X can take values 0, 1, 2.

Total ways to choose 4 items from 6:

$${}^6_4 = {}^6_2 = \frac{6 \times 5}{2} = 15$$

$$P(X = 0) = \frac{{}^2_0 {}^4_4}{{}^6_4} = \frac{1 \times 1}{15} = \frac{1}{15}$$

(i) **Probability distribution of X :** $P(X = 1) = \frac{{}^2_1 {}^4_3}{{}^6_4} = \frac{2 \times 4}{15} = \frac{8}{15}$

$$P(X = 2) = \frac{{}^2_2 {}^4_2}{{}^6_4} = \frac{1 \times 6}{15} = \frac{6}{15}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{1}{15}$	$\frac{8}{15}$	$\frac{6}{15}$

$$\begin{aligned}
 E(X) &= 0 \cdot \left(\frac{1}{15}\right) + 1 \cdot \left(\frac{8}{15}\right) + 2 \cdot \left(\frac{6}{15}\right) \\
 &= \frac{8 + 12}{15} \\
 &= \frac{20}{15} \\
 &= \frac{4}{3}
 \end{aligned}$$

(ii) Mean of the distribution:

Final Answer

(i) Probability Distribution: $P(X = 0) = \frac{1}{15}$, $P(X = 1) = \frac{8}{15}$, $P(X = 2) = \frac{6}{15}$

(ii) Mean $E(X) = \frac{4}{3}$

Question 19

Two positive numbers are selected at random (without replacement) from the first five positive integers. Let X denote the larger of the two numbers obtained. Find the mean of the distribution.

TYPE 26d Expectation – Order Statistics (Max of Multiple Draws)

Given: Set of integers $\{1, 2, 3, 4, 5\}$. 2 numbers chosen without replacement. $X = \max(a, b)$.

Concept / Formula Used: Total pairs $\binom{N}{2}$

Solution:

The first five positive integers are $\{1, 2, 3, 4, 5\}$. Total number of ways to choose 2 numbers out of 5 without replacement:

$$\binom{5}{2} = \frac{5 \times 4}{2} = 10$$

Let X denote the larger of the two numbers obtained. The possible values of X are 2, 3, 4, 5.

$X = 2$: Pairs: $(1, 2) \implies 1$ pair

$$P(X = 2) = \frac{1}{10}$$

$X = 3$: Pairs: $(1, 3), (2, 3) \implies 2$ pairs

$$P(X = 3) = \frac{2}{10}$$

Finding favorable pairs for each value of X :

$X = 4$: Pairs: $(1, 4), (2, 4), (3, 4) \implies 3$ pairs

$$P(X = 4) = \frac{3}{10}$$

$X = 5$: Pairs: $(1, 5), (2, 5), (3, 5), (4, 5) \implies 4$ pairs

$$P(X = 5) = \frac{4}{10}$$

Probability distribution table:

X	2	3	4	5
$P(X)$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$

$$\begin{aligned}
 E(X) &= \sum x_i P(x_i) \\
 &= 2 \cdot \left(\frac{1}{10}\right) + 3 \cdot \left(\frac{2}{10}\right) + 4 \cdot \left(\frac{3}{10}\right) + 5 \cdot \left(\frac{4}{10}\right) \\
 &= \frac{2 + 6 + 12 + 20}{10} \\
 &= \frac{40}{10} \\
 &= 4
 \end{aligned}$$

Calculating the mean $E(X)$:

Final Answer

The mean of the distribution is 4.

Question 20

Two numbers are selected at random (without replacement) from positive integers 2, 3, 4, 5, 6 and 7. Let X denote the larger of the two numbers obtained. Find the probability distribution of X . Also find the mean of the distribution.

TYPE 26d Expectation – Order Statistics (Max of Multiple Draws)

Given: Set of numbers $\{2, 3, 4, 5, 6, 7\}$. 2 numbers chosen without replacement. $X = \max(a, b)$.

Concept / Formula Used: Total sample space size $S = \binom{n}{2}$

Solution:

The given set of positive integers is $\{2, 3, 4, 5, 6, 7\}$, which contains 6 elements. Total ways

to select 2 numbers without replacement:

$$\binom{6}{2} = \frac{6 \times 5}{2} = 15$$

Let X denote the larger of the two numbers selected. The possible values for X are 3, 4, 5, 6, 7.

$$X = 3 : \text{ Pair: } (2, 3) \Rightarrow 1 \text{ pair} \Rightarrow P(X = 3) = \frac{1}{15}$$

$$X = 4 : \text{ Pairs: } (2, 4), (3, 4) \Rightarrow 2 \text{ pairs} \Rightarrow P(X = 4) = \frac{2}{15}$$

Determining the number of favorable pairs for each $X = x$: $X = 5$: Pairs: (2, 5), (3, 5), (4, 5) $\Rightarrow$ 3 pairs $\Rightarrow P(X = 5) = \frac{3}{15}$

$$X = 6 : \text{ Pairs: } (2, 6), (3, 6), (4, 6), (5, 6) \Rightarrow 4 \text{ pairs} \Rightarrow P(X = 6) = \frac{4}{15}$$

$$X = 7 : \text{ Pairs: } (2, 7), (3, 7), (4, 7), (5, 7), (6, 7) \Rightarrow 5 \text{ pairs} \Rightarrow P(X = 7) = \frac{5}{15}$$

Probability distribution table:

X	3	4	5	6	7
$P(X)$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{4}{15}$	$\frac{5}{15}$

$$\begin{aligned} E(X) &= 3 \cdot \left(\frac{1}{15}\right) + 4 \cdot \left(\frac{2}{15}\right) + 5 \cdot \left(\frac{3}{15}\right) + 6 \cdot \left(\frac{4}{15}\right) + 7 \cdot \left(\frac{5}{15}\right) \\ &= \frac{3 + 8 + 15 + 24 + 35}{15} \end{aligned}$$

Calculating the mean $E(X)$:

$$\begin{aligned} &= \frac{85}{15} \\ &= \frac{17}{3} \end{aligned}$$

Final Answer

Probability Distribution: $P(3) = \frac{1}{15}, P(4) = \frac{2}{15}, P(5) = \frac{3}{15}, P(6) = \frac{4}{15}, P(7) = \frac{5}{15}$

Mean = $\frac{17}{3}$ (or 5.67)

Question 21

Three numbers are selected at random (without replacement) from the first six positive integers. Let X denote the largest of the three numbers obtained. Find the probability distribution of X . Also find the mean of the distribution.

TYPE 26d Expectation – Order Statistics (Max of Multiple Draws)

Given: Set $\{1, 2, 3, 4, 5, 6\}$. 3 numbers selected without replacement. $X = \max(a, b, c)$.

Concept / Formula Used: $\binom{x-1}{2}$ $\binom{N}{3}$

Solution:

The first six positive integers are $\{1, 2, 3, 4, 5, 6\}$. Total number of ways to choose 3 numbers out of 6 without replacement:

$$\binom{6}{3} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$$

Let X denote the largest of the three numbers chosen. Since we need 3 numbers, the smallest possible maximum is 3. Thus, possible values of X are 3, 4, 5, 6.

For $X = x$, the remaining 2 numbers must be chosen from the $(x - 1)$ integers smaller

$$P(X = 3) = \frac{\binom{2}{2}}{\binom{6}{3}} = \frac{1}{20}$$

$$P(X = 4) = \frac{\binom{3}{2}}{\binom{6}{3}} = \frac{3}{20}$$

than x :

$$P(X = 5) = \frac{\binom{4}{2}}{\binom{6}{3}} = \frac{6}{20}$$

$$P(X = 6) = \frac{\binom{5}{2}}{\binom{6}{3}} = \frac{10}{20}$$

Probability distribution table:

X	3	4	5	6
$P(X)$	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{6}{20}$	$\frac{10}{20}$

$$\begin{aligned} E(X) &= 3 \cdot \left(\frac{1}{20}\right) + 4 \cdot \left(\frac{3}{20}\right) + 5 \cdot \left(\frac{6}{20}\right) + 6 \cdot \left(\frac{10}{20}\right) \\ &= \frac{3 + 12 + 30 + 60}{20} \end{aligned}$$

Calculating the mean $E(X)$:

$$\begin{aligned} &= \frac{105}{20} \\ &= \frac{21}{4} = 5.25 \end{aligned}$$

Final Answer

Probability Distribution: $P(3) = \frac{1}{20}$, $P(4) = \frac{3}{20}$, $P(5) = \frac{6}{20}$, $P(6) = \frac{10}{20}$

Mean number = $\frac{21}{4} = 5.25$

Question 22

From a lot of 30 bulbs which include 6 defective bulbs, a sample of 2 bulbs is drawn at random one by one with replacement. Find the probability distribution of the number of defective bulbs and hence, find the mean number of defective bulbs.

TYPE 26c Expectation – Binomial-Type Distribution

Given: Total bulbs = 30, Defective bulbs = 6. Draw 2 bulbs with replacement.
 X = number of defective bulbs.

Concept / Formula Used: $P(\text{Defective}) = p = \frac{6}{30} = \frac{1}{5}$ $q = 1 - p = \frac{4}{5}$
 $P(X = r) = \binom{n}{r} p^r q^{n-r}$

Solution:

Since drawing is done with replacement, trials are independent.

Probability of drawing a defective bulb in a single draw:

$$p = \frac{6}{30} = \frac{1}{5}$$

Probability of drawing a non-defective bulb:

$$q = 1 - \frac{1}{5} = \frac{4}{5}$$

Let X denote the number of defective bulbs in 2 draws. Possible values of X are 0, 1, 2.

$$P(X = 0) = q^2 = \left(\frac{4}{5}\right)^2 = \frac{16}{25}$$

Calculating probabilities: $P(X = 1) = 2pq = 2 \left(\frac{1}{5}\right) \left(\frac{4}{5}\right) = \frac{8}{25}$

$$P(X = 2) = p^2 = \left(\frac{1}{5}\right)^2 = \frac{1}{25}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{16}{25}$	$\frac{8}{25}$	$\frac{1}{25}$

$$\begin{aligned} E(X) &= \sum x_i P(x_i) \\ &= 0 \cdot \left(\frac{16}{25}\right) + 1 \cdot \left(\frac{8}{25}\right) + 2 \cdot \left(\frac{1}{25}\right) \\ &= \frac{8+2}{25} \\ &= \frac{10}{25} = \frac{2}{5} = 0.4 \end{aligned}$$

Calculating the mean number of defective bulbs:

Alternatively, using $E(X) = np$:

$$E(X) = 2 \times \frac{1}{5} = \frac{2}{5} = 0.4$$

Final Answer

Probability Distribution: $P(0) = \frac{16}{25}$, $P(1) = \frac{8}{25}$, $P(2) = \frac{1}{25}$

Mean number of defective bulbs = 0.4 (or $\frac{2}{5}$)

Question 23

A box contains 10 tickets, 2 of which carry a prize of Rs. 80 each, 5 of which carry a prize of Rs. 40 each and remaining 3 carry a prize of Rs. 20 each. If one ticket is drawn at random, find the mean value of the prize.

TYPE 26 Expectation/Mean – Direct Calculation from PMF

Given: Total tickets = 10. Tickets with Rs. 80 prize = 2, Rs. 40 prize = 5, Rs. 20 prize = 3. X = prize value.

Concept / Formula Used: $E(X) = \sum x_i P(X = x_i)$

Solution:

Let X denote the prize value of a ticket drawn at random. The possible values of X are 80, 40, 20.

$$P(X = 80) = \frac{2}{10}$$

The corresponding probabilities are: $P(X = 40) = \frac{5}{10}$

$$P(X = 20) = \frac{3}{10}$$

Probability distribution table:

X (in Rs.)	80	40	20
$P(X)$	$\frac{2}{10}$	$\frac{5}{10}$	$\frac{3}{10}$

$$\begin{aligned}
 E(X) &= \sum x_i P(x_i) \\
 &= 80 \cdot \left(\frac{2}{10}\right) + 40 \cdot \left(\frac{5}{10}\right) + 20 \cdot \left(\frac{3}{10}\right) \\
 &= \frac{160 + 200 + 60}{10} \\
 &= \frac{420}{10} \\
 &= 42
 \end{aligned}$$

Calculating the mean value of the prize $E(X)$:

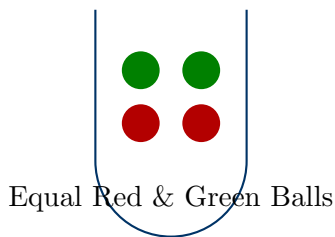
Final Answer

The mean value of the prize is Rs. 42.

Question 24

Two balls are drawn at random one by one with replacement from an urn containing equal number of red balls and green balls. Find the probability distribution of number of red balls. Also, find the mean of the random variable.

TYPE 26c Expectation – Binomial-Type Distribution



Urn with Equal Red and Green Balls

Given: Urn contains equal red and green balls. 2 balls drawn with replacement.
 X = number of red balls.

Concept / Formula Used: $P(\text{Red}) = p = \frac{1}{2}$ $P(\text{Green}) = q = \frac{1}{2}$ $E(X) = np$

Solution:

Since the urn contains an equal number of red and green balls, the probability of drawing a red ball in a single draw is:

$$p = \frac{1}{2}$$

The probability of drawing a green ball is:

$$q = 1 - p = \frac{1}{2}$$

Two balls are drawn with replacement, so trials are independent. Let X denote the number of red balls drawn. Possible values for X are 0, 1, 2.

$$P(X = 0) = q^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Calculating probabilities: $P(X = 1) = 2pq = 2 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{2}{4} = \frac{1}{2}$

$$P(X = 2) = p^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Probability distribution table:

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$E(X) = \sum x_i P(x_i)$$

$$= 0 \cdot \left(\frac{1}{4}\right) + 1 \cdot \left(\frac{1}{2}\right) + 2 \cdot \left(\frac{1}{4}\right)$$

Calculating the mean $E(X)$:

$$= 0 + \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

Final Answer

Probability Distribution: $P(X = 0) = \frac{1}{4}$, $P(X = 1) = \frac{1}{2}$, $P(X = 2) = \frac{1}{4}$

Mean number of red balls = 1

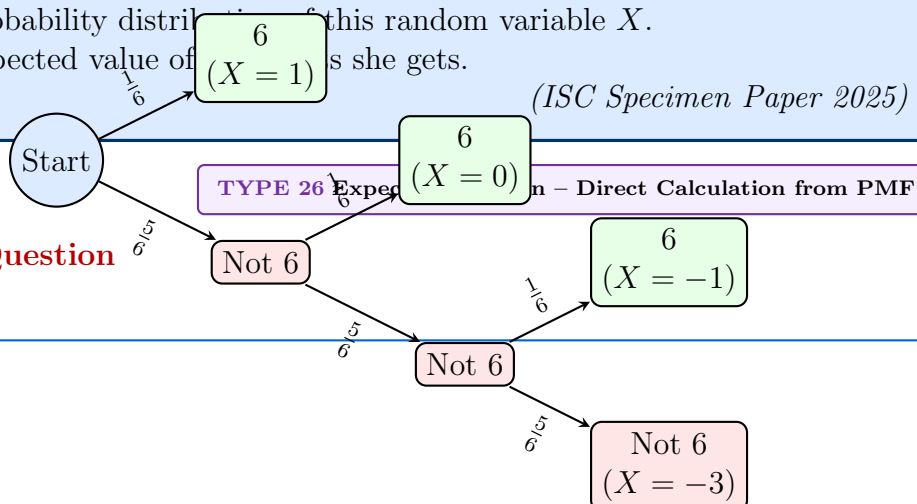
Question 25

Kiran plays a game of throwing a fair die 3 times but to quit as and when she gets a six. Kiran gets +1 point for a six and -1 for any other number.

- If X denotes the random variable “points earned” then what are the possible values X can take?
- Find the probability distribution of this random variable X .
- Find the expected value of X as she gets.

(ISC Specimen Paper 2025)

★ Likely Exam Question



Probability Tree Diagram for Kiran's Game

Given: A fair die is thrown at most 3 times. The game stops immediately when a 6 is obtained. Points awarded: +1 for a 6, -1 for any other number. X is the random variable representing the total points earned.

Concept / Formula Used: $E(X) = \sum_{i=1}^n x_i P(X = x_i)$ $P(X = x_i)$

Solution:

Let the outcome of getting a six on a throw be denoted by S and getting any other number (not six) be denoted by F . Since the die is fair, the probabilities of these outcomes on

any single throw are:

$$P(S) = \frac{1}{6}$$

$$P(F) = \frac{5}{6}$$

Part (i): Possible values of X

The game can terminate in one of the following four mutually exclusive scenarios:

1. **Six is obtained on the first throw (S):** The game stops immediately. Points earned (X) = +1
2. **Non-six on the first throw, and six on the second throw (FS):** The game stops after the second throw. $\text{Points earned } (X) = -1 + 1 = 0$
3. **Non-six on the first two throws, and six on the third throw (FFS):** The game stops after the third throw. $\text{Points earned } (X) = -1 - 1 + 1 = -1$
4. **Non-six on all three throws (FFF):** Since the maximum number of throws is 3, the game stops. $\text{Points earned } (X) = -1 - 1 - 1 = -3$

Thus, the possible values that the random variable X can take are $\{-3, -1, 0, 1\}$.

Part (ii): Probability Distribution of X

Let us calculate the probability corresponding to each value of X :

- For $X = 1$ (Outcome: S): $P(X = 1) = P(S) = \frac{1}{6}$
- For $X = 0$ (Outcome: FS): $P(X = 0) = P(F) \times P(S) = \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$
- For $X = -1$ (Outcome: FFS): $P(X = -1) = P(F) \times P(F) \times P(S) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216}$
- For $X = -3$ (Outcome: FFF): $P(X = -3) = P(F) \times P(F) \times P(F) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \frac{125}{216}$

$$\begin{aligned}
 \sum P(X = x_i) &= P(X = 1) + P(X = 0) + P(X = -1) \\
 &= \frac{1}{6} + \frac{5}{36} + \frac{25}{216} + \frac{125}{216} \\
 &= \frac{36}{216} + \frac{30}{216} + \frac{25}{216} + \frac{125}{216} \\
 &= \frac{36 + 30 + 25 + 125}{216} \\
 &= \frac{216}{216} \\
 &= 1
 \end{aligned}$$

Let us verify that the sum of all probabilities is 1:

The probability distribution table is:

X	-3	-1	0	1
$P(X)$	$\frac{125}{216}$	$\frac{25}{216}$	$\frac{5}{36}$	$\frac{1}{6}$

Part (iii): Expected Value of X

$$\begin{aligned}
 E(X) &= \sum x_i P(X = x_i) \\
 &= (-3) \times P(X = -3) + (-1) \times P(X = -1) + 0 \times P(X = 0) + 1 \times P(X = 1) \\
 &= (-3) \times \frac{125}{216} + (-1) \times \frac{25}{216} + 0 \times \frac{5}{36} + 1 \times \frac{1}{6} \\
 &= -\frac{375}{216} - \frac{25}{216} + 0 + \frac{36}{216} \\
 &= \frac{-375 - 25 + 36}{216} \\
 &= \frac{-400 + 36}{216} \\
 &= -\frac{364}{216} \\
 E(X) &= -\frac{364 \div 4}{216 \div 4} \\
 &= -\frac{91}{54} \\
 &\approx -1.69
 \end{aligned}$$

Using the standard formula for the expected value:

Dividing both the numerator and the denominator by 4:

Final Answer

(i) The possible values X can take are: $\{-3, -1, 0, 1\}$

(ii) The probability distribution of X is:

X	-3	-1	0	1
$P(X)$	$\frac{125}{216}$	$\frac{25}{216}$	$\frac{5}{36}$	$\frac{1}{6}$

(iii) The expected value of the points she gets is:

$$E(X) = -\frac{91}{54} \approx -1.69$$

Common Mistake: A common error is assuming that the game always runs for exactly 3 throws, leading to incorrect sample space calculations. Remember that the game terminates immediately as soon as a 6 is rolled.